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Fourier Analysis: An Introduction

Chapter 1. The Genesis of Fourier Analysis

1. If $z = x + iy$ is a complex number with $x, y \in \mathbb{R}$, we define

$$|z| = (x^2 + y^2)^{1/2}$$

and call this quantity the **modulus** or **absolute value** of z.

(a) What is the geometric interpretation of $|z|$?

(b) Show that if $|z| = 0$, then $z = 0$.

(c) Show that if $\lambda \in \mathbb{R}$, then $|\lambda z| = |\lambda||z|$, where $|\lambda|$ denotes the standard absolute value of a real number.

(d) If z_1 and z_2 are two complex numbers, prove that

$$|z_1 z_2| = |z_1||z_2| \qquad \text{and} \qquad |z_1 + z_2| \leq |z_1| + |z_2|.$$

(e) Show that if $z \neq 0$, then $|1/z| = 1/|z|$.

2. If $z = x + iy$ is a complex number with $x, y \in \mathbb{R}$, we define the **complex conjugate** of z by

$$\overline{z} = x - iy.$$

(a) What is the geometric interpretation of $\overline{z}$?

(b) Show that $|z|^2 = z\overline{z}$.

(c) Prove that if z belongs to the unit circle, then $1/z = \overline{z}$.

3. A sequence of complex numbers $\{w_n\}_{n=1}^{\infty}$ is said to converge if there exists $w \in \mathbb{C}$ such that

$$\lim_{n\to\infty} |w_n - w| = 0,$$

and we say that w is a limit of the sequence.

(a) Show that a converging sequence of complex numbers has a unique limit.

The sequence $\{w_n\}_{n=1}^{\infty}$ is said to be a **Cauchy sequence** if for every $\epsilon > 0$ there exists a positive integer N such that

$$|w_n - w_m| < \epsilon \qquad \text{whenever } n, m > N.$$

(b) Prove that a sequence of complex numbers converges if and only if it is a Cauchy sequence. [Hint: A similar theorem exists for the convergence of a sequence of real numbers. Why does it carry over to sequences of complex numbers?]

A series $\sum_{n=1}^{\infty} z_n$ of complex numbers is said to converge if the sequence formed by the partial sums

$$S_N = \sum_{n=1}^{N} z_n$$

converges. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of non-negative real numbers such that the series $\sum_n a_n$ converges.

(c) Show that if $\{z_n\}_{n=1}^{\infty}$ is a sequence of complex numbers satisfying $|z_n| \leq a_n$ for all n, then the series $\sum_n z_n$ converges. [Hint: Use the Cauchy criterion.]

4. For $z \in \mathbb{C}$, we define the **complex exponential** by

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

(a) Prove that the above definition makes sense, by showing that the series converges for every complex number z. Moreover, show that the convergence is uniform[5] on every bounded subset of $\mathbb{C}$.

(b) If z_1, z_2 are two complex numbers, prove that $e^{z_1} e^{z_2} = e^{z_1+z_2}$. [Hint: Use the binomial theorem to expand $(z_1 + z_2)^n$, as well as the formula for the binomial coefficients.]

(c) Show that if z is purely imaginary, that is, $z = iy$ with $y \in \mathbb{R}$, then

$$e^{iy} = \cos y + i \sin y.$$

This is Euler's identity. [Hint: Use power series.]

(d) More generally,

$$e^{x+iy} = e^x(\cos y + i \sin y)$$

whenever $x, y \in \mathbb{R}$, and show that

$$|e^{x+iy}| = e^x.$$

(e) Prove that $e^z = 1$ if and only if $z = 2\pi k i$ for some integer k.

(f) Show that every complex number $z = x + iy$ can be written in the form

$$z = re^{i\theta},$$

where r is unique and in the range $0 \le r < \infty$, and $\theta \in \mathbb{R}$ is unique up to an integer multiple of 2π. Check that

$$r = |z| \quad \text{and} \quad \theta = \arctan(y/x)$$

whenever these formulas make sense.

(g) In particular, $i = e^{i\pi/2}$. What is the geometric meaning of multiplying a complex number by i? Or by $e^{i\theta}$ for any $\theta \in \mathbb{R}$?

(h) Given $\theta \in \mathbb{R}$, show that

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \quad \text{and} \quad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

These are also called Euler's identities.

(i) Use the complex exponential to derive trigonometric identities such as

$$\cos(\theta + \vartheta) = \cos\theta\cos\vartheta - \sin\theta\sin\vartheta,$$

and then show that

$$\begin{aligned} 2\sin\theta\sin\varphi &= \cos(\theta - \varphi) - \cos(\theta + \varphi), \\ 2\sin\theta\cos\varphi &= \sin(\theta + \varphi) + \sin(\theta - \varphi). \end{aligned}$$

This calculation connects the solution given by d'Alembert in terms of traveling waves and the solution in terms of superposition of standing waves.

[5] A sequence of functions $\{f_n(z)\}_{n=1}^{\infty}$ is said to be uniformly convergent on a set S if there exists a function f on S so that for every $\epsilon > 0$ there is an integer N such that $|f_n(z) - f(z)| < \epsilon$ whenever $n > N$ and $z \in S$.

5. Verify that $f(x) = e^{inx}$ is periodic with period 2π and that

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{inx}\,dx = \begin{cases} 1 & \text{if } n = 0, \\ 0 & \text{if } n \neq 0. \end{cases}$$

Use this fact to prove that if $n, m \geq 1$ we have

$$\frac{1}{\pi}\int_{-\pi}^{\pi} \cos nx\, \cos mx\,dx = \begin{cases} 0 & \text{if } n \neq m, \\ 1 & n = m, \end{cases}$$

and similarly

$$\frac{1}{\pi}\int_{-\pi}^{\pi} \sin nx\, \sin mx\,dx = \begin{cases} 0 & \text{if } n \neq m, \\ 1 & n = m. \end{cases}$$

Finally, show that

$$\int_{-\pi}^{\pi} \sin nx\, \cos mx\,dx = 0 \qquad \text{for any } n, m.$$

[Hint: Calculate $e^{inx}e^{-imx} + e^{inx}e^{imx}$ and $e^{inx}e^{-imx} - e^{inx}e^{imx}$.]

6. Prove that if f is a twice continuously differentiable function on $\mathbb{R}$ which is a solution of the equation

$$f''(t) + c^2 f(t) = 0,$$

then there exist constants a and b such that

$$f(t) = a \cos ct + b \sin ct.$$

This can be done by differentiating the two functions $g(t) = f(t) \cos ct - c^{-1} f'(t) \sin ct$ and $h(t) = f(t) \sin ct + c^{-1} f'(t) \cos ct$.

7. Show that if a and b are real, then one can write

$$a\cos ct + b\sin ct = A\cos(ct - \varphi),$$

where $A = \sqrt{a^2 + b^2}$, and φ is chosen so that

$$\cos\varphi = \frac{a}{\sqrt{a^2 + b^2}} \quad \text{and} \quad \sin\varphi = \frac{b}{\sqrt{a^2 + b^2}}.$$

8. Suppose F is a function on (a, b) with two continuous derivatives. Show that whenever x and $x + h$ belong to (a, b), one may write

$$F(x+h) = F(x) + hF'(x) + \frac{h^2}{2}F''(x) + h^2\varphi(h)\,,$$

where $\varphi(h) \to 0$ as $h \to 0$.

Deduce that

$$\frac{F(x+h) + F(x-h) - 2F(x)}{h^2} \to F''(x) \qquad \text{as } h \to 0.$$

[Hint: This is simply a Taylor expansion. It may be obtained by noting that

$$F(x+h) - F(x) = \int_x^{x+h} F'(y)\,dy,$$

and then writing $F'(y) = F'(x) + (y-x)F''(x) + (y-x)\psi(y-x)$, where $\psi(h) \to 0$ as $h \to 0$.]

9. In the case of the plucked string, use the formula for the Fourier sine coefficients to show that

$$A_m = \frac{2h}{m^2} \frac{\sin mp}{p(\pi - p)}.$$

For what position of p are the second, fourth, ... harmonics missing? For what position of p are the third, sixth, ... harmonics missing?

10. Show that the expression of the Laplacian

$$\triangle = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

is given in polar coordinates by the formula

$$\triangle = \frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}.$$

Also, prove that

$$\left|\frac{\partial u}{\partial x}\right|^2 + \left|\frac{\partial u}{\partial y}\right|^2 = \left|\frac{\partial u}{\partial r}\right|^2 + \frac{1}{r^2}\left|\frac{\partial u}{\partial \theta}\right|^2.$$

11. Show that if $n \in \mathbb{Z}$ the only solutions of the differential equation

$$r^2F''(r) + rF'(r) - n^2F(r) = 0,$$

which are twice differentiable when $r > 0$, are given by linear combinations of r^n and r^{-n} when $n \neq 0$, and 1 and $\log r$ when $n = 0$.

[Hint: If F solves the equation, write $F(r) = g(r)r^n$, find the equation satisfied by g, and conclude that $rg'(r) + 2ng(r) = c$ where c is a constant.]

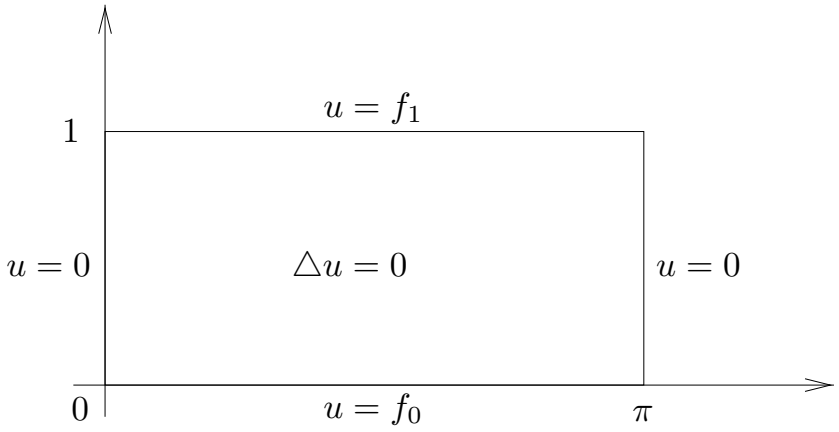


Figure 11. Dirichlet problem in a rectangle

Chapter 2. Basic Properties of Fourier Series

1. Suppose f is 2π-periodic and integrable on any finite interval. Prove that if $a, b \in \mathbb{R}$, then

$$\int_a^b f(x)\,dx = \int_{a+2\pi}^{b+2\pi} f(x)\,dx = \int_{a-2\pi}^{b-2\pi} f(x)\,dx.$$

Also prove that

$$\int_{-\pi}^{\pi} f(x+a)\,dx = \int_{-\pi}^{\pi} f(x)\,dx = \int_{-\pi+a}^{\pi+a} f(x)\,dx.$$

2. In this exercise we show how the symmetries of a function imply certain properties of its Fourier coefficients. Let f be a 2π-periodic Riemann integrable function defined on $\mathbb{R}$.

(a) Show that the Fourier series of the function f can be written as

$$f(\theta) \sim \hat{f}(0) + \sum_{n\geq 1} [\hat{f}(n) + \hat{f}(-n)] \cos n\theta + i[\hat{f}(n) - \hat{f}(-n)] \sin n\theta.$$

(b) Prove that if f is even, then $\hat{f}(n) = \hat{f}(-n)$, and we get a cosine series.

(c) Prove that if f is odd, then $\hat{f}(n) = -\hat{f}(-n)$, and we get a sine series.

(d) Suppose that $f(\theta + \pi) = f(\theta)$ for all $\theta \in \mathbb{R}$. Show that $\hat{f}(n) = 0$ for all odd n.

(e) Show that f is real-valued if and only if $\overline{\hat{f}(n)} = \hat{f}(-n)$ for all n.

3. We return to the problem of the plucked string discussed in Chapter 1. Show that the initial condition f is *equal* to its Fourier sine series

$$f(x) = \sum_{m=1}^{\infty} A_m \sin mx \qquad \text{with} \qquad A_m = \frac{2h}{m^2} \frac{\sin mp}{p(\pi - p)}.$$

[Hint: Note that $|A_m| \leq C/m^2$.]

4. Consider the 2π-periodic odd function defined on $[0, \pi]$ by $f(\theta) = \theta(\pi - \theta)$.

(a) Draw the graph of f.

(b) Compute the Fourier coefficients of f, and show that

$$f(\theta) = \frac{8}{\pi} \sum_{k \text{ odd} \geq 1} \frac{\sin k\theta}{k^3}.$$

5. On the interval $[-\pi, \pi]$ consider the function

$$f(\theta) = \begin{cases} 0 & \text{if } |\theta| > \delta, \\ 1 - |\theta|/\delta & \text{if } |\theta| \leq \delta. \end{cases}$$

Thus the graph of f has the shape of a triangular tent. Show that

$$f(\theta) = \frac{\delta}{2\pi} + 2\sum_{n=1}^{\infty} \frac{1 - \cos n\delta}{n^2 \pi \delta} \cos n\theta.$$

6. Let f be the function defined on $[-\pi, \pi]$ by $f(\theta) = |\theta|$.

(a) Draw the graph of f.

(b) Calculate the Fourier coefficients of f, and show that

$$\hat{f}(n) = \begin{cases} \dfrac{\pi}{2} & \text{if } n = 0, \\ \dfrac{-1 + (-1)^n}{\pi n^2} & \text{if } n \neq 0. \end{cases}$$

(c) What is the Fourier series of f in terms of sines and cosines?

(d) Taking $\theta = 0$, prove that

$$\sum_{n \text{ odd } \geq 1} \frac{1}{n^2} = \frac{\pi^2}{8} \qquad \text{and} \qquad \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

See also Example 2 in Section 1.1.

7. Suppose $\{a_n\}_{n=1}^N$ and $\{b_n\}_{n=1}^N$ are two finite sequences of complex numbers. Let $B_k = \sum_{n=1}^k b_n$ denote the partial sums of the series $\sum b_n$ with the convention $B_0 = 0$.

(a) Prove the **summation by parts** formula

$$\sum_{n=M}^{N} a_n b_n = a_N B_N - a_M B_{M-1} - \sum_{n=M}^{N-1} (a_{n+1} - a_n) B_n.$$

(b) Deduce from this formula Dirichlet's test for convergence of a series: if the partial sums of the series $\sum b_n$ are bounded, and $\{a_n\}$ is a sequence of real numbers that decreases monotonically to 0, then $\sum a_n b_n$ converges.

8. Verify that $\frac{1}{2i}\sum_{n\neq 0}\frac{e^{inx}}{n}$ is the Fourier series of the 2π-periodic **sawtooth** function illustrated in Figure 6, defined by $f(0)=0$, and

$$f(x)=\begin{cases} -\frac{\pi}{2}-\frac{x}{2} & \text{if } -\pi<x<0,\\ \\ \frac{\pi}{2}-\frac{x}{2} & \text{if } 0<x<\pi. \end{cases}$$

Note that this function is not continuous. Show that nevertheless, the series converges for every x (by which we mean, as usual, that the symmetric partial sums of the series converge). In particular, the value of the series at the origin, namely 0, is the average of the values of $f(x)$ as x approaches the origin from the left and the right.

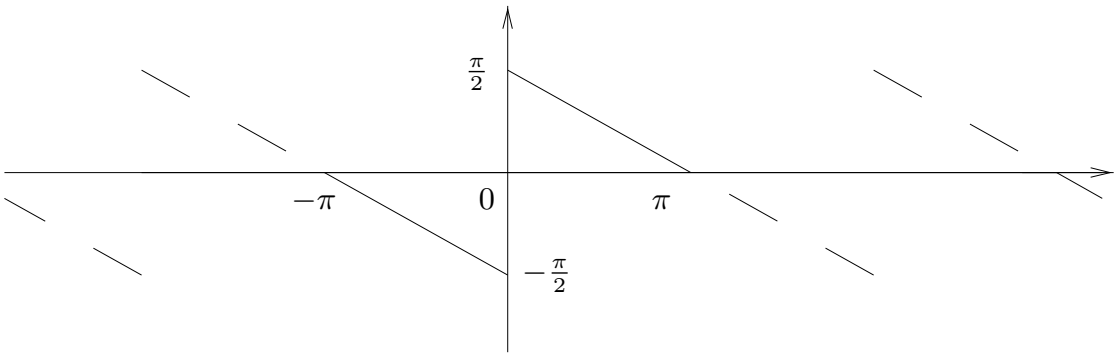


Figure 6. The sawtooth function

[Hint: Use Dirichlet's test for convergence of a series $\sum a_n b_n$.]

9. Let $f(x) = \chi_{[a,b]}(x)$ be the characteristic function of the interval $[a, b] \subset [-\pi, \pi]$, that is,

$$\chi_{[a,b]}(x) = \begin{cases} 1 & \text{if } x \in [a, b], \\ 0 & \text{otherwise.} \end{cases}$$

(a) Show that the Fourier series of f is given by

$$f(x) \sim \frac{b-a}{2\pi} + \sum_{n \neq 0} \frac{e^{-ina} - e^{-inb}}{2\pi i n} \, e^{inx}.$$

The sum extends over all positive and negative integers excluding 0.

(b) Show that if $a \neq -\pi$ or $b \neq \pi$ and $a \neq b$, then the Fourier series does not converge absolutely for any x. [Hint: It suffices to prove that for many values of n one has $|\sin n\theta_0| \geq c > 0$ where $\theta_0 = (b-a)/2$.]

(c) However, prove that the Fourier series converges at every point x. What happens if $a = -\pi$ and $b = \pi$?

10. Suppose f is a periodic function of period 2π which belongs to the class C^k. Show that

$$\hat{f}(n) = O(1/|n|^k) \qquad \text{as } |n| \to \infty.$$

This notation means that there exists a constant C such $|\hat{f}(n)| \leq C/|n|^k$. We could also write this as $|n|^k \hat{f}(n) = O(1)$, where $O(1)$ means bounded.

[Hint: Integrate by parts.]

11. Suppose that $\{f_k\}_{k=1}^{\infty}$ is a sequence of Riemann integrable functions on the interval $[0,1]$ such that

$$\int_0^1 |f_k(x) - f(x)|\, dx \to 0 \qquad \text{as } k \to \infty.$$

Show that $\hat{f}_k(n) \to \hat{f}(n)$ uniformly in n as $k \to \infty$.

12. Prove that if a series of complex numbers $\sum c_n$ converges to s, then $\sum c_n$ is Cesàro summable to s.

[Hint: Assume $s_n \to 0$ as $n \to \infty$.]

13. The purpose of this exercise is to prove that Abel summability is stronger than the standard or Cesàro methods of summation.

(a) Show that if the series $\sum_{n=1}^{\infty} c_n$ of complex numbers converges to a finite limit s, then the series is Abel summable to s. [Hint: Why is it enough to prove the theorem when $s = 0$? Assuming $s = 0$, show that if $s_N = c_1 + \cdots + c_N$, then $\sum_{n=1}^{N} c_n r^n = (1-r)\sum_{n=1}^{N} s_n r^n + s_N r^{N+1}$. Let $N \to \infty$ to show that

$$\sum c_n r^n = (1-r)\sum s_n r^n.$$

Finally, prove that the right-hand side converges to 0 as $r \to 1$.]

(b) However, show that there exist series which are Abel summable, but that do not converge. [Hint: Try $c_n = (-1)^n$. What is the Abel limit of $\sum c_n$?]

(c) Argue similarly to prove that if a series $\sum_{n=1}^{\infty} c_n$ is Cesàro summable to σ, then it is Abel summable to σ. [Hint: Note that

$$\sum_{n=1}^{\infty} c_n r^n = (1-r)^2 \sum_{n=1}^{\infty} n\sigma_n r^n,$$

and assume $\sigma = 0$.]

(d) Give an example of a series that is Abel summable but not Cesàro summable. [Hint: Try $c_n = (-1)^{n-1} n$. Note that if $\sum c_n$ is Cesàro summable, then c_n/n tends to 0.]

The results above can be summarized by the following implications about series:

$$\text{convergent} \Longrightarrow \text{Cesàro summable} \Longrightarrow \text{Abel summable},$$

and the fact that none of the arrows can be reversed.

14. This exercise deals with a theorem of Tauber which says that under an additional condition on the coefficients c_n, the above arrows can be reversed.

(a) If $\sum c_n$ is Cesàro summable to σ and $c_n = o(1/n)$ (that is, $nc_n \to 0$), then $\sum c_n$ converges to σ. [Hint: $s_n - \sigma_n = [(n-1)c_n + \cdots + c_2]/n$.]

(b) The above statement holds if we replace Cesàro summable by Abel summable. [Hint: Estimate the difference between $\sum_{n=1}^{N} c_n$ and $\sum_{n=1}^{N} c_n r^n$ where $r = 1 - 1/N$.]

15. Prove that the Fejér kernel is given by

$$F_N(x) = \frac{1}{N} \frac{\sin^2(Nx/2)}{\sin^2(x/2)}.$$

[Hint: Remember that $NF_N(x) = D_0(x) + \cdots + D_{N-1}(x)$ where $D_n(x)$ is the Dirichlet kernel. Therefore, if $\omega = e^{ix}$ we have

$$NF_N(x) = \sum_{n=0}^{N-1} \frac{\omega^{-n} - \omega^{n+1}}{1 - \omega}.]$$

16. The Weierstrass approximation theorem states: Let f be a continuous function on the closed and bounded interval $[a, b] \subset \mathbb{R}$. Then, for any $\epsilon > 0$, there exists a polynomial P such that

$$\sup_{x \in [a,b]} |f(x) - P(x)| < \epsilon.$$

Prove this by applying Corollary 5.4 of Fejér's theorem and using the fact that the exponential function e^{ix} can be approximated by polynomials uniformly on any interval.

17. In Section 5.4 we proved that the Abel means of f converge to f at all points of continuity, that is,

$$\lim_{r\to 1} A_r(f)(\theta) = \lim_{r\to 1}(P_r * f)(\theta) = f(\theta), \qquad \text{with } 0 < r < 1,$$

whenever f is continuous at θ. In this exercise, we will study the behavior of $A_r(f)(\theta)$ at certain points of discontinuity.

An integrable function is said to have a **jump discontinuity** at θ if the two limits

$$\lim_{\substack{h\,\to\,0\\ h\,>\,0}} f(\theta+h) = f(\theta^+) \qquad \text{and} \qquad \lim_{\substack{h\,\to\,0\\ h\,>\,0}} f(\theta-h) = f(\theta^-)$$

exist.

(a) Prove that if f has a jump discontinuity at θ, then

$$\lim_{r\to 1} A_r(f)(\theta) = \frac{f(\theta^+) + f(\theta^-)}{2}, \qquad \text{with } 0 \le r < 1.$$

[Hint: Explain why $\frac{1}{2\pi}\int_{-\pi}^{0} P_r(\theta)\,d\theta = \frac{1}{2\pi}\int_{0}^{\pi} P_r(\theta)\,d\theta = \frac{1}{2}$, then modify the proof given in the text.]

(b) Using a similar argument, show that if f has a jump discontinuity at θ, the Fourier series of f at θ is Cesàro summable to $\frac{f(\theta^+)+f(\theta^-)}{2}$.

18. If $P_r(\theta)$ denotes the Poisson kernel, show that the function

$$u(r,\theta) = \frac{\partial P_r}{\partial \theta},$$

defined for $0 \leq r < 1$ and $\theta \in \mathbb{R}$, satisfies:

(i) $\triangle u = 0$ in the disc.

(ii) $\lim_{r\to 1} u(r,\theta) = 0$ for each θ.

However, u is not identically zero.

19. Solve Laplace's equation $\triangle u = 0$ in the semi infinite strip

$$S = \{(x, y) : 0 < x < 1,\ 0 < y\},$$

subject to the following boundary conditions

$$\begin{cases} u(0, y) = 0 & \text{when } 0 \leq y, \\ u(1, y) = 0 & \text{when } 0 \leq y, \\ u(x, 0) = f(x) & \text{when } 0 \leq x \leq 1 \end{cases}$$

where f is a given function, with of course $f(0) = f(1) = 0$. Write

$$f(x) = \sum_{n=1}^{\infty} a_n \sin(n\pi x)$$

and expand the general solution in terms of the special solutions given by

$$u_n(x, y) = e^{-n\pi y} \sin(n\pi x).$$

Express u as an integral involving f, analogous to the Poisson integral formula (6).

20. Consider the Dirichlet problem in the annulus defined by $\{(r,\theta) : \rho < r < 1\}$, where $0 < \rho < 1$ is the inner radius. The problem is to solve

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0$$

subject to the boundary conditions

$$\begin{cases} u(1,\theta) = f(\theta), \\ u(\rho,\theta) = g(\theta), \end{cases}$$

where f and g are given continuous functions.

Arguing as we have previously for the Dirichlet problem in the disc, we can hope to write

$$u(r,\theta) = \sum c_n(r) e^{in\theta}$$

with $c_n(r) = A_n r^n + B_n r^{-n}$, $n \neq 0$. Set

$$f(\theta) \sim \sum a_n e^{in\theta} \qquad \text{and} \qquad g(\theta) \sim \sum b_n e^{in\theta}.$$

We want $c_n(1) = a_n$ and $c_n(\rho) = b_n$. This leads to the solution

$$\begin{aligned} u(r,\theta) = &\sum_{n\neq 0} \left(\frac{1}{\rho^n - \rho^{-n}}\right) \left[((\rho/r)^n - (r/\rho)^n)\, a_n + (r^n - r^{-n}) b_n\right] e^{in\theta} \\ &+ a_0 + (b_0 - a_0)\frac{\log r}{\log \rho}. \end{aligned}$$

Show that as a result we have

$$u(r,\theta) - (P_r * f)(\theta) \to 0 \qquad \text{as } r \to 1 \text{ uniformly in } \theta,$$

and

$$u(r,\theta) - (P_{\rho/r} * g)(\theta) \to 0 \qquad \text{as } r \to \rho \text{ uniformly in } \theta.$$

Chapter 3. Convergence of Fourier Series

1. Show that the first two examples of inner product spaces, namely $\mathbb{R}^d$ and $\mathbb{C}^d$, are complete.

[Hint: Every Cauchy sequence in $\mathbb{R}$ has a limit.]

2. Prove that the vector space $\ell^2(\mathbb{Z})$ is complete.

[Hint: Suppose $A_k = \{a_{k,n}\}_{n\in\mathbb{Z}}$ with $k = 1, 2, \ldots$ is a Cauchy sequence. Show that for each n, $\{a_{k,n}\}_{k=1}^{\infty}$ is a Cauchy sequence of complex numbers, therefore it converges to a limit, say b_n. By taking partial sums of $\|A_k - A_{k'}\|$ and letting $k' \to \infty$, show that $\|A_k - B\| \to 0$ as $k \to \infty$, where $B = (\ldots, b_{-1}, b_0, b_1, \ldots)$. Finally, prove that $B \in \ell^2(\mathbb{Z})$.]

3. Construct a sequence of integrable functions $\{f_k\}$ on $[0, 2\pi]$ such that

$$\lim_{k\to\infty} \frac{1}{2\pi} \int_0^{2\pi} |f_k(\theta)|^2 \, d\theta = 0$$

but $\lim_{k\to\infty} f_k(\theta)$ fails to exist for any θ.

[Hint: Choose a sequence of intervals $I_k \subset [0, 2\pi]$ whose lengths tend to 0, and so that each point belongs to infinitely many of them; then let $f_k = \chi_{I_k}$.]

4. Recall the vector space $\mathcal{R}$ of integrable functions, with its inner product and norm

$$\|f\| = \left(\frac{1}{2\pi} \int_0^{2\pi} |f(x)|^2 \, dx \right)^{1/2}.$$

(a) Show that there exist non-zero integrable functions f for which $\|f\| = 0$.

(b) However, show that if $f \in \mathcal{R}$ with $\|f\| = 0$, then $f(x) = 0$ whenever f is continuous at x.

(c) Conversely, show that if $f \in \mathcal{R}$ vanishes at all of its points of continuity, then $\|f\| = 0$.

5. Let

$$f(\theta) = \begin{cases} 0 & \text{for } \theta = 0 \\ \log(1/\theta) & \text{for } 0 < \theta \leq 2\pi, \end{cases}$$

and define a sequence of functions in $\mathcal{R}$ by

$$f_n(\theta) = \begin{cases} 0 & \text{for } 0 \leq \theta \leq 1/n \\ f(\theta) & \text{for } 1/n < \theta \leq 2\pi. \end{cases}$$

Prove that $\{f_n\}_{n=1}^{\infty}$ is a Cauchy sequence in $\mathcal{R}$. However, f does not belong to $\mathcal{R}$.

[Hint: Show that $\int_a^b (\log\theta)^2\, d\theta \to 0$ if $0 < a < b$ and $b \to 0$, by using the fact that the derivative of $\theta(\log\theta)^2 - 2\theta\log\theta + 2\theta$ is equal to $(\log\theta)^2$.]

6. Consider the sequence $\{a_k\}_{k=-\infty}^{\infty}$ defined by

$$a_k = \begin{cases} 1/k & \text{if } k \geq 1 \\ 0 & \text{if } k \leq 0. \end{cases}$$

Note that $\{a_k\} \in \ell^2(\mathbb{Z})$, but that no Riemann integrable function has k^{th} Fourier coefficient equal to a_k for all k.

7. Show that the trigonometric series

$$\sum_{n\geq 2} \frac{1}{\log n} \sin nx$$

converges for every x, yet it is not the Fourier series of a Riemann integrable function.

The same is true for $\sum \frac{\sin nx}{n^\alpha}$ for $0 < \alpha < 1$, but the case $1/2 < \alpha < 1$ is more difficult. See Problem 1.

8. Exercise 6 in Chapter 2 dealt with the sums

$$\sum_{n \text{ odd } \geq 1} \frac{1}{n^2} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

Similar sums can be derived using the methods of this chapter.

(a) Let f be the function defined on $[-\pi, \pi]$ by $f(\theta) = |\theta|$. Use Parseval's identity to find the sums of the following two series:

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^4} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{n^4}.$$

In fact, they are $\pi^4/96$ and $\pi^4/90$, respectively.

(b) Consider the 2π-periodic odd function defined on $[0, \pi]$ by $f(\theta) = \theta(\pi - \theta)$. Show that

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^6} = \frac{\pi^6}{960} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{n^6} = \frac{\pi^6}{945}.$$

Remark. The general expression when k is even for $\sum_{n=1}^{\infty} 1/n^k$ in terms of π^k is given in Problem 4. However, finding a formula for the sum $\sum_{n=1}^{\infty} 1/n^3$, or more generally $\sum_{n=1}^{\infty} 1/n^k$ with k odd, is a famous unresolved question.

9. Show that for α not an integer, the Fourier series of

$$\frac{\pi}{\sin \pi\alpha} e^{i(\pi - x)\alpha}$$

on $[0, 2\pi]$ is given by

$$\sum_{n=-\infty}^{\infty} \frac{e^{inx}}{n+\alpha}.$$

Apply Parseval's formula to show that

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\alpha)^2} = \frac{\pi^2}{(\sin \pi\alpha)^2}.$$

10. Consider the example of a vibrating string which we analyzed in Chapter 1. The displacement $u(x,t)$ of the string at time t satisfies the wave equation

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \qquad c^2 = \tau/\rho.$$

The string is subject to the initial conditions

$$u(x,0) = f(x) \qquad \text{and} \qquad \frac{\partial u}{\partial t}(x,0) = g(x),$$

where we assume that $f \in C^1$ and g is continuous. We define the total **energy** of the string by

$$E(t) = \frac{1}{2}\rho \int_0^L \left(\frac{\partial u}{\partial t}\right)^2 dx + \frac{1}{2}\tau \int_0^L \left(\frac{\partial u}{\partial x}\right)^2 dx.$$

The first term corresponds to the "kinetic energy" of the string (in analogy with $(1/2)mv^2$, the kinetic energy of a particle of mass m and velocity v), and the second term corresponds to its "potential energy."

Show that the total energy of the string is conserved, in the sense that $E(t)$ is constant. Therefore,

$$E(t) = E(0) = \frac{1}{2}\rho \int_0^L g(x)^2\,dx + \frac{1}{2}\tau \int_0^L f'(x)^2\,dx.$$

11. The inequalities of Wirtinger and Poincaré establish a relationship between the norm of a function and that of its derivative.

(a) If f is T-periodic, continuous, and piecewise C^1 with $\int_0^T f(t)\,dt = 0$, show that

$$\int_0^T |f(t)|^2\,dt \le \frac{T^2}{4\pi^2}\int_0^T |f'(t)|^2\,dt,$$

with equality if and only if $f(t) = A\sin(2\pi t/T) + B\cos(2\pi t/T)$. [Hint: Apply Parseval's identity.]

(b) If f is as above and g is just C^1 and T-periodic, prove that

$$\left|\int_0^T \overline{f(t)}g(t)\,dt\right|^2 \le \frac{T^2}{4\pi^2}\int_0^T |f(t)|^2\,dt\int_0^T |g'(t)|^2\,dt.$$

(c) For any compact interval $[a,b]$ and any continuously differentiable function f with $f(a) = f(b) = 0$, show that

$$\int_a^b |f(t)|^2\,dt \le \frac{(b-a)^2}{\pi^2}\int_a^b |f'(t)|^2\,dt.$$

Discuss the case of equality, and prove that the constant $(b-a)^2/\pi^2$ cannot be improved. [Hint: Extend f to be odd with respect to a and periodic of period $T = 2(b-a)$ so that its integral over an interval of length T is 0. Apply part a) to get the inequality, and conclude that equality holds if and only if $f(t) = A\sin(\pi\frac{t-a}{b-a})$].

12. Prove that $\displaystyle\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}$.

[Hint: Start with the fact that the integral of $D_N(\theta)$ equals 2π, and note that the difference $(1/\sin(\theta/2)) - 2/\theta$ is continuous on $[-\pi, \pi]$. Apply the Riemann-Lebesgue lemma.]

13. Suppose that f is periodic and of class C^k. Show that

$$\hat{f}(n) = o(1/|n|^k),$$

that is, $|n|^k \hat{f}(n)$ goes to 0 as $|n| \to \infty$. This is an improvement over Exercise 10 in Chapter 2.

[Hint: Use the Riemann-Lebesgue lemma.]

14. Prove that the Fourier series of a continuously differentiable function f on the circle is absolutely convergent.

[Hint: Use the Cauchy-Schwarz inequality and Parseval's identity for f'.]

15. Let f be 2π-periodic and Riemann integrable on $[-\pi, \pi]$.

(a) Show that

$$\hat{f}(n) = -\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x+\pi/n)e^{-inx}\,dx$$

hence

$$\hat{f}(n) = \frac{1}{4\pi}\int_{-\pi}^{\pi}[f(x)-f(x+\pi/n)]e^{-inx}\,dx.$$

(b) Now assume that f satisfies a Hölder condition of order α, namely

$$|f(x+h)-f(x)| \le C|h|^{\alpha}$$

for some $0 < \alpha \le 1$, some $C > 0$, and all x, h. Use part a) to show that

$$\hat{f}(n) = O(1/|n|^{\alpha}).$$

(c) Prove that the above result cannot be improved by showing that the function

$$f(x) = \sum_{k=0}^{\infty} 2^{-k\alpha}e^{i2^k x},$$

where $0 < \alpha < 1$, satisfies

$$|f(x+h)-f(x)| \le C|h|^{\alpha},$$

and $\hat{f}(N) = 1/N^{\alpha}$ whenever $N = 2^k$.

[Hint: For (c), break up the sum as follows $f(x+h)-f(x) = \sum_{2^k \le 1/|h|} + \sum_{2^k > 1/|h|}$. To estimate the first sum use the fact that $|1-e^{i\theta}| \le |\theta|$ whenever θ is small. To estimate the second sum, use the obvious inequality $|e^{ix}-e^{iy}| \le 2$.]

16. Let f be a 2π-periodic function which satisfies a Lipschitz condition with constant K; that is,

$$|f(x) - f(y)| \leq K|x - y| \qquad \text{for all } x, y.$$

This is simply the Hölder condition with $\alpha = 1$, so by the previous exercise, we see that $\hat{f}(n) = O(1/|n|)$. Since the harmonic series $\sum 1/n$ diverges, we cannot say anything (yet) about the absolute convergence of the Fourier series of f. The outline below actually proves that the Fourier series of f converges absolutely and uniformly.

(a) For every positive h we define $g_h(x) = f(x + h) - f(x - h)$. Prove that

$$\frac{1}{2\pi} \int_0^{2\pi} |g_h(x)|^2 \, dx = \sum_{n=-\infty}^{\infty} 4|\sin nh|^2 |\hat{f}(n)|^2,$$

and show that

$$\sum_{n=-\infty}^{\infty} |\sin nh|^2 |\hat{f}(n)|^2 \leq K^2 h^2.$$

(b) Let p be a positive integer. By choosing $h = \pi/2^{p+1}$, show that

$$\sum_{2^{p-1} < |n| \leq 2^p} |\hat{f}(n)|^2 \leq \frac{K^2 \pi^2}{2^{2p+1}}.$$

(c) Estimate $\sum_{2^{p-1} < |n| \leq 2^p} |\hat{f}(n)|$, and conclude that the Fourier series of f converges absolutely, hence uniformly. [Hint: Use the Cauchy-Schwarz inequality to estimate the sum.]

(d) In fact, modify the argument slightly to prove Bernstein's theorem: If f satisfies a Hölder condition of order $\alpha > 1/2$, then the Fourier series of f converges absolutely.

17. If f is a bounded monotonic function on $[-\pi, \pi]$, then

$$\hat{f}(n) = O(1/|n|).$$

[Hint: One may assume that f is increasing, and say $|f| \leq M$. First check that the Fourier coefficients of the characteristic function of $[a, b]$ satisfy $O(1/|n|)$. Now show that a sum of the form

$$\sum_{k=1}^{N} \alpha_k \chi_{[a_k, a_{k+1}]}(x)$$

with $-\pi = a_1 < a_2 < \cdots < a_N < a_{N+1} = \pi$ and $-M \leq \alpha_1 \leq \cdots \leq \alpha_N \leq M$ has Fourier coefficients that are $O(1/|n|)$ uniformly in N. Summing by parts one gets a telescopic sum $\sum(\alpha_{k+1} - \alpha_k)$ which can be bounded by $2M$. Now approximate f by functions of the above type.]

18. Here are a few things we have learned about the decay of Fourier coefficients:

(a) if f is of class C^k, then $\hat{f}(n) = o(1/|n|^k)$;

(b) if f is Lipschitz, then $\hat{f}(n) = O(1/|n|)$;

(c) if f is monotonic, then $\hat{f}(n) = O(1/|n|)$;

(d) if f is satisfies a Hölder condition with exponent α where $0 < \alpha < 1$, then $\hat{f}(n) = O(1/|n|^\alpha)$;

(e) if f is merely Riemann integrable, then $\sum |\hat{f}(n)|^2 < \infty$ and therefore $\hat{f}(n) = o(1)$.

Nevertheless, show that the Fourier coefficients of a continuous function can tend to 0 arbitrarily slowly by proving that for every sequence of nonnegative real numbers $\{\epsilon_n\}$ converging to 0, there exists a continuous function f such that $|\hat{f}(n)| \geq \epsilon_n$ for infinitely many values of n.

[Hint: Choose a subsequence $\{\epsilon_{n_k}\}$ so that $\sum_k \epsilon_{n_k} < \infty$.]

19. Give another proof that the sum $\sum_{0<|n|\le N} e^{inx}/n$ is uniformly bounded in N and $x \in [-\pi, \pi]$ by using the fact that

$$\frac{1}{2i} \sum_{0<|n|\le N} \frac{e^{inx}}{n} = \sum_{n=1}^{N} \frac{\sin nx}{n} = \frac{1}{2} \int_0^x (D_N(t) - 1)\, dt,$$

where D_N is the Dirichlet kernel. Now use the fact that $\int_0^\infty \frac{\sin t}{t}\, dt < \infty$ which was proved in Exercise 12.

20. Let $f(x)$ denote the sawtooth function defined by $f(x) = (\pi - x)/2$ on the interval $(0, 2\pi)$ with $f(0) = 0$ and extended by periodicity to all of $\mathbb{R}$. The Fourier series of f is

$$f(x) \sim \frac{1}{2i} \sum_{|n|\neq 0} \frac{e^{inx}}{n} = \sum_{n=1}^{\infty} \frac{\sin nx}{n},$$

and f has a jump discontinuity at the origin with

$$f(0^+) = \frac{\pi}{2}, \qquad f(0^-) = -\frac{\pi}{2}, \qquad \text{and hence} \qquad f(0^+) - f(0^-) = \pi.$$

Show that

$$\max_{0<x\leq \pi/N} S_N(f)(x) - \frac{\pi}{2} = \int_0^{\pi} \frac{\sin t}{t}\, dt - \frac{\pi}{2},$$

which is roughly 9% of the jump π. This result is a manifestation of Gibbs's phenomenon which states that near a jump discontinuity, the Fourier series of a function overshoots (or undershoots) it by approximately 9% of the jump.

[Hint: Use the expression for $S_N(f)$ given in Exercise 19.]

Chapter 4. Some Applications of Fourier Series

1. Let $\gamma : [a, b] \to \mathbb{R}^2$ be a parametrization for the closed curve Γ.

(a) Prove that γ is a parametrization by arc-length if and only if the length of the curve from $\gamma(a)$ to $\gamma(s)$ is precisely $s - a$, that is,

$$\int_a^s |\gamma'(t)|\, dt = s - a.$$

(b) Prove that any curve Γ admits a parametrization by arc-length. [Hint: If η is any parametrization, let $h(s) = \int_a^s |\eta'(t)|\, dt$ and consider $\gamma = \eta \circ h^{-1}$.]

2. Suppose $\gamma : [a, b] \to \mathbb{R}^2$ is a parametrization for a closed curve Γ, with $\gamma(t) = (x(t), y(t))$.

(a) Show that

$$\frac{1}{2}\int_a^b (x(s)y'(s) - y(s)x'(s))\,ds = \int_a^b x(s)y'(s)\,ds = -\int_a^b y(s)x'(s)\,ds.$$

(b) Define the **reverse parametrization** of γ by $\gamma^- : [a, b] \to \mathbb{R}^2$ with $\gamma^-(t) = \gamma(b + a - t)$. The image of γ^- is precisely Γ, except that the points $\gamma^-(t)$ and $\gamma(t)$ travel in opposite directions. Thus γ^- "reverses" the orientation of the curve. Prove that

$$\int_\gamma (x\,dy - y\,dx) = -\int_{\gamma^-} (x\,dy - y\,dx).$$

In particular, we may assume (after a possible change in orientation) that

$$\mathcal{A} = \frac{1}{2}\int_a^b (x(s)y'(s) - y(s)x'(s))\,ds = \int_a^b x(s)y'(s)\,ds.$$

3. Suppose Γ is a curve in the plane, and that there exists a set of coordinates x and y so that the x-axis divides the curve into the union of the graph of two continuous functions $y = f(x)$ and $y = g(x)$ for $0 \le x \le 1$, and with $f(x) \ge g(x)$ (see Figure 6). Let Ω denote the region between the graphs of these two functions:

$$\Omega = \{(x, y) : \ 0 \le x \le 1 \ \text{ and } \ g(x) \le y \le f(x)\}.$$

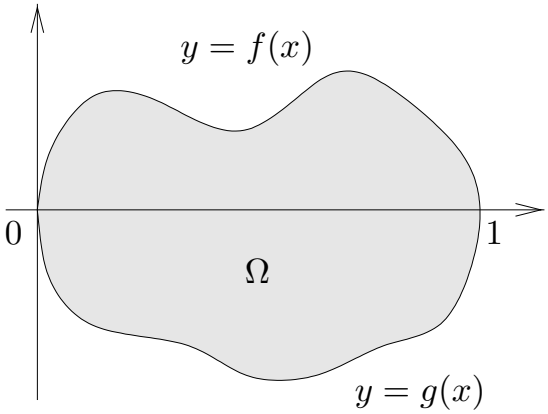


Figure 6. Simple version of the area formula

With the familiar interpretation that the integral $\int h(x)\,dx$ gives the area under the graph of the function h, we see that the area of Ω is $\int_0^1 f(x)\,dx - \int_0^1 g(x)\,dx$. Show that this definition coincides with the area formula $\mathcal{A}$ given in the text, that is,

$$\int_0^1 f(x)\,dx - \int_0^1 g(x)\,dx = \left| -\int_\Gamma y\,dx \right| = \mathcal{A}.$$

Also, note that if the orientation of the curve is chosen so that Ω "lies to the left" of Γ, then the above formula holds without the absolute value signs.

This formula generalizes to any set that can be written as a finite union of domains like Ω above.

4. Observe that with the definition of ℓ and $\mathcal{A}$ given in the text, the isoperimetric inequality continues to hold (with the same proof) even when Γ is not simple.

Show that this stronger version of the isoperimetric inequality is equivalent to Wirtinger's inequality, which says that if f is 2π-periodic, of class C^1, and satisfies $\int_0^{2\pi} f(t)\,dt = 0$, then

$$\int_0^{2\pi} |f(t)|^2\,dt \leq \int_0^{2\pi} |f'(t)|^2\,dt$$

with equality if and only if $f(t) = A\sin t + B\cos t$ (Exercise 11, Chapter 3).

[Hint: In one direction, note that if the length of the curve is 2π and γ is an appropriate arc-length parametrization, then

$$2(\pi - \mathcal{A}) = \int_0^{2\pi} \left[x'(s) + y(s)\right]^2\,ds + \int_0^{2\pi} (y'(s)^2 - y(s)^2)\,ds.$$

A change of coordinates will guarantee $\int_0^{2\pi} y(s)\,ds = 0$. For the other direction, start with a real-valued f satisfying all the hypotheses of Wirtinger's inequality, and construct g, 2π-periodic and so that the term in brackets above vanishes.]

5. Prove that the sequence $\{\gamma_n\}_{n=1}^{\infty}$, where γ_n is the fractional part of

$$\left(\frac{1+\sqrt{5}}{2}\right)^n,$$

is not equidistributed in $[0,1]$.

[Hint: Show that $U_n = \left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{1-\sqrt{5}}{2}\right)^n$ is the solution of the difference equation $U_{r+1} = U_r + U_{r-1}$ with $U_0 = 2$ and $U_1 = 1$. The U_n satisfy the same difference equation as the Fibonacci numbers.]

6. Let $\theta = p/q$ be a rational number where p and q are relatively prime integers (that is, θ is in lowest form). We assume without loss of generality that $q > 0$. Define a sequence of numbers in $[0, 1)$ by $\xi_n = \langle n\theta \rangle$ where $\langle \cdot \rangle$ denotes the fractional part. Show that the sequence $\{\xi_1, \xi_2, \ldots\}$ is equidistributed on the points of the form

$$0,\ 1/q,\ 2/q,\ \ldots,\ (q-1)/q.$$

In fact, prove that for any $0 \leq a < q$, one has

$$\frac{\#\{n:\ 1 \leq n \leq N,\ \langle n\theta \rangle = a/q\}}{N} = \frac{1}{q} + O\left(\frac{1}{N}\right).$$

[Hint: For each integer $k \geq 0$, there exists a unique integer n with $kq \leq n < (k+1)q$ and so that $\langle n\theta \rangle = a/q$. Why can one assume $k = 0$? Prove the existence of n by using the fact[1] that if p and q are relatively prime, there exist integers x, y such that $xp + yq = 1$. Next, divide N by q with remainder, that is, write $N = \ell q + r$ where $0 \leq \ell$ and $0 \leq r < q$. Establish the inequalities

$$\ell \leq \#\{n:\ 1 \leq n \leq N,\ \langle n\theta \rangle = a/q\} \leq \ell + 1.]$$

[1]The elementary results in arithmetic used in this exercise can be found at the beginning of Chapter 8.

7. Prove the second part of Weyl's criterion: if a sequence of numbers $\xi_1, \xi_2, \ldots$ in $[0, 1)$ is equidistributed, then for all $k \in \mathbb{Z} - \{0\}$

$$\frac{1}{N} \sum_{n=1}^{N} e^{2\pi i k \xi_n} \to 0 \qquad \text{as } N \to \infty.$$

[Hint: It suffices to show that $\frac{1}{N} \sum_{n=1}^{N} f(\xi_n) \to \int_0^1 f(x)\, dx$ for all continuous f. Prove this first when f is the characteristic function of an interval.]

8. Show that for any $a \neq 0$, and σ with $0 < \sigma < 1$, the sequence $\langle an^\sigma \rangle$ is equidistributed in $[0, 1)$.

[Hint: Prove that $\sum_{n=1}^{N} e^{2\pi i b n^\sigma} = O(N^\sigma) + O(N^{1-\sigma})$ if $b \neq 0$.] In fact, note the following

$$\sum_{n=1}^{N} e^{2\pi i b n^\sigma} - \int_1^N e^{2\pi i b x^\sigma}\, dx = O\left(\sum_{n=1}^{N} n^{-1+\sigma}\right).$$

9. In contrast with the result in Exercise 8, prove that $\langle a \log n \rangle$ is *not* equidistributed for any a.
[Hint: Compare the sum $\sum_{n=1}^{N} e^{2\pi i b \log n}$ with the corresponding integral.]

10. Suppose that f is a periodic function on $\mathbb{R}$ of period 1, and $\{\xi_n\}$ is a sequence which is equidistributed in $[0,1)$. Prove that:

(a) If f is continuous and satisfies $\int_0^1 f(x)\,dx = 0$, then

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} f(x+\xi_n) = 0 \qquad \text{uniformly in } x.$$

[Hint: Establish this result first for trigonometric polynomials.]

(b) If f is merely integrable on $[0,1]$ and satisfies $\int_0^1 f(x)\,dx = 0$, then

$$\lim_{N\to\infty} \int_0^1 \left| \frac{1}{N} \sum_{n=1}^{N} f(x+\xi_n) \right|^2 dx = 0.$$

11. Show that if $u(x,t) = (f * H_t)(x)$ where H_t is the heat kernel, and f is Riemann integrable, then

$$\int_0^1 |u(x,t) - f(x)|^2 \, dx \to 0 \qquad \text{as } t \to 0.$$

12. A change of variables in (8) leads to the solution

$$u(\theta,\tau) = \sum a_n e^{-n^2\tau} e^{in\theta} = (f * h_\tau)(\theta)$$

of the equation

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial \theta^2} \qquad \text{with } 0 \le \theta \le 2\pi \text{ and } \tau > 0,$$

with boundary condition $u(\theta,0) = f(\theta) \sim \sum a_n e^{in\theta}$. Here $h_\tau(\theta) = \sum_{n=-\infty}^{\infty} e^{-n^2\tau} e^{in\theta}$. This version of the heat kernel on $[0, 2\pi]$ is the analogue of the Poisson kernel, which can be written as $P_r(\theta) = \sum_{n=-\infty}^{\infty} e^{-|n|\tau} e^{in\theta}$ with $r = e^{-\tau}$ (and so $0 < r < 1$ corresponds to $\tau > 0$).

13. The fact that the kernel $H_t(x)$ is a good kernel, hence $u(x,t) \to f(x)$ at each point of continuity of f, is not easy to prove. This will be shown in the next chapter. However, one can prove directly that $H_t(x)$ is "peaked" at $x = 0$ as $t \to 0$ in the following sense:

(a) Show that $\int_{-1/2}^{1/2} |H_t(x)|^2\,dx$ is of the order of magnitude of $t^{-1/2}$ as $t \to 0$.

More precisely, prove that $t^{1/2} \int_{-1/2}^{1/2} |H_t(x)|^2\,dx$ converges to a non-zero limit as $t \to 0$.

(b) Prove that $\int_{-1/2}^{1/2} x^2 |H_t(x)|^2\,dx = O(t^{1/2})$ as $t \to 0$.

[Hint: For (a) compare the sum $\sum_{-\infty}^{\infty} e^{-cn^2 t}$ with the integral $\int_{-\infty}^{\infty} e^{-cx^2 t}\,dx$ where $c > 0$. For (b) use $x^2 \le C(\sin \pi x)^2$ for $-1/2 \le x \le 1/2$, and apply the mean value theorem to $e^{-cx^2 t}$.]

Chapter 5. The Fourier Transform on R

1. Corollary 2.3 in Chapter 2 leads to the following simplified version of the Fourier inversion formula. Suppose f is a continuous function supported on an interval $[-M, M]$, whose Fourier transform $\hat{f}$ is of moderate decrease.

(a) Fix L with $L/2 > M$, and show that $f(x) = \sum a_n(L) e^{2\pi i n x/L}$ where

$$a_n(L) = \frac{1}{L} \int_{-L/2}^{L/2} f(x) e^{-2\pi i n x/L}\, dx = \frac{1}{L} \hat{f}(n/L).$$

Alternatively, we may write $f(x) = \delta \sum_{n=-\infty}^{\infty} \hat{f}(n\delta) e^{2\pi i n \delta x}$ with $\delta = 1/L$.

(b) Prove that if F is continuous and of moderate decrease, then

$$\int_{-\infty}^{\infty} F(\xi)\, d\xi = \lim_{\substack{\delta \to 0 \\ \delta > 0}} \delta \sum_{n=-\infty}^{\infty} F(\delta n).$$

(c) Conclude that $f(x) = \displaystyle\int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi}\, d\xi$.

[Hint: For (a), note that the Fourier series of f on $[-L/2, L/2]$ converges absolutely. For (b), first approximate the integral by $\int_{-N}^{N} F$ and the sum by $\delta \sum_{|n| \le N/\delta} F(n\delta)$. Then approximate the second integral by Riemann sums.]

2. Let f and g be the functions defined by

$$f(x) = \chi_{[-1,1]}(x) = \begin{cases} 1 & \text{if } |x| \leq 1, \\ 0 & \text{otherwise,} \end{cases} \qquad \text{and} \qquad g(x) = \begin{cases} 1 - |x| & \text{if } |x| \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Although f is not continuous, the integral defining its Fourier transform still makes sense. Show that

$$\hat{f}(\xi) = \frac{\sin 2\pi\xi}{\pi\xi} \qquad \text{and} \qquad \hat{g}(\xi) = \left(\frac{\sin \pi\xi}{\pi\xi}\right)^2,$$

with the understanding that $\hat{f}(0) = 2$ and $\hat{g}(0) = 1$.

3. The following exercise illustrates the principle that the decay of $\hat{f}$ is related to the continuity properties of f.

(a) Suppose that f is a function of moderate decrease on $\mathbb{R}$ whose Fourier transform $\hat{f}$ is continuous and satisfies

$$\hat{f}(\xi) = O\left(\frac{1}{|\xi|^{1+\alpha}}\right) \qquad \text{as } |\xi| \to \infty$$

for some $0 < \alpha < 1$. Prove that f satisfies a Hölder condition of order α, that is, that

$$|f(x+h) - f(x)| \leq M|h|^{\alpha} \qquad \text{for some } M > 0 \text{ and all } x, h \in \mathbb{R}.$$

(b) Let f be a continuous function on $\mathbb{R}$ which vanishes for $|x| \geq 1$, with $f(0) = 0$, and which is equal to $1/\log(1/|x|)$ for all x in a neighborhood of the origin. Prove that $\hat{f}$ is not of moderate decrease. In fact, there is no $\epsilon > 0$ so that $\hat{f}(\xi) = O(1/|\xi|^{1+\epsilon})$ as $|\xi| \to \infty$.

[Hint: For part (a), use the Fourier inversion formula to express $f(x+h) - f(x)$ as an integral involving $\hat{f}$, and estimate this integral separately for ξ in the two ranges $|\xi| \leq 1/|h|$ and $|\xi| \geq 1/|h|$.]

4. Bump functions. Examples of compactly supported functions in $\mathcal{S}(\mathbb{R})$ are very handy in many applications in analysis. Some examples are:

(a) Suppose $a < b$, and f is the function such that $f(x) = 0$ if $x \leq a$ or $x \geq b$ and

$$f(x) = e^{-1/(x-a)}e^{-1/(b-x)} \qquad \text{if } a < x < b.$$

Show that f is indefinitely differentiable on $\mathbb{R}$.

(b) Prove that there exists an indefinitely differentiable function F on $\mathbb{R}$ such that $F(x) = 0$ if $x \leq a$, $F(x) = 1$ if $x \geq b$, and F is strictly increasing on $[a, b]$.

(c) Let $\delta > 0$ be so small that $a + \delta < b - \delta$. Show that there exists an indefinitely differentiable function g such that g is 0 if $x \leq a$ or $x \geq b$, g is 1 on $[a + \delta, b - \delta]$, and g is strictly monotonic on $[a, a + \delta]$ and $[b - \delta, b]$.

[Hint: For (b) consider $F(x) = c\int_{-\infty}^{x} f(t)\,dt$ where c is an appropriate constant.]

5. Suppose f is continuous and of moderate decrease.

(a) Prove that $\hat{f}$ is continuous and $\hat{f}(\xi) \to 0$ as $|\xi| \to \infty$.

(b) Show that if $\hat{f}(\xi) = 0$ for all ξ, then f is identically 0.

[Hint: For part (a), show that $\hat{f}(\xi) = \frac{1}{2}\int_{-\infty}^{\infty}[f(x) - f(x - 1/(2\xi))]e^{-2\pi i x\xi}\,dx$. For part (b), verify that the multiplication formula $\int f(x)\hat{g}(x)\,dx = \int \hat{f}(y)g(y)\,dy$ still holds whenever $g \in \mathcal{S}(\mathbb{R})$.]

6. The function $e^{-\pi x^2}$ is its own Fourier transform. Generate other functions that (up to a constant multiple) are their own Fourier transforms. What must the constant multiples be? To decide this, prove that $\mathcal{F}^4 = I$. Here $\mathcal{F}(f) = \hat{f}$ is the Fourier transform, $\mathcal{F}^4 = \mathcal{F} \circ \mathcal{F} \circ \mathcal{F} \circ \mathcal{F}$, and I is the identity operator $(If)(x) = f(x)$ (see also Problem 7).

7. Prove that the convolution of two functions of moderate decrease is a function of moderate decrease.

[Hint: Write

$$\int f(x-y)g(y)\,dy = \int_{|y|\le|x|/2} + \int_{|y|\ge|x|/2}.$$

In the first integral $f(x-y) = O(1/(1+x^2))$ while in the second integral $g(y) = O(1/(1+x^2))$.]

8. Prove that f is continuous, of moderate decrease, and $\int_{-\infty}^{\infty} f(y)e^{-y^2}e^{2xy}dy = 0$ for all $x \in \mathbb{R}$, then $f = 0$.

[Hint: Consider $f * e^{-x^2}$.]

9. If f is of moderate decrease, then

$$\int_{-R}^{R} \left(1 - \frac{|\xi|}{R}\right) \hat{f}(\xi) e^{2\pi i x \xi} \, d\xi = (f * \mathcal{F}_R)(x), \tag{14}$$

where the Fejér kernel on the real line is defined by

$$\mathcal{F}_R(t) = \begin{cases} R\left(\dfrac{\sin \pi t R}{\pi t R}\right)^2 & \text{if } t \neq 0, \\ R & \text{if } t = 0. \end{cases}$$

Show that $\{\mathcal{F}_R\}$ is a family of good kernels as $R \to \infty$, and therefore (14) tends uniformly to $f(x)$ as $R \to \infty$. This is the analogue of Fejér's theorem for Fourier series in the context of the Fourier transform.

10. Below is an outline of a different proof of the Weierstrass approximation theorem.

Define the **Landau** kernels by

$$L_n(x) = \begin{cases} \dfrac{(1-x^2)^n}{c_n} & \text{if } -1 \le x \le 1, \\ 0 & \text{if } |x| \ge 1, \end{cases}$$

where c_n is chosen so that $\int_{-\infty}^{\infty} L_n(x)\,dx = 1$. Prove that $\{L_n\}_{n\ge 0}$ is a family of good kernels as $n \to \infty$. As a result, show that if f is a continuous function supported in $[-1/2, 1/2]$, then $(f * L_n)(x)$ is a sequence of polynomials on $[-1/2, 1/2]$ which converges uniformly to f.

[Hint: First show that $c_n \ge 2/(n+1)$.]

11. Suppose that u is the solution to the heat equation given by $u = f * \mathcal{H}_t$ where $f \in \mathcal{S}(\mathbb{R})$. If we also set $u(x,0) = f(x)$, prove that u is continuous on the closure of the upper half-plane, and vanishes at infinity, that is,

$$u(x,t) \to 0 \qquad \text{as } |x| + t \to \infty.$$

[Hint: To prove that u vanishes at infinity, show that (i) $|u(x,t)| \leq C/\sqrt{t}$ and (ii) $|u(x,t)| \leq C/(1+|x|^2) + Ct^{-1/2}e^{-cx^2/t}$. Use (i) when $|x| \leq t$, and (ii) otherwise.]

12. Show that the function defined by

$$u(x,t) = \frac{x}{t}\mathcal{H}_t(x)$$

satisfies the heat equation for $t > 0$ and $\lim_{t\to 0} u(x,t) = 0$ for every x, but u is *not* continuous at the origin.

[Hint: Approach the origin with (x,t) on the parabola $x^2/4t = c$ where c is a constant.]

13. Prove the following uniqueness theorem for harmonic functions in the strip $\{(x, y) : 0 < y < 1,\ -\infty < x < \infty\}$: if u is harmonic in the strip, continuous on its closure with $u(x, 0) = u(x, 1) = 0$ for all $x \in \mathbb{R}$, and u vanishes at infinity, then $u = 0$.

14. Prove that the periodization of the Fejér kernel $\mathcal{F}_N$ on the real line (Exercise 9) is equal to the Fejér kernel for periodic functions of period 1. In other words,

$$\sum_{n=-\infty}^{\infty} \mathcal{F}_N(x+n) = F_N(x),$$

when $N \geq 1$ is an integer, and where

$$F_N(x) = \sum_{n=-N}^{N} \left(1 - \frac{|n|}{N}\right) e^{2\pi i n x} = \frac{1}{N} \frac{\sin^2(N\pi x)}{\sin^2(\pi x)}.$$

15. This exercise provides another example of periodization.

(a) Apply the Poisson summation formula to the function g in Exercise 2 to obtain

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\alpha)^2} = \frac{\pi^2}{(\sin \pi\alpha)^2}$$

whenever α is real, but not equal to an integer.

(b) Prove as a consequence that

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\alpha)} = \frac{\pi}{\tan \pi\alpha} \tag{15}$$

whenever α is real but not equal to an integer. [Hint: First prove it when $0 < \alpha < 1$. To do so, integrate the formula in (b). What is the precise meaning of the series on the left-hand side of (15)? Evaluate at $\alpha = 1/2$.]

16. The Dirichlet kernel on the real line is defined by

$$\int_{-R}^{R} \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi = (f * \mathcal{D}_R)(x) \qquad \text{so that} \qquad \mathcal{D}_R(x) = \widehat{\chi_{[-R,R]}}(x) = \frac{\sin(2\pi R x)}{\pi x}.$$

Also, the modified Dirichlet kernel for periodic functions of period 1 is defined by

$$D_N^*(x) = \sum_{|n|\leq N-1} e^{2\pi i n x} + \frac{1}{2}(e^{-2\pi i N x} + e^{2\pi i N x}).$$

Show that the result in Exercise 15 gives

$$\sum_{n=-\infty}^{\infty} \mathcal{D}_N(x+n) = D_N^*(x),$$

where $N \geq 1$ is an integer, and the infinite series must be summed symmetrically. In other words, the periodization of $\mathcal{D}_N$ is the modified Dirichlet kernel D_N^*.

17. The **gamma function** is defined for $s > 0$ by

$$\Gamma(s) = \int_0^\infty e^{-x} x^{s-1}\, dx.$$

(a) Show that for $s > 0$ the above integral makes sense, that is, that the following two limits exist:

$$\lim_{\substack{\delta \to 0 \\ \delta > 0}} \int_\delta^1 e^{-x} x^{s-1}\, dx \quad \text{and} \quad \lim_{A \to \infty} \int_1^A e^{-x} x^{s-1}\, dx.$$

(b) Prove that $\Gamma(s+1) = s\Gamma(s)$ whenever $s > 0$, and conclude that for every integer $n \geq 1$ we have $\Gamma(n+1) = n!$.

(c) Show that

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \quad \text{and} \quad \Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}.$$

[Hint: For (c), use $\int_{-\infty}^{\infty} e^{-\pi x^2}\, dx = 1$.]

18. The **zeta function** is defined for $s > 1$ by $\zeta(s) = \sum_{n=1}^{\infty} 1/n^s$. Verify the identity

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \frac{1}{2}\int_0^\infty t^{\frac{s}{2}-1}(\vartheta(t)-1)\,dt \qquad \text{whenever } s > 1$$

where Γ and ϑ are the gamma and theta functions, respectively:

$$\Gamma(s) = \int_0^\infty e^{-t}t^{s-1}\,dt \qquad \text{and} \qquad \vartheta(s) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 s}.$$

More about the zeta function and its relation to the prime number theorem can be found in Book II.

19. The following is a variant of the calculation of $\zeta(2m) = \sum_{n=1}^{\infty} 1/n^{2m}$ found in Problem 4, Chapter 3.

(a) Apply the Poisson summation formula to $f(x) = t/(\pi(x^2+t^2))$ and $\hat{f}(\xi) = e^{-2\pi t|\xi|}$ where $t > 0$ in order to get

$$\frac{1}{\pi}\sum_{n=-\infty}^{\infty}\frac{t}{t^2+n^2} = \sum_{n=-\infty}^{\infty} e^{-2\pi t|n|}.$$

(b) Prove the following identity valid for $0 < t < 1$:

$$\frac{1}{\pi}\sum_{n=-\infty}^{\infty}\frac{t}{t^2+n^2} = \frac{1}{\pi t} + \frac{2}{\pi}\sum_{m=1}^{\infty}(-1)^{m+1}\zeta(2m)t^{2m-1}$$

as well as

$$\sum_{n=-\infty}^{\infty} e^{-2\pi t|n|} = \frac{2}{1-e^{-2\pi t}} - 1.$$

(c) Use the fact that

$$\frac{z}{e^z-1} = 1 - \frac{z}{2} + \sum_{m=1}^{\infty}\frac{B_{2m}}{(2m)!}z^{2m},$$

where B_k are the Bernoulli numbers to deduce from the above formula,

$$2\zeta(2m) = (-1)^{m+1}\frac{(2\pi)^{2m}}{(2m)!}B_{2m}.$$

20. The following results are relevant in information theory when one tries to recover a signal from its samples.

Suppose f is of moderate decrease and that its Fourier transform $\hat{f}$ is supported in $I = [-1/2, 1/2]$. Then, f is entirely determined by its restriction to $\mathbb{Z}$. This means that if g is another function of moderate decrease whose Fourier transform is supported in I and $f(n) = g(n)$ for all $n \in \mathbb{Z}$, then $f = g$. More precisely:

(a) Prove that the following reconstruction formula holds:

$$f(x) = \sum_{n=-\infty}^{\infty} f(n)K(x-n) \qquad \text{where } K(y) = \frac{\sin \pi y}{\pi y}.$$

Note that $K(y) = O(1/|y|)$ as $|y| \to \infty$.

(b) If $\lambda > 1$, then

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{1}{\lambda} f\left(\frac{n}{\lambda}\right) K_\lambda\left(x - \frac{n}{\lambda}\right) \qquad \text{where } K_\lambda(y) = \frac{\cos \pi y - \cos \pi \lambda y}{\pi^2(\lambda - 1)y^2}.$$

Thus, if one samples f "more often," the series in the reconstruction formula converges faster since $K_\lambda(y) = O(1/|y|^2)$ as $|y| \to \infty$. Note that $K_\lambda(y) \to K(y)$ as $\lambda \to 1$.

(c) Prove that $\displaystyle\int_{-\infty}^{\infty} |f(x)|^2\, dx = \sum_{n=-\infty}^{\infty} |f(n)|^2$.

[Hint: For part (a) show that if χ is the characteristic function of I, then $\hat{f}(\xi) = \chi(\xi) \sum_{n=-\infty}^{\infty} f(n)e^{-2\pi i n\xi}$. For (b) use the function in Figure 2 instead of $\chi(\xi)$.]

21. Suppose that f is continuous on $\mathbb{R}$. Show that f and $\hat{f}$ cannot both be compactly supported unless $f = 0$. This can be viewed in the same spirit as the uncertainty principle.

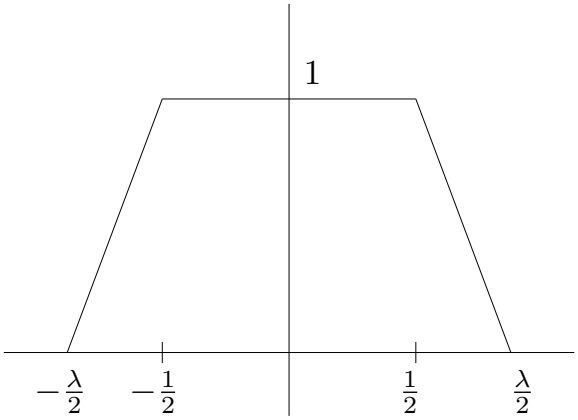


Figure 2. The function in Exercise 20

[Hint: Assume f is supported in $[0, 1/2]$. Expand f in a Fourier series in the interval $[0, 1]$, and note that as a result, f is a trigonometric polynomial.]

22. The heuristic assertion stated before Theorem 4.1 can be made precise as follows. If F is a function on $\mathbb{R}$, then we say that the preponderance of its mass is contained in an interval I (centered at the origin) if

$$\int_I x^2 |F(x)|^2 \, dx \geq \frac{1}{2} \int_{\mathbb{R}} x^2 |F(x)|^2 \, dx. \tag{16}$$

Now suppose $f \in \mathcal{S}$, and (16) holds with $F = f$ and $I = I_1$; also with $F = \hat{f}$ and $I = I_2$. Then if L_j denotes the length of I_j, we have

$$L_1 L_2 \geq \frac{1}{2\pi}.$$

A similar conclusion holds if the intervals are not necessarily centered at the origin.

23. The Heisenberg uncertainty principle can be formulated in terms of the operator $L = -\frac{d^2}{dx^2} + x^2$, which acts on Schwartz functions by the formula

$$L(f) = -\frac{d^2 f}{dx^2} + x^2 f.$$

This operator, sometimes called the **Hermite operator**, is the quantum analogue of the harmonic oscillator. Consider the usual inner product on $\mathcal{S}$ given by

$$(f, g) = \int_{-\infty}^{\infty} f(x)\overline{g(x)}\, dx \qquad \text{whenever } f, g \in \mathcal{S}.$$

(a) Prove that the Heisenberg uncertainty principle implies

$$(Lf, f) \geq (f, f) \qquad \text{for all } f \in \mathcal{S}.$$

This is usually denoted by $L \geq I$. [Hint: Integrate by parts.]

(b) Consider the operators A and A^* defined on $\mathcal{S}$ by

$$A(f) = \frac{df}{dx} + xf \qquad \text{and} \qquad A^*(f) = -\frac{df}{dx} + xf.$$

The operators A and A^* are sometimes called the **annihilation** and **creation** operators, respectively. Prove that for all $f, g \in \mathcal{S}$ we have

(i) $(Af, g) = (f, A^* g)$,
(ii) $(Af, Af) = (A^* Af, f) \geq 0$,
(iii) $A^* A = L - I$.

In particular, this again shows that $L \geq I$.

(c) Now for $t \in \mathbb{R}$, let

$$A_t(f) = \frac{df}{dx} + txf \qquad \text{and} \qquad A_t^*(f) = -\frac{df}{dx} + txf.$$

Use the fact that $(A_t^* A_t f, f) \geq 0$ to give another proof of the Heisenberg uncertainty principle which says that whenever $\int_{-\infty}^{\infty} |f(x)|^2\, dx = 1$ then

$$\left(\int_{-\infty}^{\infty} x^2 |f(x)|^2\, dx\right)\left(\int_{-\infty}^{\infty} \left|\frac{df}{dx}\right|^2\, dx\right) \geq 1/4.$$

[Hint: Think of $(A_t^* A_t f, f)$ as a quadratic polynomial in t.]

Chapter 6. The Fourier Transform on $\mathbf{R}^d$

1. Suppose that R is a rotation in the plane $\mathbb{R}^2$, and let

$$R = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

denote its matrix with respect to the standard basis vectors $e_1 = (1, 0)$ and $e_2 = (0, 1)$.

(a) Write the conditions $R^t = R^{-1}$ and $\det(R) = \pm 1$ in terms of equations in a, b, c, d.

(b) Show that there exists $\varphi \in \mathbb{R}$ such that $a + ib = e^{i\varphi}$.

(c) Conclude that if R is proper, then it can be expressed as $z \mapsto ze^{i\varphi}$, and if R is improper, then it takes the form $z \mapsto \overline{z}e^{i\varphi}$, where $\overline{z} = x - iy$.

2. Suppose that $R : \mathbb{R}^3 \to \mathbb{R}^3$ is a proper rotation.

(a) Show that $p(t) = \det(R - tI)$ is a polynomial of degree 3, and prove that there exists $\gamma \in S^2$ (where S^2 denotes the unit sphere in $\mathbb{R}^3$) with

$$R(\gamma) = \gamma.$$

[Hint: Use the fact that $p(0) > 0$ to see that there is $\lambda > 0$ with $p(\lambda) = 0$. Then $R - \lambda I$ is singular, so its kernel is non-trivial.]

(b) If $\mathcal{P}$ denotes the plane perpendicular to γ and passing through the origin, show that

$$R : \mathcal{P} \to \mathcal{P},$$

and that this linear map is a rotation.

3. Recall the formula

$$\int_{\mathbb{R}^d} F(x)\,dx = \int_{S^{d-1}} \int_0^\infty F(r\gamma) r^{d-1}\,dr\,d\sigma(\gamma).$$

Apply this to the special case when $F(x) = g(r)f(\gamma)$, where $x = r\gamma$, to prove that for any rotation R, one has

$$\int_{S^{d-1}} f(R(\gamma))\,d\sigma(\gamma) = \int_{S^{d-1}} f(\gamma)\,d\sigma(\gamma),$$

whenever f is a continuous function on the sphere S^{d-1}.

4. Let A_d and V_d denote the area and volume of the unit sphere and unit ball in $\mathbb{R}^d$, respectively.

(a) Prove the formula

$$A_d = \frac{2\pi^{d/2}}{\Gamma(d/2)}$$

so that $A_2 = 2\pi,\ A_3 = 4\pi,\ A_4 = 2\pi^2, \ldots$. Here $\Gamma(x) = \int_0^\infty e^{-t} t^{x-1}\, dt$ is the Gamma function. [Hint: Use polar coordinates and the fact that $\int_{\mathbb{R}^d} e^{-\pi|x|^2}\, dx = 1$.]

(b) Show that $d\, V_d = A_d$, hence

$$V_d = \frac{\pi^{d/2}}{\Gamma(d/2+1)}.$$

In particular $V_2 = \pi$, $V_3 = 4\pi/3, \ldots$.

5. Let A be a $d \times d$ positive definite symmetric matrix with real coefficients. Show that

$$\int_{\mathbb{R}^d} e^{-\pi(x,A(x))}\, dx = (\det(A))^{-1/2}.$$

This generalizes the fact that $\int_{\mathbb{R}^d} e^{-\pi|x|^2}\, dx = 1$, which corresponds to the case where A is the identity.

[Hint: Apply the spectral theorem to write $A = RDR^{-1}$ where R is a rotation and, D is diagonal with entries $\lambda_1, \ldots, \lambda_d$, where $\{\lambda_i\}$ are the eigenvalues of A.]

6. Suppose $\psi \in \mathcal{S}(\mathbb{R}^d)$ satisfies $\int |\psi(x)|^2\,dx = 1$. Show that

$$\left(\int_{\mathbb{R}^d} |x|^2 |\psi(x)|^2\,dx\right)\left(\int_{\mathbb{R}^d} |\xi|^2 |\hat{\psi}(\xi)|^2\,d\xi\right) \geq \frac{d^2}{16\pi^2}.$$

This is the statement of the Heisenberg uncertainty principle in d dimensions.

7. Consider the time-dependent heat equation in $\mathbb{R}^d$:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x_1^2} + \cdots + \frac{\partial^2 u}{\partial x_d^2}, \qquad \text{where } t > 0, \tag{15}$$

with boundary values $u(x,0) = f(x) \in \mathcal{S}(\mathbb{R}^d)$. If

$$\mathcal{H}_t^{(d)}(x) = \frac{1}{(4\pi t)^{d/2}} e^{-|x|^2/4t} = \int_{\mathbb{R}^d} e^{-4\pi^2 t|\xi|^2} e^{2\pi i x\cdot\xi}\, d\xi$$

is the d-dimensional **heat kernel**, show that the convolution

$$u(x,t) = (f * \mathcal{H}_t^{(d)})(x)$$

is indefinitely differentiable when $x \in \mathbb{R}^d$ and $t > 0$. Moreover, u solves (15), and is continuous up to the boundary $t = 0$ with $u(x,0) = f(x)$.

The reader may also wish to formulate the d-dimensional analogues of Theorem 2.1 and 2.3 in Chapter 5.

8. In Chapter 5, we found that a solution to the steady-state heat equation in the upper half-plane with boundary values f is given by the convolution $u = f * \mathcal{P}_y$ where the Poisson kernel is

$$\mathcal{P}_y(x) = \frac{1}{\pi}\frac{y}{x^2+y^2} \qquad \text{where } x \in \mathbb{R} \text{ and } y > 0.$$

More generally, one can calculate the d-dimensional Poisson kernel using the Fourier transform as follows.

(a) The **subordination principle** allows one to write expressions involving the function e^{-x} in terms of corresponding expressions involving the function e^{-x^2}. One form of this is the identity

$$e^{-\beta} = \int_0^\infty \frac{e^{-u}}{\sqrt{\pi u}} e^{-\beta^2/4u}\, du$$

when $\beta \geq 0$. Prove this identity with $\beta = 2\pi|x|$ by taking the Fourier transform of both sides.

(b) Consider the steady-state heat equation in the upper half-space $\{(x, y) : x \in \mathbb{R}^d,\ y > 0\}$

$$\sum_{j=1}^{d} \frac{\partial^2 u}{\partial x_j^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

with the Dirichlet boundary condition $u(x, 0) = f(x)$. A solution to this problem is given by the convolution $u(x, y) = (f * P_y^{(d)})(x)$ where $P_y^{(d)}(x)$ is the d-dimensional Poisson kernel

$$P_y^{(d)}(x) = \int_{\mathbb{R}^d} e^{2\pi i x\cdot\xi} e^{-2\pi|\xi|y}\, d\xi.$$

Compute $P_y^{(d)}(x)$ by using the subordination principle and the d-dimensional heat kernel. (See Exercise 7.) Show that

$$P_y^{(d)}(x) = \frac{\Gamma((d+1)/2)}{\pi^{(d+1)/2}} \frac{y}{(|x|^2+y^2)^{(d+1)/2}}.$$

9. A **spherical wave** is a solution $u(x,t)$ of the Cauchy problem for the wave equation in $\mathbb{R}^d$, which as a function of x is radial. Prove that u is a spherical wave if and only if the initial data $f, g \in \mathcal{S}$ are both radial.

10. Let $u(x,t)$ be a solution of the wave equation, and let $E(t)$ denote the energy of this wave

$$E(t) = \int_{\mathbb{R}^d} \left| \frac{\partial u}{\partial t}(x,t) \right|^2 + \sum_{j=1}^{d} \int_{\mathbb{R}^d} \left| \frac{\partial u}{\partial x_j}(x,t) \right|^2 \, dx.$$

We have seen that $E(t)$ is constant using Plancherel's formula. Give an alternate proof of this fact by differentiating the integral with respect to t and showing that

$$\frac{dE}{dt} = 0.$$

[Hint: Integrate by parts.]

11. Show that the solution of the wave equation

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2} + \frac{\partial^2 u}{\partial x_3^2}$$

subject to $u(x,0) = f(x)$ and $\frac{\partial u}{\partial t}(x,0) = g(x)$, where $f, g \in \mathcal{S}(\mathbb{R}^3)$, is given by

$$u(x,t) = \frac{1}{|S(x,t)|} \int_{S(x,t)} [tg(y) + f(y) + \nabla f(y) \cdot (y-x)] \, d\sigma(y),$$

where $S(x,t)$ denotes the sphere of center x and radius t, and $|S(x,t)|$ its area. This is an alternate expression for the solution of the wave equation given in Theorem 3.6. It is sometimes called **Kirchhoff's** formula.

12. Establish the identity (14) about the dual transform given in the text. In other words, prove that

$$\int_{\mathbb{R}} \int_{S^2} \mathcal{R}(f)(t,\gamma)\overline{F(t,\gamma)} d\sigma(\gamma)\, dt = \int_{\mathbb{R}^3} f(x)\overline{\mathcal{R}^*(F)(x)}\, dx \tag{16}$$

where $f \in \mathcal{S}(\mathbb{R}^3)$, $F \in \mathcal{S}(\mathbb{R} \times S^2)$, and

$$\mathcal{R}(f) = \int_{\mathcal{P}_{t,\gamma}} f \quad \text{and} \quad \mathcal{R}^*(F)(x) = \int_{S^2} F(x \cdot \gamma, \gamma)\, d\sigma(\gamma).$$

[Hint: Consider the integral

$$\int\int\int f(t\gamma + u_1 e_2 + u_2 e_2)\overline{F(t,\gamma)}\, dt\, d\sigma(\gamma)\, du_1\, du_2.$$

Integrating first in u gives the left-hand side of (16), while integrating in u and t and setting $x = t\gamma + u_1 e_2 + u_2 e_2$ gives the right-hand side.]

13. For each (t, θ) with $t \in \mathbb{R}$ and $|\theta| \leq \pi$, let $L = L_{t,\theta}$ denote the line in the (x, y)-plane given by

$$x \cos\theta + y \sin\theta = t.$$

This is the line perpendicular to the direction $(\cos\theta, \sin\theta)$ at "distance" t from the origin (we allow negative t). For $f \in \mathcal{S}(\mathbb{R}^2)$ the X-ray transform or two-dimensional Radon transform of f is defined by

$$X(f)(t, \theta) = \int_{L_{t,\theta}} f = \int_{-\infty}^{\infty} f(t\cos\theta + u\sin\theta, t\sin\theta - u\cos\theta)\, du.$$

Calculate the X-ray transform of the function $f(x, y) = e^{-\pi(x^2+y^2)}$.

14. Let X be the X-ray transform. Show that if $f \in \mathcal{S}$ and $X(f) = 0$, then $f = 0$, by taking the Fourier transform in one variable.

15. For $F \in \mathcal{S}(\mathbb{R} \times S^1)$, define the **dual X-ray transform** $X^*(F)$ by integrating F over all lines that pass through the point (x, y) (that is, those lines $L_{t,\theta}$ with $x \cos\theta + y \sin\theta = t$):

$$X^*(F)(x, y) = \int F(x \cos\theta + y \sin\theta, \theta)\, d\theta.$$

Check that in this case, if $f \in \mathcal{S}(\mathbb{R}^2)$ and $F \in \mathcal{S}(\mathbb{R} \times S^1)$, then

$$\int\int X(f)(t, \theta)\overline{F(t, \theta)}\, dt\, d\theta = \int\int f(x, y)\overline{X^*(F)(x, y)}\, dx\, dy.$$

Chapter 7. Finite Fourier Analysis

1. Let f be a function on the circle. For each $N \geq 1$ the discrete Fourier coefficients of f are defined by

$$a_N(n) = \frac{1}{N} \sum_{k=1}^{N} f(e^{2\pi ik/N}) e^{-2\pi ikn/N}, \qquad \text{for } n \in \mathbb{Z}.$$

We also let

$$a(n) = \int_0^1 f(e^{2\pi ix}) e^{-2\pi inx}\, dx$$

denote the ordinary Fourier coefficients of f.

(a) Show that $a_N(n) = a_N(n+N)$.

(b) Prove that if f is continuous, then $a_N(n) \to a(n)$ as $N \to \infty$.

2. If f is a C^1 function on the circle, prove that $|a_N(n)| \le c/|n|$ whenever $0 < |n| \le N/2$.

[Hint: Write

$$a_N(n)[1 - e^{2\pi i \ell n/N}] = \frac{1}{N} \sum_{k=1}^{N} [f(e^{2\pi i k/N}) - f(e^{2\pi i (k+\ell)/N})] e^{-2\pi i k n/N},$$

and choose ℓ so that $\ell n/N$ is nearly $1/2$.]

3. By a similar method, show that if f is a C^2 function on the circle, then

$$|a_N(n)| \le c/|n|^2, \qquad \text{whenever } 0 < |n| \le N/2.$$

As a result, prove the inversion formula for $f \in C^2$,

$$f(e^{2\pi ix}) = \sum_{n=-\infty}^{\infty} a(n) e^{2\pi inx}$$

from its finite version.
[Hint: For the first part, use the second symmetric difference

$$f(e^{2\pi i(k+\ell)/N}) + f(e^{2\pi i(k-\ell)/N}) - 2f(e^{2\pi ik/N}).$$

For the second part, if N is odd (say), write the inversion formula as

$$f(e^{2\pi ik/N}) = \sum_{|n|<N/2} a_N(n) e^{2\pi ikn/N}.]$$

4. Let e be a character on $G = \mathbb{Z}(N)$, the additive group of integers modulo N. Show that there exists a unique $0 \leq \ell \leq N-1$ so that

$$e(k) = e_\ell(k) = e^{2\pi i \ell k/N} \qquad \text{for all } k \in \mathbb{Z}(N).$$

Conversely, every function of this type is a character on $\mathbb{Z}(N)$. Deduce that $e_\ell \mapsto \ell$ defines an isomorphism from $\hat{G}$ to G.

[Hint: Show that $e(1)$ is an Nth root of unity.]

5. Show that all characters on S^1 are given by

$$e_n(x) = e^{2\pi inx} \qquad \text{with } n \in \mathbb{Z},$$

and check that $e_n \mapsto n$ defines an isomorphism from $\widehat{S^1}$ to $\mathbb{Z}$.
[Hint: If F is continuous and $F(x+y) = F(x)F(y)$, then F is differentiable. To see this, note that if $F(0) \neq 0$, then for appropriate δ, $c = \int_0^\delta F(y)\,dy \neq 0$, and $cF(x) = \int_x^{\delta+x} F(y)\,dy$. Differentiate to conclude that $F(x) = e^{Ax}$ for some A.]

6. Prove that all characters on $\mathbb{R}$ take the form

$$e_\xi(x) = e^{2\pi i \xi x} \qquad \text{with } \xi \in \mathbb{R},$$

and that $e_\xi \mapsto \xi$ defines an isomorphism from $\widehat{\mathbb{R}}$ to $\mathbb{R}$. The argument in Exercise 5 applies here as well.

7. Let $\zeta = e^{2\pi i/N}$. Define the $N \times N$ matrix $M = (a_{jk})_{1 \le j,k \le N}$ by $a_{jk} = N^{-1/2}\zeta^{jk}$.

(a) Show that M is unitary.

(b) Interpret the identity $(Mu, Mv) = (u, v)$ and the fact that $M^* = M^{-1}$ in terms of Fourier series on $\mathbb{Z}(N)$.

8. Suppose that $P(x) = \sum_{n=1}^{N} a_n e^{2\pi i n x}$.

(a) Show by using the Parseval identities for the circle and $\mathbb{Z}(N)$, that

$$\int_0^1 |P(x)|^2 \, dx = \frac{1}{N} \sum_{j=1}^{N} |P(j/N)|^2.$$

(b) Prove the reconstruction formula

$$P(x) = \sum_{j=1}^{N} P(j/N) K(x - (j/N))$$

where

$$K(x) = \frac{e^{2\pi i x}}{N} \frac{1 - e^{2\pi i N x}}{1 - e^{2\pi i x}} = \frac{1}{N}(e^{2\pi i x} + e^{2\pi i 2x} + \cdots + e^{2\pi i N x}).$$

Observe that P is completely determined by the values $P(j/N)$ for $1 \leq j \leq N$. Note also that $K(0) = 1$, and $K(j/N) = 0$ whenever j is not congruent to 0 modulo N.

9. To prove the following assertions, modify the argument given in the text.

(a) Show that one can compute the Fourier coefficients of a function on $\mathbb{Z}(N)$ when $N = 3^n$ with at most $6N \log_3 N$ operations.

(b) Generalize this to $N = \alpha^n$ where α is an integer > 1.

10. A group G is **cyclic** if there exists $g \in G$ that generates all of G, that is, if any element in G can be written as g^n for some $n \in \mathbb{Z}$. Prove that a finite abelian group is cyclic if and only if it is isomorphic to $\mathbb{Z}(N)$ for some N.

11. Write down the multiplicative tables for the groups $\mathbb{Z}^*(3)$, $\mathbb{Z}^*(4)$, $\mathbb{Z}^*(5)$, $\mathbb{Z}^*(6)$, $\mathbb{Z}^*(8)$, and $\mathbb{Z}^*(9)$. Which of these groups are cyclic?

12. Suppose that G is a finite abelian group and $e : G \to \mathbb{C}$ is a function that satisfies $e(x \cdot y) = e(x)e(y)$ for all $x, y \in G$. Prove that either e is identically 0, or e never vanishes. In the second case, show that for each x, $e(x) = e^{2\pi i r}$ for some $r \in \mathbb{Q}$ of the form $r = p/q$, where $q = |G|$.

13. In analogy with ordinary Fourier series, one may interpret finite Fourier expansions using convolutions as follows. Suppose G is a finite abelian group, 1_G its unit, and V the vector space of complex-valued functions on G.

(a) The convolution of two functions f and g in V is defined for each $a \in G$ by

$$(f * g)(a) = \frac{1}{|G|} \sum_{b \in G} f(b) g(a \cdot b^{-1}).$$

Show that for all $e \in \hat{G}$ one has $\widehat{(f * g)}(e) = \hat{f}(e)\hat{g}(e)$.

(b) Use Theorem 2.5 to show that if e is a character on G, then

$$\sum_{e \in \hat{G}} e(c) = 0 \qquad \text{whenever } c \in G \text{ and } c \neq 1_G.$$

(c) As a result of (b), show that the Fourier series $Sf(a) = \sum_{e \in \hat{G}} \hat{f}(e)e(a)$ of a function $f \in V$ takes the form

$$Sf = f * D,$$

where D is defined by

$$D(c) = \sum_{e \in \hat{G}} e(c) = \begin{cases} |G| & \text{if } c = 1_G, \\ 0 & \text{otherwise.} \end{cases} \tag{4}$$

Since $f * D = f$, we recover the fact that $Sf = f$. Loosely speaking, D corresponds to a "Dirac delta function"; it has unit mass

$$\frac{1}{|G|} \sum_{c \in G} D(c) = 1,$$

and (4) says that this mass is concentrated at the unit element in G. Thus D has the same interpretation as the "limit" of a family of good kernels. (See Section 4, Chapter 2.)

Note. The function D reappears in the next chapter as $\delta_1(n)$.

Chapter 8. Dirichlet's Theorem

1. Prove that there are infinitely many primes by observing that if there were only finitely many, $p_1, \ldots, p_N$, then

$$\prod_{j=1}^{N} \frac{1}{1-1/p_j} \geq \sum_{n=1}^{\infty} \frac{1}{n}.$$

2. In the text we showed that there are infinitely many primes of the form $4k + 3$ by a modification of Euclid's original argument. Adapt this technique to prove the similar result for primes of the form $3k + 2$, and for those of the form $6k + 5$.

3. Prove that if p and q are relatively prime, then $\mathbb{Z}^*(p) \times \mathbb{Z}^*(q)$ is isomorphic to $\mathbb{Z}^*(pq)$.

4. Let $\varphi(n)$ denote the number of positive integers $\leq n$ that are relatively prime to n. Use the previous exercise to show that if n and m are relatively prime, then

$$\varphi(nm) = \varphi(n)\varphi(m).$$

One can give a formula for the Euler phi-function as follows:

(a) Calculate $\varphi(p)$ when p is prime by counting the number of elements in $\mathbb{Z}^*(p)$.

(b) Give a formula for $\varphi(p^k)$ when p is prime and $k \geq 1$ by counting the number of elements in $\mathbb{Z}^*(p^k)$.

(c) Show that

$$\varphi(n) = n \prod_i \left(1 - \frac{1}{p_i}\right),$$

where p_i are the primes that divide n.

5. If n is a positive integer, show that

$$n = \sum_{d|n} \varphi(d),$$

where φ is the Euler phi-function.
[Hint: There are precisely $\varphi(n/d)$ integers $1 \leq m \leq n$ with $\gcd(m, n) = d$.]

6. Write down the characters of the groups $\mathbb{Z}^*(3)$, $\mathbb{Z}^*(4)$, $\mathbb{Z}^*(5)$, $\mathbb{Z}^*(6)$, and $\mathbb{Z}^*(8)$.

(a) Which ones are real, or complex?

(b) Which ones are even, or odd? (A character is even if $\chi(-1) = 1$, and odd otherwise).

7. Recall that for $|z| < 1$,

$$\log_1\left(\frac{1}{1-z}\right) = \sum_{k\geq 1} \frac{z^k}{k}.$$

We have seen that

$$e^{\log_1\left(\frac{1}{1-z}\right)} = \frac{1}{1-z}.$$

(a) Show that if $w = 1/(1-z)$, then $|z| < 1$ if and only if $\mathrm{Re}(w) > 1/2$.

(b) Show that if $\mathrm{Re}(w) > 1/2$ and $w = \rho e^{i\varphi}$ with $\rho > 0$, $|\varphi| < \pi$, then

$$\log_1 w = \log \rho + i\varphi.$$

[Hint: If $e^{\zeta} = w$, then the real part of ζ is uniquely determined and its imaginary part is determined modulo 2π.]

8. Let ζ denote the zeta function defined for $s > 1$.

(a) Compare $\zeta(s)$ with $\int_1^\infty x^{-s}\,dx$ to show that

$$\zeta(s) = \frac{1}{s-1} + O(1) \qquad \text{as } s \to 1^+.$$

(b) Prove as a consequence that

$$\sum_p \frac{1}{p^s} = \log\left(\frac{1}{s-1}\right) + O(1) \qquad \text{as } s \to 1^+.$$

9. Let χ_0 denote the trivial Dirichlet character mod q, and $p_1, \ldots, p_k$ the distinct prime divisors of q. Recall that $L(s, \chi_0) = (1 - p_1^{-s}) \cdots (1 - p_k^{-s})\zeta(s)$, and show as a consequence

$$L(s, \chi_0) = \frac{\varphi(q)}{q} \frac{1}{s-1} + O(1) \qquad \text{as } s \to 1^+.$$

[Hint: Use the asymptotics for ζ in Exercise 8.]

10. Show that if ℓ is relatively prime to q, then

$$\sum_{p\equiv\ell} \frac{1}{p^s} = \frac{1}{\varphi(q)} \log\left(\frac{1}{s-1}\right) + O(1) \qquad \text{as } s \to 1^+.$$

This is a quantitative version of Dirichlet's theorem.
[Hint: Recall (4).]

11. Use the characters for $\mathbb{Z}^*(3)$, $\mathbb{Z}^*(4)$, $\mathbb{Z}^*(5)$, and $\mathbb{Z}^*(6)$ to verify directly that $L(1,\chi) \neq 0$ for all non-trivial Dirichlet characters modulo q when $q = 3, 4, 5,$ and 6.

[Hint: Consider in each case the appropriate alternating series.]

12. Suppose χ is real and non-trivial; assuming the theorem that $L(1,\chi) \neq 0$, show directly that $L(1,\chi) > 0$.

[Hint: Use the product formula for $L(s,\chi)$.]

13. Let $\{a_n\}_{n=-\infty}^{\infty}$ be a sequence of complex numbers such that $a_n = a_m$ if $n = m \bmod q$. Show that the series

$$\sum_{n=1}^{\infty} \frac{a_n}{n}$$

converges if and only if $\sum_{n=1}^{q} a_n = 0$.

[Hint: Sum by parts.]

14. The series

$$F(\theta) = \sum_{|n|\neq 0} \frac{e^{in\theta}}{n}, \qquad \text{for } |\theta| < \pi,$$

converges for every θ and is the Fourier series of the function defined on $[-\pi, \pi]$ by $F(0) = 0$ and

$$F(\theta) = \begin{cases} i(-\pi - \theta) & \text{if } -\pi \leq \theta < 0 \\ i(\pi - \theta) & \text{if } 0 < \theta \leq \pi, \end{cases}$$

and extended by periodicity (period 2π) to all of $\mathbb{R}$ (see Exercise 8 in Chapter 2).

Show also that if $\theta \neq 0 \bmod 2\pi$, then the series

$$E(\theta) = \sum_{n=1}^{\infty} \frac{e^{in\theta}}{n}$$

converges, and that

$$E(\theta) = \frac{1}{2} \log \left(\frac{1}{2 - 2\cos\theta} \right) + \frac{i}{2} F(\theta).$$

15. To sum the series $\sum_{n=1}^{\infty} a_n/n$, with $a_n = a_m$ if $n = m \bmod q$ and $\sum_{n=1}^{q} a_n = 0$, proceed as follows.

(a) Define

$$A(m) = \sum_{n=1}^{q} a_n \zeta^{-mn} \qquad \text{where } \zeta = e^{2\pi i/q}.$$

Note that $A(q) = 0$. With the notation of the previous exercise, prove that

$$\sum_{n=1}^{\infty} \frac{a_n}{n} = \frac{1}{q} \sum_{m=1}^{q-1} A(m) E(2\pi m/q).$$

[Hint: Use Fourier inversion on $\mathbb{Z}(q)$.]

(b) If $\{a_m\}$ is odd, $(a_{-m} = -a_m)$ for $m \in \mathbb{Z}$, observe that $a_0 = a_q = 0$ and show that

$$A(m) = \sum_{1 \le n < q/2} a_n (\zeta^{-mn} - \zeta^{mn}).$$

(c) Still assuming that $\{a_m\}$ is odd, show that

$$\sum_{n=1}^{\infty} \frac{a_n}{n} = \frac{1}{2q} \sum_{m=1}^{q-1} A(m) F(2\pi m/q).$$

[Hint: Define $\tilde{A}(m) = \sum_{n=1}^{q} a_n \zeta^{mn}$ and apply the Fourier inversion formula.]

16. Use the previous exercises to show that

$$\frac{\pi}{3\sqrt{3}} = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{5} + \frac{1}{7} - \frac{1}{8} + \cdots,$$

which is $L(1, \chi)$ for the non-trivial (odd) Dirichlet character modulo 3.

Complex Analysis

Chapter 1. Preliminaries to Complex Analysis

1. Describe geometrically the sets of points z in the complex plane defined by the following relations:

(a) $|z - z_1| = |z - z_2|$ where $z_1, z_2 \in \mathbb{C}$.

(b) $1/z = \overline{z}$.

(c) $\mathrm{Re}(z) = 3$.

(d) $\mathrm{Re}(z) > c$, (resp., $\geq c$) where $c \in \mathbb{R}$.

(e) $\mathrm{Re}(az + b) > 0$ where $a, b \in \mathbb{C}$.

(f) $|z| = \mathrm{Re}(z) + 1$.

(g) $\mathrm{Im}(z) = c$ with $c \in \mathbb{R}$.

2. Let $\langle \cdot, \cdot \rangle$ denote the usual inner product in $\mathbb{R}^2$. In other words, if $Z = (x_1, y_1)$ and $W = (x_2, y_2)$, then

$$\langle Z, W \rangle = x_1 x_2 + y_1 y_2.$$

Similarly, we may define a Hermitian inner product $(\cdot, \cdot)$ in $\mathbb{C}$ by

$$(z, w) = z\overline{w}.$$

The term Hermitian is used to describe the fact that $(\cdot, \cdot)$ is not symmetric, but rather satisfies the relation

$$(z, w) = \overline{(w, z)} \qquad \text{for all } z, w \in \mathbb{C}.$$

Show that

$$\langle z, w \rangle = \frac{1}{2}[(z, w) + (w, z)] = \mathrm{Re}(z, w),$$

where we use the usual identification $z = x + iy \in \mathbb{C}$ with $(x, y) \in \mathbb{R}^2$.

3. With $\omega = se^{i\varphi}$, where $s \geq 0$ and $\varphi \in \mathbb{R}$, solve the equation $z^n = \omega$ in $\mathbb{C}$ where n is a natural number. How many solutions are there?

4. Show that it is impossible to define a total ordering on $\mathbb{C}$. In other words, one cannot find a relation $\succ$ between complex numbers so that:

(i) For any two complex numbers z, w, one and only one of the following is true: $z \succ w$, $w \succ z$ or $z = w$.

(ii) For all $z_1, z_2, z_3 \in \mathbb{C}$ the relation $z_1 \succ z_2$ implies $z_1 + z_3 \succ z_2 + z_3$.

(iii) Moreover, for all $z_1, z_2, z_3 \in \mathbb{C}$ with $z_3 \succ 0$, then $z_1 \succ z_2$ implies $z_1 z_3 \succ z_2 z_3$.

[Hint: First check if $i \succ 0$ is possible.]

5. A set$\Omega$is said to be **pathwise connected** if any two points in Ω can be joined by a (piecewise-smooth) curve entirely contained in Ω. The purpose of this exercise is to prove that an *open* set Ω is pathwise connected if and only if Ω is connected.

(a) Suppose first that Ω is open and pathwise connected, and that it can be written as $\Omega = \Omega_1 \cup \Omega_2$ where Ω_1 and Ω_2 are disjoint non-empty open sets. Choose two points $w_1 \in \Omega_1$ and $w_2 \in \Omega_2$ and let γ denote a curve in Ω joining w_1 to w_2. Consider a parametrization $z : [0, 1] \to \Omega$of this curve with $z(0) = w_1$ and $z(1) = w_2$, and let

$$t^* = \sup_{0 \le t \le 1} \{t : z(s) \in \Omega_1 \quad \text{for all } 0 \le s < t\}.$$

Arrive at a contradiction by considering the point $z(t^*)$.

(b) Conversely, suppose that Ω is open and connected. Fix a point $w \in \Omega$and let $\Omega_1 \subset \Omega$denote the set of all points that can be joined to w by a curve contained in Ω. Also, let $\Omega_2 \subset \Omega$denote the set of all points that cannot be joined to w by a curve in Ω. Prove that both Ω_1 and Ω_2 are open, disjoint and their union is Ω. Finally, since Ω_1 is non-empty (why?) conclude that Ω=Ω $_1$ as desired.

The proof actually shows that the regularity and type of curves we used to define pathwise connectedness can be relaxed without changing the equivalence between the two definitions when Ω is open. For instance, we may take all curves to be continuous, or simply polygonal lines.[2]

[2] A polygonal line is a piecewise-smooth curve which consists of finitely many straight line segments.

6. Let Ω be an open set in $\mathbb{C}$ and $z \in \Omega$. The **connected component** (or simply the **component**) ofΩcontaining z is the set $\mathcal{C}_z$ of all points w in Ω that can be joined to z by a curve entirely contained in Ω.

(a) Check first that $\mathcal{C}_z$ is open and connected. Then, show that $w \in \mathcal{C}_z$ defines an equivalence relation, that is: (i) $z \in \mathcal{C}_z$, (ii) $w \in \mathcal{C}_z$ implies $z \in \mathcal{C}_w$, and (iii) if $w \in \mathcal{C}_z$ and $z \in \mathcal{C}_\zeta$, then $w \in \mathcal{C}_\zeta$.

Thus Ω is the union of all its connected components, and two components are either disjoint or coincide.

(b) Show that Ω can have only countably many distinct connected components.

(c) Prove that if Ω is the complement of a compact set, then Ω has only one unbounded component.

[Hint: For (b), one would otherwise obtain an uncountable number of disjoint open balls. Now, each ball contains a point with rational coordinates. For (c), note that the complement of a large disc containing the compact set is connected.]

7. The family of mappings introduced here plays an important role in complex analysis. These mappings, sometimes called **Blaschke factors**, will reappear in various applications in later chapters.

(a) Let z, w be two complex numbers such that $\overline{z}w \neq 1$. Prove that

$$\left|\frac{w-z}{1-\overline{w}z}\right| < 1 \qquad \text{if } |z|<1 \text{ and } |w|<1,$$

and also that

$$\left|\frac{w-z}{1-\overline{w}z}\right| = 1 \qquad \text{if } |z|=1 \text{ or } |w|=1.$$

[Hint: Why can one assume that z is real? It then suffices to prove that

$$(r-w)(r-\overline{w}) \leq (1-rw)(1-r\overline{w})$$

with equality for appropriate r and $|w|$.]

(b) Prove that for a fixed w in the unit disc $\mathbb{D}$, the mapping

$$F : z \mapsto \frac{w-z}{1-\overline{w}z}$$

satisfies the following conditions:

(i) F maps the unit disc to itself (that is, $F : \mathbb{D} \to \mathbb{D}$), and is holomorphic.

(ii) F interchanges 0 and w, namely $F(0) = w$ and $F(w) = 0$.

(iii) $|F(z)| = 1$ if $|z| = 1$.

(iv) $F : \mathbb{D} \to \mathbb{D}$ is bijective. [Hint: Calculate $F \circ F$.]

8. Suppose U and V are open sets in the complex plane. Prove that if $f : U \to V$ and $g : V \to \mathbb{C}$ are two functions that are differentiable (in the real sense, that is, as functions of the two real variables x and y), and $h = g \circ f$, then

$$\frac{\partial h}{\partial z} = \frac{\partial g}{\partial z}\frac{\partial f}{\partial z} + \frac{\partial g}{\partial \overline{z}}\frac{\partial \overline{f}}{\partial z}$$

and

$$\frac{\partial h}{\partial \overline{z}} = \frac{\partial g}{\partial z}\frac{\partial f}{\partial \overline{z}} + \frac{\partial g}{\partial \overline{z}}\frac{\partial \overline{f}}{\partial \overline{z}}.$$

This is the complex version of the chain rule.

9. Show that in polar coordinates, the Cauchy-Riemann equations take the form

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta} \qquad \text{and} \qquad \frac{1}{r}\frac{\partial u}{\partial \theta} = -\frac{\partial v}{\partial r}.$$

Use these equations to show that the logarithm function defined by

$$\log z = \log r + i\theta \qquad \text{where } z = re^{i\theta} \text{ with } -\pi<\theta<\pi$$

is holomorphic in the region $r > 0$ and $-\pi<\theta<\pi$.

10. Show that

$$4\frac{\partial}{\partial z}\frac{\partial}{\partial \overline{z}} = 4\frac{\partial}{\partial \overline{z}}\frac{\partial}{\partial z} = \triangle\,,$$

where $\triangle$ is the **Laplacian**

$$\triangle = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}.$$

11. Use Exercise 10 to prove that if f is holomorphic in the open set Ω, then the real and imaginary parts of f are **harmonic**; that is, their Laplacian is zero.

12. Consider the function defined by

$$f(x+iy) = \sqrt{|x||y|}, \qquad \text{whenever } x, y \in \mathbb{R}.$$

Show that f satisfies the Cauchy-Riemann equations at the origin, yet f is not holomorphic at 0.

13. Suppose that f is holomorphic in an open set Ω. Prove that in any one of the following cases:

(a) $\mathrm{Re}(f)$ is constant;

(b) $\mathrm{Im}(f)$ is constant;

(c) $|f|$ is constant;

one can conclude that f is constant.

14. Suppose $\{a_n\}_{n=1}^N$ and $\{b_n\}_{n=1}^N$ are two finite sequences of complex numbers. Let $B_k = \sum_{n=1}^k b_n$ denote the partial sums of the series $\sum b_n$ with the convention $B_0 = 0$. Prove the **summation by parts** formula

$$\sum_{n=M}^{N} a_n b_n = a_N B_N - a_M B_{M-1} - \sum_{n=M}^{N-1} (a_{n+1} - a_n) B_n.$$

15. Abel's theorem. Suppose $\sum_{n=1}^{\infty} a_n$ converges. Prove that

$$\lim_{r\to 1,\ r<1} \sum_{n=1}^{\infty} r^n a_n = \sum_{n=1}^{\infty} a_n.$$

[Hint: Sum by parts.] In other words, if a series converges, then it is Abel summable with the same limit. For the precise definition of these terms, and more information on summability methods, we refer the reader to Book I, Chapter 2.

16. Determine the radius of convergence of the series $\sum_{n=1}^{\infty} a_n z^n$ when:

(a) $a_n = (\log n)^2$

(b) $a_n = n!$

(c) $a_n = \frac{n^2}{4^n + 3n}$

(d) $a_n = (n!)^3/(3n)!$ [Hint: Use Stirling's formula, which says that $n! \sim cn^{n+\frac{1}{2}}e^{-n}$ for some $c > 0$..]

(e) Find the radius of convergence of the **hypergeometric series**

$$F(\alpha, \beta, \gamma; z) = 1 + \sum_{n=1}^{\infty} \frac{\alpha(\alpha+1)\cdots(\alpha+n-1)\beta(\beta+1)\cdots(\beta+n-1)}{n!\gamma(\gamma+1)\cdots(\gamma+n-1)} z^n.$$

Here $\alpha, \beta \in \mathbb{C}$ and $\gamma \neq 0, -1, -2, \ldots$.

(f) Find the radius of convergence of the Bessel function of order r:

$$J_r(z) = \left(\frac{z}{2}\right)^r \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+r)!} \left(\frac{z}{2}\right)^{2n},$$

where r is a positive integer.

17. Show that if $\{a_n\}_{n=0}^{\infty}$ is a sequence of non-zero complex numbers such that

$$\lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} = L,$$

then

$$\lim_{n\to\infty} |a_n|^{1/n} = L.$$

In particular, this exercise shows that when applicable, the ratio test can be used to calculate the radius of convergence of a power series.

18. Let f be a power series centered at the origin. Prove that f has a power series expansion around any point in its disc of convergence.

[Hint: Write $z = z_0 + (z - z_0)$ and use the binomial expansion for z^n.]

19. Prove the following:

(a) The power series $\sum nz^n$ does not converge on any point of the unit circle.

(b) The power series $\sum z^n/n^2$ converges at every point of the unit circle.

(c) The power series $\sum z^n/n$ converges at every point of the unit circle except $z = 1$. [Hint: Sum by parts.]

20. Expand $(1-z)^{-m}$ in powers of z. Here m is a fixed positive integer. Also, show that if

$$(1-z)^{-m} = \sum_{n=0}^{\infty} a_n z^n,$$

then one obtains the following asymptotic relation for the coefficients:

$$a_n \sim \frac{1}{(m-1)!}\, n^{m-1} \qquad \text{as } n \to \infty.$$

21. Show that for $|z| < 1$, one has

$$\frac{z}{1-z^2} + \frac{z^2}{1-z^4} + \cdots + \frac{z^{2^n}}{1-z^{2^{n+1}}} + \cdots = \frac{z}{1-z},$$

and

$$\frac{z}{1+z} + \frac{2z^2}{1+z^2} + \cdots + \frac{2^k z^{2^k}}{1+z^{2^k}} + \cdots = \frac{z}{1-z}.$$

Justify any change in the order of summation.

[Hint: Use the dyadic expansion of an integer and the fact that $2^{k+1} - 1 = 1 + 2 + 2^2 + \cdots + 2^k$.]

22. Let $\mathbb{N} = \{1, 2, 3, \ldots\}$ denote the set of positive integers. A subset $S \subset \mathbb{N}$ is said to be in arithmetic progression if

$$S = \{a, a + d, a + 2d, a + 3d, \ldots\}$$

where $a, d \in \mathbb{N}$. Here d is called the step of S.

Show that $\mathbb{N}$ cannot be partitioned into a finite number of subsets that are in arithmetic progression with distinct steps (except for the trivial case $a = d = 1$). [Hint: Write $\sum_{n\in\mathbb{N}} z^n$ as a sum of terms of the type $\frac{z^a}{1-z^d}$.]

23. Consider the function f defined on $\mathbb{R}$ by

$$f(x) = \begin{cases} 0 & \text{if } x \leq 0 \ , \\ e^{-1/x^2} & \text{if } x > 0. \end{cases}$$

Prove that f is indefinitely differentiable on $\mathbb{R}$, and that $f^{(n)}(0) = 0$ for all $n \geq 1$. Conclude that f does not have a converging power series expansion $\sum_{n=0}^{\infty} a_n x^n$ for x near the origin.

24. Let γ be a smooth curve in $\mathbb{C}$ parametrized by $z(t) : [a, b] \to \mathbb{C}$. Let γ^- denote the curve with the same image as γ but with the reverse orientation. Prove that for any continuous function f on γ

$$\int_\gamma f(z)\,dz = -\int_{\gamma^-} f(z)\,dz.$$

25. The next three calculations provide some insight into Cauchy's theorem, which we treat in the next chapter.

(a) Evaluate the integrals

$$\int_\gamma z^n \, dz$$

for all integers n. Here γ is any circle centered at the origin with the positive (counterclockwise) orientation.

(b) Same question as before, but with γ any circle not containing the origin.

(c) Show that if $|a| < r < |b|$, then

$$\int_\gamma \frac{1}{(z-a)(z-b)} \, dz = \frac{2\pi i}{a-b},$$

where γ denotes the circle centered at the origin, of radius r, with the positive orientation.

26. Suppose f is continuous in a region Ω. Prove that any two primitives of f (if they exist) differ by a constant.

Chapter 2. Cauchy's Theorem and Its Applications

1. Prove that

$$\int_0^\infty \sin(x^2)\,dx = \int_0^\infty \cos(x^2)\,dx = \frac{\sqrt{2\pi}}{4}.$$

These are the **Fresnel integrals**. Here, $\int_0^\infty$ is interpreted as $\lim_{R\to\infty}\int_0^R$.

[Hint: Integrate the function e^{-z^2} over the path in Figure 14. Recall that $\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$.]

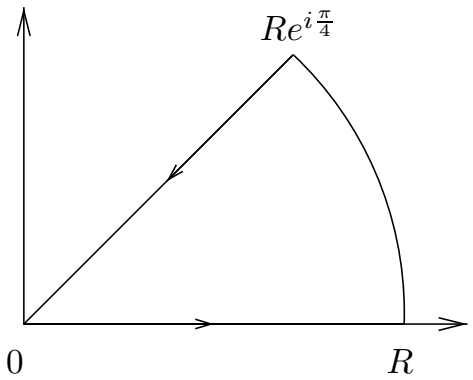


Figure 14. The contour in Exercise 1

2. Show that $\displaystyle\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}$.

[Hint: The integral equals $\frac{1}{2i}\int_{-\infty}^{\infty} \frac{e^{ix}-1}{x}\,dx$. Use the indented semicircle.]

3. Evaluate the integrals

$$\int_0^\infty e^{-ax}\cos bx\,dx \qquad \text{and} \qquad \int_0^\infty e^{-ax}\sin bx\,dx\,, \qquad a>0$$

by integrating e^{-Az}, $A=\sqrt{a^2+b^2}$, over an appropriate sector with angle ω, with $\cos\omega = a/A$.

4. Prove that for all $\xi \in \mathbb{C}$ we have $e^{-\pi\xi^2} = \displaystyle\int_{-\infty}^{\infty} e^{-\pi x^2} e^{2\pi i x\xi}\, dx$.

5. Suppose f is continuously *complex* differentiable on Ω, and $T \subset \Omega$is a triangle whose interior is also contained in Ω. Apply Green's theorem to show that

$$\int_T f(z)\,dz = 0.$$

This provides a proof of Goursat's theorem under the additional assumption that f' is continuous.

[Hint: Green's theorem says that if (F, G) is a continuously differentiable vector field, then

$$\int_T F\,dx + G\,dy = \int_{\text{Interior of } T} \left(\frac{\partial G}{\partial x} - \frac{\partial F}{\partial y}\right) dxdy.$$

For appropriate F and G, one can then use the Cauchy-Riemann equations.]

6. Let Ω be an open subset of $\mathbb{C}$ and let $T \subset \Omega$be a triangle whose interior is also contained in Ω. Suppose that f is a function holomorphic in Ω except possibly at a point w inside T. Prove that if f is bounded near w, then

$$\int_T f(z)\,dz = 0.$$

7. Suppose $f : \mathbb{D} \to \mathbb{C}$ is holomorphic. Show that the diameter $d = \sup_{z,\, w \in \mathbb{D}} |f(z) - f(w)|$ of the image of f satisfies

$$2|f'(0)| \leq d.$$

Moreover, it can be shown that equality holds precisely when f is linear, $f(z) = a_0 + a_1 z$.

Note. In connection with this result, see the relationship between the diameter of a curve and Fourier series described in Problem 1, Chapter 4, Book I.

[Hint: $2f'(0) = \frac{1}{2\pi i} \int_{|\zeta|=r} \frac{f(\zeta)-f(-\zeta)}{\zeta^2}\, d\zeta$ whenever $0 < r < 1$.]

8. If f is a holomorphic function on the strip $-1 < y < 1$, $x \in \mathbb{R}$ with

$$|f(z)| \leq A(1 + |z|)^{\eta}, \eta \qquad \text{a fixed real number}$$

for all z in that strip, show that for each integer $n \geq 0$ there exists $A_n \geq 0$ so that

$$|f^{(n)}(x)| \leq A_n(1 + |x|)^{\eta}, \qquad \text{for all } x \in \mathbb{R}.$$

[Hint: Use the Cauchy inequalities.]

9. Let Ω be a bounded open subset of $\mathbb{C}$, and $\varphi : \Omega \to \Omega$ a holomorphic function. Prove that if there exists a point $z_0 \in \Omega$ such that

$$\varphi(z_0) = z_0 \qquad \text{and} \qquad \varphi'(z_0) = 1$$

then φ is linear.

[Hint: Why can one assume that $z_0 = 0$? Write $\varphi(z) = z + a_n z^n + O(z^{n+1})$ near 0, and prove that if $\varphi_k = \varphi \circ \cdots \circ \varphi$ (where φ appears k times), then $\varphi_k(z) = z + k a_n z^n + O(z^{n+1})$. Apply the Cauchy inequalities and let $k \to \infty$ to conclude the proof. Here we use the standard O notation, where $f(z) = O(g(z))$ as $z \to 0$ means that $|f(z)| \leq C|g(z)|$ for some constant C as $|z| \to 0$.]

10. Weierstrass's theorem states that a continuous function on $[0, 1]$ can be uniformly approximated by polynomials. Can every continuous function on the closed unit disc be approximated uniformly by polynomials in the variable z?

11. Let f be a holomorphic function on the disc D_{R_0} centered at the origin and of radius R_0.

(a) Prove that whenever $0 < R < R_0$ and $|z| < R$, then

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(Re^{i\varphi}) \mathrm{Re} \left(\frac{Re^{i\varphi} + z}{Re^{i\varphi} - z} \right) \, d\varphi.$$

(b) Show that

$$\mathrm{Re} \left(\frac{Re^{i\gamma} + r}{Re^{i\gamma} - r} \right) = \frac{R^2 - r^2}{R^2 - 2Rr\cos\gamma + r^2}.$$

[Hint: For the first part, note that if $w = R^2/\overline{z}$, then the integral of $f(\zeta)/(\zeta - w)$ around the circle of radius R centered at the origin is zero. Use this, together with the usual Cauchy integral formula, to deduce the desired identity.]

12. Let u be a real-valued function defined on the unit disc $\mathbb{D}$. Suppose that u is twice continuously differentiable and harmonic, that is,

$$\triangle u(x, y) = 0$$

for all $(x, y) \in \mathbb{D}$.

(a) Prove that there exists a holomorphic function f on the unit disc such that

$$\mathrm{Re}(f) = u.$$

Also show that the imaginary part of f is uniquely defined up to an additive (real) constant. [Hint: From the previous chapter we would have $f'(z) = 2\partial u/\partial z$. Therefore, let $g(z) = 2\partial u/\partial z$ and prove that g is holomorphic. Why can one find F with $F' = g$? Prove that $\mathrm{Re}(F)$ differs from u by a real constant.]

(b) Deduce from this result, and from Exercise 11, the Poisson integral representation formula from the Cauchy integral formula: If u is harmonic in the unit disc and continuous on its closure, then if $z = re^{i\theta}$ one has

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} P_r(\theta - \varphi) u(\varphi)\, d\varphi$$

where $P_r(\gamma)$ is the Poisson kernel for the unit disc given by

$$P_r(\gamma) = \frac{1 - r^2}{1 - 2r\cos\gamma + r^2}.$$

13. Suppose f is an analytic function defined everywhere in $\mathbb{C}$ and such that for each $z_0 \in \mathbb{C}$ at least one coefficient in the expansion

$$f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$$

is equal to 0. Prove that f is a polynomial.
[Hint: Use the fact that $c_n n! = f^{(n)}(z_0)$ and use a countability argument.]

14. Suppose that f is holomorphic in an open set containing the closed unit disc, except for a pole at z_0 on the unit circle. Show that if

$$\sum_{n=0}^{\infty} a_n z^n$$

denotes the power series expansion of f in the open unit disc, then

$$\lim_{n\to\infty} \frac{a_n}{a_{n+1}} = z_0.$$

15. Suppose f is a non-vanishing continuous function on $\overline{\mathbb{D}}$ that is holomorphic in $\mathbb{D}$. Prove that if

$$|f(z)| = 1 \qquad \text{whenever } |z| = 1,$$

then f is constant.

[Hint: Extend f to all of $\mathbb{C}$ by $f(z) = 1/\overline{f(1/\overline{z})}$ whenever $|z| > 1$, and argue as in the Schwarz reflection principle.]

Chapter 3. Meromorphic Functions and the Logarithm

1. Using Euler's formula

$$\sin \pi z = \frac{e^{i\pi z} - e^{-i\pi z}}{2i},$$

show that the complex zeros of $\sin \pi z$ are exactly at the integers, and that they are each of order 1.

Calculate the residue of $1/\sin \pi z$ at $z = n \in \mathbb{Z}$.

2. Evaluate the integral

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^4}.$$

Where are the poles of $1/(1+z^4)$?

3. Show that

$$\int_{-\infty}^{\infty} \frac{\cos x}{x^2 + a^2}\, dx = \pi \frac{e^{-a}}{a}, \qquad \text{for } a > 0.$$

4. Show that

$$\int_{-\infty}^{\infty} \frac{x \sin x}{x^2 + a^2}\, dx = \pi e^{-a}, \qquad \text{for all } a > 0.$$

5. Use contour integration to show that

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x\xi}}{(1+x^2)^2}\, dx = \frac{\pi}{2}(1+2\pi|\xi|)e^{-2\pi|\xi|}$$

for all ξ real.

6. Show that

$$\int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^{n+1}} = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \cdot \pi.$$

7. Prove that

$$\int_0^{2\pi} \frac{d\theta}{(a+\cos\theta)^2} = \frac{2\pi a}{(a^2-1)^{3/2}}, \qquad \text{whenever } a > 1.$$

8. Prove that

$$\int_0^{2\pi} \frac{d\theta}{a + b\cos\theta} = \frac{2\pi}{\sqrt{a^2 - b^2}}$$

if $a > |b|$ and $a, b \in \mathbb{R}$.

9. Show that

$$\int_0^1 \log(\sin \pi x)\, dx = -\log 2.$$

[Hint: Use the contour shown in Figure 9.]

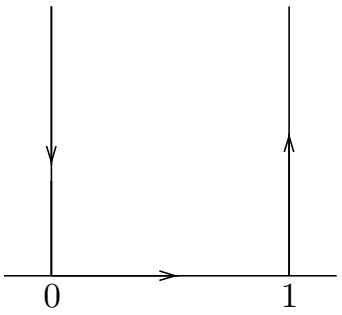


Figure 9. Contour in Exercise 9

10. Show that if $a > 0$, then

$$\int_0^\infty \frac{\log x}{x^2 + a^2}\, dx = \frac{\pi}{2a} \log a.$$

[Hint: Use the contour in Figure 10.]

11. Show that if $|a| < 1$, then

$$\int_0^{2\pi} \log|1 - ae^{i\theta}|\, d\theta = 0.$$

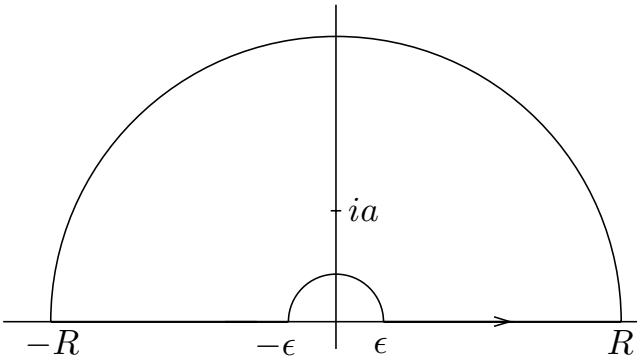


Figure 10. Contour in Exercise 10

Then, prove that the above result remains true if we assume only that $|a| \leq 1$.

12. Suppose u is not an integer. Prove that

$$\sum_{n=-\infty}^{\infty} \frac{1}{(u+n)^2} = \frac{\pi^2}{(\sin \pi u)^2}$$

by integrating

$$f(z) = \frac{\pi \cot \pi z}{(u+z)^2}$$

over the circle $|z| = R_N = N + 1/2$ (N integral, $N \geq |u|$), adding the residues of f inside the circle, and letting N tend to infinity.
Note. Two other derivations of this identity, using Fourier series, were given in Book I.

13. Suppose $f(z)$ is holomorphic in a punctured disc $D_r(z_0) - \{ z_0\}$. Suppose also that

$$|f(z)| \leq \ A|z - z_0|^{-1+\epsilon}$$

for some $\epsilon > \ 0$, and all z near z_0. Show that the singularity of f at z_0 is removable.

14. Prove that all entire functions that are also injective take the form $f(z) = az + b$ with $a, b \in \mathbb{C}$, and $a \neq 0$.

[Hint: Apply the Casorati-Weierstrass theorem to $f(1/z)$.]

15. Use the Cauchy inequalities or the maximum modulus principle to solve the following problems:

(a) Prove that if f is an entire function that satisfies

$$\sup_{|z|=R} |f(z)| \leq \ AR^k + B$$

for all $R > 0$, and for some integer $k \geq 0$ and some constants $A, B > 0$, then f is a polynomial of degree $\leq k$.

(b) Show that if f is holomorphic in the unit disc, is bounded, and converges uniformly to zero in the sector $\theta < \ \arg z < \varphi \ $ as $|z| \to \ 1$, then $f = 0$.

(c) Let $w_1, \ldots, w_n$ be points on the unit circle in the complex plane. Prove that there exists a point z on the unit circle such that the product of the distances from z to the points w_j, $1 \leq j \leq n$, is at least 1. Conclude that there exists a point w on the unit circle such that the product of the distances from w to the points w_j, $1 \leq j \leq n$, is exactly equal to 1.

(d) Show that if the real part of an entire function f is bounded, then f is constant.

16. Suppose f and g are holomorphic in a region containing the disc $|z| \leq 1$. Suppose that f has a simple zero at $z = 0$ and vanishes nowhere else in $|z| \leq 1$. Let

$$f_\epsilon(z) = f(z) + \epsilon g(z).$$

Show that if ϵ is sufficiently small, then

(a) $f_\epsilon(z)$ has a unique zero in $|z| \leq 1$, and

(b) if z_ϵ is this zero, the mapping $\epsilon \mapsto z_\epsilon$ is continuous.

17. Let f be non-constant and holomorphic in an open set containing the closed unit disc.

(a) Show that if $|f(z)| = 1$ whenever $|z| = 1$, then the image of f contains the unit disc. [Hint: One must show that $f(z) = w_0$ has a root for every $w_0 \in \mathbb{D}$. To do this, it suffices to show that $f(z) = 0$ has a root (why?). Use the maximum modulus principle to conclude.]

(b) If $|f(z)| \geq 1$ whenever $|z| = 1$ and there exists a point $z_0 \in \mathbb{D}$ such that $|f(z_0)| < 1$, then the image of f contains the unit disc.

18. Give another proof of the Cauchy integral formula

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta$$

using homotopy of curves.

[Hint: Deform the circle C to a small circle centered at z, and note that the quotient $(f(\zeta) - f(z))/(\zeta - z)$ is bounded.]

19. Prove the maximum principle for harmonic functions, that is:

(a) If u is a non-constant real-valued harmonic function in a region Ω, then u cannot attain a maximum (or a minimum) in Ω.

(b) Suppose that Ω is a region with compact closure $\overline{\Omega}$. If u is harmonic in Ω and continuous in $\overline{\Omega}$, then

$$\sup_{z\in\Omega} |u(z)| \leq \sup_{z\in\overline{\Omega}-\Omega} |u(z)|.$$

[Hint: To prove the first part, assume that u attains a local maximum at z_0. Let f be holomorphic near z_0 with $u = \mathrm{Re}(f)$, and show that f is not open. The second part follows directly from the first.]

20. This exercise shows how the mean square convergence dominates the uniform convergence of analytic functions. If U is an open subset of $\mathbb{C}$ we use the notation

$$\|f\|_{L^2(U)} = \left(\int_U |f(z)|^2 \, dxdy \right)^{1/2}$$

for the mean square norm, and

$$\|f\|_{L^\infty(U)} = \sup_{z \in U} |f(z)|$$

for the sup norm.

(a) If f is holomorphic in a neighborhood of the disc $D_r(z_0)$, show that for any $0 < s < r$ there exists a constant $C > 0$ (which depends on s and r) such that

$$\|f\|_{L^\infty(D_s(z_0))} \leq C \|f\|_{L^2(D_r(z_0))} .$$

(b) Prove that if $\{f_n\}$ is a Cauchy sequence of holomorphic functions in the mean square norm $\|\cdot\|_{L^2(U)}$, then the sequence $\{f_n\}$ converges uniformly on every compact subset of U to a holomorphic function.

[Hint: Use the mean-value property.]

21. Certain sets have geometric properties that guarantee they are simply connected.

(a) An open set $\Omega \subset \mathbb{C}$ is **convex** if for any two points in Ω, the straight line segment between them is contained in Ω. Prove that a convex open set is simply connected.

(b) More generally, an open set $\Omega \subset \mathbb{C}$ is **star-shaped** if there exists a point $z_0 \in \Omega$such that for any $z \in \Omega$, the straight line segment between z and z_0 is contained in Ω. Prove that a star-shaped open set is simply connected. Conclude that the slit plane $\mathbb{C} - \{(-\infty, 0]\}$ (and more generally any sector, convex or not) is simply connected.

(c) What are other examples of open sets that are simply connected?

22. Show that there is no holomorphic function f in the unit disc $\mathbb{D}$ that extends continuously to $\partial\mathbb{D}$ such that $f(z) = 1/z$ for $z \in \partial\mathbb{D}$.

Chapter 4. The Fourier Transform

1. Suppose f is continuous and of moderate decrease, and $\hat{f}(\xi) = 0$ for all $\xi \in \mathbb{R}$. Show that $f = 0$ by completing the following outline:

(a) For each fixed real number t consider the two functions

$$A(z) = \int_{-\infty}^{t} f(x)e^{-2\pi iz(x-t)}\,dx \qquad \text{and} \qquad B(z) = -\int_{t}^{\infty} f(x)e^{-2\pi iz(x-t)}\,dx.$$

Show that $A(\xi) = B(\xi)$ for all $\xi \in \mathbb{R}$.

(b) Prove that the function F equal to A in the closed upper half-plane, and B in the lower half-plane, is entire and bounded, thus constant. In fact, show that $F = 0$.

(c) Deduce that

$$\int_{-\infty}^{t} f(x)\,dx = 0,$$

for all t, and conclude that $f = 0$.

2. If $f \in \mathfrak{F}_a$ with $a > 0$, then for any positive integer n one has $f^{(n)} \in \mathfrak{F}_b$ whenever $0 \leq b < a$.

[Hint: Modify the solution to Exercise 8 in Chapter 2.]

3. Show, by contour integration, that if $a > 0$ and $\xi \in \mathbb{R}$ then

$$\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{a}{a^2+x^2} e^{-2\pi i x\xi}\, dx = e^{-2\pi a|\xi|},$$

and check that

$$\int_{-\infty}^{\infty} e^{-2\pi a|\xi|} e^{2\pi i\xi x}\, d\xi = \frac{1}{\pi}\frac{a}{a^2+x^2}.$$

4. Suppose Q is a polynomial of degree ≥ 2 with distinct roots, none lying on the real axis. Calculate

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x\xi}}{Q(x)}\, dx, \qquad \xi \in \mathbb{R}$$

in terms of the roots of Q. What happens when several roots coincide?

[Hint: Consider separately the cases $\xi < 0$, $\xi = 0$, and $\xi > 0$. Use residues.]

5. More generally, let $R(x) = P(x)/Q(x)$ be a rational function with (degree Q) $\geq$ (degree$P$)+2 and $Q(x) \neq 0$ on the real axis.

(a) Prove that if $\alpha_1, \ldots, \alpha_k$ are the roots of R in the upper half-plane, then there exists polynomials $P_j(\xi)$ of degree less than the multiplicity of α_j so that

$$\int_{-\infty}^{\infty} R(x)e^{-2\pi i x\xi}\, dx = \sum_{j=1}^{k} P_j(\xi)e^{-2\pi i \alpha_j \xi}, \qquad \text{when } \xi < 0.$$

(b) In particular, if $Q(z)$ has no zeros in the upper half-plane, then $\int_{-\infty}^{\infty} R(x)e^{-2\pi i x\xi}\, dx = 0$ for $\xi < 0$.

(c) Show that similar results hold in the case $\xi > 0$.

(d) Show that

$$\int_{-\infty}^{\infty} R(x)e^{-2\pi i x\xi}\, dx = O(e^{-a|\xi|}), \xi \qquad \in \mathbb{R}$$

as $|\xi| \to \infty$ for some $a > 0$. Determine the best possible a's in terms of the roots of R.

[Hint: For part (a), use residues. The powers of ξ appear when one differentiates the function $f(z) = R(z)e^{-2\pi i z\xi}$ (as in the formula of Theorem 1.4 in the previous chapter). For part (c) argue in the lower half-plane.]

6. Prove that

$$\frac{1}{\pi}\sum_{n=-\infty}^{\infty}\frac{a}{a^2+n^2}=\sum_{n=-\infty}^{\infty}e^{-2\pi a|n|}$$

whenever $a>0$. Hence show that the sum equals $\coth \pi a$.

7. The Poisson summation formula applied to specific examples often provides interesting identities.

(a) Let τ be fixed with $\mathrm{Im}(\tau) > 0$. Apply the Poisson summation formula to

$$f(z) = (\tau + z)^{-k},$$

where k is an integer ≥ 2, to obtain

$$\sum_{n=-\infty}^{\infty} \frac{1}{(\tau+n)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{m=1}^{\infty} m^{k-1} e^{2\pi i m\tau}.$$

(b) Set $k = 2$ in the above formula to show that if $\mathrm{Im}(\tau) > 0$, then

$$\sum_{n=-\infty}^{\infty} \frac{1}{(\tau+n)^2} = \frac{\pi^2}{\sin^2(\pi\tau)}.$$

(c) Can one conclude that the above formula holds true whenever τ is any complex number that is not an integer?

[Hint: For (a), use residues to prove that $\hat{f}(\xi) = 0$, if $\xi < 0$, and

$$\hat{f}(\xi) = \frac{(-2\pi i)^k}{(k-1)!}\, \xi^{k-1} e^{2\pi i \xi\tau}, \qquad \text{when } \xi > 0.]$$

8. Suppose $\hat{f}$ has compact support contained in $[-M, M]$ and let $f(z) = \sum_{n=0}^{\infty} a_n z^n$. Show that

$$a_n = \frac{(2\pi i)^n}{n!} \int_{-M}^{M} \hat{f}(\xi)\xi^n \, d\xi,$$

and as a result

$$\limsup_{n\to\infty} (n!|a_n|)^{1/n} \leq 2\pi M.$$

In the converse direction, let f be *any* power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ with $\limsup_{n\to\infty}(n!|a_n|)^{1/n} \leq 2\pi M$. Then, f is holomorphic in the complex plane, and for every $\epsilon > 0$ there exists $A_\epsilon > 0$ such that

$$|f(z)| \leq A_\epsilon e^{2\pi(M+\epsilon)|z|}.$$

9. Here are further results similar to the Phragmén-Lindelöf theorem.

(a) Let F be a holomorphic function in the right half-plane that extends continuously to the boundary, that is, the imaginary axis. Suppose that $|F(iy)| \leq 1$ for all $y \in \mathbb{R}$, and

$$|F(z)| \leq Ce^{c|z|^\gamma}$$

for some $c, C > 0$ and $\gamma < 1$. Prove that $|F(z)| \leq 1$ for all z in the right half-plane.

(b) More generally, let S be a sector whose vertex is the origin, and forming an angle of π/β. Let F be a holomorphic function in S that is continuous on the closure of S, so that $|F(z)| \leq 1$ on the boundary of S and

$$|F(z)| \leq Ce^{c|z|^\alpha} \text{ for all } z \in S$$

for some $c, C > 0$ and $0 < \alpha < \beta$. Prove that $|F(z)| \leq 1$ for all $z \in S$.

10. This exercise generalizes some of the properties of $e^{-\pi x^2}$ related to the fact that it is its own Fourier transform.

Suppose $f(z)$ is an entire function that satisfies

$$|f(x+iy)| \leq \ ce^{-ax^2+by^2}$$

for some $a, b, c > 0$. Let

$$\hat{f}(\zeta) = \int_{-\infty}^{\infty} f(x)e^{-2\pi i x\zeta}\, dx.$$

Then, $\hat{f}$ is an entire function of ζ that satisfies

$$|\hat{f}(\xi+i\eta)| \leq \ c'e^{-a'\xi^2+b'\eta^2}$$

for some $a', b', c' > 0$.

[Hint: To prove $\hat{f}(\xi) = O(e^{-a'\xi^2})$, assume $\xi > \ 0$ and change the contour of integration to $x - iy$ for some $y > 0$ fixed, and $-\infty < x < \infty$. Then

$$\hat{f}(\xi) = O(e^{-2\pi y\xi}e^{by^2}).$$

Finally, choose $y = d\xi$ where d is a small constant.]

11. One can give a neater formulation of the result in Exercise 10 by proving the following fact.

Suppose $f(z)$ is an entire function of strict order 2, that is,

$$f(z) = O(e^{c_1|z|^2})$$

for some $c_1 > 0$. Suppose also that for x real,

$$f(x) = O(e^{-c_2|x|^2})$$

for some $c_2 > 0$. Then

$$|f(x+iy)| = O(e^{-ax^2+by^2})$$

for some $a, b > 0$. The converse is obviously true.

12. The principle that a function and its Fourier transform cannot both be too small at infinity is illustrated by the following theorem of Hardy.

If f is a function on $\mathbb{R}$ that satisfies

$$f(x) = O(e^{-\pi x^2}) \quad \text{and} \quad \hat{f}(\xi) = O(e^{-\pi \xi^2}),$$

then f is a constant multiple of $e^{-\pi x^2}$. As a result, if $f(x) = O(e^{-\pi A x^2})$, and $\hat{f}(\xi) = O(e^{-\pi B \xi^2})$, with $AB > 1$ and $A, B > 0$, then f is identically zero.

(a) If f is even, show that $\hat{f}$ extends to an even entire function. Moreover, if $g(z) = \hat{f}(z^{1/2})$, then g satisfies

$$|g(x)| \leq ce^{-\pi x} \quad \text{and} \quad |g(z)| \leq ce^{\pi R \sin^2(\theta/2)} \leq ce^{\pi |z|}$$

when $x \in \mathbb{R}$ and $z = Re^{i\theta}$ with $R \geq 0$ and $\theta \in \mathbb{R}$.

(b) Apply the Phragmén-Lindelöf principle to the function

$$F(z) = g(z)e^{\gamma z} \quad \text{where } \gamma = i\pi \frac{e^{-i\pi/(2\beta)}}{\sin \pi/(2\beta)}$$

and the sector $0 \leq \theta \leq \pi/\beta < \pi$, and let $\beta \to \pi$ to deduce that $e^{\pi z} g(z)$ is bounded in the closed upper half-plane. The same result holds in the lower half-plane, so by Liouville's theorem $e^{\pi z} g(z)$ is constant, as desired.

(c) If f is odd, then $\hat{f}(0) = 0$, and apply the above argument to $\hat{f}(z)/z$ to deduce that $f = \hat{f} = 0$. Finally, write an arbitrary f as an appropriate sum of an even function and an odd function.

Chapter 5. Entire Functions

1. Give another proof of Jensen's formula in the unit disc using the functions (called Blaschke factors)

$$\psi_\alpha(z) = \frac{\alpha - z}{1 - \overline{\alpha} z}.$$

[Hint: The function $f/(\psi_{z_1} \cdots \psi_{z_N})$ is nowhere vanishing.]

2. Find the order of growth of the following entire functions:

(a) $p(z)$ where p is a polynomial.

(b) e^{bz^n} for $b \neq 0$.

(c) e^{e^z}.

3. Show that if τ is fixed with $\mathrm{Im}(\tau) > 0$, then the Jacobi theta function

$$\Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}$$

is of order 2 as a function of z. Further properties of Θ will be studied in Chapter 10.

[Hint: $-n^2 t + 2n|z| \leq -n^2 t/2$ when $t > 0$ and $n \geq 4|z|/t$.]

4. Let $t > 0$ be given and fixed, and define $F(z)$ by

$$F(z) = \prod_{n=1}^{\infty} (1 - e^{-2\pi nt} e^{2\pi iz}).$$

Note that the product defines an entire function of z.

(a) Show that $|F(z)| \leq Ae^{a|z|^2}$, hence F is of order 2.

(b) F vanishes exactly when $z = -int + m$ for $n \geq 1$ and n, m integers. Thus, if z_n is an enumeration of these zeros we have

$$\sum \frac{1}{|z_n|^2} = \infty \qquad \text{but} \qquad \sum \frac{1}{|z_n|^{2+\epsilon}} < \infty.$$

[Hint: To prove (a), write $F(z) = F_1(z)F_2(z)$ where

$$F_1(z) = \prod_{n=1}^{N} (1 - e^{-2\pi nt} e^{2\pi iz}) \qquad \text{and} \qquad F_2(z) = \prod_{n=N+1}^{\infty} (1 - e^{-2\pi nt} e^{2\pi iz}).$$

Choose $N \approx c|z|$ with c appropriately large. Then, since

$$\left(\sum_{N+1}^{\infty} e^{-2\pi nt} \right) e^{2\pi|z|} \leq 1\,,$$

one has $|F_2(z)| \leq A$. However,

$$|1 - e^{-2\pi nt} e^{2\pi iz}| \leq 1 + e^{2\pi|z|} \leq 2e^{2\pi|z|}.$$

Thus $|F_1(z)| \leq 2^N e^{2\pi N|z|} \leq e^{c'|z|^2}$. Note that a simple variant of the function F arises as a factor in the triple product formula for the Jacobi theta function Θ, taken up in Chapter 10.]

5. Show that if $\alpha > 1$, then

$$F_\alpha(z) = \int_{-\infty}^{\infty} e^{-|t|^\alpha} e^{2\pi i z t}\, dt$$

is an entire function of growth order $\alpha/(\alpha - 1)$.

[Hint: Show that

$$-\frac{|t|^\alpha}{2} + 2\pi|z||t| \leq c|z|^{\alpha/(\alpha-1)}$$

by considering the two cases $|t|^{\alpha-1} \leq A|z|$ and $|t|^{\alpha-1} \geq A|z|$, for an appropriate constant A.]

6. Prove Wallis's product formula

$$\frac{\pi}{2} = \frac{2 \cdot 2}{1 \cdot 3} \cdot \frac{4 \cdot 4}{3 \cdot 5} \cdots \frac{2m \cdot 2m}{(2m-1) \cdot (2m+1)} \cdots .$$

[Hint: Use the product formula for $\sin z$ at $z = \pi/2$.]

7. Establish the following properties of infinite products.

(a) Show that if $\sum |a_n|^2$ converges, then the product $\prod(1 + a_n)$ converges to a non-zero limit if and only if $\sum a_n$ converges.

(b) Find an example of a sequence of complex numbers $\{a_n\}$ such that $\sum a_n$ converges but $\prod(1 + a_n)$ diverges.

(c) Also find an example such that $\prod(1 + a_n)$ converges and $\sum a_n$ diverges.

8. Prove that for every z the product below converges, and

$$\cos(z/2)\cos(z/4)\cos(z/8)\cdots = \prod_{k=1}^{\infty} \cos(z/2^k) = \frac{\sin z}{z}.$$

[Hint: Use the fact that $\sin 2z = 2\sin z \cos z$.]

9. Prove that if $|z| < 1$, then

$$(1+z)(1+z^2)(1+z^4)(1+z^8)\cdots = \prod_{k=0}^{\infty}(1+z^{2^k}) = \frac{1}{1-z}.$$

10. Find the Hadamard products for:

(a) $e^z - 1$;

(b) $\cos \pi z$.

[Hint: The answers are $e^{z/2} z \prod_{n=1}^{\infty}(1 + z^2/4n^2\pi^2)$ and $\prod_{n=0}^{\infty}(1 - 4z^2/(2n+1)^2)$, respectively.]

11. Show that if f is an entire function of finite order that omits two values, then f is constant. This result remains true for any entire function and is known as Picard's little theorem.

[Hint: If f misses a, then $f(z) - a$ is of the form $e^{p(z)}$ where p is a polynomial.]

12. Suppose f is entire and never vanishes, and that none of the higher derivatives of f ever vanish. Prove that if f is also of finite order, then $f(z) = e^{az+b}$ for some constants a and b.

13. Show that the equation $e^z - z = 0$ has infinitely many solutions in $\mathbb{C}$.
[Hint: Apply Hadamard's theorem.]

14. Deduce from Hadamard's theorem that if F is entire and of growth order ρ that is non-integral, then F has infinitely many zeros.

15. Prove that every meromorphic function in $\mathbb{C}$ is the quotient of two entire functions. Also, if $\{a_n\}$ and $\{b_n\}$ are two disjoint sequences having no finite limit points, then there exists a meromorphic function in the whole complex plane that vanishes exactly at $\{a_n\}$ and has poles exactly at $\{b_n\}$.

16. Suppose that

$$Q_n(z) = \sum_{k=1}^{N_n} c_k^n \, z^k$$

are given polynomials for $n = 1, 2, \ldots$. Suppose also that we are given a sequence of complex numbers $\{a_n\}$ without limit points. Prove that there exists a meromorphic function $f(z)$ whose only poles are at $\{a_n\}$, and so that for each n, the difference

$$f(z) - Q_n\left(\frac{1}{z - a_n}\right)$$

is holomorphic near a_n. In other words, f has a prescribed poles and principal parts at each of these poles. This result is due to Mittag-Leffler.

17. Given two countably infinite sequences of complex numbers $\{a_k\}_{k=0}^{\infty}$ and $\{b_k\}_{k=0}^{\infty}$, with $\lim_{k\to\infty} |a_k| = \infty$, it is always possible to find an entire function F that satisfies $F(a_k) = b_k$ for all k.

(a) Given n distinct complex numbers $a_1, \ldots, a_n$, and another n complex numbers $b_1, \ldots, b_n$, construct a polynomial P of degree $\leq n-1$ with

$$P(a_i) = b_i \qquad \text{for } i = 1, \ldots, n.$$

(b) Let $\{a_k\}_{k=0}^{\infty}$ be a sequence of distinct complex numbers such that $a_0 = 0$ and $\lim_{k\to\infty} |a_k| = \infty$, and let $E(z)$ denote a Weierstrass product associated with $\{a_k\}$. Given complex numbers $\{b_k\}_{k=0}^{\infty}$, show that there exist integers $m_k \geq 1$ such that the series

$$F(z) = \frac{b_0}{E'(z)} \frac{E(z)}{z} + \sum_{k=1}^{\infty} \frac{b_k}{E'(a_k)} \frac{E(z)}{z - a_k} \left(\frac{z}{a_k} \right)^{m_k}$$

defines an entire function that satisfies

$$F(a_k) = b_k \qquad \text{for all } k \geq 0.$$

This is known as the Pringsheim interpolation formula.

Chapter 6. The Gamma and Zeta Functions

1. Prove that

$$\Gamma(s) = \lim_{n\to\infty} \frac{n^s n!}{s(s+1)\cdots(s+n)}$$

whenever $s \neq 0, -1, -2, \ldots$.
[Hint: Use the product formula for $1/\Gamma$, and the definition of the Euler constant γ.]

2. Prove that

$$\prod_{n=1}^{\infty} \frac{n(n+a+b)}{(n+a)(n+b)} = \frac{\Gamma(a+1)\Gamma(\ b+1)}{\Gamma(a+b+1)}$$

whenever a and b are positive. Using the product formula for $\sin \pi s$, give another proof that $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$.

3. Show that Wallis's product formula can be written as

$$\sqrt{\frac{\pi}{2}} = \lim_{n\to\infty} \frac{2^{2n}(n!)^2}{(2n+1)!}(2n+1)^{1/2}.$$

As a result, prove the following identity:

$$\Gamma(s)\Gamma(s+1/2) = \sqrt{\pi}2^{1-2s}\Gamma(2s).$$

4. Prove that if we take

$$f(z) = \frac{1}{(1-z)^\alpha}, \qquad \text{for } |z| < 1$$

(defined in terms of the principal branch of the logarithm), where α is a fixed complex number, then

$$f(z) = \sum_{n=0}^{\infty} a_n(\alpha) z^n$$

with

$$a_n(\alpha) \sim \frac{1}{\Gamma(\alpha)}\, n^{\alpha-1} \qquad \text{as } n \to \infty.$$

5. Use the fact that $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$ to prove that

$$|\Gamma(1/2 + it)| = \sqrt{\frac{2\pi}{e^{\pi t} + e^{-\pi t}}}, \qquad \text{whenever } t \in \mathbb{R}.$$

6. Show that

$$1 + \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{2n-1} - \frac{1}{2}\log n \to \frac{\gamma}{2} + \log 2,$$

where γ is Euler's constant.

7. The **Beta function** is defined for $\mathrm{Re}(\alpha) > 0$ and $\mathrm{Re}(\beta) > 0$ by

$$B(\alpha, \beta) = \int_0^1 (1-t)^{\alpha-1} t^{\beta-1}\, dt.$$

(a) Prove that $B(\alpha, \beta) = \dfrac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$.

(b) Show that $B(\alpha, \beta) = \displaystyle\int_0^\infty \frac{u^{\alpha-1}}{(1+u)^{\alpha+\beta}}\, du$.

[Hint: For part (a), note that

$$\Gamma(\alpha)\Gamma(\beta) = \int_0^\infty \int_0^\infty t^{\alpha-1} s^{\beta-1} e^{-t-s}\, dt ds,$$

and make the change of variables $s = ur$, $t = u(1-r)$.]

8. The Bessel functions arise in the study of spherical symmetries and the Fourier transform. See Chapter 6 in Book I. Prove that the following power series identity holds for Bessel functions of real order $\nu > -1/2$:

$$J_\nu(x) = \frac{(x/2)^\nu}{\Gamma(\nu + 1/2)\sqrt{\pi}} \int_{-1}^{1} e^{ixt}(1-t^2)^{\nu-(1/2)}\, dt = \left(\frac{x}{2}\right)^\nu \sum_{m=0}^{\infty} \frac{(-1)^m \left(\frac{x^2}{4}\right)^m}{m!\Gamma(\nu + m + 1)}$$

whenever $x > 0$. In particular, the Bessel function J_ν satisfies the ordinary differential equation

$$\frac{d^2 J_\nu}{dx^2} + \frac{1}{x}\frac{dJ_\nu}{dx} + \left(1 - \frac{\nu^2}{x^2}\right) J_\nu = 0.$$

[Hint: Expand the exponential e^{ixt} in a power series, and express the remaining integrals in terms of the gamma function, using Exercise 7.]

9. The hypergeometric series $F(\alpha, \beta, \gamma; z)$ was defined in Exercise 16 of Chapter 1. Show that

$$F(\alpha, \beta, \gamma; z) = \frac{\Gamma(\gamma)}{\Gamma(\beta)\Gamma(\gamma - \beta)} \int_0^1 t^{\beta-1}(1-t)^{\gamma-\beta-1}(1-zt)^{-\alpha}\, dt.$$

Here $\alpha > 0, \beta > 0, \gamma > \beta$, and $|z| < 1$.

Show as a result that the hypergeometric function, initially defined by a power series convergent in the unit disc, can be continued analytically to the complex plane slit along the half-line $[1, \infty)$.

Note that

$$\begin{aligned} \log(1-z) &= -zF(1, 1, 2; z), \\ e^z &= \lim_{\beta\to\infty} F(1, \beta, 1; z/\beta), \\ (1-z)^{-\alpha} &= F(\alpha, 1, 1; z). \end{aligned}$$

[Hint: To prove the integral identity, expand $(1 - zt)^{-\alpha}$ as a power series.]

10. An integral of the form

$$F(z) = \int_0^\infty f(t)t^{z-1}\,dt$$

is called a **Mellin transform**, and we shall write $\mathcal{M}(f)(z) = F(z)$. For example, the gamma function is the Mellin transform of the function e^{-t}.

(a) Prove that

$$\mathcal{M}(\cos)(z) = \int_0^\infty \cos(t)t^{z-1}\,dt = \Gamma(z)\cos\left(\pi\frac{z}{2}\right) \qquad \text{for } 0 < \mathrm{Re}(z) < 1,$$

and

$$\mathcal{M}(\sin)(z) = \int_0^\infty \sin(t)t^{z-1}\,dt = \Gamma(z)\sin\left(\pi\frac{z}{2}\right) \qquad \text{for } 0 < \mathrm{Re}(z) < 1.$$

(b) Show that the second of the above identities is valid in the larger strip $-1 < \mathrm{Re}(z) < 1$, and that as a consequence, one has

$$\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2} \qquad \text{and} \qquad \int_0^\infty \frac{\sin x}{x^{3/2}}\,dx = \sqrt{2\pi}.$$

This generalizes the calculation in Exercise 2 of Chapter 2.

[Hint: For the first part, consider the integral of the function $f(w) = e^{-w}w^{z-1}$ around the contour illustrated in Figure 1. Use analytic continuation to prove the second part.]

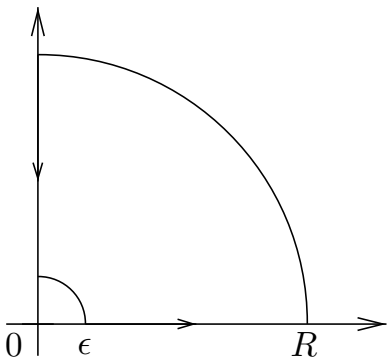


Figure 1. The contour in Exercise 10

11. Let $f(z) = e^{az}e^{-e^z}$ where $a > 0$. Observe that in the strip $\{x + iy : \ |y| < \pi \}$ the function $f(x + iy)$ is exponentially decreasing as $|x|$ tends to infinity. Prove that

$$\hat{f}(\xi) = \Gamma(\ a + i\xi), \qquad \text{for all } \xi \in \mathbb{R}.$$

12. This exercise gives two simple observations about $1/\Gamma$.

(a) Show that $1/|\Gamma(s)|$ is not $O(e^{c|s|})$ for any $c > 0$. [Hint: If $s = -k - 1/2$, where k is a positive integer, then $|1/\Gamma(s)| \geq k!/\pi$.]

(b) Show that there is no entire function $F(s)$ with $F(s) = O(e^{c|s|})$ that has simple zeros at $s = 0, -1, -2, \ldots, -n, \ldots$, and that vanishes nowhere else.

13. Prove that

$$\frac{d^2 \log \Gamma(s)}{ds^2} = \sum_{n=0}^{\infty} \frac{1}{(s+n)^2}$$

whenever s is a positive number. Show that if the left-hand side is interpreted as $(\Gamma'/\Gamma)'$, then the above formula also holds for all complex numbers s with $s \neq 0, -1, -2, \ldots$.

14. This exercise gives an asymptotic formula for $\log n!$. A more refined asymptotic formula for $\Gamma(s)$ as $s \to \infty$ (Stirling's formula) is given in Appendix A.

(a) Show that

$$\frac{d}{dx}\int_x^{x+1} \log\Gamma(t)\,dt = \log x, \qquad \text{for } x > 0,$$

and as a result

$$\int_x^{x+1} \log\Gamma(t)\,dt = x\log x - x + c.$$

(b) Show as a consequence that $\log\Gamma(n) \sim n\log n$ as $n \to \infty$. In fact, prove that $\log\Gamma(n) \sim n\log n + O(n)$ as $n \to \infty$. [Hint: Use the fact that $\log\Gamma(x)$ is monotonically increasing for all large x.]

15. Prove that for $\mathrm{Re}(s) > 1$,

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1}}{e^x - 1}\, dx.$$

[Hint: Write $1/(e^x - 1) = \sum_{n=1}^{\infty} e^{-nx}$.]

16. Use the previous exercise to give another proof that $\zeta(s)$ is continuable in the complex plane with only singularity a simple pole at $s = 1$.

[Hint: Write

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^1 \frac{x^{s-1}}{e^x - 1}\, dx + \frac{1}{\Gamma(s)} \int_1^\infty \frac{x^{s-1}}{e^x - 1}\, dx.$$

The second integral defines an entire function, while

$$\int_0^1 \frac{x^{s-1}}{e^x - 1}\, dx = \sum_{m=0}^{\infty} \frac{B_m}{m!(s + m - 1)},$$

where B_m denotes the mth Bernoulli number defined by

$$\frac{x}{e^x - 1} = \sum_{m=0}^{\infty} \frac{B_m}{m!} x^m.$$

Then $B_0 = 1$, and since $z/(e^z - 1)$ is holomorphic for $|z| < 2\pi$, we must have $\limsup_{m\to\infty} |B_m/m!|^{1/m} = 1/2\pi$.]

17. Let f be an indefinitely differentiable function on $\mathbb{R}$ that has compact support, or more generally, let f belong to the Schwartz space.[4] Consider

$$I(s) = \frac{1}{\Gamma(s)} \int_0^\infty f(x) x^{-1+s}\, dx.$$

(a) Observe that $I(s)$ is holomorphic for $\mathrm{Re}(s) > 0$. Prove that I has an analytic continuation as an entire function in the complex plane.

(b) Prove that $I(0) = 0$, and more generally

$$I(-n) = (-1)^n f^{(n+1)}(0) \qquad \text{for all } n \geq 0.$$

[Hint: To prove the analytic continuation, as well as the formulas in the second part, integrate by parts to show that $I(s) = \frac{(-1)^k}{\Gamma(s+k)} \int_0^\infty f^{(k)}(x) x^{s+k-1}\, dx$.]

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[4]The Schwartz space on $\mathbb{R}$ is denoted by $\mathcal{S}$ and consists of all indefinitely differentiable functions f, so that f and all its derivatives decay faster than any polynomials. In other words, $\sup_{x\in\mathbb{R}} |x|^m |f^{(\ell)}(x)| < \infty$ for all integers $m, \ell \geq 0$. This space appeared in the study of the Fourier transform in Book I.

Chapter 7. The Zeta Function and Prime Number Theorem

1. Suppose that $\{a_n\}_{n=1}^{\infty}$ is a sequence of real numbers such that the partial sums

$$A_n = a_1 + \cdots + a_n$$

are bounded. Prove that the Dirichlet series

$$\sum_{n=1}^{\infty} \frac{a_n}{n^s}$$

converges for $\mathrm{Re}(s) > 0$ and defines a holomorphic function in this half-plane.

[Hint: Use summation by parts to compare the original (non-absolutely convergent) series to the (absolutely convergent) series $\sum A_n(n^{-s} - (n+1)^{-s})$. An estimate for the term in parentheses is provided by the mean value theorem. To prove that the series is analytic, show that the partial sums converge uniformly on every compact subset of the half-plane $\mathrm{Re}(s) > 0$.]

2. The following links the multiplication of Dirichlet series with the divisibility properties of their coefficients.

(a) Show that if $\{a_m\}$ and $\{b_k\}$ are two bounded sequences of complex numbers, then

$$\left(\sum_{m=1}^{\infty} \frac{a_m}{m^s}\right)\left(\sum_{k=1}^{\infty} \frac{b_k}{k^s}\right) = \sum_{n=1}^{\infty} \frac{c_n}{n^s} \qquad \text{where } c_n = \textstyle\sum_{mk=n} a_m b_k.$$

The above series converge absolutely when $\mathrm{Re}(s) > 1$.

(b) Prove as a consequence that one has

$$(\zeta(s))^2 = \sum_{n=1}^{\infty} \frac{d(n)}{n^s} \qquad \text{and} \qquad \zeta(s)\zeta(s-a) = \sum_{n=1}^{\infty} \frac{\sigma_a(n)}{n^s}$$

for $\mathrm{Re}(s) > 1$ and $\mathrm{Re}(s-a) > 1$, respectively. Here $d(n)$ equals the number of divisors of n, and $\sigma_a(n)$ is the sum of the ath powers of divisors of n. In particular, one has $\sigma_0(n) = d(n)$.

3. In line with the previous exercise, we consider the Dirichlet series for $1/\zeta$.

(a) Prove that for $\mathrm{Re}(s) > 1$

$$\frac{1}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s},$$

where $\mu(n)$ is the **Möbius function** defined by

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1\,, \\ (-1)^k & \text{if } n = p_1 \cdots p_k, \text{ and the } p_j \text{ are distinct primes}\,, \\ 0 & \text{otherwise}\,. \end{cases}$$

Note that $\mu(nm) = \mu(n)\mu(m)$ whenever n and m are relatively prime. [Hint: Use the Euler product formula for $\zeta(s)$.]

(b) Show that

$$\sum_{k|n} \mu(k) = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{otherwise.} \end{cases}$$

4. Suppose $\{a_n\}_{n=1}^{\infty}$ is a sequence of complex numbers such that $a_n = a_m$ if $n \equiv m \mod q$ for some positive integer q. Define the **Dirichlet L-series** associated to $\{a_n\}$ by

$$L(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s} \qquad \text{for } \mathrm{Re}(s) > 1.$$

Also, with $a_0 = a_q$, let

$$Q(x) = \sum_{m=0}^{q-1} a_{q-m} e^{mx}.$$

Show, as in Exercises 15 and 16 of the previous chapter, that

$$L(s) = \frac{1}{\Gamma(s)} \int_0^{\infty} \frac{Q(x) x^{s-1}}{e^{qx} - 1}\, dx, \qquad \text{for } \mathrm{Re}(s) > 1.$$

Prove as a result that $L(s)$ is continuable into the complex plane, with the only possible singularity a pole at $s = 1$. In fact, $L(s)$ is regular at $s = 1$ if and only if $\sum_{m=0}^{q-1} a_m = 0$. Note the connection with the Dirichlet $L(s, \chi)$ series, taken up in Book I, Chapter 8, and that as a consequence, $L(s, \chi)$ is regular at $s = 1$ if and only if χ is a non-trivial character.

5. Consider the following function

$$\tilde{\zeta}(s) = 1 - \frac{1}{2^s} + \frac{1}{3^s} - \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^s}.$$

(a) Prove that the series defining $\tilde{\zeta}(s)$ converges for $\mathrm{Re}(s) > 0$ and defines a holomorphic function in that half-plane.

(b) Show that for $s > 1$ one has $\tilde{\zeta}(s) = (1 - 2^{1-s})\zeta(s)$.

(c) Conclude, since $\tilde{\zeta}$ is given as an alternating series, that ζ has no zeros on the segment $0 < \sigma < 1$. Extend this last assertion to $\sigma = 0$ by using the functional equation.

6. Show that for every $c > 0$

$$\lim_{N\to\infty} \frac{1}{2\pi i} \int_{c-iN}^{c+iN} a^s \, \frac{ds}{s} = \begin{cases} 1 & \text{if } a > 1, \\ 1/2 & \text{if } a = 1, \\ 0 & \text{if } 0 \le a < 1. \end{cases}$$

The integral is taken over the vertical segment from $c - iN$ to $c + iN$.

7. Show that the function

$$\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$$

is real when s is real, or when $\mathrm{Re}(s) = 1/2$.

8. The function ζ has infinitely many zeros in the critical strip. This can be seen as follows.

(a) Let

$$F(s) = \xi(1/2 + s), \qquad \text{where} \qquad \xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s).$$

Show that $F(s)$ is an even function of s, and as a result, there exists G so that $G(s^2) = F(s)$.

(b) Show that the function $(s-1)\zeta(s)$ is an entire function of growth order 1, that is

$$|(s-1)\zeta(s)| \leq \ A_\epsilon e^{a_\epsilon |s|^{1+\epsilon}}.$$

As a consequence $G(s)$ is of growth order $1/2$.

(c) Deduce from the above that ζ has infinitely many zeros in the critical strip.

[Hint: To prove (a) and (b) use the functional equation for $\zeta(s)$. For (c), use a result of Hadamard, which states that an entire function with fractional order has infinitely many zeros (Exercise 14 in Chapter 5).]

9. Refine the estimates in Proposition 2.7 in Chapter 6 and Proposition 1.6 to show that

(a) $|\zeta(1+it)| \leq A \log |t|$,

(b) $|\zeta'(1+it)| \leq A(\log |t|)^2$,

(c) $1/|\zeta(1+it)| \leq A(\log |t|)^a$,

when $|t| \geq 2$ (with $a = 7$).

10. In the theory of primes, a better approximation to $\pi(x)$ (instead of $x/\log x$) turns out to be $\mathrm{Li}(x)$ defined by

$$\mathrm{Li}(x) = \int_2^x \frac{dt}{\log t}.$$

(a) Prove that

$$\mathrm{Li}(x) = \frac{x}{\log x} + O\left(\frac{x}{(\log x)^2}\right) \qquad \text{as } x \to \infty,$$

and that as a consequence

$$\pi(x) \sim \mathrm{Li}(x) \qquad \text{as } x \to \infty.$$

[Hint: Integrate by parts in the definition of $\mathrm{Li}(x)$ and observe that it suffices to prove

$$\int_2^x \frac{dt}{(\log t)^2} = O\left(\frac{x}{(\log x)^2}\right).$$

To see this, split the integral from 2 to $\sqrt{x}$ and from $\sqrt{x}$ to x.]

(b) Refine the previous analysis by showing that for every integer $N > 0$ one has the following asymptotic expansion

$$\mathrm{Li}(x) = \frac{x}{\log x} + \frac{x}{(\log x)^2} + 2\frac{x}{(\log x)^3} \cdots + (N-1)!\frac{x}{(\log x)^N} + O\left(\frac{x}{(\log x)^{N+1}}\right)$$

as $x \to \infty$.

11. Let

$$\varphi(x) = \sum_{p \leq x} \log p$$

where the sum is taken over all primes $\leq x$. Prove that the following are equivalent as $x \to \infty$:

(i) $\varphi(x) \sim x$,

(ii) $\pi(x) \sim x/\log x$,

(iii) $\psi(x) \sim x$,

(iv) $\psi_1(x) \sim x^2/2$.

12. If p_n denotes the nth prime, the prime number theorem implies that $p_n \sim n \log n$ as $n \to \infty$.

(a) Show that $\pi(x) \sim x/\log x$ implies that

$$\log \pi(x) + \log \log x \sim \log x.$$

(b) As a consequence, prove that $\log \pi(x) \sim \log x$, and take $x = p_n$ to conclude the proof.

Chapter 8. Conformal Mappings

1. A holomorphic mapping $f : U \to V$ is a **local bijection** on U if for every $z \in U$ there exists an open disc $D \subset U$ centered at z, so that $f : D \to f(D)$ is a bijection.

Prove that a holomorphic map $f : U \to V$ is a local bijection on U if and only if $f'(z) \neq 0$ for all $z \in U$.

[Hint: Use Rouché's theorem as in the proof of Proposition 1.1.]

2. Supppose $F(z)$ is holomorphic near $z = z_0$ and $F(z_0) = F'(z_0) = 0$, while $F''(z_0) \neq 0$. Show that there are two curves Γ_1 and Γ_2 that pass through z_0, are orthogonal at z_0, and so that F restricted to Γ_1 is real and has a minimum at z_0, while F restricted to Γ_2 is also real but has a maximum at z_0.

[Hint: Write $F(z) = (g(z))^2$ for z near z_0, and consider the mapping $z \mapsto g(z)$ and its inverse.]

3. Suppose U and V are conformally equivalent. Prove that if U is simply connected, then so is V. Note that this conclusion remains valid if we merely assume that there exists a continuous bijection between U and V.

4. Does there exist a holomorphic surjection from the unit disc to $\mathbb{C}$?
[Hint: Move the upper half-plane "down" and then square it to get $\mathbb{C}$.]

5. Prove that $f(z) = -\frac{1}{2}(z + 1/z)$ is a conformal map from the half-disc $\{z = x + iy : |z| < 1,\ y > 0\}$ to the upper half-plane.

[Hint: The equation $f(z) = w$ reduces to the quadratic equation $z^2 + 2wz + 1 = 0$, which has two distinct roots in $\mathbb{C}$ whenever $w \neq \pm 1$. This is certainly the case if $w \in \mathbb{H}$.]

6. Give another proof of Lemma 1.3 by showing directly that the Laplacian of $u \circ F$ is zero.

[Hint: The real and imaginary parts of F satisfy the Cauchy-Riemann equations.]

7. Provide all the details in the proof of the formula for the solution of the Dirichlet problem in a strip discussed in Section 1.3. Recall that it suffices to compute the solution at the points $z = iy$ with $0 < y < 1$.

(a) Show that if $re^{i\theta} = G(iy)$, then

$$re^{i\theta} = i\frac{\cos \pi y}{1 + \sin \pi y}.$$

This leads to two separate cases: either $0 < y \leq 1/2$ and $\theta = \pi/2$, or $1/2 \leq y < 1$ and $\theta = -\pi/2$. In either case, show that

$$r^2 = \frac{1 - \sin \pi y}{1 + \sin \pi y} \qquad \text{and} \qquad P_r(\theta - \varphi) = \frac{\sin \pi y}{1 - \cos \pi y \sin \varphi}.$$

(b) In the integral $\frac{1}{2\pi}\int_0^\pi P_r(\theta - \varphi)\tilde{f}_0(\varphi)\, d\varphi$ make the change of variables $t = F(e^{i\varphi})$. Observe that

$$e^{i\varphi} = \frac{i - e^{\pi t}}{i + e^{\pi t}},$$

and then take the imaginary part and differentiate both sides to establish the two identities

$$\sin \varphi = \frac{1}{\cosh \pi t} \qquad \text{and} \qquad \frac{d\varphi}{dt} = \frac{\pi}{\cosh \pi t}.$$

Hence deduce that

$$\begin{aligned}\frac{1}{2\pi}\int_0^\pi P_r(\theta - \varphi)\tilde{f}_0(\varphi)\, d\varphi &= \frac{1}{2\pi}\int_0^\pi \frac{\sin \pi y}{1 - \cos \pi y \sin \varphi}\tilde{f}_0(\varphi)\, d\varphi \\ &= \frac{\sin \pi y}{2}\int_{-\infty}^{\infty} \frac{f_0(t)}{\cosh \pi t - \cos \pi y}\, dt.\end{aligned}$$

(c) Use a similar argument to prove the formula for the integral $\frac{1}{2\pi}\int_{-\pi}^0 P_r(\theta - \varphi)\tilde{f}_1(\varphi)\, d\varphi$.

8. Find a harmonic function u in the open first quadrant that extends continuously up to the boundary except at the points 0 and 1, and that takes on the following boundary values: $u(x, y) = 1$ on the half-lines $\{y = 0,\ x > 1\}$ and $\{x = 0,\ y > 0\}$, and $u(x, y) = 0$ on the segment $\{0 < x < 1, y = 0\}$.

[Hint: Find conformal maps $F_1, F_2, \ldots, F_5$ indicated in Figure 11. Note that $\frac{1}{\pi} \arg(z)$ is harmonic on the upper half-plane, equals 0 on the positive real axis, and 1 on the negative real axis.]

9. Prove that the function u defined by

$$u(x, y) = \operatorname{Re}\left(\frac{i+z}{i-z}\right) \qquad \text{and} \qquad u(0, 1) = 0$$

is harmonic in the unit disc and vanishes on its boundary. Note that u is not bounded in $\mathbb{D}$.

10. Let $F : \mathbb{H} \to \mathbb{C}$ be a holomorphic function that satisfies

$$|F(z)| \leq 1 \qquad \text{and} \qquad F(i) = 0.$$

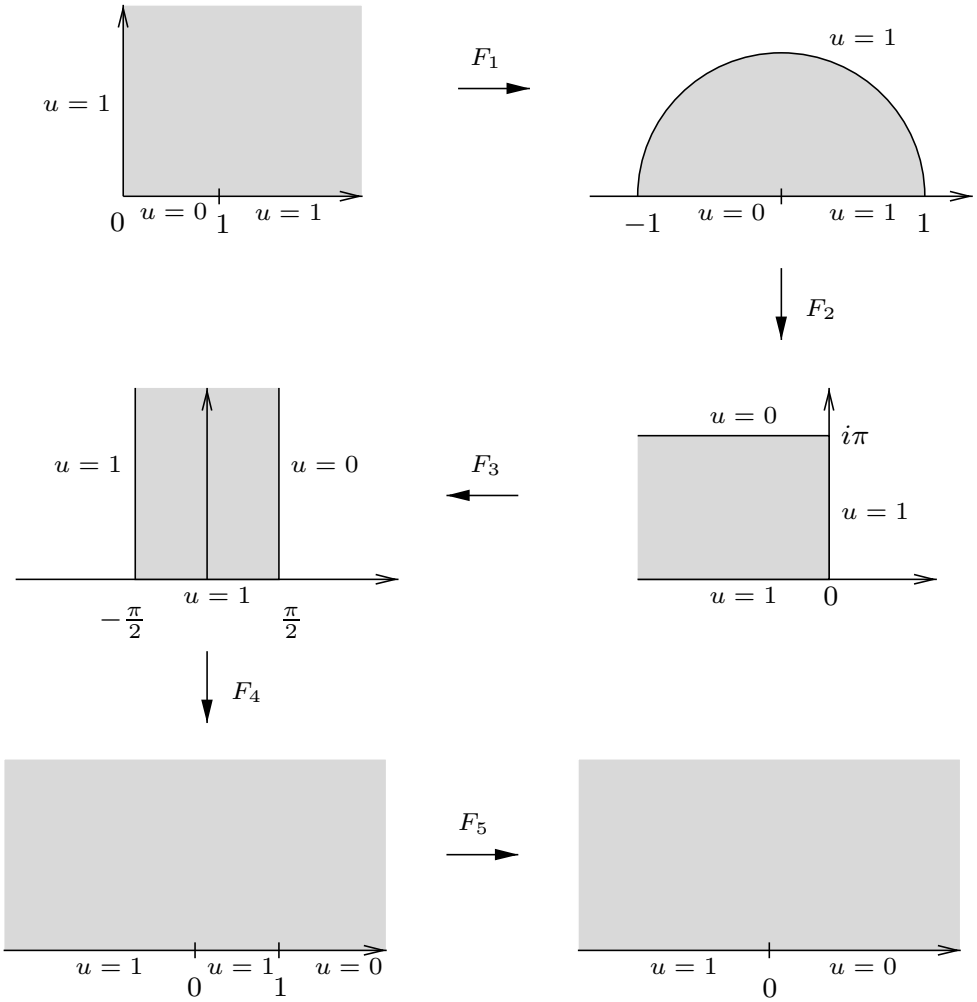


Figure 11. Successive conformal maps in Exercise 8

Prove that

$$|F(z)| \leq \left| \frac{z - i}{z + i} \right| \qquad \text{for all } z \in \mathbb{H}.$$

11. Show that if $f : D(0, R) \to \mathbb{C}$ is holomorphic, with $|f(z)| \leq M$ for some $M > 0$, then

$$\left| \frac{f(z) - f(0)}{M^2 - \overline{f(0)} f(z)} \right| \leq \frac{|z|}{MR}.$$

[Hint: Use the Schwarz lemma.]

12. A complex number $w \in \mathbb{D}$ is a **fixed point** for the map $f : \mathbb{D} \to \mathbb{D}$ if $f(w) = w$.

(a) Prove that if $f : \mathbb{D} \to \mathbb{D}$ is analytic and has two distinct fixed points, then f is the identity, that is, $f(z) = z$ for all $z \in \mathbb{D}$.

(b) Must every holomorphic function $f : \mathbb{D} \to \mathbb{D}$ have a fixed point? [Hint: Consider the upper half-plane.]

13. The **pseudo-hyperbolic distance** between two points $z, w \in \mathbb{D}$ is defined by

$$\rho(z,w) = \left| \frac{z-w}{1-\overline{w}z} \right|.$$

(a) Prove that if $f : \mathbb{D} \to \mathbb{D}$ is holomorphic, then

$$\rho(f(z), f(w)) \leq \rho(z,w) \qquad \text{for all } z, w \in \mathbb{D}.$$

Moreover, prove that if f is an automorphism of $\mathbb{D}$ then f preserves the pseudo-hyperbolic distance

$$\rho(f(z), f(w)) = \rho(z,w) \qquad \text{for all } z, w \in \mathbb{D}.$$

[Hint: Consider the automorphism $\psi_\alpha(z) = (z-\alpha)/(1-\overline{\alpha}z)$ and apply the Schwarz lemma to $\psi_{f(w)} \circ f \circ \psi_w^{-1}$.]

(b) Prove that

$$\frac{|f'(z)|}{1 - |f(z)|^2} \leq \frac{1}{1-|z|^2} \qquad \text{for all } z \in \mathbb{D}.$$

This result is called the Schwarz-Pick lemma. See Problem 3 for an important application of this lemma.

14. Prove that all conformal mappings from the upper half-plane $\mathbb{H}$ to the unit disc $\mathbb{D}$ take the form

$$e^{i\theta}\frac{z-\beta}{z-\overline{\beta}}, \theta \qquad \in \mathbb{R} \text{ and } \beta \in \mathbb{H}.$$

15. Here are two properties enjoyed by automorphisms of the upper half-plane.

(a) Suppose Φ is an automorphism of $\mathbb{H}$ that fixes three distinct points on the real axis. Then Φ is the identity.

(b) Suppose (x_1, x_2, x_3) and (y_1, y_2, y_3) are two pairs of three distinct points on the real axis with

$$x_1 < x_2 < x_3 \qquad \text{and} \qquad y_1 < y_2 < y_3.$$

Prove that there exists (a unique) automorphism Φ of $\mathbb{H}$ so that $\Phi(x_j) = y_j$, $j = 1, 2, 3$. The same conclusion holds if $y_3 < y_1 < y_2$ or $y_2 < y_3 < y_1$.

16. Let

$$f(z) = \frac{i-z}{i+z} \qquad \text{and} \qquad f^{-1}(w) = i\frac{1-w}{1+w}.$$

(a) Given $\theta \in \mathbb{R}$, find real numbers a, b, c, d such that $ad - bc = 1$, and so that for any $z \in \mathbb{H}$

$$\frac{az+b}{cz+d} = f^{-1}\left(e^{i\theta} f(z)\right).$$

(b) Given $\alpha \in \mathbb{D}$ find real numbers a, b, c, d so that $ad - bc = 1$, and so that for any $z \in \mathbb{H}$

$$\frac{az+b}{cz+d} = f^{-1}\left(\psi_\alpha(f(z))\right),$$

with ψ_α defined in Section 2.1.

(c) Prove that if g is an automorphism of the unit disc, then there exist real numbers a, b, c, d such that $ad - bc = 1$ and so that for any $z \in \mathbb{H}$

$$\frac{az+b}{cz+d} = f^{-1} \circ g \circ f(z).$$

[Hint: Use parts (a) and (b).]

17. If $\psi_\alpha(z) = (\alpha - z)/(1 - \overline{\alpha}z)$ for $|\alpha| < 1$, prove that

$$\frac{1}{\pi}\iint_{\mathbb{D}} |\psi'_\alpha|^2\, dxdy = 1 \qquad \text{and} \qquad \frac{1}{\pi}\iint_{\mathbb{D}} |\psi'_\alpha|\, dxdy = \frac{1 - |\,\alpha|^2}{|\alpha|^2} \log \frac{1}{1 - |\,\alpha|^2},$$

where in the case $\alpha = 0$ the expression on the right is understood as the limit as $|\alpha| \to\ 0$.

[Hint: The first integral can be evaluated without a calculation. For the second, use polar coordinates, and for each fixed r use contour integration to evaluate the integral in θ.]

18. Suppose that Ω is a simply connected domain that is bounded by a piecewise-smooth closed curve γ (in the terminology of Chapter 1). Then any conformal map F of $\mathbb{D}$ to Ω extends to a continuous bijection of $\overline{\mathbb{D}}$ to $\overline{\Omega}$. The proof is simply a generalization of the argument used in Theorem 4.2.

19. Prove that the complex plane slit along the union of the rays $\cup_{k=1}^{n}\{A_k + iy : y \leq 0\}$ is simply connected.

[Hint: Given a curve, first "raise" it so that it is completely contained in the upper half-plane.]

20. Other examples of elliptic integrals providing conformal maps from the upper half-plane to rectangles are given below.

(a) The function

$$\int_0^z \frac{d\zeta}{\sqrt{\zeta(\zeta-1)(\zeta-\lambda)}}, \qquad \text{with } \lambda \in \mathbb{R} \text{ and } \lambda \neq 1$$

maps the upper half-plane conformally to a rectangle, one of whose vertices is the image of the point at infinity.

(b) In the case $\lambda = -1$, the image of

$$\int_0^z \frac{d\zeta}{\sqrt{\zeta(\zeta^2-1)}}$$

is a square whose side lengths are $\frac{\Gamma^2(1/4)}{2\sqrt{2\pi}}$.

21. We consider conformal mappings to triangles.

(a) Show that

$$\int_0^z z^{-\beta_1}(1-z)^{-\beta_2}\,dz,$$

with $0 < \beta_1 < 1$, $0 < \beta_2 < 1$, and $1 < \beta_1 + \beta_2 < 2$, maps $\mathbb{H}$ to a triangle whose vertices are the images of 0, 1, and ∞, and with angles $\alpha_1\pi$, $\alpha_2\pi$, and $\alpha_3\pi$, where $\alpha_j + \beta_j = 1$ and $\beta_1 + \beta_2 + \beta_3 = 2$.

(b) What happens when $\beta_1 + \beta_2 = 1$?

(c) What happens when $0 < \beta_1 + \beta_2 < 1$?

(d) In (a), the length of the side of the triangle opposite angle $\alpha_j\pi$ is $\frac{\sin(\alpha_j\pi)}{\pi}\Gamma(\alpha_1)\Gamma(\alpha_2)\Gamma(\alpha_3)$.

22. If P is a simply connected region bounded by a polygon with vertices $a_1, \ldots, a_n$ and angles $\alpha_1\pi, \ldots, \alpha_n\pi$, and F is a conformal map of the disc $\mathbb{D}$ to P, then there exist complex numbers $B_1, \ldots, B_n$ on the unit circle, and constants c_1 and c_2 so that

$$F(z) = c_1 \int_1^z \frac{d\zeta}{(\zeta - B_1)^{\beta_1} \cdots (\zeta - B_n)^{\beta_n}} + c_2.$$

[Hint: This follows from the standard correspondence between $\mathbb{H}$ and $\mathbb{D}$ and an argument similar to that used in the proof of Theorem 4.7.]

23. If

$$F(z) = \int_1^z \frac{d\zeta}{(1-\zeta^n)^{2/n}},$$

then F maps the unit disc conformally onto the interior of a regular polygon with n sides and perimeter

$$2^{\frac{n-2}{n}} \int_0^\pi (\sin\theta)^{-2/n}\, d\theta.$$

24. The elliptic integrals K and K' defined for $0 < k < 1$ by

$$K(k) = \int_0^1 \frac{dx}{((1-x^2)(1-k^2x^2))^{1/2}} \quad \text{and} \quad K'(k) = \int_1^{1/k} \frac{dx}{((x^2-1)(1-k^2x^2))^{1/2}}$$

satisfy various interesting identities. For instance:

(a) Show that if $\tilde{k}^2 = 1 - k^2$ and $0 < \tilde{k} < 1$, then

$$K'(k) = K(\tilde{k}).$$

[Hint: Change variables $x = (1 - \tilde{k}^2 y^2)^{-1/2}$ in the integral defining $K'(k)$.]

(b) Prove that if $\tilde{k}^2 = 1 - k^2$, and $0 < \tilde{k} < 1$, then

$$K(k) = \frac{2}{1+\tilde{k}} K\left(\frac{1-\tilde{k}}{1+\tilde{k}}\right).$$

[Hint: Change variables $x = 2t/(1 + \tilde{k} + (1 - \tilde{k})t^2)$.]

(c) Show that for $0 < k < 1$ one has

$$K(k) = \frac{\pi}{2} F(1/2, 1/2, 1; k^2),$$

where F the hypergeometric series. [Hint: This follows from the integral representation for F given in Exercise 9, Chapter 6.]

Chapter 9. An Introduction to Elliptic Functions

1. Suppose that a meromorphic function f has two periods ω_1 and ω_2, with $\omega_2/\omega_1 \in \mathbb{R}$.

(a) Suppose ω_2/ω_1 is rational, say equal to p/q, where p and q are relatively prime integers. Prove that as a result the periodicity assumption is equivalent to the assumption that f is periodic with the simple period $\omega_0 = \frac{1}{q}\omega_1$. [Hint: Since p and q are relatively prime, there exist integers m and n such that $mq + np = 1$ (Corollary 1.3, Chapter 8, Book I).]

(b) If ω_2/ω_1 is irrational, then f is constant. To prove this, use the fact that $\{m - n\tau\}$ is dense in $\mathbb{R}$ whenever τ is irrational and m, n range over the integers.

2. Suppose that $a_1, \ldots, a_r$ and $b_1, \ldots, b_r$ are the zeros and poles, respectively, in the fundamental parallelogram of an elliptic function f. Show that

$$a_1 + \cdots + a_r - b_1 - \cdots - b_r = n\omega_1 + m\omega_2$$

for some integers n and m.

[Hint: If the boundary of the parallelogram contains no zeros or poles, simply integrate $zf'(z)/f(z)$ over that boundary, and observe that the integral of $f'(z)/f(z)$ over a side is an integer multiple of $2\pi i$. If there are zeros or poles on the side of the parallelogram, translate it by a small amount to reduce the problem to the first case.]

3. In contrast with the result in Lemma 1.5, prove that the series

$$\sum_{n+m\tau\in\Lambda^*} \frac{1}{|n+m\tau|^2} \qquad \text{where } \tau \in \mathbb{H}$$

does not converge. In fact, show that

$$\sum_{1\le n^2+m^2\le R^2} 1/(n^2+m^2) = 2\pi \log R + O(1) \qquad \text{as } R \to \infty.$$

4. By rearranging the series

$$\frac{1}{z^2} + \sum_{\omega \in \Lambda^*} \left[\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2} \right],$$

show directly, without differentiation, that $\wp(z+\omega) = \wp(z)$ whenever $\omega \in \Lambda$.

[Hint: For R sufficiently large, note that $\wp(z) = \wp^R(z) + O(1/R)$, where $\wp^R(z) = z^{-2} + \sum_{0<|\omega|<R}((z+\omega)^{-2} - \omega^{-2})$. Next, observe that both $\wp^R(z+1) - \wp^R(z)$ and $\wp^R(z+\tau) - \wp^R(z)$ are $O(\sum_{R-c<|\omega|<R+c} |\omega|^{-2}) = O(1/R)$.]

5. Let $\sigma(z)$ be the canonical product

$$\sigma(z) = z \prod_{j=1}^{\infty} E_2(z/\tau_j),$$

where τ_j is an enumeration of the periods $\{n + m\tau\}$ with $(n, m) \neq (0, 0)$, and $E_2(z) = (1 - z)e^{z+z^2/2}$.

(a) Show that $\sigma(z)$ is an entire function of order 2 that has simple zeros at all the periods $n + m\tau$, and vanishes nowhere else.

(b) Show that

$$\frac{\sigma'(z)}{\sigma(z)} = \frac{1}{z} + \sum_{(n,m)\neq(0,0)} \left[\frac{1}{z - n - m\tau} + \frac{1}{n + m\tau} + \frac{z}{(n + m\tau)^2} \right],$$

and that this series converges whenever z is not a lattice point.

(c) Let $L(z) = -\sigma'(z)/\sigma(z)$. Then

$$L'(z) = \frac{(\sigma'(z))^2 - \sigma(z)\sigma''(z)}{(\sigma(z))^2} = \wp(z).$$

6. Prove that $\wp''$ is a quadratic polynomial in $\wp$.

7. Setting $\tau = 1/2$ in the expression

$$\sum_{m=-\infty}^{\infty} \frac{1}{(m+\tau)^2} = \frac{\pi^2}{\sin^2(\pi\tau)},$$

deduce that

$$\sum_{m\geq 1,\ m\ \text{odd}} \frac{1}{m^2} = \frac{\pi^2}{8} \qquad \text{and} \qquad \sum_{m\geq 1} \frac{1}{m^2} = \frac{\pi^2}{6} = \zeta(2).$$

Similarly, using $\sum 1/(m+\tau)^4$ deduce that

$$\sum_{m\geq 1,\ m\ \text{odd}} \frac{1}{m^4} = \frac{\pi^4}{96} \qquad \text{and} \qquad \sum_{m\geq 1} \frac{1}{m^4} = \frac{\pi^4}{90} = \zeta(4).$$

These results were already obtained using Fourier series in the exercises at the end of Chapters 2 and 3 in Book I.

8. Let

$$E_4(\tau) = \sum_{(n,m)\neq(0,0)} \frac{1}{(n+m\tau)^4}$$

be the Eisenstein series of order 4.

(a) Show that $E_4(\tau) \to \pi^4/45$ as $\mathrm{Im}(\tau) \to \infty$.

(b) More precisely,

$$\left| E_4(\tau) - \frac{\pi^4}{45} \right| \leq ce^{-2\pi t} \qquad \text{if } \tau = x + it \text{ and } t \geq 1.$$

(c) Deduce that

$$\left| E_4(\tau) - \tau^{-4}\frac{\pi^4}{45} \right| \leq ct^{-4}e^{-2\pi/t} \qquad \text{if } \tau = it \text{ and } 0 < t \leq 1.$$

Chapter 10. Applications of Theta Functions

1. Prove that

$$\frac{(\Theta'(z|\tau))^2 - \Theta(z|\tau)\Theta''(z|\tau)}{\Theta(z|\tau)^2} = \wp_\tau(z - 1/2 - \tau/2) + c_\tau,$$

where c_τ can be expressed in terms of the first two derivatives of $\Theta(z|\tau)$, with respect to z, at $z = 1/2 + \tau/2$. Compare this formula with the result in Exercise 5 in the previous chapter.

2. Consider the **Fibonacci numbers** $\{F_n\}_{n=0}^{\infty}$, defined by the two initial values $F_0 = 0$, $F_1 = 1$ and the recursion relation

$$F_n = F_{n-1} + F_{n-2} \qquad \text{for } n \geq 2.$$

(a) Consider the generating function $F(x) = \sum_{n=0}^{\infty} F_n x^n$ associated to $\{F_n\}$, and prove that

$$F(x) = x^2 F(x) + xF(x) + x$$

for all x in a neighborhood of 0.

(b) Show that the polynomial $q(x) = 1 - x - x^2$ can be factored as

$$q(x) = (1 - \alpha x)(1 - \beta x),$$

where α and β are the roots of the polynomial $p(x) = x^2 - x - 1$.

(c) Expand the expression for F in partial fractions and obtain

$$F(x) = \frac{x}{1 - x - x^2} = \frac{x}{(1 - \alpha x)(1 - \beta x)} = \frac{A}{1 - \alpha x} + \frac{B}{1 - \beta x},$$

where $A = 1/(\alpha - \beta)$ and $B = 1/(\beta - \alpha)$.

(d) Conclude that $F_n = A\alpha^n + B\beta^n$ for $n \geq 0$. The two roots of p are actually

$$\alpha = \frac{1 + \sqrt{5}}{2} \qquad \text{and} \qquad \beta = \frac{1 - \sqrt{5}}{2},$$

so that $A = 1/\sqrt{5}$ and $B = -1/\sqrt{5}$.

The number $1/\alpha = (\sqrt{5} - 1)/2$, which is known as the **golden mean**, satisfies the following property: given a line segment $[AC]$ of unit length (Figure 2), there exists a unique point B on this segment so that the following proportion holds

$$\frac{AC}{AB} = \frac{AB}{BC}.$$

If $\ell = AB$, this reduces to the equation $\ell^2 + \ell - 1 = 0$, whose only positive solution is the golden mean. This ratio arises also in the construction of the regular pentagon. It has played a role in architecture and art, going back to the time of ancient Greece.

3. More generally, consider the difference equation given by the initial values u_0 and u_1, and the recurrence relation $u_n = au_{n-1} + bu_{n-2}$ for $n \geq 2$. Define the generating function associated to $\{u_n\}_{n=0}^{\infty}$ by $U(x) = \sum_{n=0}^{\infty} u_n x^n$. The recurrence relation implies that $U(x)(1 - ax - bx^2) = u_0 + (u_1 - au_0)x$ in a neighborhood of

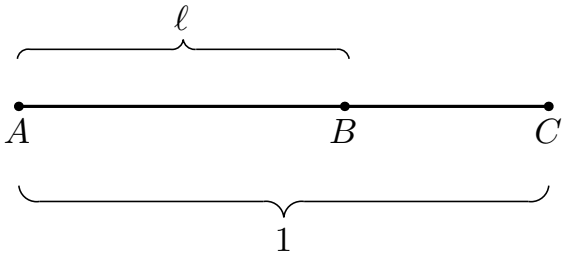


Figure 2. Appearance of the golden mean

the origin. If α and β denote the roots of the polynomial $p(x) = x^2 - ax - b$, then we may write

$$U(x) = \frac{u_0 + (u_1 - au_0)x}{(1 - \alpha x)(1 - \beta x)} = \frac{A}{1 - \alpha x} + \frac{B}{(1 - \beta x)} = A \sum_{n=0}^{\infty} \alpha^n x^n + B \sum_{n=0}^{\infty} \beta^n x^n,$$

where it is an easy matter to solve for A and B. Finally, this gives $u_n = A\alpha^n + B\beta^n$. Note that this approach yields a solution to our problem if the roots of p are distinct, namely $\alpha \neq \beta$. A variant of the formula holds if $\alpha = \beta$.

4. Using the generating formula for $p(n)$, prove the recurrence formula

$$\begin{aligned} p(n) &= p(n-1) + p(n-2) - p(n-5) - p(n-7) - \cdots \\ &= \sum_{k \neq 0} (-1)^{k+1} p\left(n - \frac{k(3k+1)}{2}\right), \end{aligned}$$

where the right-hand side is the finite sum taken over those $k \in \mathbb{Z}$, $k \neq 0$, with $k(3k+1)/2 \leq n$. Use this formula to calculate $p(5)$, $p(6)$, $p(7)$, $p(8)$, $p(9)$, and $p(10)$; check that $p(10) = 42$.

The next two exercises give elementary results related to the asymptotics of the partition function. More refined statements can be found in Appendix A.

5. Let

$$F(x) = \sum_{n=0}^{\infty} p(n)x^n = \prod_{n=1}^{\infty} \frac{1}{1-x^n}$$

be the generating function for the partitions. Show that

$$\log F(x) \sim \frac{\pi^2}{6(1-x)} \qquad \text{as } x \to 1, \text{ with } 0 < x < 1.$$

[Hint: Use $\log F(x) = \sum \log(1/(1-x^n))$ and $\log(1/(1-x^n)) = \sum (1/m)x^{nm}$, so

$$\log F(x) = \sum \frac{1}{m} \frac{x^m}{1-x^m}.$$

Use also $mx^{m-1}(1-x) < 1 - x^m < m(1-x)$.]

6. Show as a consequence of Exercise 5 that

$$e^{c_1 n^{1/2}} \leq p(n) \leq e^{c_2 n^{1/2}}$$

for two positive constants c_1 and c_2.
[Hint: $F(e^{-y}) = \sum p(n)e^{-ny} \leq Ce^{c/y}$ as $y \to 0$. So $p(n)e^{-ny} \leq ce^{c/y}$. Take $y = 1/n^{1/2}$ to get $p(n) \leq c'e^{c'n^{1/2}}$. In the opposite direction

$$\sum_{n=0}^{m} p(n)e^{-ny} \geq C(e^{c/y} - \sum_{n=m+1}^{\infty} e^{cn^{1/2}} e^{-ny}),$$

and it suffices to take $y = Am^{-1/2}$ where A is a large constant, and use the fact that the sequence $p(n)$ is increasing.]

7. Use the product formula for Θ to prove:

(a) The "triangular number" identity

$$\prod_{n=0}^{\infty}(1+x^n)(1-x^{2n+2}) = \sum_{n=-\infty}^{\infty} x^{n(n+1)/2},$$

which holds for $|x| < 1$.

(b) The "septagonal number" identity

$$\prod_{n=0}^{\infty}(1-x^{5n+1})(1-x^{5n+4})(1-x^{5n+5}) = \sum_{n=-\infty}^{\infty} (-1)^n x^{n(5n+3)/2},$$

which holds for $|x| < 1$.

8. Consider Pythagorean triples (a, b, c) with $a^2 + b^2 = c^2$, and with $a, b, c \in \mathbb{Z}$. Suppose moreover that a and b have no common factors.

(a) Show that either a or b must be odd, and the other even.

(b) Show in this case (assuming a is odd and b even) that there are integers m, n so that $a = m^2 - n^2$, $b = 2mn$, and $c = m^2 + n^2$. [Hint: Note that $b^2 = (c - a)(c + a)$, and prove that $(c - a)/2$ and $(c + a)/2$ are relatively prime integers.]

(c) Conversely, show that whenever c is a sum of two-squares, then there exist integers a and b such that $a^2 + b^2 = c^2$.

9. Use the formula for $r_2(n)$ to prove the following:

(a) If $n = p$, where p is a prime of the form $4k + 1$, then $r_2(n) = 8$. This implies that n can be written in a unique way as $n = n_1^2 + n_2^2$, except for the signs and reordering of n_1 and n_2.

(b) If $n = q^a$, where q is prime of the form $4k + 3$ and a is a positive integer, then $r_2(n) > 0$ if and only if a is even.

(c) In general, n can be represented as the sum of two squares if and only if all the primes of the form $4k + 3$ that arise in the prime decomposition of n occur with even exponents.

10. Observe the following irregularities of the functions $r_2(n)$ and $r_4(n)$ as n becomes large:

(a) $r_2(n) = 0$ for infinitely many n, while $\limsup_{n\to\infty} r_2(n) = \infty$.

(b) $r_4(n) = 24$ for infinitely many n while $\limsup_{n\to\infty} r_4(n)/n = \infty$.

[Hint: For (a) consider $n = 5^k$; for (b) consider alternatively $n = 2^k$, and $n = q^k$ where q is odd and large.]

11. Recall from Problem 2 in Chapter 2, that

$$\sum_{n=1}^{\infty} d(n)z^n = \sum_{n=1}^{\infty} \frac{z^n}{1-z^n}, \qquad |z| < 1$$

where $d(n)$ denotes the number of divisors of n.

More generally, show that

$$\sum_{n=1}^{\infty} \sigma_\ell(n)z^n = \sum_{n=1}^{\infty} \frac{n^\ell z^n}{1-z^n}, \qquad |z| < 1$$

where $\sigma_\ell(n)$ is the sum of the ℓth powers of divisors of n.

12. Here we give another identity involving θ^4, which is equivalent to the four-squares theorem.

(a) Show that for $|q| < 1$

$$\sum_{n=1}^{\infty} \frac{nq^n}{1-q^n} = \sum_{n=1}^{\infty} \frac{q^n}{(1-q^n)^2}.$$

[Hint: The left-hand side is $\sum \sigma_1(n) q^n$. Use $x/(1-x)^2 = \sum_{n=1}^{\infty} nx^n$.]

(b) Show as a result that

$$\sum_{n=1}^{\infty} \frac{nq^n}{1-q^n} - \sum_{n=1}^{\infty} \frac{4nq^{4n}}{1-q^{4n}} = \sum_{n=1}^{\infty} \frac{q^n}{(1-q^n)^2} - 4\sum_{n=1}^{\infty} \frac{q^{4n}}{(1-q^{4n})^2} = \sum \sigma_1^*(n) q^n$$

where $\sigma_1^*(n)$ is the sum of the divisors of d that are not divisible by 4.

(c) Show that the four-squares theorem is equivalent to the identity

$$\theta(\tau)^4 = 1 + 8\sum_{n=1}^{\infty} \frac{q^n}{(1+(-1)^n q^n)^2}, \qquad q = e^{\pi i \tau}.$$

Real Analysis: Measure Theory, Integration, and Hilbert Spaces

Chapter 1. Measure Theory

1. Prove that the Cantor set $\mathcal{C}$ constructed in the text is totally disconnected and perfect. In other words, given two distinct points $x, y \in \mathcal{C}$, there is a point $z \notin \mathcal{C}$ that lies between x and y, and yet $\mathcal{C}$ has no isolated points.

[Hint: If $x, y \in \mathcal{C}$ and $|x - y| > 1/3^k$, then x and y belong to two different intervals in C_k. Also, given any $x \in \mathcal{C}$ there is an end-point y_k of some interval in C_k that satisfies $x \neq y_k$ and $|x - y_k| \leq 1/3^k$.]

2. The Cantor set $\mathcal{C}$ can also be described in terms of ternary expansions.

(a) Every number in $[0,1]$ has a ternary expansion

$$x = \sum_{k=1}^{\infty} a_k 3^{-k}, \qquad \text{where } a_k = 0, 1, \text{ or } 2.$$

Note that this decomposition is not unique since, for example, $1/3 = \sum_{k=2}^{\infty} 2/3^k$.

Prove that $x \in \mathcal{C}$ if and only if x has a representation as above where every a_k is either 0 or 2.

(b) The **Cantor-Lebesgue function** is defined on $\mathcal{C}$ by

$$F(x) = \sum_{k=1}^{\infty} \frac{b_k}{2^k} \qquad \text{if } x = \textstyle\sum_{k=1}^{\infty} a_k 3^{-k}, \text{ where } b_k = a_k/2.$$

In this definition, we choose the expansion of x in which $a_k = 0$ or 2.

Show that F is well defined and continuous on $\mathcal{C}$, and moreover $F(0) = 0$ as well as $F(1) = 1$.

(c) Prove that $F : \mathcal{C} \to [0,1]$ is surjective, that is, for every $y \in [0,1]$ there exists $x \in \mathcal{C}$ such that $F(x) = y$.

(d) One can also extend F to be a continuous function on $[0,1]$ as follows. Note that if (a,b) is an open interval of the complement of $\mathcal{C}$, then $F(a) = F(b)$. Hence we may define F to have the constant value $F(a)$ in that interval.

A geometrical construction of F is described in Chapter 3.

3. Cantor sets of constant dissection. Consider the unit interval $[0,1]$, and let ξ be a fixed real number with $0 < \xi < 1$ (the case $\xi = 1/3$ corresponds to the Cantor set $\mathcal{C}$ in the text).

In stage 1 of the construction, remove the centrally situated open interval in $[0,1]$ of length ξ. In stage 2, remove two central intervals each of relative length ξ, one in each of the remaining intervals after stage 1, and so on.

Let $\mathcal{C}_\xi$ denote the set which remains after applying the above procedure indefinitely.[6]

(a) Prove that the complement of $\mathcal{C}_\xi$ in $[0,1]$ is the union of open intervals of total length equal to 1.

(b) Show directly that $m_*(\mathcal{C}_\xi) = 0$.

[Hint: After the kth stage, show that the remaining set has total length $= (1-\xi)^k$.]

[6]The set we call $\mathcal{C}_\xi$ is sometimes denoted by $\mathcal{C}_{\frac{1-\xi}{2}}$.

4. Cantor-like sets. Construct a closed set $\hat{\mathcal{C}}$ so that at the kth stage of the construction one removes 2^{k-1} centrally situated open intervals each of length ℓ_k, with

$$\ell_1 + 2\ell_2 + \cdots + 2^{k-1}\ell_k < 1.$$

(a) If ℓ_j are chosen small enough, then $\sum_{k=1}^{\infty} 2^{k-1}\ell_k < 1$. In this case, show that $m(\hat{\mathcal{C}}) > 0$, and in fact, $m(\hat{\mathcal{C}}) = 1 - \sum_{k=1}^{\infty} 2^{k-1}\ell_k$.

(b) Show that if $x \in \hat{\mathcal{C}}$, then there exists a sequence of points $\{x_n\}_{n=1}^{\infty}$ such that $x_n \notin \hat{\mathcal{C}}$, yet $x_n \to x$ and $x_n \in I_n$, where I_n is a sub-interval in the complement of $\hat{\mathcal{C}}$ with $|I_n| \to 0$.

(c) Prove as a consequence that $\hat{\mathcal{C}}$ is perfect, and contains no open interval.

(d) Show also that $\hat{\mathcal{C}}$ is uncountable.

5. Suppose E is a given set, and $\mathcal{O}_n$ is the open set:

$$\mathcal{O}_n = \{x : \ d(x, E) < 1/n\}.$$

Show:

(a) If E is compact, then $m(E) = \lim_{n\to\infty} m(\mathcal{O}_n)$.

(b) However, the conclusion in (a) may be false for E closed and unbounded; or E open and bounded.

6. Using translations and dilations, prove the following: Let B be a ball in $\mathbb{R}^d$ of radius r. Then $m(B) = v_d r^d$, where $v_d = m(B_1)$, and B_1 is the unit ball, $B_1 = \{x \in \mathbb{R}^d : \ |x| < 1\}$.

A calculation of the constant v_d is postponed until Exercise 14 in the next chapter.

7. If $\delta = (\delta_1, \ldots, \delta_d)$ is a d-tuple of positive numbers $\delta_i > 0$, and E is a subset of $\mathbb{R}^d$, we define δE by

$$\delta E = \{(\delta_1 x_1, \ldots, \delta_d x_d) : \text{ where } (x_1, \ldots, x_d) \in E\}.$$

Prove that δE is measurable whenever E is measurable, and

$$m(\delta E) = \delta_1 \cdots \delta_d m(E).$$

8. Suppose L is a linear transformation of $\mathbb{R}^d$. Show that if E is a measurable subset of $\mathbb{R}^d$, then so is $L(E)$, by proceeding as follows:

(a) Note that if E is compact, so is $L(E)$. Hence if E is an F_σ set, so is $L(E)$.

(b) Because L automatically satisfies the inequality

$$|L(x) - L(x')| \leq M|x - x'|$$

for some M, we can see that L maps any cube of side length ℓ into a cube of side length $c_d M\ell$, with $c_d = 2\sqrt{d}$. Now if $m(E) = 0$, there is a collection of cubes $\{Q_j\}$ such that $E \subset \bigcup_j Q_j$, and $\sum_j m(Q_j) < \epsilon$. Thus $m_*(L(E)) \leq c'\epsilon$, and hence $m(L(E)) = 0$. Finally, use Corollary 3.5.

One can show that $m(L(E)) = |\det L|\, m(E)$; see Problem 4 in the next chapter.

9. Give an example of an open set $\mathcal{O}$ with the following property: the boundary of the closure of $\mathcal{O}$ has positive Lebesgue measure.

[Hint: Consider the set obtained by taking the union of open intervals which are deleted at the odd steps in the construction of a Cantor-like set.]

10. This exercise provides a construction of a decreasing sequence of positive continuous functions on the interval $[0, 1]$, whose pointwise limit is *not* Riemann integrable.

Let $\hat{\mathcal{C}}$ denote a Cantor-like set obtained from the construction detailed in Exercise 4, so that in particular $m(\hat{\mathcal{C}}) > 0$. Let F_1 denote a piecewise-linear and continuous function on $[0, 1]$, with $F_1 = 1$ in the complement of the first interval removed in the construction of $\hat{\mathcal{C}}$, $F_1 = 0$ at the center of this interval, and $0 \leq F_1(x) \leq 1$ for all x. Similarly, construct $F_2 = 1$ in the complement of the intervals in stage two of the construction of $\hat{\mathcal{C}}$, with $F_2 = 0$ at the center of these intervals, and $0 \leq F_2 \leq 1$. Continuing this way, let $f_n = F_1 \cdot F_2 \cdots F_n$ (see Figure 5).

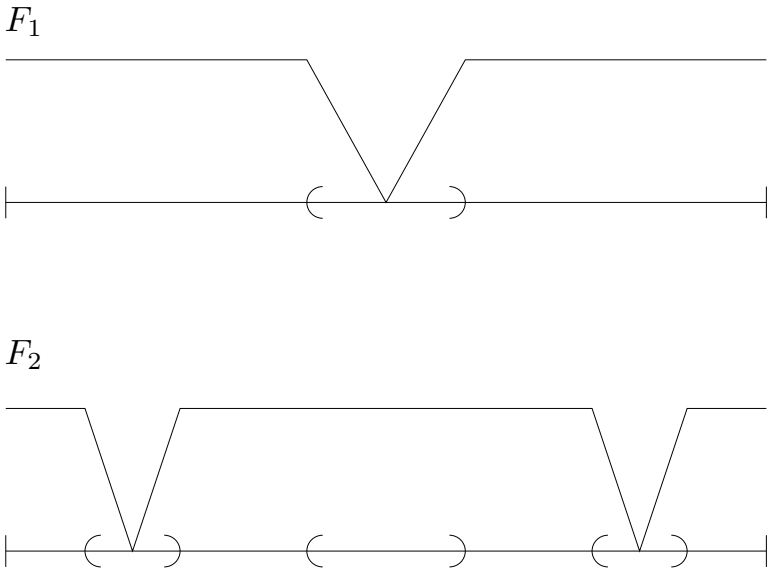


Figure 5. Construction of $\{F_n\}$ in Exercise 10

Prove the following:

(a) For all $n \geq 1$ and all $x \in [0, 1]$, one has $0 \leq f_n(x) \leq 1$ and $f_n(x) \geq f_{n+1}(x)$. Therefore, $f_n(x)$ converges to a limit as $n \to \infty$ which we denote by $f(x)$.

(b) The function f is discontinuous at every point of $\hat{\mathcal{C}}$.

[Hint: Note that $f(x) = 1$ if $x \in \hat{\mathcal{C}}$, and find a sequence of points $\{x_n\}$ so that $x_n \to x$ and $f(x_n) = 0$.]

Now $\int f_n(x)\,dx$ is decreasing, hence $\int f_n$ converges. However, a bounded function is Riemann integrable if and only if its set of discontinuities has measure zero.

(The proof of this fact, which is given in the Appendix of Book I, is outlined in Problem 4.) Since f is discontinuous on a set of positive measure, we find that f is not Riemann integrable.

11. Let A be the subset of $[0, 1]$ which consists of all numbers which do not have the digit 4 appearing in their decimal expansion. Find $m(A)$.

12. Theorem 1.3 states that every open set in $\mathbb{R}$ is the disjoint union of open intervals. The analogue in $\mathbb{R}^d$, $d \geq 2$, is generally false. Prove the following:

(a) An open disc in $\mathbb{R}^2$ is not the disjoint union of open rectangles.

[Hint: What happens to the boundary of any of these rectangles?]

(b) An open connected set Ω is the disjoint union of open rectangles if and only if Ω is itself an open rectangle.

13. The following deals with G_δ and F_σ sets.

(a) Show that a closed set is a G_δ and an open set an F_σ.
[Hint: If F is closed, consider $\mathcal{O}_n = \{x : d(x, F) < 1/n\}$.]

(b) Give an example of an F_σ which is not a G_δ.
[Hint: This is more difficult; let F be a denumerable set that is dense.]

(c) Give an example of a Borel set which is not a G_δ nor an F_σ.

14. The purpose of this exercise is to show that covering by a *finite* number of intervals will not suffice in the definition of the outer measure m_*.

The **outer Jordan content** $J_*(E)$ of a set E in $\mathbb{R}$ is defined by

$$J_*(E) = \inf \sum_{j=1}^{N} |I_j|,$$

where the inf is taken over every *finite* covering $E \subset \bigcup_{j=1}^{N} I_j$, by intervals I_j.

(a) Prove that $J_*(E) = J_*(\overline{E})$ for every set E (here $\overline{E}$ denotes the closure of E).

(b) Exhibit a countable subset $E \subset [0,1]$ such that $J_*(E) = 1$ while $m_*(E) = 0$.

15. At the start of the theory, one might define the outer measure by taking coverings by rectangles instead of cubes. More precisely, we define

$$m_*^{\mathcal{R}}(E) = \inf \sum_{j=1}^{\infty} |R_j|,$$

where the inf is now taken over all countable coverings $E \subset \bigcup_{j=1}^{\infty} R_j$ by (closed) rectangles.

Show that this approach gives rise to the same theory of measure developed in the text, by proving that $m_*(E) = m_*^{\mathcal{R}}(E)$ for every subset E of $\mathbb{R}^d$.

[Hint: Use Lemma 1.1.]

16. The Borel-Cantelli lemma. Suppose $\{E_k\}_{k=1}^{\infty}$ is a countable family of measuable subsets of $\mathbb{R}^d$ and that

$$\sum_{k=1}^{\infty} m(E_k) < \infty.$$

Let

$$\begin{aligned} E &= \{x \in \mathbb{R}^d : x \in E_k, \text{ for infinitely many } k\} \\ &= \limsup_{k\to\infty}(E_k). \end{aligned}$$

(a) Show that E is measurable.

(b) Prove $m(E) = 0$.

[Hint: Write $E = \bigcap_{n=1}^{\infty} \bigcup_{k\geq n} E_k$.]

17. Let $\{f_n\}$ be a sequence of measurable functions on $[0,1]$ with $|f_n(x)| < \infty$ for a.e x. Show that there exists a sequence c_n of positive real numbers such that

$$\frac{f_n(x)}{c_n} \to 0 \qquad \text{a.e. } x$$

[Hint: Pick c_n such that $m(\{x : |f_n(x)/c_n| > 1/n\}) < 2^{-n}$, and apply the Borel-Cantelli lemma.]

18. Prove the following assertion: Every measurable function is the limit a.e. of a sequence of continuous functions.

19. Here are some observations regarding the set operation $A + B$.

(a) Show that if either A and B is open, then $A + B$ is open.

(b) Show that if A and B are closed, then $A + B$ is measurable.

(c) Show, however, that $A + B$ might not be closed even though A and B are closed.

[Hint: For (b) show that $A + B$ is an F_σ set.]

20. Show that there exist closed sets A and B with $m(A) = m(B) = 0$, but $m(A + B) > 0$:

(a) In $\mathbb{R}$, let $A = \mathcal{C}$ (the Cantor set), $B = \mathcal{C}/2$. Note that $A + B \supset [0, 1]$.

(b) In $\mathbb{R}^2$, observe that if $A = I \times \{0\}$ and $B = \{0\} \times I$ (where $I = [0, 1]$), then $A + B = I \times I$.

21. Prove that there is a continuous function that maps a Lebesgue measurable set to a non-measurable set.

[Hint: Consider a non-measurable subset of $[0, 1]$, and its inverse image in $\mathcal{C}$ by the function F in Exercise 2.]

22. Let $\chi_{[0,1]}$ be the characteristic function of $[0,1]$. Show that there is no everywhere continuous function f on $\mathbb{R}$ such that

$$f(x) = \chi_{[0,1]}(x) \qquad \text{almost everywhere.}$$

23. Suppose $f(x, y)$ is a function on $\mathbb{R}^2$ that is separately continuous: for each fixed variable, f is continuous in the other variable. Prove that f is measurable on $\mathbb{R}^2$.

[Hint: Approximate f in the variable x by piecewise-linear functions f_n so that $f_n \to f$ pointwise.]

24. Does there exist an enumeration $\{r_n\}_{n=1}^{\infty}$ of the rationals, such that the complement of

$$\bigcup_{n=1}^{\infty}\left(r_n - \frac{1}{n}, r_n + \frac{1}{n}\right)$$

in $\mathbb{R}$ is non-empty?

[Hint: Find an enumeration where the only rationals outside of a fixed bounded interval take the form r_n, with $n = m^2$ for some integer m.]

25. An alternative definition of measurability is as follows: E is measurable if for every $\epsilon > 0$ there is a *closed* set F contained in E with $m_*(E - F) < \epsilon$. Show that this definition is equivalent with the one given in the text.

26. Suppose $A \subset E \subset B$, where A and B are measurable sets of finite measure. Prove that if $m(A) = m(B)$, then E is measurable.

27. Suppose E_1 and E_2 are a pair of compact sets in $\mathbb{R}^d$ with $E_1 \subset E_2$, and let $a = m(E_1)$ and $b = m(E_2)$. Prove that for any c with $a < c < b$, there is a compact set E with $E_1 \subset E \subset E_2$ and $m(E) = c$.

[Hint: As an example, if $d = 1$ and E is a measurable subset of $[0, 1]$, consider $m(E \cap [0, t])$ as a function of t.]

28. Let E be a subset of $\mathbb{R}$ with $m_*(E) > 0$. Prove that for each $0 < \alpha < 1$, there exists an open interval I so that

$$m_*(E \cap I) \geq \alpha\, m_*(I).$$

Loosely speaking, this estimate shows that E contains almost a whole interval.

[Hint: Choose an open set $\mathcal{O}$ that contains E, and such that $m_*(E) \geq \alpha\, m_*(\mathcal{O})$. Write $\mathcal{O}$ as the countable union of disjoint open intervals, and show that one of these intervals must satisfy the desired property.]

29. Suppose E is a measurable subset of $\mathbb{R}$ with $m(E) > 0$. Prove that the **difference set** of E, which is defined by

$$\{z \in \mathbb{R} : \ z = x - y \text{ for some } x, y \in E\},$$

contains an open interval centered at the origin.

If E contains an interval, then the conclusion is straightforward. In general, one may rely on Exercise 28.

[Hint: Indeed, by Exercise 28, there exists an open interval I so that $m(E \cap I) \geq (9/10)\, m(I)$. If we denote $E \cap I$ by E_0, and suppose that the difference set of E_0 does not contain an open interval around the origin, then for arbitrarily small a the sets E_0, and $E_0 + a$ are disjoint. From the fact that $(E_0 \cup (E_0 + a)) \subset (I \cup (I + a))$ we get a contradiction, since the left-hand side has measure $2m(E_0)$, while the right-hand side has measure only slightly larger than $m(I)$.]

A more general formulation of this result is as follows.

30. If E and F are measurable, and $m(E) > 0$, $m(F) > 0$, prove that

$$E + F = \{x + y : x \in E,\ x \in F\}$$

contains an interval.

31. The result in Exercise 29 provides an alternate proof of the non-measurability of the set $\mathcal{N}$ studied in the text. In fact, we may also prove the non-measurability of a set in $\mathbb{R}$ that is very closely related to the set $\mathcal{N}$.

Given two real numbers x and y, we shall write as before that $x \sim y$ whenever the difference $x - y$ is rational. Let $\mathcal{N}^*$ denote a set that consists of one element in each equivalence class of $\sim$. Prove that $\mathcal{N}^*$ is non-measurable by using the result in Exercise 29.

[Hint: If $\mathcal{N}^*$ is measurable, then so are its translates $\mathcal{N}_n^* = \mathcal{N}^* + r_n$, where $\{r_n\}_{n=1}^{\infty}$ is an enumeration of $\mathbb{Q}$. How does this imply that $m(\mathcal{N}^*) > 0$? Can the difference set of $\mathcal{N}^*$ contain an open interval centered at the origin?]

32. Let $\mathcal{N}$ denote the non-measurable subset of $I = [0, 1]$ constructed at the end of Section 3.

(a) Prove that if E is a measurable subset of $\mathcal{N}$, then $m(E) = 0$.

(b) If G is a subset of $\mathbb{R}$ with $m_*(G) > 0$, prove that a subset of G is non-measurable.

[Hint: For (a) use the translates of E by the rationals.]

33. Let $\mathcal{N}$ denote the non-measurable set constructed in the text. Recall from the exercise above that measurable subsets of $\mathcal{N}$ have measure zero.

Show that the set $\mathcal{N}^c = I - \mathcal{N}$ satisfies $m_*(\mathcal{N}^c) = 1$, and conclude that if $E_1 = \mathcal{N}$ and $E_2 = \mathcal{N}^c$, then

$$m_*(E_1) + m_*(E_2) \neq m_*(E_1 \cup E_2),$$

although E_1 and E_2 are disjoint.

[Hint: To prove that $m_*(\mathcal{N}^c) = 1$, argue by contradiction and pick a measurable set U such that $U \subset I$, $\mathcal{N}^c \subset U$ and $m_*(U) < 1 - \epsilon$.]

34. Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be any two Cantor sets (constructed in Exercise 3). Show that there exists a function $F : [0,1] \to [0,1]$ with the following properties:

(i) F is continuous and bijective,

(ii) F is monotonically increasing,

(iii) F maps $\mathcal{C}_1$ surjectively onto $\mathcal{C}_2$.

[Hint: Copy the construction of the standard Cantor-Lebesgue function.]

35. Give an example of a measurable function f and a continuous function Φ so that $f \circ \Phi$ is non-measurable.

[Hint: Let $\Phi : \mathcal{C}_1 \to \mathcal{C}_2$ as in Exercise 34, with $m(\mathcal{C}_1) > 0$ and $m(\mathcal{C}_2) = 0$. Let $N \subset \mathcal{C}_1$ be non-measurable, and take $f = \chi_{\Phi(N)}$.]

Use the construction in the hint to show that there exists a Lebesgue measurable set that is not a Borel set.

36. This exercise provides an example of a measurable function f on $[0,1]$ such that every function g equivalent to f (in the sense that f and g differ only on a set of measure zero) is discontinuous at *every* point.

(a) Construct a measurable set $E \subset [0,1]$ such that for any non-empty open sub-interval I in $[0,1]$, both sets $E \cap I$ and $E^c \cap I$ have positive measure.

(b) Show that $f = \chi_E$ has the property that whenever $g(x) = f(x)$ a.e x, then g must be discontinuous at every point in $[0,1]$.

[Hint: For the first part, consider a Cantor-like set of positive measure, and add in each of the intervals that are omitted in the first step of its construction, another Cantor-like set. Continue this procedure indefinitely.]

37. Suppose Γ is a curve $y = f(x)$ in $\mathbb{R}^2$, where f is continuous. Show that $m(\Gamma) = 0$.

[Hint: Cover Γ by rectangles, using the uniform continuity of f.]

38. Prove that $(a+b)^\gamma \geq a^\gamma + b^\gamma$ whenever $\gamma \geq 1$ and $a, b \geq 0$. Also, show that the reverse inequality holds when $0 \leq \gamma \leq 1$.

[Hint: Integrate the inequality between $(a+t)^{\gamma-1}$ and $t^{\gamma-1}$ from 0 to b.]

39. Establish the inequality

$$\frac{x_1 + \cdots + x_d}{d} \geq (x_1 \cdots x_d)^{1/d} \qquad \text{for all } x_j \geq 0,\ j = 1, \ldots, d \tag{10}$$

by using backward induction as follows:

(a) The inequality is true whenever d is a power of 2 ($d = 2^k$, $k \geq 1$).

(b) If (10) holds for some integer $d \geq 2$, then it must hold for $d - 1$, that is, one has $(y_1 + \cdots + y_{d-1})/(d-1) \geq (y_1 \cdots y_{d-1})^{1/(d-1)}$ for all $y_j \geq 0$, with $j = 1, \ldots, d-1$.

[Hint: For (a), if $k \geq 2$, write $(x_1 + \cdots + x_{2^k})/2^k$ as $(A + B)/2$, where $A = (x_1 + \cdots + x_{2^{k-1}})/2^{k-1}$, and apply the inequality when $d = 2$. For (b), apply the inequality to $x_1 = y_1, \ldots, x_{d-1} = y_{d-1}$ and $x_d = (y_1 + \cdots + y_{d-1})/(d-1)$.]

Chapter 2. Integration Theory

1. Given a collection of sets $F_1, F_2, \ldots, F_n$, construct another collection $F_1^*, F_2^*, \ldots, F_N^*$, with $N = 2^n - 1$, so that $\bigcup_{k=1}^n F_k = \bigcup_{j=1}^N F_j^*$; the collection $\{F_j^*\}$ is disjoint; also

$F_k = \bigcup_{F_j^* \subset F_k} F_j^*$, for every k.

[Hint: Consider the 2^n sets $F_1' \cap F_2' \cap \cdots \cap F_n'$ where each F_k' is either F_k or F_k^c.]

2. In analogy to Proposition 2.5, prove that if f is integrable on $\mathbb{R}^d$ and $\delta > 0$, then $f(\delta x)$ converges to $f(x)$ in the L^1-norm as $\delta \to 1$.

3. Suppose f is integrable on $(-\pi, \pi]$ and extended to $\mathbb{R}$ by making it periodic of period 2π. Show that

$$\int_{-\pi}^{\pi} f(x)\,dx = \int_{I} f(x)\,dx,$$

where I is any interval in $\mathbb{R}$ of length 2π.

[Hint: I is contained in two consecutive intervals of the form $(k\pi, (k+2)\pi)$.]

4. Suppose f is integrable on $[0, b]$, and

$$g(x) = \int_x^b \frac{f(t)}{t}\, dt \qquad \text{for } 0 < x \leq b.$$

Prove that g is integrable on $[0, b]$ and

$$\int_0^b g(x)\, dx = \int_0^b f(t)\, dt.$$

5. Suppose F is a closed set in $\mathbb{R}$, whose complement has finite measure, and let $\delta(x)$ denote the distance from x to F, that is,

$$\delta(x) = d(x, F) = \inf\{|x - y| : \ y \in F\}.$$

Consider

$$I(x) = \int_{\mathbb{R}} \frac{\delta(y)}{|x - y|^2}\, dy.$$

(a) Prove that δ is continuous, by showing that it satisfies the Lipschitz condition

$$|\delta(x) - \delta(y)| \leq |x - y|.$$

(b) Show that $I(x) = \infty$ for each $x \notin F$.

(c) Show that $I(x) < \infty$ for a.e. $x \in F$. This may be surprising in view of the fact that the Lispshitz condition cancels only one power of $|x - y|$ in the integrand of I.

[Hint: For the last part, investigate $\int_F I(x)\, dx$.]

6. Integrability of f on $\mathbb{R}$ does not necessarily imply the convergence of $f(x)$ to 0 as $x \to \infty$.

(a) There exists a positive continuous function f on $\mathbb{R}$ so that f is integrable on $\mathbb{R}$, but yet $\limsup_{x\to\infty} f(x) = \infty$.

(b) However, if we assume that f is uniformly continuous on $\mathbb{R}$ and integrable, then $\lim_{|x|\to\infty} f(x) = 0$.

[Hint: For (a), construct a continuous version of the function equal to n on the segment $[n, n + 1/n^3)$, $n \geq 1$.]

7. Let $\Gamma \subset \mathbb{R}^d \times \mathbb{R}$, $\Gamma = \{(x, y) \in \mathbb{R}^d \times \mathbb{R} : \ y = f(x)\}$, and assume f is measurable on $\mathbb{R}^d$. Show that Γ is a measurable subset of $\mathbb{R}^{d+1}$, and $m(\Gamma) = 0$.

8. If f is integrable on $\mathbb{R}$, show that $F(x) = \int_{-\infty}^{x} f(t)\,dt$ is uniformly continuous.

9. Tchebychev inequality. Suppose $f \geq 0$, and f is integrable. If $\alpha > 0$ and $E_\alpha = \{x : f(x) > \alpha\}$, prove that

$$m(E_\alpha) \leq \frac{1}{\alpha} \int f.$$

10. Suppose $f \geq 0$, and let $E_{2^k} = \{x : f(x) > 2^k\}$ and $F_k = \{x : 2^k < f(x) \leq 2^{k+1}\}$. If f is finite almost everywhere, then

$$\bigcup_{k=-\infty}^{\infty} F_k = \{f(x) > 0\},$$

and the sets F_k are disjoint.

Prove that f is integrable if and only if

$$\sum_{k=-\infty}^{\infty} 2^k m(F_k) < \infty, \qquad \text{if and only if} \qquad \sum_{k=-\infty}^{\infty} 2^k m(E_{2^k}) < \infty.$$

Use this result to verify the following assertions. Let

$$f(x) = \begin{cases} |x|^{-a} & \text{if } |x| \leq 1, \\ 0 & \text{otherwise,} \end{cases} \qquad \text{and} \qquad g(x) = \begin{cases} |x|^{-b} & \text{if } |x| > 1, \\ 0 & \text{otherwise.} \end{cases}$$

Then f is integrable on $\mathbb{R}^d$ if and only if $a < d$; also g is integrable on $\mathbb{R}^d$ if and only if $b > d$.

11. Prove that if f is integrable on $\mathbb{R}^d$, real-valued, and $\int_E f(x)\,dx \geq 0$ for every measurable E, then $f(x) \geq 0$ a.e. x. As a result, if $\int_E f(x)\,dx = 0$ for every measurable E, then $f(x) = 0$ a.e.

12. Show that there are $f \in L^1(\mathbb{R}^d)$ and a sequence $\{f_n\}$ with $f_n \in L^1(\mathbb{R}^d)$ such that

$$\|f - f_n\|_{L^1} \to 0,$$

but $f_n(x) \to f(x)$ for no x.

[Hint: In $\mathbb{R}$, let $f_n = \chi_{I_n}$, where I_n is an appropriately chosen sequence of intervals with $m(I_n) \to 0$.]

13. Give an example of two measurable sets A and B such that $A + B$ is not measurable.

[Hint: In $\mathbb{R}^2$ take $A = \{0\} \times [0, 1]$ and $B = \mathcal{N} \times \{0\}$.]

14. In Exercise 6 of the previous chapter we saw that $m(B) = v_d r^d$, whenever B is a ball of radius r in $\mathbb{R}^d$ and $v_d = m(B_1)$, with B_1 the unit ball. Here we evaluate the constant v_d.

(a) For $d = 2$, prove using Corollary 3.8 that

$$v_2 = 2 \int_{-1}^{1} (1 - x^2)^{1/2} \, dx,$$

and hence by elementary calculus, that $v_2 = \pi$.

(b) By similar methods, show that

$$v_d = 2v_{d-1} \int_{0}^{1} (1 - x^2)^{(d-1)/2} \, dx.$$

(c) The result is

$$v_d = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)}.$$

Another derivation is in Exercise 5 in Chapter 6 below. Relevant facts about the gamma and beta functions can be found in Chapter 6 of Book II.

15. Consider the function defined over $\mathbb{R}$ by

$$f(x) = \begin{cases} x^{-1/2} & \text{if } 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

For a fixed enumeration $\{r_n\}_{n=1}^{\infty}$ of the rationals $\mathbb{Q}$, let

$$F(x) = \sum_{n=1}^{\infty} 2^{-n} f(x - r_n).$$

Prove that F is integrable, hence the series defining F converges for almost every $x \in \mathbb{R}$. However, observe that this series is unbounded on every interval, and in fact, any function $\tilde{F}$ that agrees with F a.e is unbounded in any interval.

16. Suppose f is integrable on $\mathbb{R}^d$. If $\delta = (\delta_1, \ldots, \delta_d)$ is a d-tuple of non-zero real numbers, and

$$f^\delta(x) = f(\delta x) = f(\delta_1 x_1, \ldots, \delta_d x_d),$$

show that f^δ is integrable with

$$\int_{\mathbb{R}^d} f^\delta(x)\,dx = |\delta_1|^{-1} \cdots |\delta_d|^{-1} \int_{\mathbb{R}^d} f(x)\,dx.$$

17. Suppose f is defined on $\mathbb{R}^2$ as follows: $f(x,y) = a_n$ if $n \le x < n+1$ and $n \le y < n+1$, $(n \ge 0)$; $f(x,y) = -a_n$ if $n \le x < n+1$ and $n+1 \le y < n+2$, $(n \ge 0)$; while $f(x,y) = 0$ elsewhere. Here $a_n = \sum_{k \le n} b_k$, with $\{b_k\}$ a positive sequence such that $\sum_{k=0}^{\infty} b_k = s < \infty$.

(a) Verify that each slice f^y and f_x is integrable. Also for all x, $\int f_x(y)\,dy = 0$, and hence $\int \left(\int f(x,y)\,dy \right)\,dx = 0$.

(b) However, $\int f^y(x)\,dx = a_0$ if $0 \le y < 1$, and $\int f^y(x)\,dx = a_n - a_{n-1}$ if $n \le y < n+1$ with $n \ge 1$. Hence $y \mapsto \int f^y(x)\,dx$ is integrable on $(0,\infty)$ and

$$\int \left(\int f(x,y)\,dx \right)\,dy = s.$$

(c) Note that $\int_{\mathbb{R}\times\mathbb{R}} |f(x,y)|\,dx\,dy = \infty$.

18. Let f be a measurable finite-valued function on $[0,1]$, and suppose that $|f(x) - f(y)|$ is integrable on $[0,1] \times [0,1]$. Show that $f(x)$ is integrable on $[0,1]$.

19. Suppose f is integrable on $\mathbb{R}^d$. For each $\alpha > 0$, let $E_\alpha = \{x : |f(x)| > \alpha\}$. Prove that

$$\int_{\mathbb{R}^d} |f(x)|\, dx = \int_0^\infty m(E_\alpha)\, d\alpha.$$

20. The problem (highlighted in the discussion preceding Fubini's theorem) that certain slices of measurable sets can be non-measurable may be avoided by restricting attention to Borel measurable functions and Borel sets. In fact, prove the following:

Suppose E is a Borel set in $\mathbb{R}^2$. Then for every y, the slice E^y is a Borel set in $\mathbb{R}$.

[Hint: Consider the collection $\mathcal{C}$ of subsets E of $\mathbb{R}^2$ with the property that each slice E^y is a Borel set in $\mathbb{R}$. Verify that $\mathcal{C}$ is a σ-algebra that contains the open sets.]

21. Suppose that f and g are measurable functions on $\mathbb{R}^d$.

(a) Prove that $f(x-y)g(y)$ is measurable on $\mathbb{R}^{2d}$.

(b) Show that if f and g are integrable on $\mathbb{R}^d$, then $f(x-y)g(y)$ is integrable on $\mathbb{R}^{2d}$.

(c) Recall the definition of the convolution of f and g given by

$$(f*g)(x) = \int_{\mathbb{R}^d} f(x-y)g(y)\,dy.$$

Show that $f*g$ is well defined for a.e. x (that is, $f(x-y)g(y)$ is integrable on $\mathbb{R}^d$ for a.e. x).

(d) Show that $f*g$ is integrable whenever f and g are integrable, and that

$$\|f*g\|_{L^1(\mathbb{R}^d)} \leq \|f\|_{L^1(\mathbb{R}^d)}\,\|g\|_{L^1(\mathbb{R}^d)},$$

with equality if f and g are non-negative.

(e) The Fourier transform of an integrable function f is defined by

$$\hat{f}(\xi) = \int_{\mathbb{R}^d} f(x)e^{-2\pi i x\cdot\xi}\,dx.$$

Check that $\hat{f}$ is bounded and is a continuous function of ξ. Prove that for each ξ one has

$$\widehat{(f*g)}(\xi) = \hat{f}(\xi)\hat{g}(\xi).$$

22. Prove that if $f \in L^1(\mathbb{R}^d)$ and

$$\hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-2\pi i x\xi}\, dx,$$

then $\hat{f}(\xi) \to 0$ as $|\xi| \to \infty$. (This is the Riemann-Lebesgue lemma.)
[Hint: Write $\hat{f}(\xi) = \frac{1}{2}\int_{\mathbb{R}^d}[f(x) - f(x-\xi')]e^{-2\pi i x\xi}\, dx$, where $\xi' = \frac{1}{2}\frac{\xi}{|\xi|^2}$, and use Proposition 2.5.]

23. As an application of the Fourier transform, show that there does not exist a function $I \in L^1(\mathbb{R}^d)$ such that

$$f * I = f \qquad \text{for all } f \in L^1(\mathbb{R}^d).$$

24. Consider the convolution

$$(f * g)(x) = \int_{\mathbb{R}^d} f(x - y)g(y)\, dy.$$

(a) Show that $f * g$ is uniformly continuous when f is integrable and g bounded.

(b) If in addition g is integrable, prove that $(f * g)(x) \to 0$ as $|x| \to \infty$.

25. Show that for each $\epsilon > 0$ the function $F(\xi) = \frac{1}{(1+|\xi|^2)^\epsilon}$ is the Fourier transform of an L^1 function.
[Hint: With $K_\delta(x) = e^{-\pi|x|^2/\delta}\delta^{-d/2}$ consider $f(x) = \int_0^\infty K_\delta(x)e^{-\pi\delta}\delta^{\epsilon-1}\,d\delta$. Use Fubini's theorem to prove $f \in L^1(\mathbb{R}^d)$, and

$$\hat{f}(\xi) = \int_0^\infty e^{-\pi\delta|\xi|^2}e^{-\pi\delta}\delta^{\epsilon-1}\,d\delta,$$

and evaluate the last integral as $\pi^{-\epsilon}\Gamma(\epsilon)\frac{1}{(1+|\xi|^2)^\epsilon}$. Here $\Gamma(s)$ is the gamma function defined by $\Gamma(s) = \int_0^\infty e^{-t}t^{s-1}\,dt$.]

Chapter 3. Differentiation and Integration

1. Suppose φ is an integrable function on $\mathbb{R}^d$ with $\int_{\mathbb{R}^d} \varphi(x)\,dx = 1$. Set $K_\delta(x) = \delta^{-d}\varphi(x/\delta)$, $\delta > 0$.

(a) Prove that $\{K_\delta\}_{\delta>0}$ is a family of good kernels.

(b) Assume in addition that φ is bounded and supported in a bounded set. Verify that $\{K_\delta\}_{\delta>0}$ is an approximation to the identity.

(c) Show that Theorem 2.3 (convergence in the L^1-norm) holds for good kernels as well.

2. Suppose $\{K_\delta\}$ is a family of kernels that satisfies:

(i) $|K_\delta(x)| \le A\delta^{-d}$ for all $\delta > 0$.

(ii) $|K_\delta(x)| \le A\delta/|x|^{d+1}$ for all $\delta > 0$.

(iii) $\int_{-\infty}^{\infty} K_\delta(x)\,dx = 0$ for all $\delta > 0$.

Thus K_δ satisfies conditions (i) and (ii) of approximations to the identity, but the average value of K_δ is 0 instead of 1. Show that if f is integrable on $\mathbb{R}^d$, then

$$(f * K_\delta)(x) \to 0 \qquad \text{for a.e. } x, \text{ as } \delta \to 0.$$

3. Suppose 0 is a point of (Lebesgue) density of the set $E \subset \mathbb{R}$. Show that for each of the individual conditions below there is an infinite sequence of points $x_n \in E$, with $x_n \neq 0$, and $x_n \to 0$ as $n \to \infty$.

(a) The sequence also satisfies $-x_n \in E$ for all n.

(b) In addition, $2x_n$ belongs to E for all n.

Generalize.

4. Prove that if f is integrable on $\mathbb{R}^d$, and f is not identically zero, then

$$f^*(x) \geq \frac{c}{|x|^d}, \qquad \text{for some } c > 0 \text{ and all } |x| \geq 1.$$

Conclude that f^* is not integrable on $\mathbb{R}^d$. Then, show that the weak type estimate

$$m(\{x : \ f^*(x) > \alpha\}) \leq c/\alpha$$

for all $\alpha > 0$ whenever $\int |f| = 1$, is best possible in the following sense: if f is supported in the unit ball with $\int |f| = 1$, then

$$m(\{x : \ f^*(x) > \alpha\}) \geq c'/\alpha$$

for some $c' > 0$ and all sufficiently small α.
[Hint: For the first part, use the fact that $\int_B |f| > 0$ for some ball B.]

5. Consider the function on $\mathbb{R}$ defined by

$$f(x) = \begin{cases} \dfrac{1}{|x|(\log 1/|x|)^2} & \text{if } |x| \le 1/2, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Verify that f is integrable.

(b) Establish the inequality

$$f^*(x) \ge \frac{c}{|x|(\log 1/|x|)} \qquad \text{for some } c > 0 \text{ and all } |x| \le 1/2,$$

to conclude that the maximal function f^* is not locally integrable.

6. In one dimension there is a version of the basic inequality (1) for the maximal function in the form of an identity. We define the "one-sided" maximal function

$$f_+^*(x) = \sup_{h>0} \frac{1}{h} \int_x^{x+h} |f(y)|\, dy.$$

If $E_\alpha^+ = \{x \in \mathbb{R} : \ f_+^*(x) > \alpha\}$, then

$$m(E_\alpha^+) = \frac{1}{\alpha} \int_{E_\alpha^+} |f(y)|\, dy.$$

[Hint: Apply Lemma 3.5 to $F(x) = \int_0^x |f(y)|\, dy - \alpha x$. Then E_α^+ is the union of disjoint intervals (a_k, b_k) with $\int_{a_k}^{b_k} |f(y)|\, dy = \alpha(a_k - b_k)$.]

7. Using Corollary 1.5, prove that if a measurable subset E of $[0,1]$ satisfies $m(E \cap I) \geq \alpha\, m(I)$ for some $\alpha > 0$ and all intervals I in $[0,1]$, then E has measure 1. See also Exercise 28 in Chapter 1.

8. Suppose A is a Lebesgue measurable set in $\mathbb{R}$ with $m(A) > 0$. Does there exist a sequence $\{s_n\}_{n=1}^{\infty}$ such that the complement of $\bigcup_{n=1}^{\infty}(A + s_n)$ in $\mathbb{R}$ has measure zero?

[Hint: For every $\epsilon > 0$, find an interval I_ϵ of length ℓ_ϵ such that $m(A \cap I_\epsilon) \geq (1 - \epsilon)m(I_\epsilon)$. Consider $\bigcup_{k=-\infty}^{\infty}(A + t_k)$, with $t_k = k\ell_\epsilon$. Then vary ϵ.]

9. Let F be a closed subset in $\mathbb{R}$, and $\delta(x)$ the distance from x to F, that is,

$$\delta(x) = d(x, F) = \inf\{|x - y| : \ y \in F\}.$$

Clearly, $\delta(x + y) \leq |y|$ whenever $x \in F$. Prove the more refined estimate

$$\delta(x + y) = o(|y|) \qquad \text{for a.e. } x \in F,$$

that is, $\delta(x + y)/|y| \to 0$ for a.e. $x \in F$.
[Hint: Assume that x is a point of density of F.]

10. Construct an increasing function on $\mathbb{R}$ whose set of discontinuities is precisely $\mathbb{Q}$.

11. If $a, b > 0$, let

$$f(x) = \begin{cases} x^a \sin(x^{-b}) & \text{for } 0 < x \leq 1, \\ 0 & \text{if } x = 0. \end{cases}$$

Prove that f is of bounded variation in $[0, 1]$ if and only if $a > b$. Then, by taking $a = b$, construct (for each $0 < \alpha < 1$) a function that satisfies the Lipschitz condition of exponent α

$$|f(x) - f(y)| \leq A|x - y|^\alpha$$

but which is not of bounded variation.

[Hint: Note that if $h > 0$, the difference $|f(x + h) - f(x)|$ can be estimated by $C(x + h)^a$, or $C'h/x$ by the mean value theorem. Then, consider two cases, whether $x^{a+1} \geq h$ or $x^{a+1} < h$. What is the relationship between α and a?]

12. Consider the function $F(x) = x^2 \sin(1/x^2)$, $x \neq 0$, with $F(0) = 0$. Show that $F'(x)$ exists for every x, but F' is not integrable on $[-1, 1]$.

13. Show directly from the definition that the Cantor-Lebesgue function is not absolutely continuous.

14. The following measurability issues arose in the discussion of differentiability of functions.

(a) Suppose F is continuous on $[a, b]$. Show that

$$D^+(F)(x) = \limsup_{\substack{h \to 0 \\ h > 0}} \frac{F(x+h) - F(x)}{h}$$

is measurable.

(b) Suppose $J(x) = \sum_{n=1}^{\infty} \alpha_n j_n(x)$ is a jump function as in Section 3.3. Show that

$$\limsup_{h \to 0} \frac{J(x+h) - J(x)}{h}$$

is measurable.

[Hint: For (a), the continuity of F allows one to restrict to countably many h in taking the limsup. For (b), given $k > m$, let $F_{k,m}^N = \sup_{1/k \le |h| \le 1/m} \left| \frac{J_N(x+h) - J_N(x)}{h} \right|$, where $J_N(x) = \sum_{n=1}^{N} \alpha_n j_n(x)$. Note that each $F_{k,m}^N$ is measurable. Then, successively, let $N \to \infty$, $k \to \infty$, and finally $m \to \infty$.]

15. Suppose F is of bounded variation and continuous. Prove that $F = F_1 - F_2$, where both F_1 and F_2 are monotonic and continuous.

16. Show that if F is of bounded variation in $[a, b]$, then:

(a) $\int_a^b |F'(x)|\,dx \leq T_F(a, b)$.

(b) $\int_a^b |F'(x)|\,dx = T_F(a, b)$ if and only if F is absolutely continuous.

As a result of (b), the formula $L = \int_a^b |z'(t)|\,dt$ for the length of a rectifiable curve parametrized by z holds if and only if z is absolutely continuous.

17. Prove that if $\{K_\epsilon\}_{\epsilon>0}$ is a family of approximations to the identity, then

$$\sup_{\epsilon>0} |(f * K_\epsilon)(x)| \leq c f^*(x)$$

for some constant $c > 0$ and all integrable f.

18. Verify the agreement between the two definitions given for the Cantor-Lebesgue function in Exercise 2, Chapter 1 and in Section 3.1 of this chapter.

19. Show that if $f : \mathbb{R} \to \mathbb{R}$ is absolutely continuous, then

(a) f maps sets of measure zero to sets of measure zero.

(b) f maps measurable sets to measurable sets.

20. This exercise deals with functions F that are absolutely continuous on $[a, b]$ and are increasing. Let $A = F(a)$ and $B = F(b)$.

(a) There exists such an F that is in addition strictly increasing, but such that $F'(x) = 0$ on a set of positive measure.

(b) The F in (a) can be chosen so that there is a measurable subset $E \subset [A, B]$, $m(E) = 0$, so that $F^{-1}(E)$ is not measurable.

(c) Prove, however, that for any increasing absolutely continuous F, and E a measurable subset of $[A, B]$, the set $F^{-1}(E) \cap \{F'(x) > 0\}$ is measurable.

[Hint: (a) Let $F(x) = \int_a^x \chi_K(x)\,dx$, where K is the complement of a Cantor-like set C of positive measure. For (b), note that $F(C)$ is a set of measure zero. Finally, for (c) prove first that $m(\mathcal{O}) = \int_{F^{-1}(\mathcal{O})} F'(x)\,dx$ for any open set $\mathcal{O}$.]

21. Let F be absolutely continuous and increasing on $[a, b]$ with $F(a) = A$ and $F(b) = B$. Suppose f is any measurable function on $[A, B]$.

(a) Show that $f(F(x))F'(x)$ is measurable on $[a, b]$. Note: $f(F(x))$ need not be measurable by Exercise 20 (b).

(b) Prove the change of variable formula: If f is integrable on $[A, B]$, then so is $f(F(x))F'(x)$, and

$$\int_A^B f(y)\,dy = \int_a^b f(F(x))F'(x)\,dx.$$

[Hint: Start with the identity $m(\mathcal{O}) = \int_{F^{-1}(\mathcal{O})} F'(x)\,dx$ used in (c) of Exercise 20 above.]

22. Suppose that F and G are absolutely continuous on $[a, b]$. Show that their product FG is also absolutely continuous. This has the following consequences.

(a) Whenever F and G are absolutely continuous in $[a, b]$,

$$\int_a^b F'(x)G(x)\,dx = -\int_a^b F(x)G'(x)\,dx + [F(x)G(x)]_a^b.$$

(b) Let F be absolutely continuous in $[-\pi, \pi]$ with $F(\pi) = F(-\pi)$. Show that if

$$a_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} F(x)e^{-inx}\,dx,$$

such that $F(x) \sim \sum a_n e^{inx}$, then

$$F'(x) \sim \sum ina_n e^{inx}.$$

(c) What happens if $F(-\pi) \neq F(\pi)$? [Hint: Consider $F(x) = x$.]

23. Let F be continuous on $[a, b]$. Show the following.

(a) Suppose $(D^+F)(x) \geq 0$ for every $x \in [a, b]$. Then F is increasing on $[a, b]$.

(b) If $F'(x)$ exists for every $x \in (a, b)$ and $|F'(x)| \leq M$, then $|F(x) - F(y)| \leq M|x - y|$ and F is absolutely continuous.

[Hint: For (a) it suffices to show that $F(b) - F(a) \geq 0$. Assume otherwise. Hence with $G_\epsilon(x) = F(x) - F(a) + \epsilon(x - a)$, for sufficiently small $\epsilon > 0$ we have $G_\epsilon(a) = 0$, but $G_\epsilon(b) < 0$. Now let $x_0 \in [a, b)$ be the greatest value of x_0 such that $G_\epsilon(x_0) \geq 0$. However, $(D^+G_\epsilon)(x_0) > 0$.]

24. Suppose F is an increasing function on $[a, b]$.

(a) Prove that we can write

$$F = F_A + F_C + F_J,$$

where each of the functions F_A, F_C, and F_J is increasing and:

(i) F_A is absolutely continuous.

(ii) F_C is continuous, but $F_C'(x) = 0$ for a.e. x.

(iii) F_J is a jump function.

(b) Moreover, each component F_A, F_C, F_J is uniquely determined up to an additive constant.

The above is the **Lebesgue decomposition** of F. There is a corresponding decomposition for any F of bounded variation.

25. The following shows the necessity of allowing for general exceptional sets of measure zero in the differentiation Theorems 1.4, 3.4, and 3.11. Let E be any set of measure zero in $\mathbb{R}^d$. Show that:

(a) There exists a non-negative integrable f in $\mathbb{R}^d$, so that

$$\liminf_{\substack{m(B)\to 0 \\ x\in B}} \frac{1}{m(B)} \int_B f(y)\,dy = \infty \qquad \text{for each } x \in E.$$

(b) When $d = 1$ this may be restated as follows. There is an increasing absolutely continuous function F so that

$$D_+(F)(x) = D_-(F)(x) = \infty, \qquad \text{for each } x \in E.$$

[Hint: Find open sets $\mathcal{O}_n \supset E$, with $m(\mathcal{O}_n) < 2^{-n}$, and let $f(x) = \sum_{n=1}^{\infty} \chi_{\mathcal{O}_n}(x)$.]

26. An alternative way of defining the exterior measure $m_*(E)$ of an arbitrary set E, as given in Section 2 of Chapter 1, is to replace the coverings of E by cubes with coverings by balls. That is, suppose we define $m_*^{\mathcal{B}}(E)$ as $\inf \sum_{j=1}^{\infty} m(B_j)$, where the infimum is taken over all coverings $E \subset \bigcup_{j=1}^{\infty} B_j$ by open balls. Then $m_*(E) = m_*^{\mathcal{B}}(E)$. (Observe that this result leads to an alternate proof that the Lebesgue measure is invariant under rotations.)

Clearly $m_*(E) \leq m_*^{\mathcal{B}}(E)$. Prove the reverse inequality by showing the following. For any $\epsilon > 0$, there is a collection of balls $\{B_j\}$ such that $E \subset \bigcup_j B_j$ while $\sum_j m(B_j) \leq m_*(E) + \epsilon$. Note also that for any preassigned δ, we can choose the balls to have diameter $< \delta$.

[Hint: Assume first that E is measurable, and pick $\mathcal{O}$ open so that $\mathcal{O} \supset E$ and $m(\mathcal{O} - E) < \epsilon'$. Next, using Corollary 3.10, find balls $B_1, \ldots, B_N$ such that $\sum_{j=1}^{N} m(B_j) \leq m(E) + 2\epsilon'$ and $m(E - \bigcup_{j=1}^{N} B_j) \leq 3\epsilon'$. Finally, cover $E - \bigcup_{j=1}^{N} B_j$ by a union of cubes, the sum of whose measures is $\leq 4\epsilon'$, and replace these cubes by balls that contain them. For the general E, begin by applying the above when E is a cube.]

27. A rectifiable curve has a tangent line at almost all points of the curve. Make this statement precise.

28. A curve in $\mathbb{R}^d$ is a continuous map $t \mapsto z(t)$ of an interval $[a, b]$ into $\mathbb{R}^d$.

(a) State and prove the analogues of the conditions dealing with the rectifiability of curves and their length that are given in Theorems 3.1, 4.1, and 4.3.

(b) Define the (one-dimensional) Minkowski content $\mathcal{M}(K)$ of a compact set in $\mathbb{R}^d$ as the limit (if it exists) of

$$\frac{m(K^\delta)}{m_{d-1}(B(\delta))} \qquad \text{as } \delta \to 0,$$

where $m_{d-1}(B(\delta))$ is the measure (in $\mathbb{R}^{d-1}$) of the ball defined by $B(\delta) = \{x \in \mathbb{R}^{d-1},\ |x| < \delta\}$. State and prove analogues of Propositions 4.5 and 4.7 for curves in $\mathbb{R}^d$.

29. Let $\Gamma = \{z(t),\ a \le t \le b\}$ be a curve, and suppose it satisfies a Lipschitz condition with exponent α, $1/2 \le \alpha \le 1$, that is,

$$|z(t) - z(t')| \le A|t - t'|^{\alpha} \qquad \text{for all } t, t' \in [a, b].$$

Show that $m(\Gamma^{\delta}) = O(\delta^{2-1/\alpha})$ for $0 < \delta \le 1$.

30. A bounded function F is said to be of bounded variation on $\mathbb{R}$ if F is of bounded variation on any finite sub-interval $[a, b]$, and $\sup_{a,b} T_F(a, b) < \infty$.

Prove that such an F enjoys the following two properties:

(a) $\int_{\mathbb{R}} |F(x + h) - F(x)|\, dx \leq A|h|$, for some constant A and all $h \in \mathbb{R}$.

(b) $|\int_{\mathbb{R}} F(x)\varphi'(x)\, dx| \leq A$, where φ ranges over all C^1 functions of bounded support with $\sup_{x\in\mathbb{R}} |\varphi(x)| \leq 1$.

For the converse, and analogues in $\mathbb{R}^d$, see Problem 6^* below.

[Hint: For (a), write $F = F_1 - F_2$, where F_j are monotonic and bounded. For (b), deduce this from (a).]

31. Let F be the Cantor-Lebesgue function described in Section 3.1. Consider the curve that is the graph of F, that is, the curve given by $x(t) = t$ and $y(t) = F(t)$ with $0 \leq t \leq 1$. Prove that the length $L(\overline{x})$ of the segment $0 \leq t \leq \overline{x}$ of the curve is given by $L(\overline{x}) = \overline{x} + F(\overline{x})$. Hence the total length of the curve is 2.

32. Let $f : \mathbb{R} \to \mathbb{R}$. Prove that f satisfies the Lipschitz condition

$$|f(x) - f(y)| \leq M|x - y|$$

for some M and all $x, y \in \mathbb{R}$, if and only if f satisfies the following two properties:

(i) f is absolutely continuous.

(ii) $|f'(x)| \leq M$ for a.e. x.

Chapter 4. Hilbert Spaces: An Introduction

1. Show that properties (i) and (ii) in the definition of a Hilbert space (Section 2) imply property (iii): the Cauchy-Schwarz inequality $|(f, g)| \leq \|f\| \cdot \|g\|$ and the triangle inequality $\|f + g\| \leq \|f\| + \|g\|$.

[Hint: For the first inequality, consider $(f + \lambda g, f + \lambda g)$ as a positive quadratic function of λ. For the second, write $\|f + g\|^2$ as $(f + g, f + g)$.]

2. In the case of equality in the Cauchy-Schwarz inequality we have the following. If $|(f, g)| = \|f\| \, \|g\|$ and $g \neq 0$, then $f = cg$ for some scalar c.

[Hint: Assume $\|f\| = \|g\| = 1$ and $(f, g) = 1$. Then $f - g$ and g are orthogonal, while $f = f - g + g$. Thus $\|f\|^2 = \|f - g\|^2 + \|g\|^2$.]

3. Note that $\|f+g\|^2 = \|f\|^2 + \|g\|^2 + 2\text{Re}(f,g)$ for any pair of elements in a Hilbert space $\mathcal{H}$. As a result, verify the identity $\|f+g\|^2 + \|f-g\|^2 = 2(\|f\|^2 + \|g\|^2)$.

4. Prove from the definition that $\ell^2(\mathbb{Z})$ is complete and separable.

5. Establish the following relations between $L^2(\mathbb{R}^d)$ and $L^1(\mathbb{R}^d)$:

(a) Neither the inclusion $L^2(\mathbb{R}^d) \subset L^1(\mathbb{R}^d)$ nor the inclusion $L^1(\mathbb{R}^d) \subset L^2(\mathbb{R}^d)$ is valid.

(b) Note, however, that if f is supported on a set E of finite measure and if $f \in L^2(\mathbb{R}^d)$, applying the Cauchy-Schwarz inequality to $f\chi_E$ gives $f \in L^1(\mathbb{R}^d)$, and

$$\|f\|_{L^1(\mathbb{R}^d)} \leq m(E)^{1/2}\|f\|_{L^2(\mathbb{R}^d)}.$$

(c) If f is bounded ($|f(x)| \leq M$), and $f \in L^1(\mathbb{R}^d)$, then $f \in L^2(\mathbb{R}^d)$ with

$$\|f\|_{L^2(\mathbb{R}^d)} \leq M^{1/2}\|f\|^{1/2}_{L^1(\mathbb{R}^d)}.$$

[Hint: For (a) consider $f(x) = |x|^{-\alpha}$, when $|x| \leq 1$ or when $|x| > 1$.]

6. Prove that the following are dense subspaces of $L^2(\mathbb{R}^d)$.

(a) The simple functions.

(b) The continuous functions of compact support.

7. Suppose $\{\varphi_k\}_{k=1}^{\infty}$ is an orthonormal basis for $L^2(\mathbb{R}^d)$. Prove that the collection $\{\varphi_{k,j}\}_{1\leq k,j<\infty}$ with $\varphi_{k,j}(x,y) = \varphi_k(x)\varphi_j(y)$ is an orthonormal basis of $L^2(\mathbb{R}^d \times \mathbb{R}^d)$.

[Hint: First verify that the $\{\varphi_{k,j}\}$ are orthonormal, by Fubini's theorem. Next, for each j consider $F_j(x) = \int_{\mathbb{R}^d} F(x,y)\overline{\varphi_j(y)}\, dy$. If one assumes that $(F, \varphi_{k,j}) = 0$ for all j, then $\int F_j(x)\overline{\varphi_k(x)}\, dx = 0$.]

8. Let $\eta(t)$ be a fixed strictly positive continuous function on $[a, b]$. Define $\mathcal{H}_\eta = L^2([a, b], \eta)$ to be the space of all measurable functions f on $[a, b]$ such that

$$\int_a^b |f(t)|^2 \eta(t)\, dt < \infty.$$

Define the inner product on $\mathcal{H}_\eta$ by

$$(f, g)_\eta = \int_a^b f(t)\overline{g(t)}\eta(t)\, dt.$$

(a) Show that $\mathcal{H}_\eta$ is a Hilbert space, and that the mapping $U : f \mapsto \eta^{1/2} f$ gives a unitary correspondence between $\mathcal{H}_\eta$ and the usual space $L^2([a, b])$.

(b) Generalize this to the case when η is not necessarily continuous.

9. Let $\mathcal{H}_1 = L^2([-\pi, \pi])$ be the Hilbert space of functions $F(e^{i\theta})$ on the unit circle with inner product $(F, G) = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(e^{i\theta})\overline{G(e^{i\theta})}\, d\theta$. Let $\mathcal{H}_2$ be the space $L^2(\mathbb{R})$. Using the mapping

$$x \mapsto \frac{i - x}{i + x}$$

of $\mathbb{R}$ to the unit circle, show that:

(a) The correspondence $U : F \to f$, with

$$f(x) = \frac{1}{\pi^{1/2}(i + x)} F\left(\frac{i - x}{i + x}\right)$$

gives a unitary mapping of $\mathcal{H}_1$ to $\mathcal{H}_2$.

(b) As a result,

$$\left\{ \frac{1}{\pi^{1/2}} \left(\frac{i - x}{i + x}\right)^n \frac{1}{i + x} \right\}_{n=-\infty}^{\infty}$$

is an orthonormal basis of $L^2(\mathbb{R})$.

10. Let $\mathcal{S}$ denote a subspace of a Hilbert space $\mathcal{H}$. Prove that $(\mathcal{S}^{\perp})^{\perp}$ is the smallest closed subspace of $\mathcal{H}$ that contains $\mathcal{S}$.

11. Let P be the orthogonal projection associated with a closed subspace $\mathcal{S}$ in a Hilbert space $\mathcal{H}$, that is,

$$P(f) = f \text{ if } f \in \mathcal{S} \qquad \text{and} \qquad P(f) = 0 \text{ if } f \in \mathcal{S}^{\perp}.$$

(a) Show that $P^2 = P$ and $P^* = P$.

(b) Conversely, if P is any bounded operator satisfying $P^2 = P$ and $P^* = P$, prove that P is the orthogonal projection for some closed subspace of $\mathcal{H}$.

(c) Using P, prove that if $\mathcal{S}$ is a closed subspace of a separable Hilbert space, then $\mathcal{S}$ is also a separable Hilbert space.

12. Let E be a measurable subset of $\mathbb{R}^d$, and suppose $\mathcal{S}$ is the subspace of $L^2(\mathbb{R}^d)$ of functions that vanish for a.e. $x \notin E$. Show that the orthogonal projection P on $\mathcal{S}$ is given by $P(f) = \chi_E \cdot f$, where χ_E is the characteristic function of E.

13. Suppose P_1 and P_2 are a pair of orthogonal projections on S_1 and S_2, respectively. Then P_1P_2 is an orthogonal projection if and only if P_1 and P_2 commute, that is, $P_1P_2 = P_2P_1$. In this case, P_1P_2 projects onto $S_1 \cap S_2$.

14. Suppose $\mathcal{H}$ and $\mathcal{H}'$ are two completions of a pre-Hilbert space $\mathcal{H}_0$. Show that there is a unitary mapping from $\mathcal{H}$ to $\mathcal{H}'$ that is the identity on $\mathcal{H}_0$.

[Hint: If $f \in \mathcal{H}$, pick a Cauchy sequence $\{f_n\}$ in $\mathcal{H}_0$ that converges to f in $\mathcal{H}$. This sequence will also converge to an element f' in $\mathcal{H}'$. The mapping $f \mapsto f'$ gives the required unitary mapping.]

15. Let T be any linear transformation from $\mathcal{H}_1$ to $\mathcal{H}_2$. If we suppose that $\mathcal{H}_1$ is finite-dimensional, then T is automatically bounded. (If $\mathcal{H}_1$ is not assumed to be finite-dimensional this may fail; see Problem 1 below.)

16. Let $F_0(z) = 1/(1-z)^i$.

(a) Verify that $|F_0(z)| \leq e^{\pi/2}$ in the unit disc, but that $\lim_{r\to 1} F_0(r)$ does not exist.

[Hint: Note that $|F_0(r)| = 1$ and $F_0(r)$ oscillates between ± 1 infinitely often as $r \to 1$.]

(b) Let $\{\alpha_n\}_{n=1}^{\infty}$ be an enumeration of the rationals, and let

$$F(z) = \sum_{j=1}^{\infty} \delta^j F_0(ze^{-i\alpha_j}),$$

where δ is sufficiently small. Show that $\lim_{r\to 1} F(re^{i\theta})$ fails to exist whenever $\theta = \alpha_j$, and hence F fails to have a radial limit for a dense set of points on the unit circle.

17. Fatou's theorem can be generalized by allowing a point to approach the boundary in larger regions, as follows.

For each $0 < s < 1$ and point z on the unit circle, consider the region $\Gamma_s(z)$ defined as the smallest closed convex set that contains z and the closed disc $D_s(0)$. In other words, $\Gamma_s(z)$ consists of all lines joining z with points in $D_s(0)$. Near the point z, the region $\Gamma_s(z)$ looks like a triangle. See Figure 2.

We say that a function F defined in the open unit disc has a **non-tangential limit** at a point z on the circle, if for every $0 < s < 1$, the limit

$$\lim_{\substack{w \to z \\ w \in \Gamma_s(z)}} F(w)$$

exists.

Prove that if F is holomorphic and bounded on the open unit disc, then F has a non-tangential limit for almost every point on the unit circle.
[Hint: Show that the Poisson integral of a function f has non-tangential limits at every point of the Lebesgue set of f.]

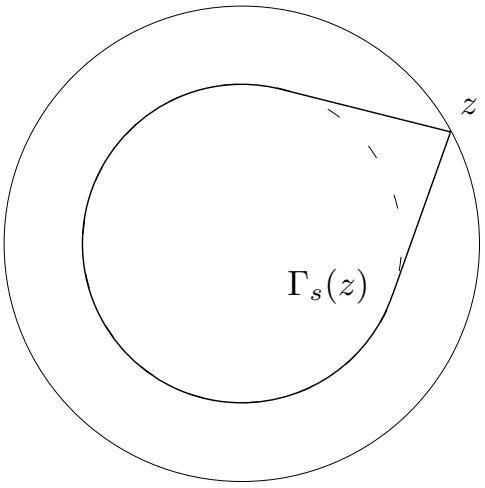


Figure 2. The region $\Gamma_s(z)$

18. Let $\mathcal{H}$ denote a Hilbert space, and $\mathcal{L}(\mathcal{H})$ the vector space of all bounded linear operators on $\mathcal{H}$. Given $T \in \mathcal{L}(\mathcal{H})$, we define the operator norm

$$\|T\| = \inf\{B : \|Tv\| \leq B\|v\|, \quad \text{for all } v \in \mathcal{H}\}.$$

(a) Show that $\|T_1 + T_2\| \leq \|T_1\| + \|T_2\|$ whenever $T_1, T_2 \in \mathcal{L}(\mathcal{H})$.

(b) Prove that

$$d(T_1, T_2) = \|T_1 - T_2\|$$

defines a metric on $\mathcal{L}(\mathcal{H})$.

(c) Show that $\mathcal{L}(\mathcal{H})$ is complete in the metric d.

19. If T is a bounded linear operator on a Hilbert space, prove that

$$\|TT^*\| = \|T^*T\| = \|T\|^2 = \|T^*\|^2.$$

20. Suppose $\mathcal{H}$ is an infinite-dimensional Hilbert space. We have seen an example of a sequence $\{f_n\}$ in $\mathcal{H}$ with $\|f_n\| = 1$ for all n, but for which no subsequence of $\{f_n\}$ converges in $\mathcal{H}$. However, show that for any sequence $\{f_n\}$ in $\mathcal{H}$ with $\|f_n\| = 1$ for all n, there exist $f \in \mathcal{H}$ and a subsequence $\{f_{n_k}\}$ such that for all $g \in \mathcal{H}$, one has

$$\lim_{k\to\infty} (f_{n_k}, g) = (f, g).$$

One says that $\{f_{n_k}\}$ **converges weakly** to f.

[Hint: Let g run through a basis for $\mathcal{H}$, and use a diagonalization argument. One can then define f by giving its series expansion with respect to the chosen basis.]

21. There are several senses in which a sequence of bounded operators $\{T_n\}$ can converge to a bounded operator T (in a Hilbert space $\mathcal{H}$). First, there is convergence in the norm, that is, $\|T_n - T\| \to 0$, as $n \to \infty$. Next, there is a weaker convergence, which happens to be called **strong convergence**, that requires that $T_n f \to Tf$, as $n \to \infty$, for every vector $f \in \mathcal{H}$. Finally, there is **weak convergence** (see also Exercise 20) that requires $(T_n f, g) \to (Tf, g)$ for every pair of vectors $f, g \in \mathcal{H}$.

(a) Show by examples that weak convergence does not imply strong convergence, nor does strong convergence imply convergence in the norm.

(b) Show that for any bounded operator T there is a sequence $\{T_n\}$ of bounded operators of finite rank so that $T_n \to T$ strongly as $n \to \infty$.

22. An operator T is an **isometry** if $\|Tf\| = \|f\|$ for all $f \in \mathcal{H}$.

(a) Show that if T is an isometry, then $(Tf, Tg) = (f, g)$ for every $f, g \in \mathcal{H}$. Prove as a result that $T^*T = I$.

(b) If T is an isometry and T is surjective, then T is unitary and $TT^* = I$.

(c) Give an example of an isometry that is not unitary.

(d) Show that if T^*T is unitary then T is an isometry.

[Hint: Use the fact that $(Tf, Tf) = (f, f)$ for f replaced by $f \pm g$ and $f \pm ig$.]

23. Suppose $\{T_k\}$ is a collection of bounded operators on a Hilbert space $\mathcal{H}$, with $\|T_k\| \leq 1$ for all k. Suppose also that

$$T_k T_j^* = T_k^* T_j = 0 \qquad \text{for all } k \neq j.$$

Let $S_N = \sum_{k=-N}^{N} T_k$.

Show that $S_N(f)$ converges as $N \to \infty$, for every $f \in \mathcal{H}$. If $T(f)$ denotes the limit, prove that $\|T\| \leq 1$.

A generalization is given in Problem 8^* below.

[Hint: Consider first the case when only finitely many of the T_k are non-zero, and note that the ranges of the T_k are mutually orthogonal.]

24. Let $\{e_k\}_{k=1}^{\infty}$ denote an orthonormal set in a Hilbert space $\mathcal{H}$. If $\{c_k\}_{k=1}^{\infty}$ is a sequence of positive real numbers such that $\sum c_k^2 < \infty$, then the set

$$A = \{\sum_{k=1}^{\infty} a_k e_k : \ |a_k| \leq c_k\}$$

is compact in $\mathcal{H}$.

25. Suppose T is a bounded operator that is diagonal with respect to a basis $\{\varphi_k\}$, with $T\varphi_k = \lambda_k \varphi_k$. Then T is compact if and only if $\lambda_k \to 0$.

[Hint: If $\lambda_k \to 0$, then note that $\|P_n T - T\| \to 0$, where P_n is the orthogonal projection on the subspace spanned by $\varphi_1, \varphi_2, \ldots, \varphi_n$.]

26. Suppose w is a measurable function on $\mathbb{R}^d$ with $0 < w(x) < \infty$ for a.e. x, and K is a measurable function on $\mathbb{R}^{2d}$ that satisfies:

(i) $\displaystyle\int_{\mathbb{R}^d} |K(x,y)|w(y)\,dy \le Aw(x)$ for almost every $x \in \mathbb{R}^d$, and

(ii) $\displaystyle\int_{\mathbb{R}^d} |K(x,y)|w(x)\,dx \le Aw(y)$ for almost every $y \in \mathbb{R}^d$.

Prove that the integral operator defined by

$$Tf(x) = \int_{\mathbb{R}^d} K(x,y)f(y)\,dy, \qquad x \in \mathbb{R}^d$$

is bounded on $L^2(\mathbb{R}^d)$ with $\|T\| \le A$.

Note as a special case that if $\int |K(x,y)|\,dy \le A$ for all x, and $\int |K(x,y)|\,dx \le A$ for all y, then $\|T\| \le A$.

[Hint: Show that if $f \in L^2(\mathbb{R}^d)$, then

$$\int |K(x,y)|\,|f(y)|\,dy \le A^{1/2}w(x)^{1/2}\left[\int |K(x,y)|\,|f(y)|^2 w(y)^{-1}\,dy\right]^{1/2}.]$$

27. Prove that the operator

$$Tf(x) = \frac{1}{\pi} \int_0^\infty \frac{f(y)}{x+y}\, dy$$

is bounded on $L^2(0,\infty)$ with norm $\|T\| \leq 1$.
[Hint: Use Exercise 26 with an appropriate w.]

28. Suppose $\mathcal{H} = L^2(B)$, where B is the unit ball in $\mathbb{R}^d$. Let $K(x, y)$ be a measurable function on $B \times B$ that satisfies $|K(x, y)| \leq A|x - y|^{-d+\alpha}$ for some $\alpha > 0$, whenever $x, y \in B$. Define

$$Tf(x) = \int_B K(x, y) f(y) dy.$$

(a) Prove that T is a bounded operator on $\mathcal{H}$.

(b) Prove that T is compact.

(c) Note that T is a Hilbert-Schmidt operator if and only if $\alpha > d/2$.

[Hint: For (b), consider the operators T_n associated with the truncated kernels $K_n(x, y) = K(x, y)$ if $|x - y| \geq 1/n$ and 0 otherwise. Show that each T_n is compact, and that $\|T_n - T\| \to 0$ as $n \to \infty$.]

29. Let T be a compact operator on a Hilbert space $\mathcal{H}$, and assume $\lambda \neq 0$.

(a) Show that the range of $\lambda I - T$ defined by

$$\{g \in \mathcal{H} : g = (\lambda I - T)f, \text{ for some } f \in \mathcal{H}\}$$

is closed. [Hint: Suppose $g_j \to g$, where $g_j = (\lambda I - T)f_j$. Let V_λ denote the eigenspace of T corresponding to λ, that is, the kernel of $\lambda I - T$. Why can one assume that $f_j \in V_\lambda^\perp$? Under this assumption prove that $\{f_j\}$ is a bounded sequence.]

(b) Show by example that this may fail when $\lambda = 0$.

(c) Show that the range of $\lambda I - T$ is all of $\mathcal{H}$ if and only if the null-space of $\overline{\lambda} I - T^*$ is trivial.

30. Let $\mathcal{H} = L^2([-\pi, \pi])$ with $[-\pi, \pi]$ identified as the unit circle. Fix a bounded sequence $\{\lambda_n\}_{n=-\infty}^{\infty}$ of complex numbers, and define an operator Tf by

$$Tf(x) \sim \sum_{n=-\infty}^{\infty} \lambda_n a_n e^{inx} \qquad \text{whenever} \qquad f(x) \sim \sum_{n=-\infty}^{\infty} a_n e^{inx}.$$

Such an operator is called a **Fourier multiplier operator**, and the sequence $\{\lambda_n\}$ is called the **multiplier sequence**.

(a) Show that T is a bounded operator on $\mathcal{H}$ and $\|T\| = \sup_n |\lambda_n|$.

(b) Verify that T commutes with translations, that is, if we define $\tau_h(x) = f(x - h)$ then

$$T \circ \tau_h = \tau_h \circ T \qquad \text{for every } h \in \mathbb{R}.$$

(c) Conversely, prove that if T is any bounded operator on $\mathcal{H}$ that commutes with translations, then T is a Fourier multiplier operator. [Hint: Consider $T(e^{inx})$.]

31. Consider a version of the sawtooth function defined on $[-\pi, \pi)$ by[5]

$$K(x) = i(sgn(x)\pi - x),$$

and extended to $\mathbb{R}$ with period 2π. Suppose $f \in L^1([-\pi, \pi])$ is extended to $\mathbb{R}$ with period 2π, and define

$$\begin{aligned} Tf(x) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} K(x-y) f(y)\, dy \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} K(y) f(x-y)\, dy. \end{aligned}$$

(a) Show that $F(x) = Tf(x)$ is absolutely continuous, and if $\int_{-\pi}^{\pi} f(y)dy = 0$, then $F'(x) = if(x)$ a.e. x.

(b) Show that the mapping $f \mapsto Tf$ is compact and symmetric on $L^2([-\pi, \pi])$.

(c) Prove that $\varphi(x) \in L^2([-\pi, \pi])$ is an eigenfunction for T if and only if $\varphi(x)$ is (up to a constant multiple) equal to e^{inx} for some integer $n \neq 0$ with eigenvalue $1/n$, or $\varphi(x) = 1$ with eigenvalue 0.

(d) Show as a result that $\{e^{inx}\}_{n\in\mathbb{Z}}$ is an orthonormal basis of $L^2([-\pi, \pi])$.

Note that in Book I, Chapter 2, Exercise 8, it is shown that the Fourier series of K is

$$K(x) \sim \sum_{n\neq 0} \frac{e^{inx}}{n}.$$

[5]The symbol $sgn(x)$ denotes the sign function: it equals 1 or -1 if x is positive or negative respectively, and 0 if $x = 0$.

32. Consider the operator $T : L^2([0,1]) \to L^2([0,1])$ defined by

$$T(f)(t) = tf(t).$$

(a) Prove that T is a bounded linear operator with $T = T^*$, but that T is not compact.

(b) However, show that T has no eigenvectors.

33. Let $\mathcal{H}$ be a Hilbert space with basis $\{\varphi_k\}_{k=1}^{\infty}$. Verify that the operator T defined by

$$T(\varphi_k) = \frac{1}{k}\,\varphi_{k+1}$$

is compact, but has no eigenvectors.

34. Let K be a Hilbert-Schmidt kernel which is real and symmetric. Then, as we saw, the operator T whose kernel is K is compact and symmetric. Let $\{\varphi_k(x)\}$ be the eigenvectors (with eigenvalues λ_k) that diagonalize T. Then:

(a) $\sum_k |\lambda_k|^2 < \infty$.

(b) $K(x,y) \sim \sum \lambda_k \varphi_k(x)\varphi_k(y)$ is the expansion of K in the basis $\{\varphi_k(x)\varphi_k(y)\}$.

(c) Suppose T is a compact operator which is symmetric. Then T is of Hilbert-Schmidt type if and only if $\sum_n |\lambda_n|^2 < \infty$, where $\{\lambda_n\}$ are the eigenvalues of T counted according to their multiplicities.

35. Let $\mathcal{H}$ be a Hilbert space. Prove the following variants of the spectral theorem.

(a) If T_1 and T_2 are two linear symmetric and compact operators on $\mathcal{H}$ that commute (that is, $T_1T_2 = T_2T_1$), show that they can be diagonalized simultaneously. In other words, there exists an orthonormal basis for $\mathcal{H}$ which consists of eigenvectors for both T_1 and T_2.

(b) A linear operator on $\mathcal{H}$ is **normal** if $TT^* = T^*T$. Prove that if T is normal and compact, then T can be diagonalized.

[Hint: Write $T = T_1 + iT_2$ where T_1 and T_2 are symmetric, compact and commute.]

(c) If U is unitary, and $U = \lambda I - T$, where T is compact, then U can be diagonalized.

Chapter 5. Hilbert Spaces: Several Examples

1. Suppose $f \in L^2(\mathbb{R}^d)$ and $k \in L^1(\mathbb{R}^d)$.

(a) Show that $(f * k)(x) = \int f(x-y)k(y)\,dy$ converges for a.e. x.

(b) Prove that $\|f * k\|_{L^2(\mathbb{R}^d)} \leq \|f\|_{L^2(\mathbb{R}^d)}\|k\|_{L^1(\mathbb{R}^d)}$.

(c) Establish $\widehat{(f * k)}(\xi) = \hat{k}(\xi)\hat{f}(\xi)$ for a.e. ξ.

(d) The operator $Tf = f * k$ is a Fourier multiplier operator with multiplier $m(\xi) = \hat{k}(\xi)$.

[Hint: See Exercise 21 in Chapter 2.]

2. Consider the **Mellin transform** defined initially for continuous functions f of compact support in $\mathbb{R}^+ = \{t \in \mathbb{R} : \ t > 0\}$ and $x \in \mathbb{R}$ by

$$\mathcal{M}f(x) = \int_0^\infty f(t)t^{ix-1}dt.$$

Prove that $(2\pi)^{-1/2}\mathcal{M}$ extends to a unitary operator from $L^2(\mathbb{R}^+, dt/t)$ to $L^2(\mathbb{R})$. The Mellin transform serves on $\mathbb{R}^+$, with its multiplicative structure, the same purpose as the Fourier transform on $\mathbb{R}$, with its additive structure.

3. Let $F(z)$ be a bounded holomorphic function in the half-plane. Show in two ways that $\lim_{y\to 0} F(x+iy)$ exists for a.e. x.

(a) By using the fact that $F(z)/(z+i)$ is in $H^2(\mathbb{R}^2_+)$.

(b) By noting that $G(z) = F\left(i\frac{1-z}{1+z}\right)$ is a bounded holomorphic function in the unit disc, and using Exercise 17 in the previous chapter.

4. Consider $F(z) = e^{i/z}/(z+i)$ in the upper half-plane. Note that $F(x+iy) \in L^2(\mathbb{R})$, for each $y > 0$ and $y = 0$. Observe also that $F(z) \to 0$ as $|z| \to 0$. However, $F \notin H^2(\mathbb{R}^2_+)$. Why?

5. For $a < b$, let $S_{a,b}$ denote the strip $\{z = x + iy, \quad a < y < b\}$. Define $H^2(S_{a,b})$ to consist of the holomorphic functions F in $S_{a,b}$ so that

$$\|F\|^2_{H^2(S_{a,b})} = \sup_{a<y<b} \int_{\mathbb{R}^2} |F(x+iy)|^2 \, dx < \infty.$$

Define $H^2(S_{a,\infty})$ and $H^2(S_{-\infty,b})$ to be the obvious variants of the Hardy spaces for the half-planes $\{z = x + iy, \ y > a\}$ and $\{z = x + iy, \ y < b\}$, respectively.

(a) Show that $F \in H^2(S_{a,b})$ if and only if F can be written as

$$F(z) = \int_{\mathbb{R}} f(\xi) e^{-2\pi i z \xi} \, d\xi,$$

with $\int_{\mathbb{R}} |f(\xi)|^2 (e^{4\pi a \xi} + e^{4\pi b \xi}) \, d\xi < \infty$.

(b) Prove that every $F \in H^2(S_{a,b})$ can be decomposed as $F = G_1 + G_2$, where $G_{\in} H^2(S_{a,\infty})$ and $G_2 \in H^2(S_{-\infty,b})$.

(c) Show that $\lim_{a<y<b, y\to a} F(x+iy) = F_a(x)$ exists in the L^2-norm and also almost everywhere, with a similar result for $\lim_{a<y<b, y\to b} F(x+iy)$.

6. Suppose Ω is an open set in $\mathbb{C} = \mathbb{R}^2$, and let $\mathcal{H}$ be the subspace of $L^2(\Omega)$ consisting of holomorphic functions on Ω. Show that $\mathcal{H}$ is a closed subspace of $L^2(\Omega)$, and hence is a Hilbert space with inner product

$$(f, g) = \int_\Omega f(z)\overline{g}(z)\, dx\, dy, \qquad \text{where } z = x + iy.$$

[Hint: Prove that for $f \in \mathcal{H}$, we have $|f(z)| \leq \frac{c}{d(z,\Omega^c)}\|f\|$ for $z \in \Omega$, where $c = \pi^{-1/2}$, using the mean-value property (9). Thus if $\{f_n\}$ is a Cauchy sequence in $\mathcal{H}$, it converges uniformly on compact subsets of Ω.]

7. Following up on the previous exercise, prove:

(a) If $\{\varphi_n\}_{n=0}^{\infty}$ is an orthonormal basis of $\mathcal{H}$, then

$$\sum_{n=0}^{\infty} |\varphi_n(z)|^2 \leq \frac{c^2}{d(z, \Omega^c)} \qquad \text{for } z \in \Omega.$$

(b) The sum

$$B(z, w) = \sum_{n=0}^{\infty} \varphi_n(z)\overline{\varphi_n}(w)$$

converges absolutely for $(z, w) \in \Omega \times \Omega$, and is independent of the choice of the orthonormal basis $\{\varphi_n\}$ of $\mathcal{H}$.

(c) To prove (b) it is useful to characterize the function $B(z, w)$, called the **Bergman kernel**, by the following property. Let T be the linear transformation on $L^2(\Omega)$ defined by

$$Tf(z) = \int_{\Omega} B(z, w) f(w)\, du\, dv, \qquad w = u + iv.$$

Then T is the orthogonal projection of $L^2(\Omega)$ to $\mathcal{H}$.

(d) Suppose that Ω is the unit disc. Then $f \in \mathcal{H}$ exactly when $f(z) = \sum_{n=0}^{\infty} a_n z^n$, with

$$\sum_{n=0}^{\infty} |a_n|^2 (n+1)^{-1} < \infty.$$

Also, the sequence $\{\frac{z^n(n+1)}{\pi^{1/2}}\}_{n=0}^{\infty}$ is an orthonormal basis of $\mathcal{H}$. Moreover, in this case

$$B(z, w) = \frac{1}{\pi(1 - z\overline{w})^2}.$$

8. Continuing with Exercise 6, suppose Ω is the upper half-plane $\mathbb{R}^2_+$. Then every $f \in \mathcal{H}$ has a representation

$$f(z) = \sqrt{4\pi} \int_0^\infty \hat{f}_0(\xi) e^{2\pi i \xi z} \, d\xi, \qquad z \in \mathbb{R}^2_+, \tag{34}$$

where $\int_0^\infty |\hat{f}_0(\xi)|^2 \frac{d\xi}{\xi} < \infty$. Moreover, the mapping $\hat{f}_0 \to f$ given by (34) is a unitary mapping from $L^2((0,\infty), \frac{d\xi}{\xi})$ to $\mathcal{H}$.

9. Let H be the Hilbert transform. Verify that

(a) $H^* = -H$, $H^2 = -I$, and H is unitary.

(b) If τ_h denotes the translation operator, $\tau_h(f)(x) = f(x-h)$, then H commutes with τ_h, $\tau_h H = H\tau_h$.

(c) If δ_a denotes the dilation operator, $\delta_a(f)(x) = f(ax)$ with $a > 0$, then H commutes with δ_a, $\delta_a H = H\delta_a$.

A converse is given in Problem 5 below.

10. Let $f \in L^2(\mathbb{R})$ and let $u(x,y)$ be the Poisson integral of f, that is $u = (f * \mathcal{P}_y)(x)$, as given in (10) above. Let $v(x,y) = (Hf * \mathcal{P}_y)(x)$, the Poisson integral of the Hilbert transform of f. Prove that:

(a) $F(x+iy) = u(x,y) + iv(x,y)$ is analytic in the half-plane $\mathbb{R}^2_+$, so that u and v are conjugate harmonic functions. We also have $f = \lim_{y\to 0} u(x,y)$ and $Hf = \lim_{y\to 0} v(x,y)$.

(b) $F(z) = \frac{1}{\pi i}\int_{\mathbb{R}} f(t)\frac{dt}{t-z}$.

(c) $v(x,y) = f * \mathcal{Q}_y$, where $\mathcal{Q}_y(x) = \frac{1}{\pi}\frac{x}{x^2+y^2}$ is the **conjugate Poisson kernel**.

[Hint: Note that $\frac{i}{\pi z} = \mathcal{P}_y(x) + i\mathcal{Q}_y(x)$, $z = x+iy$.]

11. Show that

$$\left\{ \frac{1}{\pi^{1/2}(i+z)} \left(\frac{i-z}{i+z} \right)^n \right\}_{n=0}^{\infty}$$

is an orthonormal basis of $H^2(\mathbb{R}^2_+)$.

Note that $\left\{ \frac{1}{\pi^{1/2}(i+x)} \left(\frac{i-x}{i+x} \right)^n \right\}_{n=0}^{\infty}$ is an orthonormal basis of $L^2(\mathbb{R})$; see Exercise 9 in the previous chapter.

[Hint: It suffices to show that if $F \in H^2(\mathbb{R}^2_+)$ and

$$\int_{-\infty}^{\infty} F(x) \frac{(x+i)^n}{(x-i)^{n+1}}\, dx = 0 \qquad \text{for } n = 0,\, 1,\, 2,\, \ldots,$$

then $F = 0$. Use the Cauchy integral formula to prove that

$$\left(\frac{d}{dz} \right)^n (F(z)(z+i)^n)|_{z=i} = 0,$$

and thus $F^{(n)}(i) = 0$ for $n = 0,\, 1,\, 2,\, \ldots$.]

12. We consider whether the inequality

$$\|u\|_{L^2(\Omega)} \leq c\|L(u)\|_{L^2(\Omega)}$$

can hold for open sets Ω that are unbounded.

(a) Assume $d \geq 2$. Show that for each constant coefficient partial differential operator L, there are unbounded connected open sets Ω for which the above holds for all $u \in C_0^\infty(\Omega)$.

(b) Show that $\|u\|_{L^2(\mathbb{R}^d)} \leq c\|L(u)\|_{L^2(\mathbb{R}^d)}$ for all $u \in C_0^\infty(\mathbb{R}^d)$ if and only if $|P(\xi)| \geq c > 0$ all ξ, where P is the characteristic polynomial of L.

[Hint: For (a) consider first $L = (\partial/\partial x_1)^n$ and a strip $\{x : \ -1 < x_1 < 1\}$.]

13. Suppose L is a linear partial differential operator with constant coefficients. Show that when $d \geq 2$, the linear space of solutions u of $L(u) = 0$ with $u \in C^\infty(\mathbb{R}^d)$ is not finite-dimensional.

[Hint: Consider the zeroes ζ of $P(\zeta)$, $\zeta \in \mathbb{C}^d$, where P is the characteristic polynomial of L.]

14. Suppose F and G are two integrable functions on a bounded interval $[a, b]$. Show that G is the weak derivative of F if and only if F can be corrected on a set of measure 0, such that F is absolutely continuous and $F'(x) = G(x)$ for almost every x.

[Hint: If G is the weak derivative of F, use an approximation to show that

$$\int_a^b G(x)\varphi(x)dx = -\int_a^b F(x)\varphi'(x)dx$$

holds for the function φ illustrated in Figure 9.]

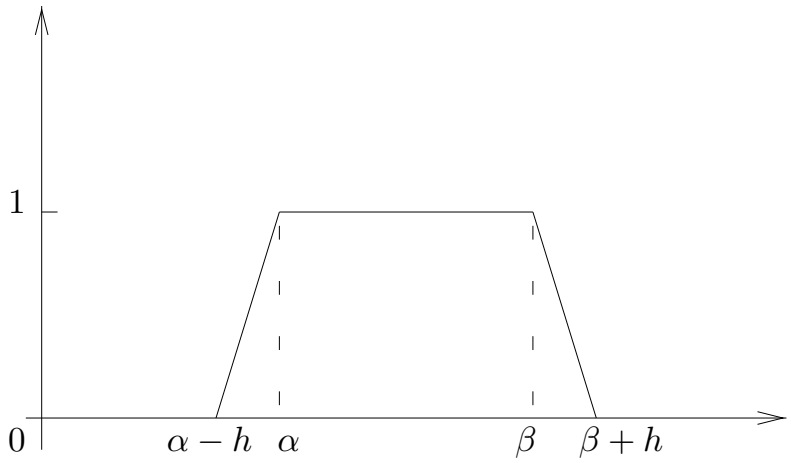


Figure 9. The function φ in Exercise 14

15. Suppose $f \in L^2(\mathbb{R}^d)$. Prove that there exists $g \in L^2(\mathbb{R}^d)$ such that

$$\left(\frac{\partial}{\partial x}\right)^{\alpha} f(x) = g(x)$$

in the weak sense, if and only if

$$(2\pi i\xi)^{\alpha} \hat{f}(\xi) = \hat{g}(\xi) \in L^2(\mathbb{R}^d).$$

16. Sobolev embedding theorem. Suppose n is the smallest integer $> d/2$. If

$$f \in L^2(\mathbb{R}^d) \quad \text{and} \quad \left(\frac{\partial}{\partial x}\right)^\alpha f \in L^2(\mathbb{R}^d)$$

in the weak sense, for all $1 \leq |\alpha| \leq n$, then f can be modified on a set of measure zero so that f is continuous and bounded.
[Hint: Express f in terms of $\hat{f}$, and show that $\hat{f} \in L^1(\mathbb{R}^d)$ by the Cauchy-Schwarz inequality.]

17. The conclusion of the Sobolev embedding theorem fails when $n = d/2$. Consider the case $d = 2$, and let $f(x) = (\log 1/|x|)^\alpha \eta(x)$, where η is a smooth cutoff function with $\eta = 1$ for x near the origin, but $\eta(x) = 0$ if $|x| \geq 1/2$. Let $0 < \alpha < 1/2$.

(a) Verify that $\partial f/\partial x_1$ and $\partial f/\partial x_2$ are in L^2 in the weak sense.

(b) Show that f cannot be corrected on a set of measure zero such that the resulting function is continuous at the origin.

18. Consider the linear partial differential operator

$$L = \sum_{|\alpha| \le n} a_\alpha \left(\frac{\partial}{\partial x} \right)^\alpha .$$

Then

$$P(\xi) = \sum_{|\alpha| \le n} a_\alpha (2\pi i \xi)^\alpha$$

is called the **characteristic polynomial** of L. The differential operator L is said to be **elliptic** if

$$|P(\xi)| \ge c|\xi|^n \qquad \text{for some } c > 0 \text{ and all } \xi \text{ sufficiently large.}$$

(a) Check that L is elliptic if and only if $\sum_{|\alpha|=n} a_\alpha (2\pi\xi)^\alpha$ vanishes only when $\xi = 0$.

(b) If L is elliptic, prove that for some $c > 0$ the inequality

$$\left\| \left(\frac{\partial}{\partial x} \right)^\alpha \varphi \right\|_{L^2(\mathbb{R}^d)} \le c \left(\|L\varphi\|_{L^2(\mathbb{R}^d)} + \|\varphi\|_{L^2(\mathbb{R}^d)} \right)$$

holds for all $\varphi \in C_0^\infty(\Omega)$ and $|\alpha| \le n$.

(c) Conversely, if (b) holds then L is elliptic.

19. Suppose u is harmonic in the punctured unit disc $\mathbb{D}^* = \{z \in \mathbb{C} : 0 < |z| < 1\}$.

(a) Show that if u is also continuous at the origin, then u is harmonic throughout the unit disc.

[Hint: Show that u is weakly harmonic.]

(b) Prove that the Dirichlet problem for the punctured unit disc is in general not solvable.

20. Let F be a continuous function on the closure $\overline{\mathbb{D}}$ of the unit disc. Assume that F is in C^1 on the (open) disc $\mathbb{D}$, and $\int_{\mathbb{D}} |\nabla F|^2 < \infty$.

Let $f(e^{i\theta})$ denote the restriction of F to the unit circle, and write $f(e^{i\theta}) \sim \sum_{n=-\infty}^{\infty} a_n e^{in\theta}$. Prove that $\sum_{n=-\infty}^{\infty} |n|\,|a_n|^2 < \infty$.

[Hint: Write $F(re^{i\theta}) \sim \sum_{n=-\infty}^{\infty} F_n(r) e^{in\theta}$, with $F_n(1) = a_n$. Express $\int_{\mathbb{D}} |\nabla F|^2$ in polar coordinates, and use the fact that

$$\frac{1}{2}|F(1)|^2 \leq L^{-1} \int_{1/2}^{1} |F'(r)|^2\, dr + L \int_{1/2}^{1} |F(r)|^2\, dr,$$

for $L \geq 2$; apply this to $F = F_n$, $L = |n|$.]

Chapter 6. Abstract Measure and Integration Theory

1. Let X be a set and $\mathcal{M}$ a non-empty collection of subsets of X. Prove that if $\mathcal{M}$ is closed under complements and countable unions of disjoint sets, then $\mathcal{M}$ is a σ-algebra.

[Hint: Any countable union of sets can be written as a countable union of disjoint sets.]

2. Let $(X, \mathcal{M}, \mu)$ be a measure space. One can define the **completion** of this space as follows. Let $\overline{\mathcal{M}}$ be the collection of sets of the form $E \cup Z$, where $E \in \mathcal{M}$, and $Z \subset F$ with $F \in \mathcal{M}$ and $\mu(F) = 0$. Also, define $\overline{\mu}(E \cup Z) = \mu(E)$. Then:

(a) $\overline{\mathcal{M}}$ is the smallest σ-algebra containing $\mathcal{M}$ and all subsets of elements of $\mathcal{M}$ of measure zero.

(b) The function $\overline{\mu}$ is a measure on $\overline{\mathcal{M}}$, and this measure is complete.

[Hint: To prove $\overline{\mathcal{M}}$ is a σ-algebra it suffices to see that if $E_1 \subset \overline{\mathcal{M}}$, then $E_1^c \subset \overline{\mathcal{M}}$. Write $E_1 = E \cup Z$ with $Z \subset F$, E and F in $\mathcal{M}$. Then $E_1^c = (E \cup F)^c \cup (F - Z)$.]

3. Consider the exterior Lebesgue measure m_* introduced in Chapter 1. Prove that a set E in $\mathbb{R}^d$ is Carathéodory measurable if and only if E is Lebesgue measurable in the sense of Chapter 1.

[Hint: If E is Lebesgue measurable and A is any set, choose a G_δ set G such that $A \subset G$ and $m_*(A) = m(G)$. Conversely, if E is Carathéodory measurable and $m_*(E) < \infty$, choose a G_δ set G with $E \subset G$ and $m_*(E) = m_*(G)$. Then $G - E$ has exterior measure 0.]

4. Let r be a rotation of $\mathbb{R}^d$. Using the fact that the mapping $x \mapsto r(x)$ preserves Lebesgue measure (see Problem 4 in Chapter 2 and Exercise 26 in Chapter 3), show that it induces a measure-preserving map of the sphere S^{d-1} with its measure $d\sigma$.

A converse is stated in Problem 4.

5. Use the polar coordinate formula to prove the following:

(a) $\int_{\mathbb{R}^d} e^{-\pi|x|^2}\,dx = 1$, when $d = 2$. Deduce from this that the same identity holds for all d.

(b) $\left(\int_0^\infty e^{-\pi r^2} r^{d-1}\,dr\right)\sigma(S^{d-1}) = 1$, and as a result, $\sigma(S^{d-1}) = 2\pi^{d/2}/\Gamma(d/2)$.

(c) If B is the unit ball, $v_d = m(B) = \pi^{d/2}/\Gamma(d/2+1)$, since this quantity equals $\left(\int_0^1 r^{d-1}\,dr\right)\sigma(S^{d-1})$. (See Exercise 14 in Chapter 2.)

6. A version of Green's formula for the unit ball B in $\mathbb{R}^d$ can be stated as follows. Suppose u and v are a pair of functions that are in $C^2(\overline{B})$. Then one has

$$\int_B (v\triangle u - u\triangle v)\, dx = \int_{S^{d-1}} \left(v\frac{\partial u}{\partial n} - u\frac{\partial v}{\partial n} \right) d\sigma.$$

Here S^{d-1} is the unit sphere with $d\sigma$ the measure defined in Section 3.2, and $\partial u/\partial n$, $\partial v/\partial n$ denote the directional derivatives of u and v (respectively) along the inner normals to S^{d-1}.

Show that the above can be derived from Lemma 4.5 of the previous chapter by taking $\eta = \eta_\epsilon^+$ and letting $\epsilon \to 0$.

7. There is an alternate version of the mean-value property given in (21) of Chapter 5. It can be stated as follows. Suppose u is harmonic in Ω, and B is any ball of center x_0 and radius r whose closure is contained in Ω. Then

$$u(x_0) = c \int_{S^{d-1}} u(x_0 + ry)\, d\sigma(y), \qquad \text{with } c^{-1} = \sigma(S^{d-1}).$$

Conversely, a continuous function satisfying this mean-value property is harmonic.

[Hint: This can be proved as a direct consequence of the corresponding result for averages over balls (Theorem 4.27 in Chapter 5), or can be deduced from Exercise 6.]

8. The fact that the Lebesgue measure is uniquely characterized by its translation invariance can be made precise by the following assertion: If μ is a Borel measure on $\mathbb{R}^d$ that is translation-invariant, and is finite on compact sets, then μ is a multiple of Lebesgue measure m. Prove this theorem by proceeding as follows.

(a) Suppose Q_a denotes a translate of the cube $\{x : 0 < x_j \leq a,\ j = 1, 2, \ldots, d\}$ of side length a. If we let $\mu(Q_1) = c$, then $\mu(Q_{1/n}) = cn^{-d}$ for each integer n.

(b) As a result μ is absolutely continuous with respect to m, and there is a locally integrable function f such that

$$\mu(E) = \int_E f\, dx.$$

(c) By the differentiation theorem (Corollary 1.7 in Chapter 3) it follows that $f(x) = c$ a.e., and hence $\mu = cm$.

[Hint: Q_1 can be written as a disjoint union of n^d translates of $Q_{1/n}$.]

9. Let $C([a,b])$ denote the vector space of continuous functions on the closed and bounded interval $[a,b]$. Suppose we are given a Borel measure μ on this interval, with $\mu([a,b]) < \infty$. Then

$$f \mapsto \ell(f) = \int_a^b f(x)\, d\mu(x)$$

is a linear functional on $C([a,b])$, with ℓ positive in the sense that $\ell(f) \geq 0$ if $f \geq 0$.

Prove that, conversely, for any linear functional ℓ on $C([a,b])$ that is positive in the above sense, there is a unique finite Borel measure μ so that $\ell(f) = \int_a^b f\, d\mu$ for $f \in C([a,b])$.

[Hint: Suppose $a = 0$ and $u \geq 0$. Define $F(u)$ by $F(u) = \lim_{\epsilon \to 0} \ell(f_\epsilon)$, where

$$f_\epsilon(x) = \begin{cases} 1 & \text{for } 0 \leq x \leq u, \\ 0 & \text{for } u + \epsilon \leq x, \end{cases}$$

and f_ϵ is linear between u and $u + \epsilon$. (See Figure 3.) Then F is increasing and right-continuous, and $\ell(f)$ can be written as $\int_a^b f(x)\, dF(x)$ via Theorem 3.5.]

The result also holds if $[a,b]$ is replaced by a closed infinite interval; we then assume that ℓ is defined on the continuous functions of bounded support, and obtain that the resulting μ is finite on all bounded intervals.

A generalization is given in Problem 5.

10. Suppose ν, ν_1, ν_2 are signed measures on $(X, \mathcal{M})$ and μ a (positive) measure on $\mathcal{M}$. Using the symbols $\perp$ and $\ll$ defined in Section 4.2, prove:

(a) If $\nu_1 \perp \mu$ and $\nu_2 \perp \mu$, then $\nu_1 + \nu_2 \perp \mu$.

(b) If $\nu_1 \ll \mu$ and $\nu_2 \ll \mu$, then $\nu_1 + \nu_2 \ll \mu$.

(c) $\nu_1 \perp \nu_2$ implies $|\nu_1| \perp |\nu_2|$.

(d) $\nu \ll |\nu|$.

(e) If $\nu \perp \mu$ and $\nu \ll \mu$, then $\nu = 0$.

11. Suppose that F is an increasing normalized function on $\mathbb{R}$, and let $F = F_A + F_C + F_J$ be the decomposition of F in Exercise 24 in Chapter 3; here F_A is

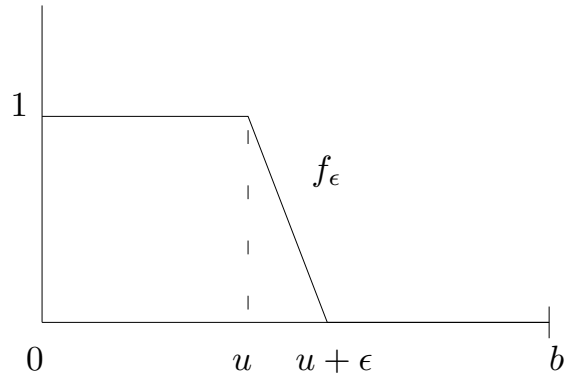


Figure 3. The function f_ϵ in Exercise 9

absolutely continuous, F_C is continuous with $F_C' = 0$ a.e, and F_J is a pure jump function. Let $\mu = \mu_A + \mu_C + \mu_J$ with μ, μ_A, μ_C, and μ_J the Borel measures associated to F, F_A, F_C, and F_J, respectively. Verify that:

(i) μ_A is absolutely continuous with respect to Lebesgue measure and $\mu_A(E) = \int_E F'(x)\,dx$ for every Lebesgue measurable set E.

(ii) As a result, if F is absolutely continuous, then $\int f\,d\mu = \int f\,dF = \int f(x)F'(x)\,dx$ whenever f and fF' are integrable.

(iii) $\mu_C + \mu_J$ and Lebesgue measure are mutually singular.

12. Suppose $\mathbb{R}^d - \{0\}$ is represented as $\mathbb{R}_+ \times S^{d-1}$, with $\mathbb{R}_+ = \{0 < r < \infty\}$. Then every open set in $\mathbb{R}^d - \{0\}$ can be written as a countable union of open rectangles of this product.

[Hint: Consider the countable collection of rectangles of the form

$$\{r_j < r < r'_k\} \times \{\gamma \in S^{d-1} : \ |\gamma - \gamma_\ell| < 1/n\}.$$

Here r_j and r'_k range over all positive rationals, and $\{\gamma_\ell\}$ is a countable dense set of S^{d-1}.]

13. Let m_j be the Lebesgue measure for the space $\mathbb{R}^{d_j}$, $j = 1, 2$. Consider the product $\mathbb{R}^d = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ $(d = d_1 + d_2)$, with m the Lebesgue measure on $\mathbb{R}^d$. Show that m is the completion (in the sense of Exercise 2) of the product measure $m_1 \times m_2$.

14. Suppose $(X_j, \mathcal{M}_j, \mu_j)$, $1 \leq j \leq k$, is a finite collection of measure spaces. Show that parallel with the case $k = 2$ considered in Section 3 one can construct a product measure $\mu_1 \times \mu_2 \times \cdots \times \mu_k$ on $X = X_1 \times X_2 \times \cdots \times X_k$. In fact, for any set $E \subset X$ such that $E = E_1 \times E_2 \times \cdots \times E_k$, with $E_j \subset \mathcal{M}_j$ for all j, define $\mu_0(E) = \prod_{j=1}^{k} \mu_j(E_j)$. Verify that μ_0 extends to a premeasure on the algebra $\mathcal{A}$ of finite disjoint unions of such sets, and then apply Theorem 1.5.

15. The product theory extends to infinitely many factors, under the requisite assumptions. We consider measure spaces $(X_j, \mathcal{M}_j, \mu_j)$ with $\mu_j(X_j) = 1$ for all but finitely many j. Define a **cylinder set** E as

$$\{x = (x_j),\ x_j \in E_j,\ E_j \in \mathcal{M}_j, \text{but } E_j = X_j \text{ for all but finitely many } j\}.$$

For such a set define $\mu_0(E) = \prod_{j=1}^{\infty} \mu_j(E_j)$. If $\mathcal{A}$ is the algebra generated by the cylinder sets, μ_0 extends to a premeasure on $\mathcal{A}$, and we can apply Theorem 1.5 again.

16. Consider the d-dimensional torus $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$. Identify $\mathbb{T}^d$ as $\mathbb{T}^1 \times \cdots \times \mathbb{T}^1$ (d factors) and let μ be the product measure on $\mathbb{T}^d$ given by $\mu = \mu_1 \times \mu_2 \times \cdots \times \mu_d$, where μ_j is Lebesgue measure on X_j identified with the circle $\mathbb{T}$. That is, if we represent each point in X_j uniquely as x_j with $0 < x_j \leq 1$, then the measure μ_j is the induced Lebesgue measure on $\mathbb{R}^1$ restricted to $(0, 1]$.

(a) Check that the completion μ is Lebesgue measure induced on the cube $Q = \{x : \ 0 < x_j \leq 1, \ j = 1, \ldots, d\}$.

(b) For each function f on Q let $\tilde{f}$ be its extension to $\mathbb{R}^d$ which is periodic, that is, $\tilde{f}(x + z) = \tilde{f}(x)$ for every $z \in \mathbb{Z}^d$. Then f is measurable on $\mathbb{T}^d$ if and only if $\tilde{f}$ is measurable on $\mathbb{R}^d$, and f is continuous on $\mathbb{T}^d$ if and only if $\tilde{f}$ is continuous on $\mathbb{R}^d$.

(c) Suppose f and g are integrable on $\mathbb{T}^d$. Show that the integral defining $(f * g)(x) = \int_{\mathbb{T}^d} f(x - y)g(y)\, dy$ is finite for a.e. x, that $f * g$ is integrable over $\mathbb{T}^d$, and that $f * g = g * f$.

(d) For any integrable function f on $\mathbb{T}^d$, write

$$f \sim \sum_{n \in \mathbb{Z}^d} a_n e^{2\pi i n \cdot x}$$

to mean that $a_n = \int_{\mathbb{T}^d} f(x) e^{-2\pi i n \cdot x}\, dx$. Prove that if g is also integrable, and $g \sim \sum_{n \in \mathbb{Z}^d} b_n e^{2\pi i n \cdot x}$, then

$$f * g \sim \sum_{n \in \mathbb{Z}^d} a_n b_n e^{2\pi i n \cdot x}.$$

(e) Verify that $\{e^{2\pi i n \cdot x}\}_{n \in \mathbb{Z}^d}$ is an orthonormal basis for $L^2(\mathbb{T}^d)$. As a result $\|f\|_{L^2(\mathbb{T}^d)} = \sum_{n \in \mathbb{Z}^d} |a_n|^2$.

(f) Let f be any continuous periodic function on $\mathbb{T}^d$. Then f can be uniformly approximated by finite linear combinations of the exponentials $\{e^{2\pi i n \cdot x}\}_{n \in \mathbb{Z}^d}$.

[Hint: For (e), reduce to the case $d = 1$ by Fubini's theorem. To prove (f) let $g(x) = g_\epsilon(x) = \epsilon^{-d}$, if $0 < x_j \leq \epsilon$, $j = 1, \ldots, d$, and $g_\epsilon(x) = 0$ elsewhere in Q. Then $(f * g_\epsilon)(x) \to f(x)$ uniformly as $\epsilon \to 0$. However $(f * g_\epsilon)(x) = \sum a_n b_n e^{2\pi i n x}$ with $b_n = \int_{\mathbb{T}^d} g_\epsilon(x) e^{-2\pi i n \cdot x}\, dx$, and $\sum |a_n b_n| < \infty$.]

17. By reducing to the case $d = 1$, show that each "rotation" $x \mapsto x + \alpha$ of the torus $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$ is measure preserving, for any $\alpha \in \mathbb{R}^d$.

18. Suppose τ is a measure-preserving transformation on a measure space (X, μ) with $\mu(X) = 1$. Recall that a measurable set E is invariant if $\tau^{-1}(E)$ and E differ by a set of measure zero. A sharper notion is to require that $\tau^{-1}(E)$ equal E. Prove that if E is any invariant set, there is a set E' so that $E' = \tau^{-1}(E')$, and E and E' differ by a set of measure zero.

[Hint: Let $E' = \limsup_{n\to\infty}\{\tau^{-n}(E)\} = \bigcap_{n=0}^{\infty}\left(\bigcup_{k\geq n}\tau^{-k}(E)\right)$.]

19. Let τ be a measure-preserving transformation on (X, μ) with $\mu(X) = 1$. Then τ is ergodic if and only if whenever ν is absolutely continuous with respect to μ and ν is invariant (that is, $\nu(\tau^{-1}(E)) = \nu(E)$ for all measurable sets E), then $\nu = c\mu$, with c a constant.

20. Suppose τ is a measure-preserving transformation on (X, μ). If

$$\mu(\tau^{-n}(E) \cap F) \to \mu(E)\mu(F)$$

as $n \to \infty$ for all measurable sets E and F, then $(T^n f, g) \to (f, 1)(1, g)$ whenever $f, g \in L^2(X)$ with $(Tf)(x) = f(\tau(x))$. Thus τ is mixing.

[Hint: By linearity the hypothesis implies the conclusion whenever f and g are simple functions.]

21. Let $\mathbb{T}^d$ be the torus, and $\tau : x \mapsto x + \alpha$ the mapping arising in Exercise 17. Then τ is ergodic if and only if $\alpha = (\alpha_1, \ldots, \alpha_d)$ with $\alpha_1, \alpha_2, \ldots, \alpha_d$, and 1 are linearly independent over the rationals. To do this show that:

(a) $\dfrac{1}{m} \displaystyle\sum_{k=0}^{m-1} f(\tau^k(x)) \to \int_{\mathbb{T}^d} f(x)\, dx$ as $m \to \infty$, for each $x \in \mathbb{T}^d$, whenever f is continuous and periodic and α satisfies the hypothesis.

(b) Prove as a result that in this case τ is uniquely ergodic.

[Hint: Use (f) in Exercise 16.]

22. Let $X = \prod_{i=1}^{\infty} X_i$, where each (X_i, μ_i) is identical to (X_1, μ_1), with $\mu_1(X_1) = 1$, and let μ be the corresponding product measure defined in Exercise 15. Define the **shift** $\tau : X \to X$ by $\tau((x_1, x_2, \ldots)) = (x_2, x_3, \ldots)$ for $x = (x_i) \in \prod_{i=1}^{\infty} X_i$.

(a) Verify that τ is a measure-preserving transformation.

(b) Prove that τ is ergodic by showing that it is mixing.

(c) Note that in general τ is not uniquely ergodic.

If we define the corresponding shift on the two-sided infinite product, then τ is also a measure-preserving isomorphism.

[Hint: For (b) note that $\mu(\tau^{-n}(E \cap F)) = \mu(E)\mu(F)$ whenever E and F are cylinder sets and n is sufficiently large. For (c) note that, for example, if we fix a point $\overline{x} \in X_1$, the set $E = \{(x_i) : \ x_j = \overline{x} \text{ all } j\}$ is invariant.]

23. Let $X = \prod_{i=1}^{\infty} Z(2)$, where each factor is the two-point space $Z(2) = \{0, 1\}$ with $\mu_1(0) = \mu_1(1) = 1/2$, and suppose μ denotes the product measure on X. Consider the mapping $D : X \to [0, 1]$ given by $D(\{a_j\}) \to \sum_{j=1}^{\infty} \frac{a_j}{2^j}$. Then there are denumerable sets $Z_1 \subset X$ and $Z_2 \subset [0, 1]$, such that:

(a) D is a bijection from $X - Z_1$ to $[0, 1] - Z_2$.

(b) A set E in X is measurable if and only if $D(E)$ is measurable in $[0, 1]$, and $\mu(E) = m(D(E))$, where m is Lebesgue measure on $[0, 1]$.

(c) The shift map on $\prod_{i=1}^{\infty} Z(2)$ then becomes the doubling map of example (b) in Section 5.4.

24. Consider the following generalization of the doubling map. For each integer m, $m \geq 2$, we define the map τ_m of $(0,1]$ by $\tau(x) = mx \mod 1$.

(a) Verify that τ is measure-preserving for Lebesgue measure.

(b) Show that τ is mixing, hence ergodic.

(c) Prove as a consequence that almost every number x is normal in the scale m, in the following sense. Consider the m-adic expansion of x,

$$x = \sum_{j=1}^{\infty} \frac{a_j}{m^j}, \qquad \text{where each } a_j \text{ is an integer } 0 \leq a_j \leq m-1.$$

Then x is **normal** if for each integer k, $0 \leq k \leq m-1$,

$$\frac{\#\{j : \ a_j = k, \ 1 \leq j \leq n\}}{N} \to \frac{1}{m} \qquad \text{as } N \to \infty.$$

Note the analogy with the equidistribution statements in Section 2, Chapter 4, of Book I.

25. Show that the mean ergodic theorem still holds if we replace the assumption that T is an isometry by the assumption that T is a **contraction**, that is, $\|Tf\| \leq \|f\|$ for all $f \in \mathcal{H}$.

[Hint: Prove that T is a contraction if and only if T^* is a contraction, and use the identity $(f, T^*f) = (Tf, f)$.]

26. There is an L^2 version of the maximal ergodic theorem. Suppose τ is a measure-preserving transformation on (X, μ). Here we do not assume that $\mu(X) < \infty$. Then

$$f^*(x) = \sup \frac{1}{m} \sum_{k=0}^{m-1} |f(\tau^k(x))|$$

satisfies

$$\|f^*\|_{L^2(X)} \leq c\|f\|_{L^2(X)}, \qquad \text{whenever } f \in L^2(X).$$

The proof is the same as outlined in Problem 6, Chapter 5 for the maximal function on $\mathbb{R}^d$. With this, extend the pointwise ergodic theorem to the case where $\mu(X) = \infty$, as follows:

(a) Show that $\lim_{m\to\infty} \frac{1}{m} \sum_{k=0}^{m-1} f(\tau^k(x))$ converges for a.e. x to $P(f)(x)$ for every $f \in L^2(X)$, because this holds for a dense subspace of $L^2(X)$.

(b) Prove that the conclusion holds for every $f \in L^1(X)$, because it holds for the dense subspace $L^1(X) \cap L^2(X)$.

27. We saw that if $\|f_n\|_{L^2} \leq 1$, then $\frac{f_n(x)}{n} \to 0$ as $n \to \infty$ for a.e. x. However, show that the analogue where one replaces the L^2-norm by the L^1-norm fails, by constructing a sequence $\{f_n\}$, $f_n \in L^1(X)$, $\|f_n\|_{L^1} \leq 1$, but with $\limsup_{n\to\infty} \frac{f_n(x)}{n} = \infty$ for a.e. x.

[Hint: Find intervals $I_n \subset [0,1]$, so that $m(I_n) = 1/(n \log n)$ but $\limsup_{n\to\infty}\{I_n\} = [0,1]$. Then take $f_n(x) = n \log n \chi_{I_n}$.]

28. We know by the Borel-Cantelli lemma that if $\{E_n\}$ is a collection of measurable sets in a measure a space (X, μ) and $\sum_{n=1}^{\infty} \mu(E_n) < \infty$ then $E = \limsup_{n\to\infty}\{E_n\}$ has measure zero.

In the opposite direction, if τ is a mixing measure-preserving transformation on X (with $\mu(X) = 1$), then whenever $\sum_{n=1}^{\infty} \mu(E_n) = \infty$, there are integers $m = m_n$ so that if $E'_n = \tau^{-m_n}(E_n)$, then $\limsup_{n\to\infty}(E'_n) = X$, except for a set of measure 0.

Chapter 7. Hausdorff Measure and Fractals

1. Show that the measure m_α is not σ-finite on $\mathbb{R}^d$ if $\alpha < d$.

2. Suppose E_1 and E_2 are two compact subsets of $\mathbb{R}^d$ such that $E_1 \cap E_2$ contains at most one point. Show directly from the definition of the exterior measure that if $0 < \alpha \leq d$, and $E = E_1 \cup E_2$, then

$$m_\alpha^*(E) = m_\alpha^*(E_1) + m_\alpha^*(E_2).$$

[Hint: Suppose $E_1 \cap E_2 = \{x\}$, let B_ϵ denote the open ball centered at x and of diameter ϵ, and let $E^\epsilon = E \cap B_\epsilon^c$. Show that

$$m_\alpha^*(E^\epsilon) \geq \mathcal{H}_\alpha^\epsilon(E) \geq m_\alpha^*(E) - \mu(\epsilon) - \epsilon^\alpha,$$

where $\mu(\epsilon) \to 0$. Hence $m_\alpha^*(E^\epsilon) \to m_\alpha^*(E)$.]

3. Prove that if $f : [0,1] \to \mathbb{R}$ satisfies a Lipschitz condition of exponent $\gamma > 1$, then f is a constant.

4. Suppose $f : [0,1] \to [0,1] \times [0,1]$ is surjective and satisfies a Lipschitz condition

$$|f(x) - f(y)| \leq C|x - y|^{\gamma}.$$

Prove that $\gamma \leq 1/2$ directly, without using Theorem 2.2.

[Hint: Divide $[0,1]$ into N intervals of equal length. The image of each sub-interval is contained in a ball of volume $O(N^{-2\gamma})$, and the union of all these balls must cover the square.]

5. Let $f(x) = x^k$ be defined on $\mathbb{R}$, where k is a positive integer and let E be a Borel subset of $\mathbb{R}$.

(a) Show that if $m_\alpha(E) = 0$ for some α, then $m_\alpha(f(E)) = 0$.

(b) Prove that $\dim(E) = \dim f(E)$.

6. Let $\{E_k\}$ be a sequence of Borel sets in $\mathbb{R}^d$. Show that if $\dim E_k \le \alpha$ for some α and all k, then

$$\dim \bigcup_k E_k \le \alpha.$$

7. Prove that the $(\log 2/\log 3)$-Hausdorff measure of the Cantor set is precisely equal to 1.

[Hint: Suppose we have a covering of $\mathcal{C}$ by finitely many closed intervals $\{I_j\}$. Then there exists another covering of $\mathcal{C}$ by intervals $\{I'_\ell\}$ each of length 3^{-k} for some k, such that $\sum_j |I_j|^\alpha \geq \sum_\ell |I'_\ell|^\alpha \geq 1$, where $\alpha = \log 2/\log 3$.]

8. Show that the Cantor set of constant dissection, $\mathcal{C}_\xi$, in Exercise 3 of Chapter 1 has strict Hausdorff dimension $\log 2/\log(2/(1-\xi))$.

9. Consider the set $\mathcal{C}_{\xi_1} \times \mathcal{C}_{\xi_2}$ in $\mathbb{R}^2$, with $\mathcal{C}_\xi$ as in the previous exercise. Show that $\mathcal{C}_{\xi_1} \times \mathcal{C}_{\xi_2}$ has strict Hausdorff dimension $\dim(\mathcal{C}_{\xi_1}) + \dim(\mathcal{C}_{\xi_2})$.

10. Construct a Cantor-like set (as in Exercise 4, Chapter 1) that has Lebesgue measure zero, yet Hausdorff dimension 1.

[Hint: Choose $\ell_1, \ell_2, \ldots, \ell_k, \ldots$ so that $1 - \sum_{j=1}^{k} 2^{j-1}\ell_j$ tends to zero sufficiently slowly as $k \to \infty$.]

11. Let $\mathcal{D} = \mathcal{D}_\mu$ be the Cantor dust in $\mathbb{R}^2$ given as the product $\mathcal{C}_\xi \times \mathcal{C}_\xi$, with $\mu = (1 - \xi)/2$.

(a) Show that for any real number λ, the set $\mathcal{C}_\xi + \lambda\mathcal{C}_\xi$ is similar to the projection of $\mathcal{D}$ on the line in $\mathbb{R}^2$ with slope $\lambda = \tan\theta$.

(b) Note that among the Cantor sets $\mathcal{C}_\xi$, the value $\xi = 1/2$ is critical in the construction of the Besicovitch set in Section 4.4. In fact, prove that with $\xi > 1/2$, then $\mathcal{C}_\xi + \lambda\mathcal{C}_\xi$ has Lebesgue measure zero for *every* λ. See also Problem 10 below.

[Hint: $m_\alpha(\mathcal{C}_\xi + \lambda\mathcal{C}_\xi) < \infty$ for $\alpha = \dim \mathcal{D}_\mu$.]

12. Define a primitive one-dimensional "measure" $\tilde{m}_1$ as

$$\tilde{m}_1 = \inf \sum_{k=1}^{\infty} \text{diam } F_k, \qquad E \subset \bigcup_{k=1}^{\infty} F_k.$$

This is akin to the one-dimensional exterior measure m_α^*, $\alpha = 1$, except that no restriction is placed on the size of the diameters F_k.

Suppose I_1 and I_2 are two *disjoint* unit segments in $\mathbb{R}^d$, $d \geq 2$, with $I_1 = I_2 + h$, and $|h| < \epsilon$. Then observe that $\tilde{m}_1(I_1) = \tilde{m}_1(I_2) = 1$, while $\tilde{m}_1(I_1 \cup I_2) \leq 1 + \epsilon$. Thus

$$\tilde{m}_1(I_1 \cup I_2) < \tilde{m}_1(I_1) + \tilde{m}_1(I_2) \qquad \text{when } \epsilon < 1;$$

hence $\tilde{m}_1$ fails to be additive.

13. Consider the von Koch curve $\mathcal{K}^\ell$, $1/4 < \ell < 1/2$, as defined in Section 2.1. Prove for it the analogue of Theorem 2.7: the function $t \mapsto \mathcal{K}^\ell(t)$ satisfies a Lipschitz condition of exponent $\gamma = \log(1/\ell)/\log 4$. Moreover, show that the set $\mathcal{K}^\ell$ has strict Hausdorff dimension $\alpha = 1/\gamma$.

[Hint: Show that if $\mathcal{O}$ is the shaded open triangle indicated in Figure 14, then $\mathcal{O} \supset S_0(\mathcal{O}) \cup S_1(\mathcal{O}) \cup S_2(\mathcal{O}) \cup S_3(\mathcal{O})$, where $S_0(x) = \ell x$, $S_1(x) = \rho_\theta(\ell x) + a$, $S_2(x) = \rho_\theta^{-1}(\ell x) + c$, and $S_3(x) = \ell x + b$, with ρ_θ the rotation of angle θ. Note that the sets $S_j(\mathcal{O})$ are disjoint.]

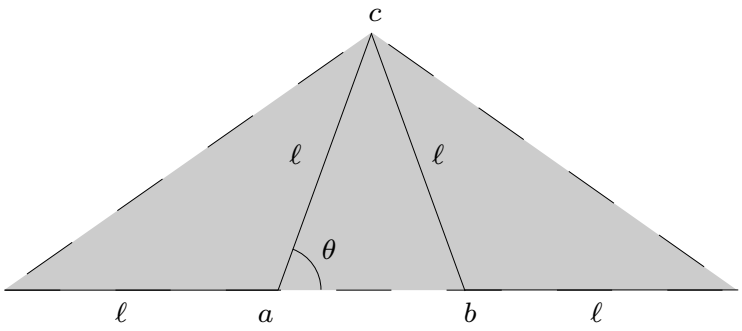


Figure 14. The open set $\mathcal{O}$ in Exercise 13

14. Show that if $\ell < 1/2$, the von Koch curve $t \mapsto \mathcal{K}^\ell(t)$ in Exercise 13 is a simple curve.

[Hint: Observe that if $t = \sum_{j=1}^{\infty} a_j/4^j$, with $a_j = 0, 1, 2$, or 3, then

$$\{\mathcal{K}(t)\} = \bigcap_{j=1}^{\infty} S_{a_j}\left(\cdots S_{a_2}\left(S_{a_1}(\overline{\mathcal{O}})\right)\right).]$$

15. Note that if we take $\ell = 1/2$ in the definition of the von Koch curve in Exercise 13 we get a "space-filling" curve, one that fills the right triangle whose vertices are $(0, 0)$, $(1, 0)$, and $(1/2, 1/2)$. The first three steps of the construction are as in Figure 15, with the intervals traced out in the indicated order.

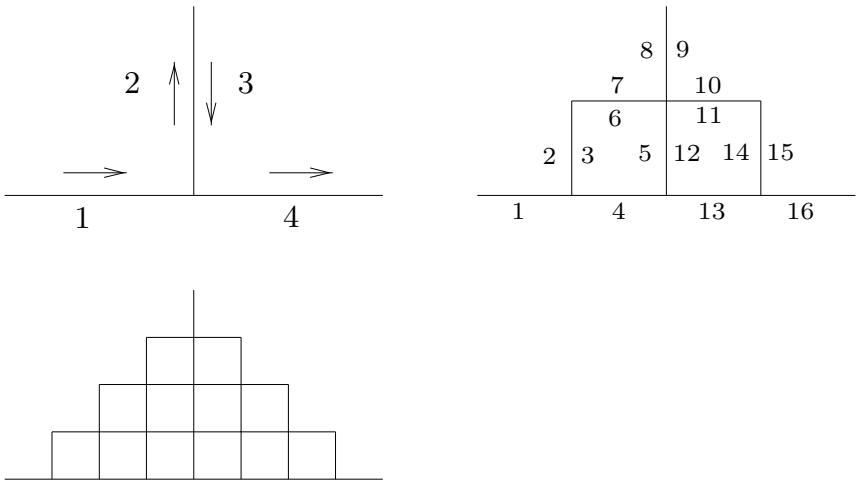


Figure 15. The first three steps of the von Koch curve when $\ell = 1/2$

16. Prove that the von Koch curve $t \mapsto \mathcal{K}^{\ell}(t)$, $1/4 < \ell \leq 1/2$ is continuous but nowhere differentiable.

[Hint: If $\mathcal{K}'(t)$ exists for some t, then

$$\lim_{n\to\infty} \frac{\mathcal{K}(u_n) - \mathcal{K}(v_n)}{u_n - v_n}$$

must exist, where $u_n \leq t \leq v_n$, and $u_n - v_n \to 0$. Choose $u_n = k/4^n$ and $v_n = (k+1)/4^n$.]

17. For a compact set E in $\mathbb{R}^d$, define $\#(\epsilon)$ to be the least number of balls of radius ϵ that cover E. Note that we always have $\#(\epsilon) = O(\epsilon^{-d})$ as $\epsilon \to 0$, and $\#(\epsilon) = O(1)$ if E is finite.

One defines the **covering dimension** of E, denoted by $\dim_C(E)$, as $\inf \beta$ such that $\#(\epsilon) = O(\epsilon^{-\beta})$, as $\epsilon \to 0$. Show that $\dim_C(E) = \dim_M(E)$, where $\dim_M$ is the Minkowski dimension discussed in Section 2.1, by proving the following inequalities for all $\delta > 0$:

(i) $m(E^\delta) \leq c\#(\delta)\delta^d$.

(ii) $\#(\delta) \leq c' m(E^\delta)\delta^{-d}$.

[Hint: To prove (ii), use Lemma 1.2 in Chapter 3 to find a collection of disjoint balls $B_1, B_2, \ldots, B_N$ of radius $\delta/3$, each centered at E, such that their "triples" $\tilde{B}_1, \tilde{B}_2, \ldots, \tilde{B}_N$ (of radius δ) cover E. Then $\#(\delta) \leq N$, while $Nm(B_j) = cN\delta^d \leq m(E^\delta)$, since the balls B_j are disjoint and are contained in E^δ.]

18. Let E be a compact set in $\mathbb{R}^d$.

(a) Prove that $\dim(E) \leq \dim_M(E)$, where dim and $\dim_M$ are the Hausdorff and Minkowski dimensions, respectively.

(b) However, prove that if $E = \{0, 1/\log 2, 1/\log 3, \ldots, 1/\log n, \ldots\}$, then $\dim_M E = 1$, yet $\dim E = 0$.

19. Show that there is a constant c_d, dependent only on the dimension d, such that whenever E is a compact set,

$$m(E^{2\delta}) \leq c_d m(E^{\delta}).$$

[Hint: Consider the maximal function f^*, with $f = \chi_{E^{\delta}}$, and take $c_d = 6^d$.]

20. Show that if F is the self-similar set considered in Theorem 2.12, then it has the same Minkowski dimension as Hausdorff dimension.

[Hint: Each F_k is the union of m^k balls of radius cr^k. In the converse direction one sees by Lemma 2.13 that if $\epsilon = r^k$, then each ball of radius ϵ can contain at most c' vertices of the kth generation. So it takes at least m^k/c' such balls to cover F.]

21. From the unit interval, remove the second and fourth quarters (open intervals). Repeat this process in the remaining two closed intervals, and so on. Let F be the limiting set, so that

$$F = \{x : \ x = \sum_{k=1}^{\infty} a_k/4^k \quad a_k = 0 \text{ or } 2\}.$$

Prove that $0 < m_{1/2}(F) < \infty$.

22. Suppose F is the self-similar set arising in Theorem 2.9.

(a) Show that if $m \leq 1/r^d$, then $m_d(F_i \cap F_j) = 0$ if $i \neq j$.

(b) However, if $m \geq 1/r^d$, prove that $F_i \cap F_j$ is not empty for some $i \neq j$.

(c) Prove that under the hypothesis of Theorem 2.12

$$m_\alpha(F_i \cap F_j) = 0, \qquad \text{with } \alpha = \log m / \log(1/r), \text{ whenever } i \neq j.$$

23. Suppose $S_1, \ldots, S_m$ are similarities with ratio r, $0 < r < 1$. For each set E, let

$$\tilde{S}(E) = S_1(E) \cup \cdots \cup S_m(E),$$

and suppose F denotes the unique non-empty compact set with $\tilde{S}(F) = F$.

(a) If $\overline{x} \in F$, show that the set of points $\{\tilde{S}^n(\overline{x})\}_{n=1}^{\infty}$ is dense in F.

(b) Show that F is **homogeneous** in the following sense: if $x_0 \in F$ and B is any open ball centered at x_0, then $F \cap B$ contains a set similar to F.

24. Suppose E is a Borel subset of $\mathbb{R}^d$ with $\dim E < 1$. Prove that E is totally disconnected, that is, any two distinct points in E belong to different connected components.

[Hint: Fix $x, y \in E$, and show that $f(t) = |t - x|$ is Lipschitz of order 1, and hence $\dim f(E) < 1$. Conclude that $f(E)$ has a dense complement in $\mathbb{R}$. Pick r in the complement of $f(E)$ so that $0 < r < f(y)$, and use the fact that $E = \{t \in E : |t - x| < r\} \cup \{t \in E : |t - x| > r\}$.]

25. Let $F(t)$ be an arbitrary non-negative measurable function on $\mathbb{R}$, and $\gamma \in S^{d-1}$. Then there exists a measurable set E in $\mathbb{R}^d$, such that $F(t) = m_{d-1}(E \cap \mathcal{P}_{t,\gamma})$.

26. Theorem 4.1 can be refined for $d \geq 4$ as follows.

Define $C^{k,\alpha}$ to be the class of functions $F(t)$ on $\mathbb{R}$ that are C^k and for which $F^{(k)}(t)$ satisfies a Lipschitz condition of exponent α.

If E has finite measure, then for a.e. $\gamma \in S^{d-1}$ the function $m(E \cap \mathcal{P}_{t,\gamma})$ is in $C^{k,\alpha}$ for $k = (d-3)/2$, $\alpha < 1/2$, if d is odd, $d \geq 3$; and for, $k = (d-4)/2$, $\alpha < 1$, if d is even, $d \geq 4$.

27. Show that the modification of the inequality (2) of Theorem 4.5 fails if we drop $\|f\|_{L^2(\mathbb{R}^d)}$ from the right-hand side.

[Hint: Consider $\mathcal{R}^*(f_\epsilon)$, with f_ϵ defined by $f_\epsilon(x) = (|x| + \epsilon)^{-d+\delta}$, for $|x| \leq 1$, with δ fixed, $0 < \delta < 1$, and $\epsilon \to 0$.]

28. Construct a compact set $E \subset \mathbb{R}^d$, $d \geq 3$, such that $m_d(E) = 0$, yet E contains translates of any segment of unit length in $\mathbb{R}^d$. (While particular examples of such sets can be easily obtained from the case $d = 2$, the determination of the least Hausdorff dimension among all such sets is an open problem.)

Functional Analysis: Introduction to Further Topics in Analysis

Chapter 1. L^p Spaces and Banach Spaces

1. Consider $L^p = L^p(\mathbb{R}^d)$ with Lebesgue measure. Let $f_0(x) = |x|^{-\alpha}$ if $|x| < 1$, $f_0(x) = 0$ for $|x| \geq 1$; also let $f_\infty(x) = |x|^{-\alpha}$ if $|x| \geq 1$, $f_\infty(x) = 0$ when $|x| < 1$. Show that:

(a) $f_0 \in L^p$ if and only if $p\alpha < d$.

(b) $f_\infty \in L^p$ if and only if $d < p\alpha$.

(c) What happens if in the definitions of f_0 and f_∞ we replace $|x|^{-\alpha}$ by $|x|^{-\alpha}/(\log(2/|x|))$ for $|x| < 1$, and $|x|^{-\alpha}$ by $|x|^{-\alpha}/(\log(2|x|))$ for $|x| \geq 1$?

2. Consider the spaces $L^p(\mathbb{R}^d)$, when $0 < p < \infty$.

(a) Show that if $\|f+g\|_{L^p} \leq \|f\|_{L^p} + \|g\|_{L^p}$ for all f and g, then necessarily $p \geq 1$.

(b) Consider $L^p(\mathbb{R})$ where $0 < p < 1$. Show that there are no bounded linear functionals on this space. In other words, if ℓ is a linear function $L^p(\mathbb{R}) \to \mathbb{C}$ that satisfies

$$|\ell(f)| \leq M\|f\|_{L^p(\mathbb{R})} \qquad \text{for all } f \in L^p(\mathbb{R}) \text{ and some } M > 0,$$

then $\ell = 0$.

[Hint: For (a), prove that if $0 < p < 1$ and $x, y > 0$, then $x^p + y^p > (x+y)^p$. For (b), let F be defined by $F(x) = \ell(\chi_x)$, where χ_x is the characteristic function of $[0, x]$, and consider $F(x) - F(y)$.]

3. If $f \in L^p$ and $g \in L^q$, both not identically equal to zero, show that equality holds in Hölder's inequality (Theorem 1.1) if and only if there exist two non-zero constants $a, b \geq 0$ such that $a|f(x)|^p = b|g(x)|^q$ for a.e. x.

4. Suppose X is a measure space and $0 < p < 1$.

(a) Prove that $\|fg\|_{L^1} \geq \|f\|_{L^p}\|g\|_{L^q}$. Note that q, the conjugate exponent of p, is negative.

(b) Suppose f_1 and f_2 are non-negative. Then $\|f_1 + f_2\|_{L^p} \geq \|f_1\|_{L^p} + \|f_2\|_{L^p}$.

(c) The function $d(f, g) = \|f - g\|_{L^p}^p$ for $f, g \in L^p$ defines a metric on $L^p(X)$.

5. Let X be a measure space. Using the argument to prove the completeness of $L^p(X)$, show that if the sequence $\{f_n\}$ converges to f in the L^p norm, then a subsequence of $\{f_n\}$ converges to f almost everywhere.

6. Let $(X, \mathcal{F}, \mu)$ be a measure space. Show that:

(a) The simple functions are dense in $L^\infty(X)$ if $\mu(X) < \infty$, and;

(b) The simple functions are dense in $L^p(X)$ for $1 \leq p < \infty$.

[Hint: For (a), use $E_{\ell,j} = \{x \in X : \frac{M\ell}{j} \leq f(x) < \frac{M(\ell+1)}{j}\}$ where $-j \leq \ell \leq j$, and $M = \|f\|_{L^\infty}$. Then consider the functions f_j that equal $M\ell/j$ on $E_{\ell,j}$. For (b) use a construction similar to that in (a).]

7. Consider the L^p spaces, $1 \leq p < \infty$, on $\mathbb{R}^d$ with Lebesgue measure. Prove that:

(a) The family of continuous functions with compact support is dense in L^p, and in fact:

(b) The family of indefinitely differentiable functions with compact support is dense in L^p.

The cases of L^1 and L^2 are in Theorem 2.4, Chapter 2 of Book III, and Lemma 3.1, Chapter 5 of Book III.

8. Suppose $1 \leq p < \infty$, and that $\mathbb{R}^d$ is equipped with Lebesgue measure. Show that if $f \in L^p(\mathbb{R}^d)$, then

$$\|f(x+h) - f(x)\|_{L^p} \to 0 \qquad \text{as } |h| \to 0.$$

Prove that this fails when $p = \infty$.

[Hint: By the previous exercise, the continuous functions with compact support are dense in $L^p(\mathbb{R}^d)$ for $1 \leq p < \infty$. See also Theorem 2.4 and Proposition 2.5 in Chapter 2 of Book III.]

9. Suppose X is a measure space and $1 \leq p_0 < p_1 \leq \infty$.

(a) Consider $L^{p_0} \cap L^{p_1}$ equipped with

$$\|f\|_{L^{p_0} \cap L^{p_1}} = \|f\|_{L^{p_0}} + \|f\|_{L^{p_1}}.$$

Show that $\|\cdot\|_{L^{p_0} \cap L^{p_1}}$ is a norm, and that $L^{p_0} \cap L^{p_1}$ (with this norm) is a Banach space.

(b) Suppose $L^{p_0} + L^{p_1}$ is defined as the vector space of measurable functions f on X that can be written as a sum $f = f_0 + f_1$ with $f_0 \in L^{p_0}$ and $f_1 \in L^{p_1}$. Consider

$$\|f\|_{L^{p_0} + L^{p_1}} = \inf \left\{ \|f_0\|_{L^{p_0}} + \|f_1\|_{L^{p_1}} \right\},$$

where the infimum is taken over all decompositions $f = f_0 + f_1$ with $f_0 \in L^{p_0}$ and $f_1 \in L^{p_1}$. Show that $\|\cdot\|_{L^{p_0} + L^{p_1}}$ is a norm, and that $L^{p_0} + L^{p_1}$ (with this norm) is a Banach space.

(c) Show that $L^p \subset L^{p_0} + L^{p_1}$ if $p_0 \leq p \leq p_1$.

10. A measure space (X, μ) is **separable** if there is a countable family of measurable subsets $\{E_k\}_{k=1}^{\infty}$ so that if E is any measurable set of finite measure, then

$$\mu(E \triangle E_{n_k}) \to 0 \qquad \text{as } k \to 0$$

for an appropriate subsequence $\{n_k\}$ which depends on E. Here $A \triangle B$ denotes the symmetric difference of the sets A and B, that is,

$$A \triangle B = (A - B) \cup (B - A).$$

(a) Verify that $\mathbb{R}^d$ with the usual Lebesgue measure is separable.

(b) The space $L^p(X)$ is **separable** if there exists a countable collection of elements $\{f_n\}_{n=1}^{\infty}$ in L^p that is dense. Prove that if the measure space X is separable, then L^p is separable when $1 \leq p < \infty$.

11. In light of the previous exercise, prove the following:

(a) Show that the space $L^\infty(\mathbb{R})$ is not separable by constructing for each $a \in \mathbb{R}$ an $f_a \in L^\infty$, with $\|f_a - f_b\| \geq 1$, if $a \neq b$.

(b) Do the same for the dual space of $L^\infty(\mathbb{R})$.

12. Suppose the measure space (X, μ) is separable as defined in Exercise 10. Let $1 \leq p < \infty$ and $1/p + 1/q = 1$. A sequence $\{f_n\}$ with $f_n \in L^p$ is said to converge to $f \in L^p$ **weakly** if

$$\int f_n g \, d\mu \to \int f g \, d\mu \qquad \text{for every } g \in L^q. \tag{22}$$

(a) Verify that if $\|f - f_n\|_{L^p} \to 0$, then f_n converges to f weakly.

(b) Suppose $\sup_n \|f_n\|_{L^p} < \infty$. Then, to verify weak convergence it suffices to check (22) for a dense subset of functions g in L^q.

(c) Suppose $1 < p < \infty$. Show that if $\sup_n \|f_n\|_{L^p} < \infty$, then there exists $f \in L^p$, and a subsequence $\{n_k\}$ so that f_{n_k} converges weakly to f.

Part (c) is known as the "weak compactness" of L^p for $1 < p < \infty$, which fails when $p = 1$ as is seen in the exercise below.

[Hint: For (b) use Exercise 10 (b).]

13. Below are some examples illustrating weak convergence.

(a) $f_n(x) = \sin(2\pi nx)$ in $L^p([0,1])$. Show that $f_n \to 0$ weakly.

(b) $f_n(x) = n^{1/p}\chi(nx)$ in $L^p(\mathbb{R})$. Then $f_n \to 0$ weakly if $p > 1$, but not when $p = 1$. Here χ denotes the characteristic function of $[0,1]$.

(c) $f_n(x) = 1 + \sin(2\pi nx)$ in $L^1([0,1])$. Then $f_n \to 1$ weakly also in $L^1([0,1])$, $\|f_n\|_{L^1} = 1$, but $\|f_n - 1\|_{L^1}$ does not converge to zero. Compare with Problem 6 part (d).

14. Suppose X is a measure space, $1 < p < \infty$, and suppose $\{f_n\}$ is a sequence of functions with $\|f_n\|_{L^p} \leq M < \infty$.

(a) Prove that if $f_n \to f$ a.e. then $f_n \to f$ weakly.

(b) Show that the above result may fail if $p = 1$.

(c) Show that if $f_n \to f_1$ a.e. and $f_n \to f_2$ weakly, then $f_1 = f_2$ a.e.

15. Minkowski's inequality for integrals. Suppose (X_1, μ_1) and (X_2, μ_2) are two measure spaces, and $1 \leq p \leq \infty$. Show that if $f(x_1, x_2)$ is measurable on $X_1 \times X_2$ and non-negative, then

$$\left\| \int f(x_1, x_2)\, d\mu_2 \right\|_{L^p(X_1)} \leq \int \|f(x_1, x_2)\|_{L^p(X_1)}\, d\mu_2.$$

Extend this statement to the case when f is complex-valued and the right-hand side of the inequality is finite.

[Hint: For $1 < p < \infty$, use a combination of Hölder's inequality, and its converse in Lemma 4.2.]

16. Prove that if $f_j \in L^{p_j}(X)$, where X is a measure space, $j = 1, \ldots, N$, and $\sum_{j=1}^{N} 1/p_j = 1$ with $p_j \geq 1$, then

$$\| \prod_{j=1}^{N} f_j \|_{L^1} \leq \prod_{j=1}^{N} \|f\|_{L^{p_j}} .$$

This is the multiple Hölder inequality.

17. The **convolution** of f and g on $\mathbb{R}^d$ equipped with the Lebesgue measure is defined by

$$(f * g)(x) = \int_{\mathbb{R}^d} f(x-y)g(y)\,dy.$$

(a) If $f \in L^p$, $1 \le p \le \infty$, and $g \in L^1$, then show that for almost every x the integrand $f(x-y)g(y)$ is integrable in y, hence $f * g$ is well defined. Moreover, $f * g \in L^p$ with

$$\|f * g\|_{L^p} \le \|f\|_{L^p}\|g\|_{L^1}.$$

(b) A version of (a) applies when g is replaced by a finite Borel measure μ: if $f \in L^p$, with $1 \le p \le \infty$, define

$$(f * \mu)(x) = \int_{\mathbb{R}^d} f(x-y)\,d\mu(y),$$

and show that $\|f * \mu\|_{L^p} \le \|f\|_{L^p}|\mu|(\mathbb{R}^d)$.

(c) Prove that if $f \in L^p$ and $g \in L^q$, where p and q are conjugate exponents, then $f * g \in L^\infty$ with $\|f * g\|_{L^\infty} \le \|f\|_{L^p}\|g\|_{L^q}$. Moreover, the convolution $f * g$ is uniformly continuous on $\mathbb{R}$, and if $1 < p < \infty$, then $\lim_{|x|\to\infty}(f * g)(x) = 0$.

[Hint: For (a) and (b) use the Minkowski inequality for integrals in Exercise 15. For part (c), use Exercise 8.]

18. We consider the L^p spaces with **mixed norm**, in a special case that is useful is several contexts.

We take as our underlying space the product space $\{(x,t)\} = \mathbb{R}^d \times \mathbb{R}$, with the product measure $dx\,dt$, where dx and dt are Lebesgue measures on $\mathbb{R}^d$ and $\mathbb{R}$ respectively. We define $L^r_t(L^p_x) = L^{p,r}$, with $1 \le p \le \infty$, $1 \le r \le \infty$, to be the space of equivalence classes of jointly measurable functions $f(x,t)$ for which the norm

$$\|f\|_{L^{p,r}} = \left(\int_{\mathbb{R}} \left(\int_{\mathbb{R}^d} |f(x,t)|^p \, dx \right)^{\frac{r}{p}} \, dt \right)^{\frac{1}{r}}$$

is finite (when $p < \infty$ and $r < \infty$), and an obvious variant when $p = \infty$ or $r = \infty$.

(a) Verify that $L^{p,r}$ with this norm is complete, and hence is a Banach space.

(b) Prove the general form of Hölder's inequality in this context

$$\int_{\mathbb{R}^d \times \mathbb{R}} |f(x,t)g(x,t)| \, dx\, dt \le \|f\|_{L^{p,r}} \|g\|_{L^{p',r'}},$$

with $1/p + 1/p' = 1$ and $1/r + 1/r' = 1$.

(c) Show that if f is integrable over all sets of finite measure, then

$$\|f\|_{L^{p,r}} = \sup \left| \int_{\mathbb{R}^d \times \mathbb{R}} f(x,t)g(x,t) \, dxdt \right|,$$

with the sup taken over all g that are simple and $\|g\|_{L^{p',r'}} \le 1$.

(d) Conclude that the dual space of $L^{p,r}$ is $L^{p',r'}$, if $1 \le p < \infty$, and $1 \le r < \infty$.

19. Young's inequality. Suppose $1 \le p, q, r \le \infty$. Prove the following on $\mathbb{R}^d$:

$$\|f * g\|_{L^q} \le \|f\|_{L^p} \|g\|_{L^r} \qquad \text{whenever } 1/q = 1/p + 1/r - 1.$$

Here, $f * g$ denotes the convolution of f and g as defined in Exercise 17.
[Hint: Assume $f, g \ge 0$, and use the decomposition

$$f(y)g(x-y) = f(y)^a g(x-y)^b [f(y)^{1-a} g(x-y)^{1-b}]$$

for appropriate a and b, together with Exercise 16 to find that

$$\left| \int f(y)g(x-y)\, dy \right| \le \|f\|_{L^p}^{1-q/p} \|g\|_{L^q}^{1-q/r} \left(\int |f(y)|^p |g(x-y)|^r \, dy \right)^{\frac{1}{q}} .]$$

20. Suppose X is a measure space, $0 < p_0 < p < p_1 \leq \infty$, and $f \in L^{p_0}(X) \cap L^{p_1}(X)$. Then $f \in L^p(X)$ and

$$\|f\|_{L^p} \leq \|f\|_{L^{p_0}}^{1-t} \|f\|_{L^{p_1}}^{t}, \qquad \text{if } t \text{ is chosen so that } \frac{1}{p} = \frac{1-t}{p_0} + \frac{t}{p_1}.$$

21. Recall the definition of a convex function. (See Problem 4, Chapter 3, in Book III.) Suppose φ is a non-negative convex function on $\mathbb{R}$ and f is real-valued and integrable on a measure space X, with $\mu(X) = 1$. Then we have **Jensen's inequality**:

$$\varphi\left(\int_X f\,d\mu\right) \leq \int_X \varphi(f)\,d\mu.$$

Note that if $\varphi(t) = |t|^p$, $1 \leq p$, then φ is convex and the above can be obtained from Hölder's inequality. Another interesting case is $\varphi(t) = e^{at}$.

[Hint: Since φ is convex, one has, $\varphi(\sum_{j=1}^N a_j x_j) \leq \sum_{j=1}^N a_j \varphi(x_j)$, whenever a_j, x_j are real, $a_j \geq 0$, and $\sum_{j=1}^N a_j = 1$.]

22. Another inequality of Young. Suppose φ and ψ are both continuous, *strictly* increasing functions on $[0, \infty)$ that are inverses of each other, that is, $(\varphi \circ \psi)(x) = x$ for all $x \geq 0$. Let

$$\Phi(x) = \int_0^x \varphi(u)\, du \qquad \text{and} \qquad \Psi(x) = \int_0^x \psi(u)\, du.$$

(a) Prove: $ab \leq \Phi(a) + \Psi(b)$ for all $a, b \geq 0$.

In particular, if $\varphi(x) = x^{p-1}$ and $\psi(y) = y^{q-1}$ with $1 < p < \infty$ and $1/p + 1/q = 1$, then we get $\Phi(x) = x^p/p$, $\Psi(y) = y^q/q$, and

$$A^\theta B^{1-\theta} \leq \theta A + (1-\theta)B \qquad \text{for all } A, B \geq 0 \text{ and } 0 \leq \theta \leq 1.$$

(b) Prove that we have equality in Young's inequality only if $b = \varphi(a)$ (that is, $a = \psi(b)$).

[Hint: Consider the area ab of the rectangle whose vertices are $(0,0)$, $(a,0)$, $(0,b)$ and (a,b), and compare it to areas "under" the curves $y = \Phi(x)$ and $x = \Psi(y)$.]

23. Let (X, μ) be a measure space and suppose $\Phi(t)$ is a continuous, convex, and increasing function on $[0, \infty)$, with $\Phi(0) = 0$. Define

$$L^{\Phi} = \{f \text{ measurable} : \int_X \Phi(|f(x)|/M)\, d\mu < \infty \text{ for some } M > 0\},$$

and

$$\|f\|_{\Phi} = \inf_{M>0} \int_X \Phi(|f(x)|/M)\, d\mu \leq 1.$$

Prove that:

(a) L^{Φ} is a vector space.

(b) $\|\cdot\|_{L^{\Phi}}$ is a norm.

(c) L^{Φ} is complete in this norm.

The Banach spaces L^{Φ} are called **Orlicz spaces**. Note that in the special case $\Phi(t) = t^p$, $1 \leq p < \infty$, then $L^{\Phi} = L^p$.

[Hint: Observe that if $f \in L^{\Phi}$, then $\lim_{N\to\infty} \int_X \Phi(|f|/N)\, d\mu = 0$. Also, use the fact that there exists $A > 0$ so that $\Phi(t) \geq At$ for all $t \geq 0$.]

24. Let $1 \le p_0 < p_1 < \infty$.

(a) Consider the Banach space $L^{p_0} \cap L^{p_1}$ with norm $\|f\|_{L^{p_0}\cap L^{p_1}} = \|f\|_{L^{p_0}} + \|f\|_{L^{p_1}}$. (See Exercise 9.) Let

$$\Phi(t) = \begin{cases} t^{p_0} & \text{if } 0 \le t \le 1, \\ t^{p_1} & \text{if } 1 \le t < \infty. \end{cases}$$

Show that L^Φ with its norm is equivalent to the space $L^{p_0} \cap L^{p_1}$. In other words, there exist $A, B > 0$, so that

$$A\|f\|_{L^{p_0}\cap L^{p_1}} \le \|f\|_{L^\Phi} \le B\|f\|_{L^{p_0}\cap L^{p_1}}.$$

(b) Similarly, consider the Banach space $L^{p_0} + L^{p_1}$ with its norm as defined in Exercise 9. Let

$$\Psi(t) = \int_0^t \psi(u)\,du \qquad \text{where} \qquad \psi(u) = \begin{cases} u^{p_1-1} & \text{if } 0 \le u \le 1, \\ u^{p_0-1} & \text{if } 1 \le u < \infty. \end{cases}$$

Show that L^Ψ with its norm is equivalent to the space $L^{p_0} + L^{p_1}$.

25. Show that a Banach space $\mathcal{B}$ is a Hilbert space if and only if the parallelogram law holds

$$\|f+g\|^2 + \|f-g\|^2 = 2(\|f\|^2 + \|g\|^2).$$

As a consequence, prove that if $L^p(\mathbb{R}^d)$ with the Lebesgue measure is a Hilbert space, then necessarily $p = 2$.

[Hint: For the first part, in the real case, let $(f,g) = \frac{1}{4}(\|f+g\|^2 + \|f-g\|^2)$.]

26. Suppose $1 < p_0, p_1 < \infty$ and $1/p_0 + 1/q_0 = 1$ and $1/p_1 + 1/q_1 = 1$. Show that the Banach spaces $L^{p_0} \cap L^{p_1}$ and $L^{q_0} + L^{q_1}$ are duals of each other up to an equivalence of norms. (See Exercise 9 for the relevant definitions of these spaces. Also, Problem 5^* gives a generalization of this result.)

27. The purpose of this exercise is to prove that the unit ball in L^p is strictly convex when $1 < p < \infty$, in the following sense. Here L^p is the space of real-valued functions whose p^{th} power are integrable. Suppose $\|f_0\|_{L^p} = \|f_1\|_{L^p} = 1$, and let

$$f_t = (1-t)f_0 + tf_1$$

be the straight-line segment joining the points f_0 and f_1. Then $\|f_t\|_{L^p} < 1$ for all t with $0 < t < 1$, unless $f_0 = f_1$.

(a) Let $f \in L^p$ and $g \in L^q$, $1/p + 1/q = 1$, with $\|f\|_{L^p} = 1$ and $\|g\|_{L^q} = 1$. Then

$$\int fg\, d\mu = 1$$

only when $f(x) = \operatorname{sign} g(x)|g(x)|^{q-1}$.

(b) Suppose $\|f_{t'}\|_{L^p} = 1$ for some $0 < t' < 1$. Find $g \in L^q$, $\|g\|_{L^q} = 1$, so that

$$\int f_{t'} g\, d\mu = 1$$

and let $F(t) = \int f_t g\, d\mu$. Observe as a result that $F(t) = 1$ for all $0 \le t \le 1$. Conclude that $f_t = f_0$ for all $0 \le t \le 1$.

(c) Show that the strict convexity fails when $p = 1$ or $p = \infty$. What can be said about these cases?

A stronger assertion is given in Problem 6*.

[Hint: To prove (a) show that the case of equality in $A^\theta B^{1-\theta} \le \theta A + (1-\theta)B$, for $A, B > 0$ and $0 < \theta < 1$ holds only when $A = B$.]

28. Verify the completeness of $\Lambda^{\alpha}(\mathbb{R}^d)$ and $L^p_k(\mathbb{R}^d)$.

29. Consider further the spaces $\Lambda^\alpha(\mathbb{R}^d)$.

(a) Show that when $\alpha > 1$ the only functions in $\Lambda^\alpha(\mathbb{R}^d)$ are the constants.

(b) Motivated by (a), one defines $C^{k,\alpha}(\mathbb{R}^d)$ to be the class of functions f on $\mathbb{R}^d$ whose partial derivatives of order less than or equal to k belong to $\Lambda^\alpha(\mathbb{R}^d)$. Here k is an integer and $0 < \alpha \leq 1$. Show that this space, endowed with the norm

$$\|f\|_{C^{k,\alpha}} = \sum_{|\beta| \leq k} \left\| \partial_x^\beta f \right\|_{\Lambda^\alpha(\mathbb{R}^d)},$$

is a Banach space.

30. Suppose $\mathcal{B}$ is a Banach space and $\mathcal{S}$ is a closed linear subspace of $\mathcal{B}$. The subspace $\mathcal{S}$ defines an equivalence relation $f \sim g$ to mean $f - g \in \mathcal{S}$. If $\mathcal{B}/\mathcal{S}$ denotes the collection of these equivalence classes, then show that $\mathcal{B}/\mathcal{S}$ is a Banach space with norm $\|f\|_{\mathcal{B}/\mathcal{S}} = \inf(\|f'\|_{\mathcal{B}}, \quad f' \sim f)$.

31. If Ω is an open subset of $\mathbb{R}^d$ then one definition of $L^p_k(\Omega)$ can be taken to be the quotient Banach space $\mathcal{B}/\mathcal{S}$, as defined in the previous exercise, with $\mathcal{B} = L^p_k(\mathbb{R}^d)$ and $\mathcal{S}$ the subspace of those functions which vanish a.e. on Ω. Another possible space, that we will denote by $L^p_k(\Omega^0)$, consists of the closure in $L^p_k(\mathbb{R}^d)$ of all f that have compact support in Ω. Observe that the natural mapping of $L^p_k(\Omega^0)$ to

$L^p_k(\Omega)$ has norm equal to 1. However, this mapping is in general not surjective. Prove this in the case when Ω is the unit ball and $k \geq 1$.

32. A Banach space is said to be separable if it contains a countable dense subset. In Exercise 11 we saw an example of a Banach space $\mathcal{B}$ that is separable, but where $\mathcal{B}^*$ is not separable. Prove, however, that in general when $\mathcal{B}^*$ is separable, then $\mathcal{B}$ is separable. Note that this gives another proof that in general L^1 is not the dual of L^∞.

33. Let V be a vector space over the complex numbers $\mathbb{C}$, and suppose there exists a real-valued function p on V satisfying:

$$\begin{cases} p(\alpha v) = |\alpha| p(v), & \text{if } \alpha \in \mathbb{C}, \text{ and } v \in V, \\ p(v_1 + v_2) \leq p(v_1) + p(v_2), & \text{if } v_1 \text{ and } v_2 \in V. \end{cases}$$

Prove that if V_0 is a subspace of V and ℓ_0 a linear functional on V_0 which satisfies $|\ell_0(f)| \leq p(f)$ for all $f \in V_0$, then ℓ_0 can be extended to a linear functional ℓ on V that satisfies $|\ell(f)| \leq p(f)$ for all $f \in V$.

[Hint: If $u = \mathrm{Re}(\ell_0)$, then $\ell_0(v) = u(v) - iu(iv)$. Apply Theorem 5.2 to u.]

34. Suppose $\mathcal{B}$ is a Banach space and $\mathcal{S}$ a closed proper subspace, and assume $f_0 \notin \mathcal{S}$. Show that there is a continuous linear functional ℓ on $\mathcal{B}$, so that $\ell(f) = 0$ for $f \in \mathcal{S}$, and $\ell(f_0) = 1$. The linear functional ℓ can be chosen so that $\|\ell\| = 1/d$ where d is the distance from f_0 to $\mathcal{S}$.

35. A linear functional ℓ on a Banach space $\mathcal{B}$ is continuous if and only if $\{f \in \mathcal{B} : \ell(f) = 0\}$ is closed.

[Hint: This is a consequence of Exercise 34.]

36. The results in Section 5.4 can be extended to d-dimensions.

(a) Show that there exists an extended-valued non-negative function $\hat{m}$ defined on all subsets of $\mathbb{R}^d$ so that (i) $\hat{m}$ is finitely additive; (ii) $\hat{m}(E) = m(E)$ whenever E is Lebesgue measurable, where m is Lebesgue measure; and $\hat{m}(E + h) = \hat{m}(E)$ for all sets E and every $h \in \mathbb{R}^d$. Prove this is as a consequence of (b) below.

(b) Show that there is an "integral" I, defined on all bounded functions on $\mathbb{R}^d/\mathbb{Z}^d$, so that $I(f) \geq 0$ whenever $f \geq 0$; the map $f \mapsto I(f)$ is linear; $I(f) = \int_{\mathbb{R}^d/\mathbb{Z}^d} f\,dx$ whenever f is measurable; and $I(f_h) = I(f)$ where $f_h(x) = f(x - h)$, and $h \in \mathbb{R}^d$.

Chapter 2. L^p Spaces in Harmonic Analysis

1. Show that an inequality

$$\|\{a_n\}\|_{L^q} \le A\|f\|_{L^p}, \qquad \text{for all } f \in L^p,$$

with $a_n = \frac{1}{2\pi}\int_0^{2\pi} f(\theta)e^{-in\theta}\,d\theta$, is possible only if $1/p + 1/q \le 1$.
[Hint: Let $D_N(\theta) = \sum_{|n|\le N} e^{in\theta}$ be the Dirichlet kernel. Then $\|D_N\|_{L^p} \approx N^{1-1/p}$ as $N \to \infty$, if $p > 1$ and $\|D_N\|_{L^1} \approx \log N$.]

2. The following are simple generalizations of the Hausdorff-Young inequalities.

(a) Suppose $\{\varphi_n\}$ is an orthonormal sequence on $L^2(X,\mu)$. Assume also that $|\varphi_n(x)| \le M$ for all n. If $a_n = \int f\overline{\varphi_n}\,d\mu$, then $\|a_n\|_{L^q} \le M^{(2/p)-1}\|f\|_{L^p(X)}$, $1 \le p \le 2$, $1/p + 1/q = 1$.

(b) Suppose $f \in L^p$ on the torus $\mathbb{T}^d$, and $a_n = \int_{\mathbb{T}^d} f(x)e^{-2\pi i n\cdot x}\,dx$, $n \in \mathbb{Z}^d$. Then $\|\{a_n\}\|_{L^q(\mathbb{Z}^d)} \le \|f\|_{L^p(\mathbb{T}^d)}$, where $1/q \le 1 - 1/p$.

3. Check that an inequality of the form $\|\hat{f}\|_{L^q(\mathbb{R}^d)} \leq A\|f\|_{L^p(\mathbb{R}^d)}$ (holding for all simple functions f) is possible if and only if $1/p + 1/q = 1$.
[Hint: Let $f_r(x) = f(rx)$, $r > 0$. Then $\hat{f}_r(\xi) = \hat{f}(\xi/r)r^{-d}$.]

4. Prove that another necessary condition for the inequality in the previous exercise is that $p \leq 2$. In fact the estimate

$$\int_{|\xi|\leq 1} |\hat{f}(\xi)|\, d\xi \leq A\|f\|_{L^p}$$

can hold only if $p \leq 2$.
[Hint: Let $f^s(x) = s^{-d/2}e^{-\pi|x|^2/s}$, $s = \sigma + it$, $\sigma > 0$. Then $(\widehat{f^s})(\xi) = e^{-\pi s|\xi|^2}$. Note that $\|f^s\|_{L^p} \leq ct^{d(1/p-1/2)}$ when $\sigma = 1$, and let $t \to \infty$.]

5. Let ψ be the conformal map of the strip $0 < \mathrm{Re}(z) < 1$ to the upper half-plane defined by $\psi(z) = e^{i\pi z}$. Check that $\Phi(z) = e^{-i\psi(z)}$ is continuous on the closure of the strip, $|\Phi(z)| = 1$ on the boundary lines, but $\Phi(z)$ is unbounded in the strip.

6. Extend the Riesz convexity theorem (in Section 2) to the $L^{p,r}$ spaces discussed in Exercise 18 of Chapter 1. We assume T is a linear transformation from simple functions to locally integrable functions. Suppose

$$\|T(f)\|_{L^{q_0,s_0}} \leq M_0\|f\|_{L^{p_0,r_0}}, \qquad \text{and} \qquad \|T(f)\|_{L^{q_1,s_1}} \leq M_1\|f\|_{L^{p_1,r_1}}$$

for all simple f. Prove that as a consequence $\|T(f)\|_{L^{q,s}} \leq M_\theta\|f\|_{L^{p,r}}$ where $\frac{1}{p} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}$, $\frac{1}{r} = \frac{1-\theta}{r_0} + \frac{\theta}{r_1}$, $\frac{1}{q} = \frac{1-\theta}{q_0} + \frac{\theta}{q_1}$, $\frac{1}{s} = \frac{1-\theta}{s_0} + \frac{\theta}{s_1}$, and $0 \leq \theta \leq 1$.

[Hint: Suppose f and g are a pair of simple functions with $\|f\|_{L^{p,r}} \leq 1$ and $\|f\|_{L^{q',s'}} \leq 1$. Define

$$f_z = |f(x,t)|^{p\alpha(z)} \frac{f(x,t)}{|f(x,t)|} \|f(\cdot,t)\|_{L^p(dx)}^{r\beta(z)-p\alpha(z)},$$

where $\alpha(z) = \frac{1-z}{p_0} + \frac{z}{p_1}$, $\beta(z) = \frac{1-z}{r_0} + \frac{z}{r_1}$. Note that when $z = \theta$, then $f_z = f$. Also

$$\|f_{1+it}\|_{L^{p_1,r_1}} \leq 1 \qquad \text{and} \qquad \|f_{0+it}\|_{L^{p_0,r_0}} \leq 1.$$

Make an analogous definition for g_z and consider $\int T(f_z)g_z\,dxdt$.]

7. Suppose f is a bounded function on $\mathbb{R}$ with compact support. Then $H(f) \in L^1(\mathbb{R})$ if and only if $\int f\,dx = 0$.
[Hint: If $a = \int f\,dx$, then $H(f)(x) = \frac{a}{\pi x} + O(1/x^2)$ as $|x| \to \infty$.]

8. Suppose T is a bounded linear transformation mapping the space of real-valued L^p functions into itself with

$$\|T(f)\|_{L^p} \leq M\|f\|_{L^p}.$$

(a) Let T' be the extension of T to complex-valued functions: $T'(f_1 + if_2) = T(f_1) + iT(f_2)$. Then T' has the same bound: $\|T'(f)\|_{L^p} \leq M\|f\|_{L^p}$.

(b) More generally, fix any N, then

$$\|(\sum_{j=1}^{N} |T(f_j)|^2)^{1/2}\|_{L^p} \leq M\|(\sum_{j=1}^{N} |f_j|^2)^{1/2}\|_{L^p}.$$

[Hint: For part (b), let ξ denote a unit vector in $\mathbb{R}^N$, and let $F_\xi = \sum_{j=1}^{N} \xi_j f_j$, $\xi = (\xi_1, \ldots, \xi_N)$. Then $\int |(TF_\xi)(x)|^p \leq M^p \int |F_\xi(x)|^p$. Integrate this inequality for ξ on the unit sphere.]

9. Show the identity of the following two classes of harmonic functions u in the upper half-plane $\mathbb{R}^2_+ = \{z : \ x + iy, \ y > 0\}$.

(a) The harmonic functions u that are continuous in the closure $\overline{\mathbb{R}^2_+}$, and that vanish at infinity (that is, $u(x, y) \to 0$, if $|x| + y \to \infty$).

(b) The functions representable as $u(x, y) = (f * \mathcal{P}_y)(x)$, where $\mathcal{P}_y(x)$ is the Poisson kernel $\frac{1}{\pi} \frac{y}{x^2+y^2}$, and f is a continuous function on $\mathbb{R}$ that vanishes at infinity (that is, $f(x) \to 0$, as $|x| \to \infty$).

[Hint: To show that (a) implies (b), let $f(x) = u(x, 0)$. Then $\mathcal{D}(x, y) = u(x, y) - (f * \mathcal{P}_y)(x)$ is harmonic in $\overline{\mathbb{R}^2_+}$, continuous on $\overline{\mathbb{R}^2_+}$, vanishes at infinity, and moreover $\mathcal{D}(x, 0) = 0$. Thus by the maximum principle, $\mathcal{D}(x, y) = 0$.]

10. Suppose $f \in L^p(\mathbb{R})$. Verify that:

(a) $\|f * \mathcal{P}_y\|_{L^p(\mathbb{R})} \leq \|f\|_{L^p}$, $1 \leq p \leq \infty$.

(b) $f * \mathcal{P}_y \to f$, as $y \to 0$, in the L^p norm, when $1 \leq p < \infty$.

11. Assume $f \in L^p(\mathbb{R})$, $1 < p < \infty$. Prove that:

(a) $f * \mathcal{Q}_y = H(f) * \mathcal{P}_y$, where H, $\mathcal{P}_y$ and $\mathcal{Q}_y$ are respectively the Hilbert transform, the Poisson kernel and the conjugate Poisson kernel.

(b) $f * \mathcal{Q}_y \to H(f)$ in the L^p norm, as $y \to 0$.

(c) $H_\epsilon(f) \to H(f)$ in the L^p norm, as $\epsilon \to 0$.

[Hint: Verify (a) first for $f \in L^2$ by noting that the Fourier transform of both sides equals $\hat{f}(\xi)\frac{\text{sign}(\xi)}{\xi}e^{-2\pi|\xi|y}$.]

12. In $\mathbb{R}^d$, suppose $f(x) = |x|^{-d}(\log 1/|x|)^{-1-\delta}$ if $|x| \leq 1/2$, $f(x) = 0$ otherwise. Then observe that $f^*(x) \geq c|x|^{-d}(\log 1/|x|)^{-\delta}$, if $|x| \leq 1/2$. Hence if $0 < \delta \leq 1$, we have $f \in L^1(\mathbb{R}^d)$ but $f^*(x)$ is not integrable over the unit ball.

13. Prove that the basic distribution function inequality (28) for the maximal function can essentially be reversed, that is, there is a constant A so that

$$m(\{x:\ f^*(x) > \alpha\}) \geq (A/\alpha) \int_{|f(x)|>\alpha} |f(x)|\, dx.$$

[Hint: Write $E_\alpha = \{x:\ f^*(x) > \alpha\}$ as $\bigcup_{j=1}^{\infty} Q_j$, with Q_j closed cubes satisfying (37), with $\Omega = E_\alpha$. For each Q_j let B_j be the smallest ball so that $Q_j \subset B_j$, and $\overline{B_j}$ intersects E_α^c. First $m(B_j) \leq cm(Q_j)$, then $\frac{1}{m(B_j)} \int_{B_j} |f|\, dx \leq \alpha$. Thus $m(Q_j) \geq \frac{c^{-1}}{\alpha} \int_{B_j} |f(x)|\, dx \geq \frac{c^{-1}}{\alpha} \int_{Q_j} |f(x)|\, dx$. Now add in j, and use the fact that $\{x:\ |f(x)| > \alpha\} \subset \{x:\ f^*(x) > \alpha\}$.]

14. Deduce the following important consequence from (28) and the previous exercise. Suppose f is an integrable function on $\mathbb{R}^d$, and B_1, B_2 are a pair of balls with $\overline{B_1} \subset B_2$.

(a) f^* is integrable on B_1 if $|f| \log(1 + |f|)$ is integrable on B_2.

(b) In the converse direction, whenever f^* is integrable on B_1 then $|f| \log(1 + |f|)$ is also integrable there.

[Hint: Integrate the inequalities in α, for $\alpha \geq 1$.]

15. Consider the weak-type space, consisting of all functions f for which $m(\{x : |f(x)| > \alpha\}) \leq \frac{A}{\alpha}$ for some A and all $\alpha > 0$. One might hope to define a norm on this space by taking the "norm" of f to be the least A for which the above inequality holds. Denote this quantity by $\mathcal{N}(f)$.

(a) Show, however, that $\mathcal{N}$ is not a genuine norm; moreover there is no norm $\|\cdot\|$ on this space so that $\|f\|$ is equivalent with $\mathcal{N}(f)$.

(b) Prove also that this space has no non-trivial bounded linear functionals.

[Hint: Consider $\mathbb{R}$. The function $f(x) = 1/|x|$ has $\mathcal{N}(f) = 2$. But if $f_N(x) = \frac{1}{N}[f(x+1) + f(x+2) + \cdots + f(x+N)]$, then $\mathcal{N}(f_N) \geq c \log N$.]

16. Prove that the space $\mathbf{H}^1_r$ is complete as follows. Let $\{f_n\}$ be a Cauchy sequence in $\mathbf{H}^1_r$. Then since $\{f_n\}$ is also Cauchy in L^1, there is an L^1 function f so that $f = \lim_{n\to\infty} f_n$ in the L^1 norm. Now for an appropriate sub-sequence $\{n_k\}$, write $f = f_{n_1} + \sum_{k=1}^{\infty}(f_{n_{k+1}} - f_{n_k})$.

17. Consider the function f defined by $f(x) = 1/(x(\log x)^2)$ for $0 < x \leq 1/2$ and $f(x) = 0$ if $x > 1/2$, and extended to $x < 0$ by $f(x) = -f(-x)$. Then f is integrable on $\mathbb{R}$, with $\int f = 0$, hence f is a multiple of a 1-atom in the terminology of Section 5.2.

Verify that $M(f) \geq c/(|x| \log |x|)$ for $|x| \leq 1/2$, hence $M(f) \notin L^1$, thus by Theorem 6.1 we know that $f \notin \mathbf{H}^1_r$.

18. Show that there exists a $c > 1$ so that every $f \in L^1(\mathbb{R}^d)$ can be written as $f(x) = \sum_{k=1}^{\infty} \lambda_k \mathfrak{a}_k(x)$, with $\sum |\lambda_k| \leq c\|f\|_{L^1}$, where the $\mathfrak{a}_k$ are "faux" atoms: each $\mathfrak{a}_k$ is supported in a ball B_k; $|\mathfrak{a}_k(x)| \leq 1/m(B_k)$ for all x; but $\mathfrak{a}_k$ does not necessarily satisfy the cancelation condition $\int \mathfrak{a}_k(x)\,dx = 0$.

[Hint: Let $f_n = \mathbb{E}_n(f)$, where $\mathbb{E}_n$ replaces f by its average over each dyadic cube of the nth-generation. Then $\|f_n - f\|_{L^1} \to 0$. Pick $\{n_k\}$ so that $\|f_{n_{k+1}} - f_{n_k}\|_{L^1} < 1/2^k$, and write $f = f_{n_1} + \sum_{k=1}^{\infty}(f_{n_{k+1}} - f_{n_k})$.]

19. The following illustrates two senses in which $\mathbf{H}^1_r$ is near L^1, but yet different.

(a) Suppose $f_0(x)$ is a positive decreasing function on $(0,\infty)$ that is integrable on $(0,\infty)$. Then show that there is a function $f \in \mathbf{H}^1_r(\mathbb{R})$ so that $|f(x)| \geq f_0(|x|)$.

(b) However if $f \in \mathbf{H}^1_r(\mathbb{R}^d)$, and f is positive on an open set, then its size must be "smaller" on that open set than a general integrable function. In fact, prove that if $f \in \mathbf{H}^1_r$, and $f \geq 0$ in a ball B_1, then $f \log(1+f)$ must be integrable over any proper sub-ball $B_0 \subset B_1$.

[Hint: For (a) take $f(x) = \text{sign}(x) f_0(|x|)$, and find an atomic decomposition for f. For (b) use Exercise 14, together with the maximal theorem in Section 6, with Φ positive.]

20. When $f \in L^1(\mathbb{R}^d)$ we know that its Fourier transform $\hat{f}$ is bounded and $\hat{f}(\xi)$ tends to 0 as $|\xi| \to \infty$ (the Riemann-Lebesgue lemma), but no better assertion about the "smallness" of $\hat{f}$ can be made. (For the analogous result for Fourier series, see Chapter 3 in Book III.) Show, however, that for $f \in \mathbf{H}^1_r$ we have

$$\int_{\mathbb{R}^d} |\hat{f}(\xi)| \frac{d\xi}{|\xi|^d} \leq A\|f\|_{\mathbf{H}^1_r}.$$

[Hint: Verify this for atoms.]

21. Prove that if $|f(x)| \le A(1+|x|)^{-d-1}$, and $\int_{\mathbb{R}^d} f(x)\,dx = 0$, then $f \in \mathbf{H}^1_r(\mathbb{R}^d)$. [Hint: While this is elementary, it is a little tricky. Write $f = \sum_{k=0}^{\infty} f_k$, where $f_0(x) = f(x)$ if $|x| \le 1$, 0 elsewhere, and $f_k(x) = f(x)$ if $2^{k-1} < |x| \le 2^k$ and 0 elsewhere, and $k \ge 1$. Let $c_k = \int f_k\,dx$, $s_k = \sum_{j \ge k} c_j$, then $s_0 = 0$. Fix a bounded function η supported in $|x| \le 1$, with $\int \eta(x)\,dx = 1$. Now write $f(x) = \sum_{k=0}^{\infty}(f_k - c_k\eta_k) + \sum_{k=0}^{\infty} c_k\eta_k$, where $\eta_k(x) = 2^{-kd}\eta(2^{-k}x)$ and $\int \eta_k = 1$. The first sum is clearly a sum of multiples of atoms (which are $O(2^{-k})$) supported on the balls $|x| \le 2^k$. That the second sum is similar can be seen by rewriting it as $\sum_{k=1}^{\infty} s_k(\eta_k - \eta_{k-1})$.]

22. Let f be the atom on $\mathbb{R}$ supported in $|x| \leq 1/2$ given by $f(x) = \text{sign}(x)$. Apply to f the maximal function f_0^* defined by

$$f_0^*(x) = \sup_{\epsilon>0} |(f * \chi_\epsilon)(x)|,$$

where χ is the characteristic function of $|x| \leq 1/2$ and $\chi_\epsilon(x) = \epsilon^{-1}\chi(x/\epsilon)$.

Verify that $|f_0^*(x)| \geq 1/(2|x|)$ if $|x| \geq 1/2$ hence $f_0^* \notin L^1$. Thus the maximal function f_0^*, defined in terms of χ, cannot be used to characterize the real Hardy space $\mathbf{H}_r^1$.

23. Verify the following examples related to BMO:

(a) $\log|x| \in \mathrm{BMO}(\mathbb{R}^d)$.

(b) If $f(x) = \log x$, when $x > 0$, and $= 0$ when $x \leq 0$, then $f \notin \mathrm{BMO}(\mathbb{R})$.

(c) If $\delta \geq 0$, $(\log|x|)^\delta \in \mathrm{BMO}(\mathbb{R}^d)$ if and only if $\delta \leq 1$.

[Hint: With $f(x) = \log|x|$, note that $f(rx) = f(x) + c_r$ and so we may assume the ball B has radius 1 in testing the condition (43). For (b), test f on small intervals centered at the origin.]

24. Using Exercises 19 (a) and 23, give examples of $f \in \mathbf{H}^1_r$ and $g \in \mathrm{BMO}$ so that $|f(x)g(x)|$ is not integrable over $\mathbb{R}^d$.

Chapter 3. Distributions: Generalized Functions

1. Suppose F is a distribution on Ω and $F = f$, with f a C^k function in Ω. Show that $\partial_x^\alpha F$, taken in the sense of distributions, agrees with $\partial_x^\alpha f$ for each $|\alpha| \leq k$.

2. The following represent converses to the previous exercise.

(a) Suppose f and g are continuous functions on $(a,b) \subset \mathbb{R}$ and $\frac{df}{dx}$ (taken in the sense of distributions) agrees with g. Show that for every $x \in (a,b)$, $(f(x+h) - f(x))/h \to g(x)$ as $h \to 0$.

(b) If f and g are merely assumed to be in $L^1(a,b)$ with $\frac{df}{dx} = g$ in the sense of distributions, then f is absolutely continuous and $(f(x+h) - f(x))/h \to g(x)$ as $h \to 0$ for a.e. x.

As a result, if f is a continuous but nowhere differentiable function on $\mathbb{R}$, then the distribution derivative of f is not a locally integrable function on any sub-interval.

(c) Generalize (a) as follows: Suppose $k \geq 1$ is an integer, and that f is a continuous function on an open set Ω. If for each multi-index α with $|\alpha| \leq k$, the distribution $\partial_x^\alpha f$ equals a continuous function g_α, then f is of class C^k and $\partial_x^\alpha f = g_\alpha$ as functions, for all $|\alpha| \leq k$.

[Hint: To see (a), let $x_0 \in (a,b)$, $h > 0$, and let η be a test function on (a,b) so that $\int \eta = 1$. With $\delta > 0$, define $\eta^\delta(x) = \delta^{-1}\eta(x/\delta)$ and

$$\varphi(x) = \int_{-\infty}^{x} \eta^\delta(x_0 + h - y) - \eta^\delta(x_0 - y)\, dy.$$

Then $\int f(x) \frac{d}{dx}\varphi(x)\, dx = -\int g(x)\varphi(x)\, dx$ and let $\delta, h \to 0$.

For (b), show as a first step that, up to a constant, f equals the indefinite integral of g, almost everywhere. Then use Theorem 3.8 in Chapter 3, Book III, about the differentiability almost everywhere of an absolutely continuous function.]

3. Show that a bounded function f on $\mathbb{R}^d$ satisfies a Lipschitz condition (also known as a Hölder condition of exponent 1)

$$|f(x) - f(y)| \leq C|x - y|, \qquad \text{for all } x, y \in \mathbb{R}^d,$$

if and only if $f \in L^\infty$ and all the first order partial derivatives $\partial f/\partial x^j$, $1 \leq j \leq d$, belong to L^∞ in the sense of distributions.

[Hint: Let $f_n = f * \psi_n$, where ψ_n is an approximation to the identity as in Corollary 1.2. Then $\partial f_n/\partial x_j \in L^\infty$ uniformly in n.]

4. Suppose F is a distribution on Ω.

(a) There exist $f_n \in C^\infty$, each of compact support in Ω, so that $f_n \to F$ in the sense of distributions.

(b) If F is supported in the compact set C, then for every $\epsilon > 0$ we can choose the f_n so that their supports are in the ϵ-neighborhood of C.

5. Let f be locally integrable on $\mathbb{R}^d$. Then the "support" of f in the measure-theoretic sense is the set $E = \{x : f(x) \neq 0\}$. Note that E is essentially determined only modulo sets of measure zero.

Show that the support of f, as a distribution, is equal to the intersection of all closed sets C such that $E - C$ has measure zero.

6. Assume that Ω is a region in $\mathbb{R}^d$ defined by $\Omega = \{x \in \mathbb{R}^d : \ x_d > \varphi(x')\}$, with $x = (x', x_d) \in \mathbb{R}^{d-1} \times \mathbb{R}$, and φ a C^1 function. Suppose f is a function that is continuous in $\overline{\Omega}$ and whose first derivatives are also continuous in $\overline{\Omega}$, with $f|_{\partial\Omega} = 0$.

Let $\tilde{f}$ be the extension of f to $\mathbb{R}^d$ defined by $\tilde{f}(x) = f(x)$ if $x \in \overline{\Omega}$, and $\tilde{f}(x) = 0$ if $x \notin \overline{\Omega}$. Then $\frac{\partial \tilde{f}}{\partial x_j}$, taken in the sense of distributions, is the function which is $\frac{\partial f}{\partial x_j}$ in $\overline{\Omega}$, and zero in $\overline{\Omega}^c$. (Note that it is not necessarily true that $\frac{\partial \tilde{f}}{\partial x_j}$ is continuous.) [Hint: Show that $-\int_\Omega f(x) \frac{\partial \psi}{\partial x_j}\, dx = \int_\Omega \frac{\partial f}{\partial x_j} \psi \, dx$ for all C^∞ functions ψ of compact support in $\mathbb{R}^d$.]

7. Show that the distribution F is tempered if and only if there is an integer N, and a constant A, so that for all $R \geq 1$,

$$|F(\varphi)| \leq AR^N \sup_{|x|\leq R,\ 0\leq|\alpha|\leq N} |\partial_x^\alpha \varphi(x)|\,,$$

for all $\varphi \in \mathcal{D}$ supported in $|x| \leq R$.

8. Suppose F is a homogeneous distribution of degree λ. Show that F is tempered.

[Hint: Fix $\eta \in \mathcal{D}$, $\eta(x) = 1$, for $|x| \leq 1$, η supported in $|x| \leq 2$. Let $\eta_R(x) = \eta(x/R)$. Find N so that $|\eta_1 F(\varphi)| \leq c\|\varphi\|_N$. Then deduce that $|(\eta_R F)(\varphi)| \leq cR^{N+|\lambda|}\|\varphi\|_N$.]

9. Check that on the real line, $f(x) = e^x$, considered as a distribution, is not tempered.

[Hint: Show that the criterion in Exercise 7 fails for every N.]

10. Verify that $\mathcal{D}$ is dense in $\mathcal{S}$.
[Hint: Fix $\eta \in \mathcal{D}$ so that $\eta = 1$ in a neighborhood of the origin. Let $\eta_k(x) = \eta(x/k)$ and consider $\varphi_k = \eta_k \varphi$.]

11. Suppose that $\varphi_1, \varphi_2 \in \mathcal{S}$.

(a) Verify that $\varphi_1 \cdot \varphi_2$ belongs to $\mathcal{S}$.

(b) Using the Fourier transform, prove that $\varphi_1 * \varphi_2 \in \mathcal{S}$.

(c) Show directly from the definition of convolution that $\varphi_1 * \varphi_2 \in \mathcal{S}$.

12. Prove that if F_1 is a distribution of compact support and $\varphi \in \mathcal{S}$, then $F_1 * \varphi \in \mathcal{S}$.

[Hint: For each N, there exists a constant c_N so that

$$\|\psi_y^{\sim}\|_N \leq c_N(1+|y|)^N \|\psi\|_N.]$$

13. Use the previous exercise to prove that if F_1 and F are distributions with F_1 having compact support and F being tempered then:

(a) $F * F_1$ is tempered, and;

(b) $(F * F_1)^\wedge = F_1^\wedge F^\wedge$, ($F_1^\wedge$ is C^∞ and slowly increasing.)

14. Check that $f(x) = \frac{1}{2}|x|$ is a fundamental solution for $\frac{d^2}{dx^2}$ on $\mathbb{R}$.

15. A d-dimensional generalization of the identity for the Heaviside function is the identity

$$\delta = \sum_{j=1}^{d} \left(\frac{\partial}{\partial x_j} \right) h_j,$$

with $h_j(x) = \frac{1}{A_d} \frac{x_j}{|x|^d}$, and $A_d = 2\pi^{d/2}/\Gamma(d/2)$ denotes the area of the unit sphere in $\mathbb{R}^d$.
[Hint: When $d > 2$, write $\delta = \sum_{j=1}^{d} \frac{\partial}{\partial x_j} \left(\frac{\partial}{\partial x_j} C_d |x|^{-d+2} \right)$.]

16. Consider the complex plane $\mathbb{C} = \mathbb{R}^2$, with $z = x + iy$.

(a) Note that the Cauchy-Riemann operator

$$\partial_{\overline{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)$$

is elliptic.

(b) Show that the locally integrable function $1/(\pi z)$ is a fundamental solution for $\partial_{\overline{z}}$.

(c) Suppose f is continuous in Ω, and $\partial_{\overline{z}} f = 0$ in the sense of distributions. Then f is analytic.

[Hint: For (b), use Theorem 2.9, and note that $\triangle = 4\partial_{\overline{z}}\partial_z$, where $\partial_z = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)$.]

17. Suppose $f(z)$ is a meromorphic function on $\Omega \subset \mathbb{C}$. Prove:

(a) $\log|f(z)|$ is locally integrable.

(b) $\triangle(\log|f(z)|)$ taken in the sense of distributions is equal to $2\pi\sum_j m_j\delta_j - 2\pi\sum_k m'_k\delta_k$. Here the δ_j are the delta functions placed at the distinct zeroes of f, namely $\delta_j(\varphi) = \varphi(z_j)$, and the δ_k are placed at the poles z'_k of f; also m_j, and m'_k are the respective multiplicities.

[Hint: $\frac{1}{2\pi}\log|z|$ is a fundamental solution of $\triangle$.]

18. Prove that a distribution F is homogeneous of degree λ if and only if

$$\sum_{j=1}^{d} x_j \frac{\partial F}{\partial x_j} = \lambda F.$$

[Hint: For the converse, consider $\Phi(a) = F(\varphi^a)$ for $a > 0$, $\varphi \in \mathcal{D}$. Then $\Phi(a)$ is C^∞ for $a > 0$, and $\frac{d\Phi(a)}{da} = \frac{\lambda}{a}\Phi(a)$.]

19. Prove the following facts about distributions in $\mathbb{R}$.

(a) Given a distribution F, there exists a distribution F_1 so that

$$\frac{d}{dx}F_1 = F.$$

(b) Show that F_1 is unique modulo an additive constant.

[Hint: For (a) fix $\varphi_0 \in \mathcal{D}$, with $\int \varphi_0 = 1$, and note that each $\varphi \in \mathcal{D}$ can be written uniquely as $\varphi = \frac{d\psi}{dx} + a\varphi_0$ for some $\psi \in \mathcal{D}$ and a constant a. Then define $F_1(\varphi) = F(\psi)$. For (b), use the fact that d/dx is elliptic.]

20. Show that if $\lambda_1, \ldots, \lambda_d$ are distinct complex exponents and $\sum_{j=1}^{n}(a_j x^{\lambda_j} + b_j x^{\lambda_j} \log x) = 0$ for all $x > 0$, then $a_j = b_j = 0$ for all $1 \leq j \leq n$.

[Hint: Proceed as in the proof of Lemma 2.6, and use the fact that $\int_1^R x^{-1+i\mu_j}\, dx$ is equal to $\log R$ if $\mu_j = 0$ and that this integral is $O(1)$ if μ_j is real and $\neq 0$.]

21. Let $F(x,t) = \mathcal{H}_t(x)$, for $t > 0$, and $F(x,t) = 0$, when $t \leq 0$, as in Theorem 2.10. Prove directly that

$$\hat{F}(\xi,\tau) = \frac{1}{4\pi^2|\xi|^2 + 2\pi i\tau}$$

where $(\xi,\tau) \in \mathbb{R}^d \times \mathbb{R}$, with ξ dual to x, and τ dual to t.
[Hint: Use the two identities

$$\int_0^\infty e^{-4\pi^2|\xi|^2 t} e^{-2\pi i\tau t}\, dt = \frac{1}{4\pi^2|\xi|^2 + 2\pi i\tau} \qquad \text{for } |\xi| > 0$$

and

$$\int_{\mathbb{R}^d} \mathcal{H}_t(x) e^{-2\pi i x\cdot\xi}\, dx = e^{-4\pi^2|\xi|^2 t} \qquad \text{for } t > 0.]$$

22. Suppose f is a locally integrable function defined on $\mathbb{R}$, and let u be the function defined by $u(x,t) = f(x-t)$, for $(x,t) \in \mathbb{R}^2$. Verify that u, taken as a distribution, satisfies the wave equation

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}.$$

More generally, let F be any distribution on $\mathbb{R}$. Construct U (in analogy to $f(x-t)$) as follows. If φ is in $\mathcal{D}(\mathbb{R}^2)$, $\mathbb{R}^2 = \{(x,t)\}$, set $U(\varphi) = \int_{\mathbb{R}} (F * \varphi(x,\cdot))(x)\,dx$. Then U satisfies

$$\frac{\partial^2 U}{\partial x^2} = \frac{\partial^2 U}{\partial t^2}.$$

Note that U is invariant under the translations (h,h), for $h \in \mathbb{R}$.

23. Show that in $\mathbb{R}^3$ the function

$$F(x) = \frac{-1}{4\pi|x|} e^{-|x|}$$

is a fundamental solution of the operator $\triangle - I$. The function F is the "Yukawa potential" in the theory of elementary particles. In contrast to the "Newtonian potential" $-1/(4\pi|x|)$, the fundamental solution of $\triangle$, the function F has a very rapid decay at infinity and it thus accounts for the short-range forces in the theory. [Hint: Let F be the inverse Fourier transform of $-(1 + 4\pi^2|\xi|^2)^{-1}$. Going to polar coordinates in $\mathbb{R}^3$, one then uses the identity

$$\int_{|\xi|=1} e^{2\pi i \xi \cdot x} \, d\sigma(\xi) = \frac{2\sin(2\pi|x|)}{|x|},$$

together with the Fourier transform of the conjugate Poisson kernel, given by (18) of the previous chapter.]

24. The following statements deal with the uniqueness of the fundamental solutions of the Laplacian.

(a) Up to an additive constant, the unique fundamental solutions of $\triangle$ in $\mathbb{R}^d$, $d \geq 2$, that are rotationally invariant, are the ones given in Theorems 2.8 and 2.9.

(b) The unique fundamental solution of $\triangle$ in $\mathbb{R}^d$, $d \geq 3$, that vanishes at infinity is the one given in Theorem 2.8.

25. A distribution F defined on $\Omega \subset \mathbb{R}$ is **positive** if $F(\varphi) \geq 0$ for all $\varphi \in \mathcal{D}$ supported in Ω, with $\varphi \geq 0$. Show that F is positive if and only if $F(\varphi) = \int \varphi \, d\mu$ for some Borel measure $d\mu$ on Ω that is finite on compact subsets.

26. Recall that a real-valued function on (a, b) is **convex** if $f(x_0(1-t)+x_1t) \leq (1-t)f(x_0)+tf(x_1)$, for $x_0, x_1 \in (a, b)$, $0 \leq t \leq 1$. (See also Problem 4 in Chapter 3, Book III.) A function f on $\Omega \subset \mathbb{R}^d$ is convex if the restriction of f to any line segment in Ω is convex.

(a) Suppose f is continuous on (a, b). Then it is convex if and only if the distribution $\frac{d^2 f}{dx^2}$ is positive.

(b) If f is continuous on $\Omega \subset \mathbb{R}^d$, it is convex if and only if for each $\xi = (\xi_1, \ldots, \xi_d) \subset \mathbb{R}^d$ the distribution $\sum_{1 \leq i,j, \leq d} \xi_j \xi_j \frac{\partial^2 f}{\partial x_i \partial x_j}$ is positive.

[Hint: For (a), let $\varphi \in \mathcal{D}$, $\varphi \geq 0$, $\int \varphi \, dx = 1$ and set $\varphi_\epsilon(x) = \epsilon^{-1}\varphi(x/\epsilon)$. Consider $f_\epsilon = f * \varphi_\epsilon$.]

27. Every distribution F of compact support in $\mathbb{R}^d$ is of **finite order** in the following sense: for each such F, there exists an integer M and continuous functions F_α of compact support, so that

$$F = \sum_{|\alpha| \leq M} \partial_x^\alpha F_\alpha.$$

Moreover if F is supported in C, then for every $\epsilon > 0$ we may take F_α to be supported in an ϵ-neighborhood of C. Prove this by carrying out the following three steps.

(a) Pick N so that $|F(\varphi)| \leq c\|\varphi\|_N$, for all $\varphi \in \mathcal{S}$, and choose M_0 so that $2M_0 > d + N$. Let Q be the inverse Fourier transform of $1/(1 + 4\pi^2|\xi|^2)^{M_0}$, and observe that Q is a fundamental solution of $(1 - \triangle)^{M_0}$, and Q is of class C^N.

(b) For each ϵ, construct Q_ϵ corresponding to Q, so that $(1 - \triangle)^{M_0} Q_\epsilon = \delta + r_\epsilon$, where Q_ϵ is supported in the ϵ-neighborhood of the origin (as in Corollary 2.13). Prove that $F * Q_\epsilon$ is a continuous function, using the fact that $|F(\varphi)| \leq c\|\varphi\|_N$.

(c) Hence $F = (1 - \triangle)^{M_0}(Q_\epsilon * F) - F * r_\epsilon$, and the result is proved with $M = 2M_0$.

28. One can characterize tempered distributions F whose Fourier transforms have compact support.

We already know by Proposition 1.6 that such an F must in fact be a function f that is C^∞ and slowly increasing. A precise characterization when $d = 1$ is given in the statement below.

The Fourier transform of a tempered distribution F is supported in the interval $[-M, M]$ if and only F equals a function f that is C^∞, slowly increasing, and having an analytic extension to the complex plane as an entire function of exponential type $2\pi M$; that is, for every $\epsilon > 0$, $|f(z)| \leq A_\epsilon e^{2\pi(M+\epsilon)|z|}$, where $z = x + iy$.

(An analogous assertion holds in higher dimensions.)

[Hint: Assume $\hat{F}$ is supported in $[-M, M]$. Using Exercise 27 allows us to write $\hat{F} = \sum_{|\alpha| N} \partial_x^\alpha(g_\alpha)$, where g_α are continuous and supported in $[-M - \epsilon, M + \epsilon]$, and thus reduce to the case when $\hat{F}$ is a continuous function.

To prove the converse, consider $f_\delta = f\gamma_\delta$ where $\gamma_\delta(x) = \frac{1}{\delta}\int e^{-2\pi i x\xi}\eta(\xi/\delta)\,d\xi$ with $\eta \in C^\infty$, supported in $|\xi| \leq 1$ and such that $\int \eta = 1$. Then $\gamma_\delta(z)$ is of exponential type $2\pi\delta$ and is rapidly decreasing on the real axis. Thus apply the simpler version of the result given in Theorem 3.3, Chapter 4 in Book II to the function f_δ, and let $\delta \to 0$.]

29. In this exercise, we consider the L^2 Sobolev spaces.

The space L^2_m consists of the functions $f \in L^2(\mathbb{R}^d)$ whose derivatives $\partial_x^\alpha f$ taken in the sense of distributions, are in $L^2(\mathbb{R}^d)$ for all $|\alpha| \le m$. This space is sometimes denoted by $H_m(\mathbb{R}^d)$. Note that this is the special case for $p = 2$ of the Sobolev space given as an example in Section 3 of Chapter 1. However, here we use a slightly different (but equivalent) norm, which makes L^2_m into a Hilbert space.

On L^2_m we define the inner product

$$(f, g)_m = \sum_{|\alpha| \le m} (\partial_x^\alpha f, \partial_x^\alpha g)_0,$$

with $(f, g)_0 = \int_{\mathbb{R}^d} f(x)\overline{g(x)}\, dx$. Then, L^2_m with the norm $\|f\|_{L^2_m} = (f, f)_m^{1/2}$ is a Hilbert space.

(a) Verify that $f \in L^2_m$ if and only if $\hat{f}(\xi)(1 + |\xi|)^m \in L^2$, and that the norms $\|f\|_{L^2_m}$ and $\|\hat{f}(\xi)(1 + |\xi|)^m\|_{L^2}$ are equivalent.

(b) If $m > d/2$, then f can be corrected on a set of measure zero, so that f becomes continuous and is in fact in C^k, for $k < m - d/2$. This is a version of the Sobolev embedding theorem.

[Hint: $\xi^\alpha \hat{f} \in L^1(\mathbb{R}^d)$ if $|\alpha| < m - d/2$.]

30. The following observation is useful in connection with the L^2 theory of Calderón-Zygmund distributions on $\mathbb{R}^d$.

(a) The Fourier transform of the distribution $\left[\frac{1}{|x|^d}\right]$ equals $c_1 \log|\xi| + c_2$, with $c_1 \neq 0$.

(b) Prove the following consequence of (a). Suppose k is a homogeneous function of degree $-d$ that is C^∞ away from the origin and with

$$\int_{|x|=1} k(x)\, d\sigma(x) \neq 0.$$

If K is any distribution that agrees with k away from the origin, then the Fourier transform of K is not a bounded function. Another way of stating this is that the operator T, defined by $T(\varphi) = K * \varphi$ initially defined for $\varphi \in \mathcal{D}$, does not extend to a bounded operator on $L^2(\mathbb{R}^d)$.

31. Suppose k is a C^∞ function homogenenous of degree $-d$, not identically equal to zero, and

$$\int_{|x|=1} k(x)\, d\sigma(x) = 0.$$

If K is the principal value distribution defined by k, that is, $K = \mathrm{pv}(k)$, then K is a Calderón-Zygmund distribution but the operator T given by $Tf = f * K$ is not bounded on L^1 or L^∞.

The special case of the Hilbert transform is in Exercise 7, Chapter 2.

[Hint: If $\varphi \in \mathcal{D}$, then $T\varphi(x) = ck(x) + O(|x|^{-d-1})$ as $|x| \to \infty$, where $c = \int \varphi$.]

32. The cancelation condition (21) for the Calderón-Zygmund distributions for some $n > 1$ implies the condition for $n = 1$. Show this by first proving the following fact: Whenever K satisfies (20) and (21) for some $n \geq 1$, then for every $1 \leq j \leq d$, the distribution $x_j \cdot K$ equals the locally integrable function $x_j k$.

[Hint: The distribution $x_j K - x_j k$ is supported at the origin. Then use Theorem 1.7 to test $x_j K - x_j k$ against φ_r as $r \to 0$ for suitable φ, to conclude that this difference vanishes. Next, write any $C^{(1)}$-normalized bump function as $\varphi(x) = \eta(x) + \sum_{j=1}^{d} x_j \varphi_j(x)$ where η and the φ_j are multiples of $C^{(n)}$ and $C^{(0)}$-normalized bump functions respectively, and use the above fact.]

33. Suppose k is a C^∞ function in $\mathbb{R}^d - \{0\}$, that satisfies the differential inequalities (20). Then there is a Calderón-Zygmund distribution K which has k as its associated function if and only if $\sup_{0<a<b} \left| \int_{a<|x|<b} k(x)\,dx \right| < \infty$.

[Hint: In one direction, note that $|K(\eta_b - \eta_a)| \leq 2A$, where $\eta(x) = 1$ if $|x| \leq 1/2$, and $\eta(x) = 0$ if $|x| \geq 1$, with $\eta \in C^\infty$. In the other direction, define

$$K(\varphi) = \int_{|x|\leq 1} k(x)(\varphi(x) - \varphi(0))\,dx + \int_{|x|\geq 1} k(x)\varphi(x)\,dx$$

and verify that the conditions (20) and (21) hold for K.]

34. Suppose H is a Calderón-Zygmund distribution and η belongs to $\mathcal{S}$. Verify that ηK is a Calderón-Zygmund distribution.

Chapter 4. Applications of the Baire Category Theorem

1. Below are some examples of generic sets and sets of the first category.

(a) Let $\{x_j\}_{j=1}^{\infty}$ denote an enumeration of the rational numbers in $\mathbb{R}$, and consider the sets

$$U_n = \bigcup_{j=1}^{\infty}(x_j - \frac{1}{n2^j},\ x_j + \frac{1}{n2^j}), \qquad \text{and} \qquad U = \bigcap_{n=1}^{\infty} U_n.$$

Show that U is generic but has Lebesgue measure zero.

(b) Use a Cantor-like set (as described, for example, in Exercise 4, Chapter 1 of Book III) to give an example of a subset of the first category that has full Lebesgue measure in $[0, 1]$. Note that automatically this subset will be uncountable and dense. Also, its complement is generic and has measure zero, giving an alternative to the set U in (a).

2. Suppose F is a closed subset and $\mathcal{O}$ an open subset of a complete metric space.

(a) Show that F is of the first category if and only if F has empty interior.

(b) Show that $\mathcal{O}$ is of the first category if and only if $\mathcal{O}$ is empty.

(c) Consequently, prove that F is generic if and only if $F = X$; and $\mathcal{O}$ is generic if and only if $\mathcal{O}^c$ contains no interior.

[Hint: For (a), argue by contradiction, assuming that a closed ball $\overline{B}$ is contained in F. Apply the category theorem to the complete metric space $\overline{B}$.]

3. Show that the conclusion of the Baire category theorem continues to hold if X_0 is a metric space that arises as an open subset of a complete metric space X.

[Hint: Apply the Baire category theorem to the closure of X_0 in X.]

4. Prove that every continuous function on $[0, 1]$ can be approximated uniformly by continuous nowhere differentiable functions. Do so by either:

(a) using Theorem 1.5.

(b) using only the fact that a continuous nowhere differentiable function exists.

5. Let X be a complete metric space. We recall that a set is a G_δ in X if it is a countable intersection of open sets. Also, a set is an F_σ in X if it is a countable union of closed sets.

(a) Show that a dense G_δ is generic.

(b) Hence a countable dense set is an F_σ, but not a G_δ.

(c) Prove the following partial converse to (a). If E is a generic set, then there exists $E_0 \subset E$ with E_0 a dense G_δ.

6. The function

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is irrational} \\ 1/q & \text{if } x = p/q \text{ is rational and expressed in lowest form} \end{cases}$$

is continuous precisely at the irrationals. In contrast to this, prove that there is no function on $\mathbb{R}$ that is continuous precisely at the rationals.

[Hint: Show that the set of points where a function is continuous is a G_δ (see the proof of Theorem 1.3), and apply Exercise 5.]

7. Let E be a subset of $[0,1]$, and let I be any closed non-trivial interval in $[0,1]$.

(a) Suppose E is of the first category in $[0,1]$. Show that for every I, the set $E \cap I$ is of the first category in I.

(b) Suppose E is generic in $[0,1]$. Show that for every I, the set $E \cap I$ is generic in I.

(c) Construct a set E in $[0,1]$ so that for all I, the set $E \cap I$ is neither of the first category nor generic in I.

[Hint: Consider the Cantor set in $[0,1]$; then in each open interval of its complement place a scaled copy of the Cantor set; continue this process indefinitely. For a related measure theoretic result, see Exercise 36 in Chapter 1 of Book III.]

8. A **Hamel basis** for a vector space X is a collection $\mathcal{H}$ of vectors in X, such that any $x \in X$ can be written as a unique *finite* linear combination of elements in $\mathcal{H}$.

Prove that a Banach space cannot have a countable Hamel basis.

[Hint: Show that otherwise the Banach space would be of the first category in itself.]

9. Consider $L^p([0,1])$ with Lebesgue measure. Note that if $f \in L^p$ with $p > 1$, then $f \in L^1$. Show that the set of $f \in L^1$ so that $f \notin L^p$, is generic.

A more general result can be found in Problem 1.

[Hint: Consider the set $E_N = \{f \in L^1 : \int_I |f| \leq Nm(I)^{1-1/p}$ for all intervals $I\}$. Note that each E_N is closed and that $L^p \subset \bigcup_N E_N$. Finally, show that E_N is nowhere dense by considering $f_0 + \epsilon g$ where $g(x) = x^{-(1-\delta)}$ with $0 < \delta < 1 - 1/p$.]

10. Consider $\Lambda^\alpha(\mathbb{R})$, with $0 < \alpha < 1$. Show that the set of nowhere differentiable functions is a generic set in $\Lambda^\alpha(\mathbb{R})$.

Note however that functions corresponding to the case $\alpha = 1$, that is, Lipschitz functions, are almost everywhere differentiable. (See Exercise 32 in Chapter 3 of Book III.)

11. Consider the Banach space $X = C([0,1])$ over the reals, with the sup-norm on X. Let $\mathcal{M}$ be the collection of functions that are not monotonic (increasing or decreasing) in any interval $[a,b]$, where $0 \leq a < b \leq 1$. Prove that $\mathcal{M}$ is generic in X.

[Hint: Let $\mathcal{M}_{[a,b]}$ denote the subset of X consisting of functions that are not monotonic in $[a,b]$. Then $\mathcal{M}_{[a,b]}$ is dense in X, while $\mathcal{M}^c_{[a,b]}$ is closed.]

12. Suppose X, Y and Z are Banach spaces, and $T : X \times Y \to Z$ is a mapping such that:

(i) For each $x \in X$, the mapping $y \mapsto T(x, y)$ is linear and continuous on Y.

(ii) For each $y \in Y$, the mapping $x \mapsto T(x, y)$ is linear and continuous on X.

Prove that T is (jointly) continuous on $X \times Y$, and in fact,

$$\|T(x, y)\|_Z \leq C \|x\|_X \|y\|_Y$$

for some $C > 0$ and all $x \in X$ and $y \in Y$.

13. Let $(X, \mathcal{F}, \mu)$ be a measure space, and let $\{f_n\}$ a sequence of functions in $L^p(X, \mu)$. We know from Exercise 12 in Chapter 1, that if $1 < p < \infty$, and $\sup_n \|f_n\|_{L^p} < \infty$, then some subsequence of $\{f_n\}$ converges weakly in L^p. In other words, there exist a subsequence $\{f_{n_k}\}$ of $\{f_n\}$, and an $f \in L^p$, so that if q denotes the conjugate exponent of p, that is $1/p + 1/q = 1$, then

$$\int_X f_{n_k}(x)g(x)\ d\mu(x) \to \int_X f(x)g(x)\ d\mu(x) \qquad \text{for every } g \in L^q.$$

More generally, we say that a sequence $\{f_n\}$ in L^p is **weakly bounded** if

$$\sup_n \left| \int_X f_n(x)g(x)\ d\mu(x) \right| < \infty \qquad \text{for all } g \in L^q.$$

Prove that if $1 < p < \infty$, and $\{f_n\}$ is a sequence of functions in L^p that is weakly bounded, then

$$\sup_n \|f_n\|_{L^p} < \infty.$$

In particular this holds if $\{f_n\}$ converges weakly in L^p.

[Hint: Apply the uniform boundedness principle to $\ell_n(g) = \int_X f_n(x)g(x)\ d\mu(x)$.]

14. Suppose X is a complete metric space with respect to a metric d, and $T : X \to X$ a continuous function. An element x^* in X is **universal** for T if the orbit set $\{T^n(x^*)\}_{n=1}^{\infty}$ is dense in X. Here $T^n = T \circ T \circ \cdots \circ T$ denotes n compositions of T.

Show that the set of universal elements for T in X is either empty or generic.

[Hint: Suppose x^* is universal for T, let $\{x_j\}$ be a dense set of elements in X, and let $F_{j,k,N} = \{x \in X : \ d(T^n x, y_j) < 1/k \ \text{ for some } n \geq N\}$. Show that $F_{j,k,N}$ is open and dense.]

15. Let $\overline{B}$ denote the closure of the unit ball in $\mathbb{R}^d$, and consider the metric space $\mathcal{C}$ of compact subsets of $\overline{B}$ with the Hausdorff distance. (See Section 5.) Show that the following two collections are generic.

(a) The subsets of Lebesgue measure zero.

(b) The subsets that are nowhere dense.

[Hint: For (a) show that the collection of sets C so that $m(C) < 1/n$ is open and dense. In fact for such a set, $C^c \supset \bigcup_{j=1}^{M} Q_j$, where Q_j are disjoint open cubes so that $\sum |Q_j| > 1 - 1/n$. Now shrink the Q_j. For (b) fix an open set $\mathcal{O}$ and show that the collection $\mathcal{C}_{\mathcal{O}}$ of sets in $\mathcal{C}$ that contain $\mathcal{O}$ is closed and nowhere dense.]

Chapter 5. Rudiments of Probability Theory

1. Consider $S_N(x) = \sum_{n=1}^{N} r_n(x)$, with N odd.

(a) Calculate $m(\{x : \ S_N(x) = k\})$, and show that as k varies over the integers, the maximum is attained at $k = -1$ and $k = 1$.

(b) Adapt the proof when N is even to show that for odd N,

$$m(\{x : \ a < \frac{S_N(x)}{N^{1/2}} < b\}) \to \frac{1}{\sqrt{2\pi}} \int_a^b e^{-t^2/2}\, dt, \qquad \text{as } N \to \infty.$$

2. Find three functions f_1, f_2, and f_3, so that any pair are mutually independent, but the three are not.

[Hint: Let $f_1 = r_1$, $f_2 = r_2$, and express f_3 in terms of r_1 and r_2.]

3. The collection $\{r_n\}$ of mutually independent functions on $[0,1]$ cannot be much enlarged and still remain mutually independent. In fact, prove that if we adjoin a function f to the collection $\{r_n\}$, then the resulting collection is also mutually independent only when f is constant.

[Hint: See also Exercise 16.]

4. Suppose μ and ν are two finite measures on a space X that agree on a collection of sets $\mathcal{C}$. If $\mathcal{C}$ contains X and is closed under finite intersections, then show that $\mu = \nu$ on the σ-algebra generated by $\mathcal{C}$.

[Hint: The equality $\mu = \nu$ holds on finite unions of sets in $\mathcal{C}$ because

$$\mu(\bigcup_{j=1}^{k} C_j) = \sum_{j=1}^{k} \mu(C_j) - \sum_{i<j} \mu(C_i \cap C_j) + \\ + \sum_{i<j<\ell} \mu(C_i \cap C_j \cap C_\ell) + \cdots + (-1)^{k-1} \mu(\bigcap_{j=1}^{k} C_j).]$$

5. Prove that the $\mathbb{R}^d$-valued functions $f_1, \ldots, f_n$ are mutually independent if and only if their joint distribution measure equals the product of the individual distribution measures:

$$\mu_{f_1,\ldots,f_n} = \mu_{f_1} \times \cdots \times \mu_{f_n} \qquad \text{as measures on } \mathbb{R}^{nd} = \mathbb{R}^d \times \cdots \times \mathbb{R}^d.$$

[Hint: Check the equality on cylinder sets in $\mathbb{R}^{nd}$ and use the previous exercise.]

6. Suppose $\{f_n\}$ is a sequence of mutually independent functions on the probability space (X, m). Prove that there exists a probability space (X', m'), with X' an infinite product, $X' = \prod_{n=1}^{\infty} X_n$, (X_n, m_n) probability spaces, and m' the product measure of the m_n, so that the following holds: there are functions $\{g_n\}$ on X' so that $\{f_n\}$ and $\{g_n\}$ have the same joint distributions, but each function g_n depends only on the nth coordinate of X'.

[Hint: Take $(X_n, m_m) = (X, m)$ for each n and define g_n in terms of the f_n accordingly, and use the previous exercise.]

7. Show that if $\mathcal{B}_0, \ldots, \mathcal{B}_n$ are mutually independent algebras, then for each $k < n$, the algebras $\bigvee_{j=0}^{k} \mathcal{B}_j$ and $\mathcal{B}_n$ are mutually independent.

Prove this by noting the following. First, use induction to show that if $B_j \in \mathcal{B}_j$, then $B_0 \cap \cdots \cap B_k$ is independent of B_n. Now fix $B \in \mathcal{B}_n$, and consider the two finite measures $\mu(E) = m(E \cap B)$ and $\nu(E) = m(E)m(B)$, and the collection $\mathcal{C}$ of sets that are of the form $E = B_0 \cap \cdots \cap B_k$, where $B_j \in \mathcal{B}_j$. Then apply Exercise 4.

8. Verify the following further facts about probability distribution measures.

(a) Suppose $f = (f_1, \ldots, f_k)$ with each f_j an $\mathbb{R}^d$-valued function. Let μ be the probability distribution measure of f, and let L be a linear transformation of $\mathbb{R}^{dk}$ to itself. Suppose that μ is the probability distribution measure of f. Then the distribution measure of $L(f)$ is μ_L, where $\mu_L(A) = \mu(L^{-1}A)$ for every Borel set $A \subset \mathbb{R}^{dk}$.

(b) Suppose the distribution measure of f_j is Gaussian with covariance matrix $\sigma_j^2 I$, $1 \leq j \leq k$. Assume also that the $\{f_j\}$ are mutually independent. Then the distribution measure of $c_1 f_1 + \cdots + c_k f_k$ is Gaussian with covariance matrix $(c_1^2\sigma_1^2 + \cdots + c_k^2\sigma_k^2)I$.

[Hint: For (b), compute the Fourier transforms (the characteristic functions) of the measures in question.]

9. Consider the space $L^2(\Omega, \mathbb{R}^d)$ of square-integrable $\mathbb{R}^d$-valued functions on the probability space (Ω, P). A closed subspace $\mathcal{G}$ of this space is called a **Gaussian subspace** if it is spanned by a sequence $\{f_n\}$ of mutually independent functions, each having a Gaussian distribution measure with mean zero, and covariance $\{\sigma_n^2 I\}$.

Prove that if $F_1, F_2, \ldots, F_k$ are mutually orthogonal elements of $\mathcal{G}$, then they are mutually independent. Note that the converse of this is immediate.

[Hint: Consider the case when $\mathcal{G}$ is finite-dimensional, and $\mathcal{G}$ is spanned by $f_1, \ldots, f_N$. One may suppose, after multiplication by appropriate scalars, that the f_j and F_j each have L^2 norm equal to 1. So there exists an orthogonal linear transformation L so that $L(f_j) = F_j$. Then apply Exercises 5 and 8.]

10. Consider the following two types of convergence of a sequence $\{f_n\}$ to a limit f on a probability space:

(i) $f_n \to f$ almost everywhere,

(ii) $f_n \to f$ in terms of weak convergence of their distribution measures.

Prove that (i) implies (ii), but that this implication cannot be reversed.

[Hint: Recall that if φ is continuous and bounded, then $\int \varphi(f)\, d\mu = \int \varphi\, d\mu_f$, where μ_f is the distribution measure of f, and apply the dominated convergence theorem.]

11. On $[0, 1]$ with Lebesgue measure, construct a function f whose distribution measure is normal.

[Hint: Consider the "error function" $\mathrm{Erf}(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2}\, dt$ and its inverse function.]

12. Prove the identity (6), which says that if G is a non-negative continuous function on $\mathbb{R}$ (or continuous and bounded), and f is a real-valued measurable function (on a probability space (X, m)) with distribution measure $\mu = \mu_f$, then

$$\int_X G(f)(x)\, dm = \int_{\mathbb{R}} G(t)\, d\mu(t).$$

[Hint: Note that if f is bounded, then $\sum_k G(k/n)\, m(\{k/n < f < (k+1)/n\})$ converges to both integrals as $n \to \infty$.]

13. The Rademacher sequence $\{r_n\}$ is far from complete on $L^2([0,1])$. In fact it cannot be completed by adjoining any finite collection of functions. Prove this in two ways.

(a) By considering the functions $\{r_n r_m\}$ for $n < m$.

(b) By using the L^p inequality of Lemma 1.8.

See also Exercise 16.

14. Consider the power series

$$\sum_{n=1}^{\infty} \pm a_n z^n = \sum_{n=1}^{\infty} r_n(t) a_n z^n = F(z,t),$$

where $\sum |a_n|^2 = \infty$ and $\limsup |a_n|^{1/n} \leq 1$.

Show that for almost every t, the function $F(z,t)$ cannot be analytically continued outside the unit disc.

[Hint: Argue as in Theorem 1.7 part (b) using Abel summation rather than Cesàro summation.]

15. Show that the $L^2([0,1])$ span of $\{r_n\}$ can be characterized as the subspace of L^2 consisting of those f for which

$$\mathbb{E}_N(f) = S_N(f), \qquad \text{for all } N,$$

where $\mathbb{E}_N$ are the conditional expectations corresponding to the dyadic intervals of length 2^{-N}.

16. A natural completion of the collection of Rademacher functions are the **Walsh-Paley functions**. One defines this collection on $[0,1]$, denoted by $\{w_n\}$, in the following way.

First one sets $w_0(t) = 1$, $w_1(t) = r_1(t)$, $w_2(t) = r_2(t)$ and $w_3(t) = r_1(t)r_2(t)$. More generally, if $n \geq 1$, $n = 2^{k_1} + 2^{k_2} + \cdots + 2^{k_\ell}$, where $0 \leq k_1 < \cdots < k_\ell$, then one defines

$$w_n(t) = \prod_{j=1}^{\ell} r_{k_j+1}(t).$$

In particular $w_{2^{k-1}} = r_k$.

(a) Prove that $\{w_n\}_{n=0}^{\infty}$ is a complete orthonormal system on $L^2([0,1])$.

(b) Verify the following additional interesting property of the Walsh-Paley functions: they are the continuous characters of the compact abelian group $\mathbb{Z}_2^\infty$ (thought of as the product of the two-point abelian groups $\mathbb{Z}_2$).

[Hint: Equip the group $\mathbb{Z}_2^\infty$ with the addition $x + y$ defined by $(x+y)_j = x_j + y_j \mod 2$ if $x = (x_j)$ and $y = (y_j)$. Then $r_k(x+y) = r_k(x)r_k(y)$.

Consider also the "Dirichlet kernel" $K_N(t) = \sum_{k=0}^{N-1} w_k(t)$, and show that if $N = 2^n$, then $K_N(t) = \prod_{j=1}^{n}(1 + r_j(t))$, hence $K_{2^n}(t) = 2^n$ if $0 \leq t \leq 2^{-n}$ and 0 otherwise. As a result, using the convolution $\int f(y)K_N(x+y)\,dy$, note that if $f \sim \sum a_k w_k$, then $\sum_{k<2^n} a_k w_k = \mathbb{E}_n(f)$, where $\mathbb{E}_n$ was defined in Section 1.6. See also Problem 2^*.]

17. The inequality in Lemma 1.8 may be strengthened as follows. Let $F(t) = \sum_{n=1}^{\infty} a_n r_n(t)$, with a_n real and $\sum_{n=1}^{\infty} a_n^2 = 1$. Then

(a) $\int_0^1 e^{\mu|F(t)|}\, dt \leq 2e^{\mu^2}$, for all $0 \leq \mu$.

(b) As a result, for some $c > 0$, $\int_0^1 e^{c|F(t)|^2}\, dt < \infty$.

[Hint: Part (a) implies that $m(\{t : |F(t)| > \alpha\}) \leq 2e^{\mu^2 - \mu\alpha}$. Choose $\mu = \alpha/2$, and obtain (b) with $c < 1/4$.]

18. Prove that there exists an f in $L^p([0, 2\pi])$, for all $p < \infty$, $f \sim \sum_{-\infty}^{\infty} c_n e^{in\theta}$, so that $\sum |c_n|^q = \infty$, for all $q < 2$. Hence the Hausdorff-Young inequality in Section 2.1 of Chapter 2 fails for $p > 2$.

[Hint: Use Theorem 1.7.]

19. Suppose $F(t) = \sum_{n=1}^{\infty} a_n r_n(t)$ with $\sum a_n^2 < \infty$.

(a) Prove directly that there exists a constant A so that

$$\|F\|_{L^4} \leq A\|F\|_{L^2}.$$

(b) Show as a result that there is a constant A' so that $\|F\|_{L^2} \leq A'\|F\|_{L^1}$.

(c) Conclude that $\|F\|_{L^p} \leq A_p\|F\|_{L^1}$, for $1 \leq p < \infty$.

[Hint: For (a) write out $\int_0^1 F^4(t)\,dt$ as a sum and use the orthogonality of $r_n(t)r_m(t)$. For (b) use Hölder's inequality. For (c) use Lemma 1.8.]

20. Suppose $\{A_n\}$ is a sequence of subsets of the probability space X.

(a) If $\sum m(A_n) < \infty$, then $m(\limsup_{n\to\infty} A_n) = 0$, where $\limsup_{n\to\infty} A_n$ is defined as $\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k$.

(b) However if $\sum m(A_n) = \infty$, and the sets $\{A_n\}$ are mutually independent, then $m(\limsup_{n\to\infty} A_n) = 1$.

This dichotomy is often referred to as the Borel-Cantelli lemma. (See also Book III.)
[Hint: In the case (b), $m\left(\bigcap_{k=r}^{n} A_k^c\right) = \prod_{k=r}^{n}(1 - m(A_k))$.]

21. Except for a countable set (the dyadic rationals) it is possible to assign a unique dyadic expansion to each real number α in $[0,1]$, that is,

$$\alpha = \sum_{j=1}^{\infty} \frac{x_j}{2^j} \qquad \text{with } x_j = 0 \text{ or } 1.$$

Given such a number α let $\#_N(\alpha)$ denote the number of 1's that appear among the first N terms in the dyadic expansion of α. We say that α is **normal**, if its dyadic expansion contains a density of 1's equal to the density of 0's, that is,

$$\lim_{N\to\infty} \frac{\#_N(\alpha)}{N} = 1/2.$$

(a) Prove that (with respect to the Lebesgue measure) almost every number in $[0,1]$ is normal.

(b) More generally, given an integer $q \geq 2$ consider the q-expansion of a real number α in $[0,1]$,

$$\alpha = \sum_{j=1}^{\infty} \frac{x_j}{q^j} \qquad \text{with } x_j = 0, 1, \ldots, q-1. \tag{21}$$

Again, ignoring a countable subset, this expansion is unique. For a given real number α and for each $0 \leq p \leq q-1$, define $\#_{p,N}(\alpha)$ to equal the number of j's with $0 \leq j \leq N$ for which $x_j = p$ in the q-expansion of α. A number that satisfies

$$\lim_{N\to\infty} \frac{\#_{p,N}(\alpha)}{N} = 1/q,$$

for every $0 \leq p \leq q-1$ is said to be **normal** to base q.

Show that almost all real numbers in $[0,1]$ satisfy this property.

[Hint: Consider the infinite product $\prod \mathbb{Z}_q$ with each factor given the uniform measure. Under (21) the product measure corresponds to the Lebesgue measure on $[0,1]$. Now apply the law of large numbers as in Theorem 2.1.]

22. A sequence $\{f_n\}_{n=0}^{\infty}$ of functions on X is called a (discrete) **stationary process** if for every N the joint probability distribution of $f_r, f_{r+1}, \ldots, f_{r+N}$ is independent of r.

Consider the probability space Y constructed as in the proof of Theorem 2.1. Show that whenever the sequence $\{f_n\}$ is a stationary process, then it has the same joint distribution as the sequence $\{g_0(\tau^n(y))\}$, where g_0 is a suitable function on Y and τ is the shift. Hence the ergodic theorem is equally applicable in this more general situation.

23. Prove that the conditions in Theorem 2.1 are sharp in the following sense. If $\{f_n\}_{n=0}^{\infty}$ are mutually independent and identically distributed, but $\int_X |f_0(x)|\, dm = \infty$, then for almost every x, the averages $\frac{1}{N}\sum_{n=0}^{N-1} f_n(x)$ fail to converge to a limit as $N \to \infty$.

[Hint: Let $A_n = \{x : \ |f_n(x)| > n\}$. The sets A_n are independent. However, $\sum_{n=0}^{\infty} m(A_n) = \sum_{n=0}^{\infty} m(\{x : \ |f_0(x)| > n\}) \approx \int_X |f_0(x)|\, dm = \infty$. Then use Exercise 20.]

24. The following are examples of conditional expectations.

(a) Suppose $X = \bigcup A_n$ is a finite (or countable) partition of X, with $m(A_n) > 0$ whenever A_n is non-empty. Let $\mathcal{A}$ be the algebra generated by the sets $\{A_n\}$. Then $\mathbb{E}_{\mathcal{A}}(f)(x) = \frac{1}{m(A_n)} \int_{A_n} f \, dm$ whenever $x \in A_n$.

(b) Let $X = X_1 \times X_2$, with the measure m on X being the product of the measures m_i on X_i. Let $\mathcal{A} = \{A \times X_2\}$, where A ranges over arbitrary measurable sets of X_1. Then $\mathbb{E}_{\mathcal{A}}(f)(x_1, x_2) = \int_{X_2} f(x_1, y) \, dm_2(y)$.

25. In the following four exercises $\{s_n\}$ will denote a martingale sequence corresponding to the increasing sequence of algebras $\mathcal{A}_n$ and their conditional expectations $\mathbb{E}_n$.

Suppose $s_n = \mathbb{E}_n(s_\infty)$ with $s_\infty \in L^2$. Then $\{s_n\}$ converges in L^2.

[Hint: Note that if $f_n = s_n - s_{n-1}$ then the f_n's are mutually orthogonal and $s_n - s_0 = \sum_{k=1}^{n} f_k$.]

26. Prove the following.

(a) If $s_\infty \in L^p$, then $s_n = \mathbb{E}_n(s_\infty) \in L^p$, and $\|s_n\|_{L^p} \leq \|s_\infty\|_{L^p}$ for all p with $1 \leq p \leq \infty$.

(b) Conversely, if $\{s_n\}$ is martingale and $\sup_n \|s_n\|_{L^p} < \infty$, then there exists $s_\infty \in L^p$, so that $s_n = \mathbb{E}_n(s_\infty)$, when $1 < p \leq \infty$.

(c) Show, however, that the conclusion in (b) may fail when $p = 1$.

[Hint: For (a) argue as in the proof of Lemma 2.5(a). For (b), use Lemma 2.5 and also the weak compactness of L^p, $p > 1$, as in Exercise 12 in Chapter 1. For (c), let $X = [0, 1]$ with Lebesgue measure, and consider $s_n(x) = 2^n$ for $0 \leq x \leq 2^{-n}$, $s_n(x) = 0$ otherwise.]

27. Suppose that $s_n = \mathbb{E}_n(s_\infty)$, with s_∞ integrable on X.

(a) Show that s_n converges in the L^1 norm as $n \to \infty$.

(b) Moreover $s_n \to s_\infty$ in L^1 if and only if s_∞ is measurable with respect to the algebra $\mathcal{A}_\infty = \bigvee_{n=1}^{\infty} \mathcal{A}_n$.

[Hint: For (a) use Exercises 25 and 26 (a). Then $\lim s_n = \mathbb{E}_{\mathcal{A}_\infty}(s_\infty)$, and use the previous exercise.]

28. Suppose that $s_n = \mathbb{E}_n(s_\infty)$, and $s_\infty \in L^1$.

(a) Show that

$$m(\{x : \sup_n |s_n(x)| > \alpha\}) \leq \frac{1}{\alpha} \int_{|s_\infty(x)|>\alpha} |s_\infty(x)|\, dx.$$

(b) Prove as a result $\|\sup_n |s_n|\|_{L^p} \leq A_p \|s_\infty\|_{L^p}$ if $s_\infty \in L^p$ and $1 < p \leq \infty$.

[Hint: For (a), note that when $s_\infty \geq 0$ this is a consequence of (15). To deduce (b) adapt the argument in the proof of Theorem 4.1 in Chapter 2 for the maximal function f^*.]

29. The results for real-valued martingale sequences $\{s_n\}$ discussed in Section 2.2 go through if we assume that the s_n take their values in $\mathbb{R}^d$. Verify in particular that the following consequences of identity (14) hold:

(a) $|s_k| \leq \mathbb{E}_k(|s_n|)$, if $k \leq n$.

(b) $m(\{x : \sup_n |s_n(x)| > \alpha\}) \leq \frac{1}{\alpha} \int_{|s_\infty(x)|>\alpha} |s_\infty(x)|\, dx$.

Here $|\cdot|$ denotes the Euclidean norm in $\mathbb{R}^d$.

[Hint: To prove (a), note that $(s_k, v) = \mathbb{E}_k((s_n, v))$, where $(\cdot, \cdot)$ is the inner product on $\mathbb{R}^d$, and v is any fixed vector in $\mathbb{R}^d$. Then take the supremum over unit vectors v. The conclusion (b) is a consequence of (a) and part (a) of Exercise 28.]

30. The ideas regarding conditional expectations extend to spaces (X, m) whose total measures are not necessarily finite. Consider the following example: $X = \mathbb{R}^d$, with m the Lebesgue measure. For each $n \in \mathbb{Z}$, let $\mathcal{A}_n$ be the algebra generated by all dyadic cubes of side-length 2^{-n}. The dyadic cubes are the open cubes, whose vertices are points of $2^{-n}\mathbb{Z}^d$, and have side-length 2^{-n}. Clearly $\mathcal{A}_n \subset \mathcal{A}_{n+1}$ for all n. Let f be integrable on $\mathbb{R}^d$, and set $\mathbb{E}_n(f) = \mathbb{E}_{\mathcal{A}_n}(f)$, with

$$\mathbb{E}_n(f)(x) = \frac{1}{m(Q)} \int_Q f \, dm$$

whenever $x \in Q$, with Q a dyadic cube of length 2^{-n}.

(a) Show that the maximal inequality in Theorem 2.10 extends to this case.

(b) If $f \geq 0$, then $\sup_{n\in\mathbb{Z}} \mathbb{E}_n(f)(x) \leq cf^*(x)$ for an appropriate constant c, with f^* the Hardy-Littlewood maximal function discussed in Chapter 2.

(c) Show by example that the converse inequality $f^*(x) \leq c' \sup_{n\in\mathbb{Z}} \mathbb{E}_n(f)(x)$ fails. Prove however that a substitute result holds

$$m(\{x : f^*(x) > \alpha\}) \leq c_1 m(\{x : \sup_{n\in\mathbb{Z}} \mathbb{E}_n(f)(x) > c_2\alpha\})$$

for all $\alpha > 0$. Here c_1 and c_2 are appropriate constants.

31. Let $\{\mu_N\}_{N=1}^{\infty}$ and ν be probability measures on $\mathbb{R}^d$. Prove the following are equivalent as $N \to \infty$.

(a) $\hat{\mu}_N(\xi) \to \hat{\nu}(\xi)$, all $\xi \in \mathbb{R}^d$.

(b) $\mu_N \to \nu$ weakly.

(c) In $\mathbb{R}$, $\mu_N((a,b)) \to \nu((a,b))$ for all open intervals (a,b), if we assume the measure ν is continuous.

(d) In $\mathbb{R}^d$, $\mu_N(\mathcal{O}) \to \nu(\mathcal{O})$ for all open sets $\mathcal{O}$, if we assume the measure ν is absolutely continuous with respect to Lebesgue measure.

[Hint: In $\mathbb{R}$, the equivalence of (a), (b) and (c) is implicit in the argument given in the proofs of Lemma 2.15 and Corollary 2.16. To show that (a) implies (d) in the case when $\mathcal{O}$ is an open cube, generalize the argument given in the text to $\mathbb{R}^d$. Then, prove that the analog of (d) holds for closed cubes. Finally, use the fact that any open set is an almost disjoint union of closed cubes. To show that (d) implies (b), approximate a continuous function φ of compact support uniformly by step functions that are constant on cubes.]

32. The proof of Theorem 2.17 requires the following calculation. Suppose σ is a strictly positive definite symmetric matrix with σ^{-1} denoting its inverse. Let ν_{σ^2} be the measure on $\mathbb{R}^d$ with density equal to $\frac{1}{(2\pi)^{d/2}(\det \sigma)} e^{-\frac{|\sigma^{-1}(x)|^2}{2}}$, $x \in \mathbb{R}^d$. Then $\hat{\nu}_{\sigma^2}(\xi) = e^{-2\pi^2|\sigma(\xi)|^2}$.

[Hint: Verify this by making an orthogonal change of variables that puts σ in a diagonal form. This reduces the d-dimensional integral in question to a product of corresponding 1-dimensional integrals.]

33. For the d-dimensional random walk $\{s_n(x)\}$ considered in Section 2.6, find the limit of the distribution measures of $s_n(x)/n^{1/2}$ as $n \to \infty$.

34. If k is a lattice point in $\mathbb{Z}^d$ and $d = 1$ or 2, show that for almost every path, the random walk visits k infinitely often, that is,

$$m(\{x : \ s_n(x) = k \quad \text{for infinitely many } n\} = 1.$$

[Hint: There exists ℓ_0 so that $m(\{s_{\ell_0} = -k\}) > 0$. If the conclusion fails, then there exists r_0 so that $m(\{s_n \neq k, \text{ for all } n \geq r_0\}) > 0$. Then note that

$$\{s_n \neq 0, \text{ all } n \geq \ell_0 + r_0\} \supset \{s_{\ell_0} = -k\} \cap \{s_n - s_{\ell_0} \neq k, \text{all } n \geq \ell_0 + r_0\},$$

and that the sets on the right-hand side are independent.]

35. Prove that if $d \geq 3$, then the random walk s_n satisfies $\lim_{n\to\infty} |s_n| = \infty$ almost everywhere.

[Hint: It is sufficient to prove that for any fixed $R > 0$ the set

$$B = \{x : \liminf_{n\to\infty} |s_n(x)| \leq R\}$$

has measure 0. To this end, for each lattice point k, define

$$B(k,\ell) = \{x : \ s_\ell(x) = k, \text{ and } s_n(x) = k \text{ for infinitely many } n\}.$$

Clearly, $B \subset \bigcup_{\ell,\ |k|\leq R} B(k,\ell)$. But $d \geq 3$, so $m(B(k,\ell)) = 0$).]

Chapter 6. An Introduction to Brownian Motion

1. Show that if $t > 0$, then the distribution measure of $S_t^{(N)}$ converges weakly to the Gaussian ν_t with mean zero and variance t as $N \to \infty$. More generally, if $t > s \geq 0$, then the distribution measure of $S_t^{(N)} - S_s^{(N)}$ converges weakly to the Gaussian ν_{t-s} with mean zero and covariance matrix $(t-s)I$.

[Hint: Using the notation in the remark following Theorem 2.17 in Chapter 5, and setting $f_k = \mathfrak{r}_k$, one has $S_t^{(N)} - S_{N,t} = \frac{(Nt-[Nt])}{N^{1/2}}\mathfrak{r}_{[Nt]+1}$.]

2. Let $(\mathcal{P}, d)$ be the metric space defined in Section 2. Verify:

(a) The space is complete.

(b) The space is separable.

[Hint: For (b), let $e_1, \ldots, e_d$ be a basis for $\mathbb{R}^d$, and consider the polynomials $p(t) = e_1 p_1(t) + \cdots + e_d p_d(t)$, where the p_j have rational coefficients.]

3. Show that the metric space $(\mathcal{P}, d)$ is not σ-compact.

[Hint: Assume the contrary. Then the Baire category theorem implies that there exists a compact set that has a non-empty interior. As a result there exists an open ball whose closure is compact. However, consider for example the ball of radius 1 centered at 0, and a sequence of continuous piecewise linear functions $\{f_n\}$ with $f_n(0) = 1$, $f_n(x) = 0$, when $x \geq 1/n$.]

4. Suppose X is a compact metric space. Show that:

(a) X is separable.

(b) $C(X)$ is separable.

[Hint: For each m, find a finite collection $\mathcal{B}_m$ of open balls, each of radius $1/m$, so that the collection $\mathcal{B}_m$ covers X. For (a) take the centers of the balls in $\bigcup_{m=1}^{\infty} \mathcal{B}_m$. For (b), consider $\{\eta_k^{(m)}\}$ the partition of unity corresponding to the covering of X by $\mathcal{B}_m$ (as given, for example, in Chapter 1). Show that the finite linear combinations of the $\eta_k^{(m)}$ with rational coefficients are dense in $C(X)$.]

5. Let X be a metric space, $K \subset X$ a compact subset, and f a continuous function on K. There there is a continuous function F on X, so that

$$F|_K = f, \qquad \text{and} \qquad \sup_{x \in X} |F(x)| = \sup_{x \in K} |f(x)|.$$

[Hint: The argument given in Lemma 4.11, Chapter 5 of Book III for $X = \mathbb{R}^d$ can be copied over in this general setting.]

6. Suppose K is a compact subset of $\mathcal{P}$. Show that for each $T > 0$, there exists a function $w_T(h)$, defined for $h \in (0, 1]$ with $w_T(h) \to 0$ as $h \to 0$ and such that

$$\sup_{\mathsf{p} \in K} \sup_{0 \le t \le T} |\mathsf{p}(t+h) - \mathsf{p}(t)| \le w_T(h), \qquad \text{for } h \in (0, 1].$$

[Hint: Fix $T > 0$ and $\epsilon > 0$. Each p is uniformly continuous on closed intervals, so there exists $\delta = \delta(\mathsf{p}) > 0$ so that $\sup_{0 \le t \le T} |\mathsf{p}(t+h) - \mathsf{p}(t)| \le \epsilon$ whenever $0 < h \le \delta$. Now use the fact that since K is compact, the covering $K \subset \bigcup_{\mathsf{p}} \{\mathsf{p}' \in \mathcal{P} : d(\mathsf{p}', \mathsf{p}) < \epsilon\}$ has a finite subcover.]

7. Suppose $\mu_N \to \mu$ weakly. Show as a result:

(a) $\liminf_{N\to\infty} \mu_N(\mathcal{O}) \geq \mu(\mathcal{O})$ for any open set $\mathcal{O}$.

(b) $\lim_{N\to\infty} \mu_N(\mathcal{O}) = \mu(\mathcal{O})$, $\mathcal{O}$ an open set, if $\mu(\overline{\mathcal{O}} - \mathcal{O}) = 0$.

[Hint: $\mu(\mathcal{O}) = \sup_f \{\int f\, d\mu,$ where $0 \leq f \leq 1$ and $\text{supp}(f) \subset \mathcal{O}\}$.]

8. Given the Wiener measure W in $\mathcal{P}$, we have a realization of Brownian motion (satisfying B-1, B-2 and B-3) with $B_t(\omega) = \mathsf{p}(t)$, $\Omega = \mathcal{P}$, and $P = W$. Conversely, suppose we start with $\{B_t\}$ satisfying B-1, B-2 and B-3. For any cylindrical set $C = \{\mathsf{p} \in \mathcal{P} : (\mathsf{p}(t_1), \ldots, \mathsf{p}(t_k)) \in A\}$ in $\mathcal{P}$, define $W^o(C) = P(\{\omega : (B_{t_1}(\omega), \ldots, B_{t_k}(\omega)) \in A\})$. Verify that W^o, initially defined on the cylindrical sets, extends to the Wiener measure on $\mathcal{P}$.

9. This exercise deals with the degree to which the Brownian motion process is uniquely determined by the properties B-1, B-2 and B-3.

Let us say that such a process is "strict" if in addition to the above it satisfies the following two conditions:

(i) $B_t(\omega_1) = B_t(\omega_2)$ for all t implies $\omega_1 = \omega_2$.

(ii) The collection of measurable sets of (Ω, P) is exactly $\mathcal{A}_\infty$, which is the σ-algebra generated by the $\mathcal{A}_t$, with $t < \infty$.

Now given any Brownian motion process B_t on (Ω, P) it induces a strict process $B_t^\#$ on $(\Omega^\#, P^\#)$ as follows: Let $\Omega^\#$ denote the collection of equivalence classes on Ω under the equivalence relation $\omega_1 \sim \omega_2$ if $B_t(\omega_1) = B_t(\omega_2)$ for all t. We also denote by $\{\omega\}$ the equivalence class to which ω belongs. On $\Omega^\#$ define $B_t^\#(\{\omega\}) = B_t(\omega)$, and $P^\#(\{A\}) = P(A)$ if $A \in \mathcal{A}_\infty$. Verify:

(a) $B_t^\#$ is a strict Brownian motion on $(\Omega^\#, P^\#)$.

(b) The process $(\mathcal{P}, W)$ constructed in Section 3 is a strict Brownian motion.

(c) If (B_t^1, Ω^1, P^1) and (B_t^2, Ω^2, P^2) are a pair of Brownian motion processes, then up to subsets of sets of measure zero, there is a bijection $\Phi : (\Omega^1)^\# \to (\Omega^2)^\#$, so that $(P^2)^\#(\Phi(A)) = (P^1)^\#(A)$ and $(B_t^2)^\#(\Phi(\omega)) = (B_t^1)^\#(\omega)$.

10. Prove the following version of Khinchin's inequality (Lemma 1.8 in the previous chapter). Suppose $\{f_n\}$ are identically distributed $\mathbb{R}^d$-valued functions that are bounded, have mean zero, and are mutually independent. Then for any $p < \infty$, we have

$$\|\sum a_n f_n\|_{L^p} \leq A_p \left(\sum |a_n|^2\right)^{1/2}.$$

[Hint: One can reduce to the case $d = 1$. Assuming $\sum |a_n|^2 \leq 1$, write $\int e^{\sum a_n f_n} = \prod_n \int e^{a_n f_n}$ and use $e^u = 1 + u + O(u^2)$ if $|u| \leq M$. The result is that the first integral above is majorized by $\prod(1 + M^2 a_n^2)$ if $|f_n| \leq M$ for all n.]

11. Prove the following variant of Lemma 3.2. Suppose $\{f_k\}_{k=1}^{\infty}$ is a sequence of identically distributed, mutually independent, $\mathbb{R}^d$-valued functions on a probability space (X, m) each having mean zero and the identity as its covariance matrix. If $s_n = \sum_{k=1}^{n} f_k$, then

$$\limsup_{n\to\infty} m(\{x : \sup_{1\le k\le n} |s_k(x)| > \lambda n^{1/2}\}) = O(\lambda^{-p}), \qquad \text{for } p > 0.$$

[Hint: If ν_n denotes the distribution measure of $s_n/n^{1/2}$ and $\alpha = \lambda n^{1/2}$, then the right-hand side of (9) equals $\frac{1}{\lambda}\int_{|t|>\lambda} |t|\, d\nu_n(t)$. With $\lambda \ge 1$ fix $M \ge 1$ and write this last integral as the sum of two terms $\lambda^{-1}\int_{|t|>\lambda^M} + \lambda^{-1}\int_{\lambda^M \ge |t|>\lambda}$. Using the fact that $\int \frac{|s_n|^2}{n}\, dm = 1$, the first term is $O(\lambda^{-1-M})$. By the central limit theorem $\lim_{n\to\infty} \lambda^{-1}\int_{\lambda^M\ge|t|>\lambda} |t|\, d\nu_n(t) = O\left(\lambda^{-1}\int_{|t|>\lambda} |t| e^{-|t|^2/2}\, dt\right)$ so the limit of the second term is also $O(\lambda^{-1-M})$.]

12. Prove that almost everywhere

$$|B_t(\omega)| = O(t^{1/2+\epsilon}), \qquad \text{as } t \to \infty,$$

for every $\epsilon > 0$. This is the analog of the strong law of large numbers given in Corollary 2.9 of the previous chapter.

[Hint: If $B_T^*(\omega)$ denotes $\sup_{0\le t\le T} |B_t(\omega)|$, then the maximal inequality (14) gives $W(\{B_T^* > \alpha\}) \le \frac{1}{\alpha}\|B_T\|_{L^1} = c'\frac{T^{1/2}}{\alpha}$. If $E_k = \{B_{2^k}^* > 2^{\frac{k}{2}(1+\epsilon)}\}$, then we have $\sum_{k\ge 0} W(E_k) = O\left(\sum_{k\ge 0} 2^{-\frac{k}{2}\epsilon}\right) < \infty$.]

13. If B_t is a Brownian motion process then so is $B'_t = tB_{1/t}$.

[Hint: Note the continuity of almost all paths of B'_t at the origin follows from the previous exercise. To verify property B-2, use Exercise 29 in the previous chapter.]

14. Show that $\limsup_{t\to 0} \frac{|B_t(\omega)|}{t^{1/2}} = \infty$ almost everywhere; hence almost all Brownian paths are not Hölder $1/2$.

Also show that $\limsup_{t\to \infty} \frac{|B_t(\omega)|}{t^{1/2}} = \infty$ almost everywhere; hence almost all Brownian paths exit every ball.

[Hint: By the previous exercise it suffices to check the result when $t \to 0$. Consider $d = 1$. Then

$$W(\{|B_\alpha - B_\beta| > \gamma\}) = \frac{1}{\sqrt{2\pi(\beta-\alpha)}} \int_{|u|>\gamma} e^{-\frac{u^2}{2(\beta-\alpha)}}\, du, \qquad \text{if } \beta > \alpha.$$

Thus

$$W(\{|B_{2^{-k}} - B_{2^{-k+1}}| > 2^{-k/2}\mu_k\}) \geq \frac{1}{\sqrt{2\pi}} \int_{|u|\geq \mu_k} e^{-u^2/2}\, du \geq c_1 e^{-c_2\mu_k^2}.$$

Now choose $\mu_k \to \infty$ so slowly that $\sum_{k\geq 0} e^{-c_2\mu_k^2} = \infty$ and apply the Borel-Cantelli lemma (Exercise 20 in the previous chapter).]

15. Calculate the (joint) probability distribution measure of $(B_{t_1}, B_{t_2}, \ldots, B_{t_k})$.
[Hint: Use Exercise 8 (a) in the previous chapter.]

16. Show that the following generalization of the fact that $\mathcal{A}_{0+} = \mathcal{A}_0$ holds: if we define $\mathcal{A}_{t+}$ to be $\bigcap_{s>t} \mathcal{A}_s$, then $\mathcal{A}_{t+} = \mathcal{A}_t$.

17. The previous exercise gives the right-continuity of the collection $\{\mathcal{A}_s\}$. Prove the following left-continuity for every $t > 0$, $\mathcal{A}_t = \mathcal{A}_{t-}$, where $\mathcal{A}_{t-}$ is the σ-algebra generated by all $\mathcal{A}_s$ for $s < t$.

[Hint: Consider first cylindrical sets in $\mathcal{A}_t$.]

18. Let σ be a stopping time. Show that:

(a) σ is $\mathcal{A}_\sigma$-measurable.

(b) $B_{\sigma(\omega)}(\omega)$ is $\mathcal{A}_\sigma$-measurable.

(c) $\mathcal{A}_\sigma$ is the σ-algebra determined by the stopped process $\hat{B}_t$ with $\hat{B}_t(\omega) = B_{t\wedge\sigma(\omega)}(\omega)$.

[Hint: For (a), note that $\{\sigma(\omega) \le \alpha\} \cap \{\sigma(\omega) \le t\} = \{\sigma(\omega) \le \min(\alpha, t)\}$. For (b), show first that for any Borel subset E of $\mathbb{R}^d$ and $t \ge 0$, one has $\{B_{\sigma(\omega)}(\omega) \in E\} \cap \{\sigma \le t\} \in \mathcal{A}_t$ whenever σ takes on only discrete values. Then approximate σ by $\sigma^{(n)}$ as in the proof of Theorem 5.3.]

19. Let u be a bounded Borel measurable function on a bounded open set $\mathcal{R} \subset \mathbb{R}^d$. Suppose that u satisfies the mean-value property on spheres, that is, (21).

(a) Show that if B is a ball contained in $\mathcal{R}$ and centered at x, then

$$u(x) = \frac{1}{m(B)} \int_B u(y)\, dy,$$

where m is the Lebesgue measure on $\mathbb{R}^d$.

(b) As a result, the function u is continuous in $\mathcal{R}$ and the argument in Section 4.1, Chapter 5 of Book III shows that the function u is harmonic in $\mathcal{R}$.

[Hint: For (b), show that locally, $u(x) = (u * \varphi)(x)$, where φ is a smooth radial function supported on an appropriately small ball and with $\int \varphi = 1$.]

20. An bounded open set $\mathcal{R}$ has a **Lipschitz boundary** if $\partial\mathcal{R}$ can be covered by finitely many balls, so that for each such ball B, the set $\partial\mathcal{R} \cap B$ can (possibly after a rotation and translation) be written as $x_d = \varphi(x_1, \ldots, x_{d-1})$, where φ is a function that satisfies a Lipschitz condition.

Verify that if $\mathcal{R}$ has a Lipschitz boundary, then it satisfies the outside cone condition. Thus, in particular, if $\mathcal{R}$ is of class C^1 (in the sense of Section 4 in Chapter 7) then $\mathcal{R}$ satisfies the outside cone condition.

So in these cases the Dirichlet problem is uniquely solvable.

21. Suppose $\mathcal{R}_1$ and $\mathcal{R}_2$ are two open and bounded sets in $\mathbb{R}^d$, with $\overline{\mathcal{R}}_1 \subset \mathcal{R}_2$. Let μ_1^x and μ_2^x denote the harmonic measures of $\mathcal{R}_1$ and $\mathcal{R}_2$ respectively, as defined at the beginning of Section 5. Show that the following generalization of the mean-value property (21) holds: whenever $x \in \mathcal{R}_1$, then

$$\mu_2^x = \int_{\partial \mathcal{R}_1} \mu_2^y \, d\mu_1^x(y),$$

in the sense that $\mu_2^x(E) = \int_{\partial \mathcal{R}_1} \mu_2^y(E) \, d\mu_1^x(y)$ for any Borel set $E \subset \partial \mathcal{R}_2$.

Chapter 7. A Glimpse into Several Complex Variables

1. Suppose f is holomorphic in a polydisc $\mathbb{P}_r(z^0)$, and assume that f vanishes in a neighborhood of z^0. Then $f = 0$ throughout $\mathbb{P}_r(z^0)$.
[Hint: Expand $f(z) = \sum a_\alpha (z - z^0)^\alpha$ in $\mathbb{P}_r(z^0)$, using Proposition 1.1, and note that all a_α are zero.]

2. Show that:

(a) If f is holomorphic in a pair $\mathbb{P}_\sigma(z^0)$ and $\mathbb{P}_\tau(z^0)$ of polydiscs centered at z^0 with $\sigma = (\sigma_1, \ldots, \sigma_n)$ and $\tau = (\tau_1, \ldots, \tau_n)$, then f extends to be holomorphic in $\mathbb{P}_r(z^0)$, wherever $r = (r_1, \ldots, r_n)$ and $r_j \leq \sigma_j^{1-\theta}\tau_j^\theta$, $1 \leq j \leq n$, for some $0 \leq \theta \leq 1$.

(b) If $S = \{s = (s_1, \ldots, s_n), \quad s_j = \log r_j, \quad \text{where } f \text{ is holomorphic in } \mathbb{P}_r(z^0)\}$, then S is a convex set.

[Hint: Consider $\sum a_\alpha(z - z^0)$ that represents f both in $\mathbb{P}_\sigma(z^0)$ and $\mathbb{P}_\tau(z^0)$.]

3. Given Ω any open subset of $\mathbb{C}^1$, construct a holomorphic function f in Ω that cannot be continued analytically outside Ω.

[Hint: Given any sequence of points $\{z_j\}$ in Ω, which does not have a limit point in Ω, there exists an analytic function in Ω vanishing exactly at those z_j.]

4. Suppose Ω is a bounded region in $\mathbb{C}^n$, $n > 1$, and f is holomorphic in Ω. Suppose Z, the zero set of f, is non-empty. Then $\overline{Z}$ intersects $\partial\Omega$, that is, $\overline{Z} \cap \partial\Omega$ is not empty.

[Hint: Let w be a point in $\overline{\Omega}^c$. Let $z^0 \in Z$ be a point furthest from w. Define γ to be the unit vector in the direction from z_0 to w, and let ν be another unit vector so that both ν and $i\nu$ are perpendicular to γ. Consider the one-variable function $h_\epsilon(\zeta)$ given by $h_\epsilon(\zeta) = f(z^0 - \epsilon\gamma + \zeta\nu)$. Then for $\epsilon > 0$, the function $h_\epsilon(\zeta)$ does not vanish in a fixed neighborhood of $\zeta = 0$.]

5. Suppose f is continuous and has compact support in $\mathbb{C}^1$.

(a) Show that $u = f * \Phi$ in Proposition 3.1 belongs to $\mathrm{Lip}(\alpha)$, for every $\alpha < 1$.

(b) Show that u is not necessarily in C^1.

[Hint: For (b) consider $f(z) = z(\log(1/|z|))^\epsilon$ but modified away from the origin to have compact support.]

6. Verify the identity in $\mathbb{C}^1$

$$F(z) = \frac{1}{2\pi i}\int_{\partial\Omega}\frac{F(\zeta)}{\zeta - z}\,d\zeta - \frac{1}{\pi}\int_{\Omega}\frac{(\partial F/\partial\overline{\zeta})(\zeta)}{(\zeta - z)}\,dm(\zeta)$$

for appropriate regions Ω and C^1 functions F. Use this identity to give an alternative proof of Proposition 3.1.

7. Prove the following. The necessary and sufficient condition that the solution $u(z) = \frac{1}{\pi} \int \frac{f(\zeta)}{\zeta - z}\, dm(\zeta)$ of $\partial u/\partial \overline{z} = f$ in $\mathbb{C}^1$, have compact support when f has compact support, is that

$$\int_{\mathbb{C}} \zeta^n f(\zeta)\, dm(\zeta) = 0, \qquad \text{for all } n \geq 0.$$

[Hint: In one direction, note that $\frac{\partial}{\partial \overline{z}}(z^n u(z)) = z^n f(z)$. For the converse, observe that for large z, $u(z) = \sum_{n=0}^{\infty} a_n z^{-n-1}$, with $a_n = \frac{1}{\pi} \int \zeta^n f(\zeta)\, dm(\zeta)$.]

8. Suppose Ω is a region in $\mathbb{R}^d$ with a defining function ρ that is of class C^k.

(a) If F is a C^k function defined on $\mathbb{R}^d$ and $F = 0$ on $\partial\Omega$, show that $F = a\rho$, with $a \in C^{k-1}$.

(b) Suppose $F_1 = F_2$ on $\partial\Omega$. Show that if X is any tangential vector field then

$$X(F_1)|_{\partial\Omega} = X(F_2)|_{\partial\Omega}.$$

[Hint: Write $F_1 - F_2 = a\rho$.]

9. Verify that the extension F given by Theorem 4.1 is the unique solution to the Dirichlet problem for Ω with boundary data F_0.

10. Use the region $\{z \in \mathbb{C}^n : \ \rho < |z| < 1\}$ to show that the connectedness hypotheses in Theorem 3.3 and Theorem 4.1 are necessary.

11. That the connectedness properties in the hypotheses of Theorems 3.3 and 4.1 are related can be seen as follows. Suppose Ω is a bounded region with C^1 boundary. For $\epsilon > 0$, let Ω_ϵ be the "collar" defined by $\{z : \ d(z, \partial\Omega) < \epsilon\}$, and let $\Omega_\epsilon^- = \Omega_\epsilon \cap \Omega$. Then for sufficiently small ϵ the following are equivalent:

(i) $\overline{\Omega}^c$ is connected,

(ii) Ω_ϵ is connected,

(iii) Ω_ϵ^- is connected.

[Hint: For instance to see why (ii) or (iii) implies (i), suppose P_1 and P_2 are two points in $\overline{\Omega}^c$, and let Γ_1 and Γ_2 denote the connected components of $\overline{\Omega}^c$ which contain P_1 and P_2 respectively. Connect P_1 to a point Q_1 on $\partial\Omega \cap \overline{\Gamma}_1$, and P_2 to a point Q_2 on $\partial\Omega \cap \overline{\Gamma}_2$. Since Ω_ϵ^- is connected one can then connect Q_1 to Q_2 by a path in $\overline{\Omega}^c$.

Conversely, to show that (i) implies (iii) for example, let A be a point in Ω and B a point in $\overline{\Omega}^c$. If P_0 and P_1 belong to $\partial\Omega$, let γ_0 be any path starting at A traveling in Ω, passing through P_0, then traveling in Ω^c ending at B. Similarly, let γ_1 be path connecting A to B passing through P_1. These paths can be constructed because both Ω and $\overline{\Omega}^c$ are connected. Then, since $\mathbb{C}^n$ is simply-connected, deform the path γ_0 into γ_1, and denote such transformation by $s \mapsto \gamma_s$ with $0 \leq s \leq 1$. To conclude, consider the intersection of γ_s with $\partial\Omega$.]

12. Let Ω be a simply connected bounded region in $\mathbb{C}^1$ with a boundary of class C^1. Suppose F_0 is a given continuous function on $\partial\Omega$. Show that a necessary and sufficient condition that there is an F, holomorphic in Ω, continuous on $\overline{\Omega}$ so that $F = F_0$ on $\partial\Omega$ is that $\int_{\partial\Omega} z^n F_0(z)\, dz = 0$, for $n = 0, 1, 2, \ldots$.

[Hint: One direction is clear from Cauchy's theorem. For the converse define $F^{\pm}(z) = \frac{1}{2\pi i}\int_{\partial\Omega} \frac{F_0(\zeta)}{\zeta - z}\, d\zeta$, according to whether $z \in \Omega$ or $z \in \overline{\Omega}^c$. Now the hypothesis implies that $F^+(z) = 0$, $z \in \overline{\Omega}^c$. Also $F^-(z) - F^+(\tilde{z}) \to F_0(\zeta)$, $z \to \zeta$ if $\zeta \in \partial\Omega$, $z \in \Omega$, the segment $[z, \zeta]$ is normal to the tangent line of $\partial\Omega$ at ζ, and $\tilde{z}$ is the reflection of z across that line. That is, $\frac{\tilde{z}+z}{2} = \zeta$, $\tilde{z} \in \overline{\Omega}^c$. The convergence asserted is related to the expression of the delta function given by $i\pi\delta = \frac{1}{2}\left(\frac{1}{x - i0} - \frac{1}{x + i0}\right)$, in Section 2 of Chapter 3.]

13. Show that with an additional change of variables, that is, introducing complex coordinates, the canonical representations (16) and (17) of the boundary can be simplified to state

$$y_n = \sum_{j=1}^{n-1} \lambda_j |z_j|^2 + o(|z'|^2), \qquad \text{for } z' \to 0.$$

[Hint: Consider the change of variables $z_n \mapsto z_n - z_n(c_1 z_1 + \cdots + c_{n-1} z_{n-1} + D z_n)$, $z_j \mapsto z_j$, $1 \leq j \leq n-1$, for suitable constants $c_1, \ldots, c_{n-1}$.]

14. The fact that when $n = 1$ there are no local holomorphic invariants at boundary points is indicated by the following fact. Suppose γ is a C^k curve in $\mathbb{C}^1$. Then for every $z^0 \in \gamma$, there is a holomorphic bijection Φ of a neighborhood of z^0 to a neighborhood of the origin, so that $\Phi(\gamma)$ is the curve $\{y = \varphi(x)\}$, with $\varphi(x) = o(x^k)$ as $x \to 0$.

[Hint: Suppose $y = a_2x^2 + \cdots + a_kx^k + o(x^k)$ as $x \to 0$, and consider Φ^{-1} defined by $\Phi^{-1}(z) = z + i\left(\sum_{j=2}^k a_j z^j\right)$.]

15. Consider the hypersurface M in $\mathbb{C}^3$ given by $M = \{\text{Im}(z_3) = |z_1|^2 - |z_2|^2\}$. Show that M has the remarkable property that any holomorphic function F defined in a neighborhood of M continues analytically into all of $\mathbb{C}^3$.

[Hint: Use Theorem 7.5 to find a fixed ball B centered at the origin so that F continues into all of B. Then rescale.]

16. That the maximum principle of Theorem 6.1 does not hold in the case $n = 1$ can be seen as follows. Start with $f(e^{i\theta}) \in C^\infty$, so that $f \geq 0$, $f(e^{i\theta}) = 0$ for $|\theta| \leq \pi/2$, $f(e^{i\theta}) = 1$ for $3\pi/4 \leq |\theta| \leq \pi$. Write $f(e^{i\theta}) = \sum_{n=0}^{\infty} a_n e^{in\theta} + \sum_{-\infty}^{n=-1} \overline{a}_n e^{in\theta}$, $G(z) = \sum_{n=0}^{\infty} a_n z^n$, and $F_N(z) = e^{NG(z)}$. Verify that F_N is continuous in the closed disc $|z| \leq 1$, $|F_N(e^{i\theta})| = 1$, for $|\theta| \leq \pi/2$ but $|F_N(z)| \geq c_1 e^{c_2 N(1-|z|)}$ in the closed disc, for two positive constants c_1 and c_2.

[Hint: $G(z) = u + iv$ where $u(r, \theta) = f * P_r$, with P_r the Poisson kernel.]

17. Verify the following:

(a) The inverse of the mapping of $\mathcal{U}$ to the unit ball given in the Appendix is $z_n = i\left(\frac{1-w_n}{1+w_n}\right)$, and $z_k = \frac{w_k}{1+w_n}$, $k = 1, \ldots, n-1$.

(b) For each $(\zeta, t) \in \mathbb{C}^{n-1} \times \mathbb{R}$ consider the following "translation" on $\mathbb{C}^n$, $r_{(\zeta,t)}$ given by

$$r_{(\zeta,t)}(z', z_n) = (z' + \zeta, z_n + t + 2i(z' \cdot \overline{\zeta}) + i|\zeta|^2).$$

Then $r_{(\zeta,t)}$ maps $\mathcal{U}$ and $\partial\mathcal{U}$ to themselves, respectively. Composing these mappings leads to the composition formula

$$(\zeta, t) \cdot (\zeta', t') = (\zeta + \zeta', t + t' + 2\mathrm{Im}(\zeta \cdot \overline{\zeta}')).$$

Under this law $\mathbb{C}^{n-1} \times \mathbb{R}$ becomes the "Heisenberg group."

(c) $\mathcal{U}$ (as well as $\partial\mathcal{U}$) is invariant under the "non-isotropic" dilations $(z', z_n) \to (\delta z', \delta^2 z_n)$, $\delta > 0$.

(d) Both $\mathcal{U}$ and $\partial\mathcal{U}$ are invariant under the mappings $(z', z_n) \mapsto (u(z'), z_n)$, where u is a unitary mapping of $\mathbb{C}^{n-1}$.

18. Define $\mathcal{H}_\lambda$ to be the space of functions f holomorphic in $\mathbb{C}^{n-1}$, for which

$$\int_{\mathbb{C}^{n-1}} |f(z)|^2 e^{-4\pi\lambda|z|^2}\, dm(z) = \|f\|^2_{\mathcal{H}_\lambda} < \infty.$$

Show that:

(a) $\mathcal{H}_\lambda$ is trivial if $\lambda \leq 0$.

(b) $\mathcal{H}_\lambda$ is complete in the indicated norm, so $\mathcal{H}_\lambda$ is a Hilbert space.

(c) Define $P_\lambda(f)(z) = \int_{\mathbb{C}^{n-1}} f(w) K_\lambda(z,w) e^{-4\pi\lambda|w|^2}\, dm(w)$, where $K_\lambda(z,w) = (4\lambda)^{n-1} e^{4\pi\lambda z\cdot\overline{w}}$.

Then P_λ is the orthogonal projection of $L^2(e^{-4\pi\lambda|w|^2} dm(w))$ to $\mathcal{H}_\lambda$.

[Hint: Show that convergence in the norm $\mathcal{H}_\lambda$ implies uniform convergence on compact subsets of $\mathbb{C}^{n-1}$, using Lemma 8.2.]

19. Prove:

(a) The space $\mathcal{H}$ in Section 8.1, is complete, and hence is a Hilbert space.

(b) Show that the Cauchy integral $f \mapsto C(f)$ gives the orthogonal projection from $L^2(\partial\mathcal{U}, d\beta)$ to the linear space of functions F_0 that arise as $\lim_{\epsilon\to 0} F_\epsilon$, for $F \in H^2(\mathcal{U})$.

[Hint: For (a), use the previous exercise.]

Chapter 8. Oscillatory Integrals in Fourier Analysis

1. Use spherical coordinates to show that in $\mathbb{R}^d$

$$\int_{S^{d-1}} e^{-2\pi i x\cdot\xi}\, d\sigma = c_d \int_{-1}^{1} e^{-2\pi i|\xi|u}(1-u^2)^{\frac{d-3}{2}}\, du,$$

with c_d the area of the unit sphere S^{d-2} in $\mathbb{R}^{d-1}$. Then deduce formula (3) from Problem 2 in Chapter 6, Book I.

2. Let the hypersurface M contain a neighborhood of a hyperplane (for example $\{x_d = 0\}$). Show that in this case $\widehat{d\mu}(\xi) \neq O(|\xi|^{-\epsilon})$, as $|\xi| \to \infty$ for any $\epsilon > 0$.

3. Principle of stationary phase when $d = 1$. Consider

$$I(\lambda) = \int_{-\infty}^{\infty} e^{i\lambda\Phi(x)}\psi(x)\,dx,$$

where ψ is a C^∞ function of compact support and $x = 0$ is the only critical point of Φ in the support of ψ, while $\Phi''(0) \neq 0$. Then for every positive integer N,

$$I(\lambda) = \frac{e^{i\lambda\Phi(0)}}{\lambda^{1/2}}\left(a_0 + a_1\lambda^{-1} + \cdots + a_N\lambda^{-N}\right) + O(\lambda^{-N-1/2}), \qquad \text{as } \lambda \to \infty.$$

The a_k are determined by $\Phi''(0), \ldots, \Phi^{(2k+2)}(0)$, and $\psi(0), \ldots, \psi^{(2k)}(0)$. In particular $a_0 = \left(\frac{2\pi}{-i\Phi''(0)}\right)^{1/2}\psi(0)$.

Prove this in two steps:

(a) Consider first the special case when $\varphi(x) = x^2$ dealt with by (8).

(b) Pass to the case of general φ by a change of variables that brings $\varphi(x)$ to x^2 or $-x^2$.

4. Suppose Φ is of class C^k in an interval $[a, b]$ with $k \geq 2$. Assume that $|\Phi^{(k)}(x)| \geq 1$ throughout the interval. Prove the following generalization of Proposition 2.3

$$\left| \int_a^b e^{i\lambda\Phi(x)}\, dx \right| \leq c_k \lambda^{-1/k}.$$

[Hint: Suppose $\Phi^{(k-1)}(x_0) = 0$, and argue by induction as in the proof of Proposition 2.3.]

5. Consider the curve $\gamma(t) = (t, t^k)$ in $\mathbb{R}^2$, with k an integer ≥ 2. Its curvature vanishes nowhere when $k = 2$ and only at the origin, of order $k - 2$, when $k > 2$. Let $d\mu$ be defined by $\int_{\mathbb{R}^2} f\, d\mu = \int_{\mathbb{R}} f(t, t^k)\psi(t)\, dt$, where ψ is a C^∞ function of compact support, and $\psi(0) \neq 0$. Then prove:

(a) $|\widehat{d\mu}(\xi)| = O(|\xi|^{-1/k})$.

(b) However, this decay estimate is optimal, that is, $|\widehat{d\mu}(0, \xi_2)| \geq c|\xi_2|^{-1/k}$ if ξ_2 is large.

[Hint: For (a) use Exercise 4. For (b) consider for example the case when k is even and verify that $\int_{-\infty}^{\infty} e^{i\lambda x^k} e^{-x^k}\, dx = c_\lambda(1 - i\lambda)^{-1/k}$.]

6. Show that the (L^p, L^q) results for the averaging operator A given by Corollary 4.2 are optimal by proving the following (in, say, the case of the sphere in $\mathbb{R}^3$):

(a) Suppose $f(x)$ vanishes for small x and $f(x) \geq |x|^{-r}$, for $|x| \geq 1$. Then observe $A(f)(x) \geq c|x|^{-r}$ and thus we must always have $q \geq p$. This restriction corresponds to the side of the triangle joining $(0,0)$ and $(1,1)$.

(b) Next let $f = \chi_{B_\delta}$, where B_δ is the ball of radius δ. Note that if δ is small $A(\chi_{B_\delta}) \geq c\delta^2$ for $|1 - |x|| < \delta/2$. So $\|f\|_{L^p} \approx \delta^{3/p}$ while $\|A(f)\|_{L^q} \gtrsim \delta^2\delta^{1/q}$. Hence the inequality $\|A(f)\|_{L^q} \leq c\|f\|_{L^p}$ implies $2 + 1/q \geq 3/p$, which corresponds to the side of the triangle joining $(3/4, 1/4)$ and $(1,1)$.

(c) For the third inequality, use duality and (b).

7. By refining the argument given in Exercise 6 (b) one can show that the smoothing of degree $(d-1)/2$ asserted in Proposition 1.1 fails when $p \neq 2$.

In the case $p < 2$ and $d = 3$, this can be seen by taking $\delta > 0$ small and setting $f = \varphi_\delta$, where $\varphi_\delta = \varphi(x/\delta)$ and φ is a non-negative smooth function of compact support. Here $\|\varphi_\delta\|_{L^p} \approx c\delta^{3/p}$, while $\|\nabla A(\varphi_\delta)\|_{L^p} \gtrsim \delta\delta^{1/p}$. Hence the inequality $\|A(\varphi_\delta)\|_{L^p_1(\mathbb{R}^3)} \leq C\|\varphi_\delta\|_{L^p(\mathbb{R}^3)}$ fails for small delta when $p < 2$.

[Hint: If $c_1 > 0$ is sufficiently small, then $\delta^2 \lesssim A(\varphi_\delta)$ and $|\nabla A(\varphi_\delta)| \gtrsim \delta$, whenever $|1 - |x|| \leq c_1\delta$.]

8. Let M be a (local) hypersurface given in coordinates $(x', x_d) \in \mathbb{R}^{d-1} \times \mathbb{R}$ as $\{x_d = \varphi(x')\}$. Suppose F is any continuous function of small support defined in a a neighborhood of M and set $f = F|_M$.

(a) Show that $\lim_{\epsilon \to 0} \frac{1}{2\epsilon} \int_{d(x,M)<\epsilon} F\, dx$ exists and equals $\int_{\mathbb{R}^{d-1}} f(x', \varphi(x'))(1 + |\nabla_{x'} \varphi|^2)^{1/2}\, dx'$. This limit defines the induced Lebesgue measure $d\sigma$ and equals $\int_M f\, d\sigma$.

(b) Suppose ρ is any defining function of M. Show that

$$\lim_{\epsilon \to 0} \frac{1}{2\epsilon} \int_{|\rho|<\epsilon} F\, dx = \int_M f \frac{d\sigma}{|\nabla \rho|}.$$

(c) Suppose h is a Schwartz function on $\mathbb{R}$ with $\int_{\mathbb{R}} h(u)\, du = 1$. Then

$$\lim_{\epsilon \to 0} \epsilon^{-1} \int_{\mathbb{R}^d} h(\rho/\epsilon) F\, dx = \int_M f \frac{d\sigma}{|\nabla \rho|}.$$

[Hint: For (c), assume h is even and let $I(t) = \int_{|\rho(x)|<t} F(x)\, dx$. Then

$$\epsilon^{-1} \int h(\rho/\epsilon) F\, dx = \epsilon^{-1} \int_0^\infty h(u/\epsilon) \frac{dI_u}{du}\, du = -\epsilon^{-1} \int_0^\infty (u/\epsilon) h'(u/\epsilon) \left(\frac{1}{u} I_u \right) du.$$

Now use the fact that $-\int_0^\infty u h'(u)\, du = 1/2$ and $\frac{I_u}{2u} \to \int_M f \frac{d\sigma}{|\nabla \rho|}$, as $u \to 0$.]

9. Observe the following Euclidean-invariance properties of the principal curvatures of a hypersurface M in $\mathbb{R}^d$. For each $h \in \mathbb{R}^d$ consider the translate $M + h$ of M; also for each rotation r of $\mathbb{R}^d$, the rotated surface $r(M)$; and for each $\delta \in \mathbb{R}$, $\delta \neq 0$, the dilated surface δM. Denote by $\{\lambda_j(x)\}$ the principal curvatures of M at x.

(a) Show that $\{\lambda_j(x-h)\}$, $\{\lambda_j(r^{-1}(x))\}$, and $\{\delta^{-2}\lambda_j(x/\delta)\}$ are the principal curvatures of $M+h$, $r(M)$, δM at the points $x+h$, $r(x)$ and δx, respectively.

(b) Consider the cone $\{x_d^2 = |x'|^2,\, x \neq 0\}$ with defining function $\rho = |x'|^2 - x_d^2$. Using (a), show that at x there are $d-2$ principal curvatures that equal x_d^{-2}, and one that vanishes.

10. Let $f_0(r) = r^{-1/2}(\log r)^{-\delta}$, $0 < \delta < 1$, when $r \geq 2$, and $f_0(r) = 0$ otherwise.

(a) Prove that $\int |J_k(2\pi\rho r)| f_0(r)\, dr = \infty$ for every $\rho > 0$.

(b) Show as a result that, if $p \geq 2d/(d+1)$, then (31) cannot hold for any q when M is the sphere.

11. One can prove that the conjectured condition $q \leq \left(\frac{d-1}{d+1}\right) p'$ for (L^p, L^q) restriction cannot hold in a larger range, by the following argument given in the case $d = 2$.

(a) Suppose the inequality (31) holds for some p and q. Show that as a result

$$\int_{1-\delta \leq |\xi| \leq 1} |\hat{f}(\xi)|^q \, d\xi \leq c' \delta \|f\|_{L^p}^q \qquad \text{for small } \delta.$$

(b) Next choose $\hat{f}(\xi_1, \xi_2) = \eta((\xi_1 - 1)/\delta)\eta(\xi_2/\delta)$ when $\eta(u) = 1$ if $|u| \geq 1$. That is, $\hat{f}(\xi)$ dominates the characteristic function of a rectangle of approximate side lengths δ and $\delta^{1/2}$ that fits inside the annulus $1 - \delta \leq |\xi| \leq 1$. Use this to obtain a contradiction $q > \left(\frac{d-1}{d+1}\right) p'$ by letting $\delta \to 0$.

12. Connect the operator $e^{it\triangle}$ and the Fourier transform as follows. Let m_t be the multiplication operator $m_t : f(x) \mapsto \frac{1}{(4\pi it)^d} e^{-\frac{i|x|^2}{4t}} f(x)$.

(a) Show that $e^{it\triangle}(f) = i^{-d} m_t (f m_t)^\wedge$ when $t = 1/4\pi$.

(b) Generalize this identity to any $t \neq 0$ by rescaling.

13. Let $\mathrm{Ai}(u) = \lim_{N\to\infty} \frac{1}{2\pi} \int_{-N}^{N} e^{i\left(\frac{v^3}{3}+uv\right)}\, dv$.

(a) Show that this limit exists for every $u \in \mathbb{R}$.

(b) Prove that $|\mathrm{Ai}(u)| \leq c(1+|u|)^{-1/4}$.

(c) Moreover, show that $\mathrm{Ai}(u)$ is rapidly decreasing as $u \to \infty$, for $u > 0$.

[Hint: Write $\Phi(r) = \frac{r^3}{3} + ru$, and apply the estimates in Section 2. For (a) use the fact that $\Phi'(r) \to \infty$ as $|r| \to \infty$. For (b), use the fact that $|\Phi'(r)| \geq |u|/2$, when $|r| \leq (\frac{1}{2}|u|)^{1/2}$, while $|\Phi''(r)| \geq 2|r|$ when $|r| > (\frac{1}{2}|u|)^{1/2}$.]

14. Suppose $F \in L^2(\mathbb{R}^d \times \mathbb{R})$ and $S(F)(x,t) = i\int_0^t e^{i(t-s)\triangle}F(\cdot,s)\,ds$. Prove that:

(a) For each t, $S(F)(\cdot,t) \in L^2(\mathbb{R}^d)$, and

$$\|S(F)(\cdot,t)\|_{L^2(\mathbb{R}^d)} \leq |t|^{1/2}\|F\|_{L^2(\mathbb{R}^d\times\mathbb{R})}.$$

(b) If $F(\cdot,t) = e^{it\triangle}G(\cdot,t)$, then

$$\|G(0,t_1) - G(0,t_2)\|_{L^2(\mathbb{R}^d)} \leq |t_1 - t_2|^{1/2}\|G\|_{L^2(\mathbb{R}^d\times\mathbb{R})}.$$

(c) As a result, $t \mapsto F(0,t)$ is continuous in the $L^2(\mathbb{R}^d)$ norm.

[Hint: For (a) and (b) use the unitarity of $e^{it\triangle}$ and Schwarz's inequality. For (c), approximate F by C^∞ functions of compact support, using (b) and (c).]

15. Suppose u is a smooth solution of (54) that decays sufficiently quickly as $|x| \to \infty$. Show that both $\int_{\mathbb{R}^d} |u|^2 \, dx$, and $\int_{\mathbb{R}^d} (\frac{1}{2}|\nabla u|^2 - \frac{\sigma}{\lambda+1}|u|^\lambda) \, dx$ are independent of t.

[Hint: For the first, note that $\int_{\mathbb{R}^d} \triangle u \, v \, dx = \int_{\mathbb{R}^d} u \triangle v \, dx$. For the second, observe that $\frac{\partial}{\partial t} \int_{\mathbb{R}^d} |\nabla u|^2 \, dx = -\int_{\mathbb{R}^d} \left(\frac{\partial u}{\partial t} \triangle \overline{u} + \frac{\partial \overline{u}}{\partial t} \triangle u \right) dx$.]

16. The following is a converse of Propositions 6.6 and 6.8. Suppose $u(\cdot, t)$ is in $L^2(\mathbb{R}^d)$ for each t, with $t \mapsto u(\cdot, t)$ continuous in the L^2 norm, and $u(\cdot, 0) = 0$. Assume that $\frac{1}{i}\frac{\partial u}{\partial t} - \triangle u = F$ as distributions, with $F \in L^2(\mathbb{R}^d \times \mathbb{R})$. Then show that $u = S(F)$.

[Hint: Use the following fact. If $H(\cdot, t)$ is in $L^2(\mathbb{R}^d)$ for each t, with $t \mapsto H(\cdot, t)$ continuous in the L^2 norm, $H(\cdot, 0) = 0$, and $\frac{\partial H}{\partial t} = 0$ in the sense of distributions, then $H = 0$. Apply this to $H(\cdot, t) = e^{-it\triangle}(u(\cdot, t) - S(F)(\cdot, t))$.]

17. A solution u of the non-linear Schrödinger equation (54) is uniquely determined by its initial data f. Moreover the solution depends continuously on this data. These are two features of the "well-posedness" of the problem and can be stated as follows. Assume $\lambda = \frac{d+4}{d}$ and $q = \frac{2d+4}{d}$.

(a) Suppose u and v are two strong solutions defined for $|t| < a$, having the same initial data $f \in L^2(\mathbb{R}^d)$. Show that $u = v$.

(b) Given $f \in L^2(\mathbb{R}^d)$, prove that there are $\epsilon > 0$ and $a > 0$ (depending on f) so that if $\|f - g\|_{L^2} < \epsilon$, and u and v are strong solutions of (54) with initial data f and g respectively, then

$$\|u - v\|_{L^q} \leq c\|f - g\|_{L^2(\mathbb{R}^d)}.$$

Here $L^q = L^q(\mathbb{R}^d \times \{|t| < a\})$.

[Hint: Adapt the argument in Theorem 6.9, and for (a) proceed as follows: note that for small $\ell > 0$

$$\|u\|_{L^q(\mathbb{R}^d \times I)} < \delta \qquad \text{and} \qquad \|v\|_{L^q(\mathbb{R} \times I)} < \delta$$

for all intervals I of length $\leq 2\ell$. Thus

$$\|u - v\|_{L^q} \leq \|\mathcal{M}(u) - \mathcal{M}(v)\|_{L^q} \leq \frac{1}{2}\|u - v\|_{L^q},$$

with $L^q = L^q(\mathbb{R}^d \times \{|t| < \ell\})$, and so $u = v$ for $0 \leq t \leq \ell$. Now use the t-translation invariance to apply the same argument for $u(\cdot, t + \ell)$ and $v(\cdot, t + \ell)$, and so on.

For (b) note that by choosing a and ϵ sufficiently small $\|e^{it\triangle} f\|_{L^q} < \delta/4$ and then $\|e^{it\triangle} g\|_{L^q} < \delta/2$, where $L^q = L^q(\mathbb{R}^d \times \{|t| < a\})$. Now the iteration argument shows that the solutions u and v satisfy $\|u\|_{L^q}, \|v\|_{L^q} < \delta$. Also $\|u - v\|_{L^q} \leq \|S(|u|^{\lambda-1}u - |v|^{\lambda-1}v)\|_{L^q} + c\|f - g\|_{L^2}$. But $\|S(|u|^{\lambda-1}u - |v|^{\lambda-1}v)\|_{L^q} \leq \frac{1}{2}\|u - v\|_{L^q}$, so this proves (b).]

18. Consider the Radon transform $\mathcal{R}_B$ defined by

$$\mathcal{R}_B(f)(x', x_d) = \int_{\mathbb{R}^{d-1}} f(y', x_d - B(x', y'))\, dy'$$

$x = (x', x_d) \in \mathbb{R}^{d-1} \times \mathbb{R}$, where B is a fixed non-degenerate bilinear form on $\mathbb{R}^{d-1} \times \mathbb{R}^{d-1}$. We write $B(x', y') = C(x') \cdot y'$, and assume that the dimension d is odd.

Verify that:

(a) $\| \left(\frac{\partial}{\partial x_d}\right)^{\frac{d-1}{2}} \mathcal{R}_B(f)\|^2_{L^2(\mathbb{R}^d)} = c_B \|f\|^2_{L^2}$ for every $f \in \mathcal{S}$, with $c_B = \frac{2(2\pi)^{d-1}}{|\det(C)|}$.

(b) If $(\mathcal{R}_B)^*$ is the (formal) adjoint of $\mathcal{R}_B$, then $\mathcal{R}_B^* = \mathcal{R}_{B^*}$, with $B^*(x, y) = -B(x, y)$. Also $\frac{\partial}{\partial x_d}\mathcal{R}_B = \mathcal{R}_B \frac{\partial}{\partial x_d}$.

(c) Deduce from (a) and (b) the inversion formula

$$\left(i\frac{\partial}{\partial x_d}\right)^{d-1} \mathcal{R}_B^* \mathcal{R}_B(f) = c_B f.$$

19. We take the Radon transform $\mathcal{R}_B$ as in the previous exercise (with the dimension d odd) and consider a localized version of it, $\mathcal{R}'_B$, given by

$$\mathcal{R}'_B = \eta' \mathcal{R}_B(\eta f)$$

where η and η' are a pair of C^∞ functions of compact support. Show that:

(a) $\|\mathcal{R}'_B(f)\|_{L^2} \leq c\|f\|_{L^2}$.

(b) $\left(\frac{\partial}{\partial x}\right)^\alpha \mathcal{R}'_B(f)$ is a finite linear combination of terms of the form $\left(\frac{\partial}{\partial x_d}\right)^\ell (\eta'_\ell \mathcal{R}_B(\eta_\ell(f)))$ with $0 \leq \ell \leq |\alpha|$.

(c) Deduce from the above and part (a) of the previous exercise that $f \mapsto \mathcal{R}'_B(f)$ is a bounded linear transformation from L^2 to $L^2_{\frac{d-1}{2}}$.

20. The averaging operator from Section 7 satisfies the L^p, L^q conclusions stated for the operator A in Corollary 4.2. Prove this by proceeding according to the following steps.

First, recall that $\mathcal{A} = \sum_{k=0}^{\infty} \mathcal{A}_k$, with $\mathcal{A}_k$ given by (65) in Section 7.4, with the sum convergent in the L^2 norm. Now fix r and consider

$$T_s = (1 - 2^{1-s})e^{s^2} \sum_{k=0}^{r} 2^{-ks} \mathcal{A}_k.$$

Note that $T_0 = -\sum_{k=0}^{r} \mathcal{A}_k$, and so it will suffice to make $L^p \to L^q$ estimates for T_0 that are independent of r. Now prove:

(a) $\|T_s(f)\|_{L^2(\mathbb{R}^d)} \leq M\|f\|_{L^2(\mathbb{R}^d)}$ if $\mathrm{Re}(s) = -\frac{d-1}{2}$.

(b) $\|T_s(f)\|_{L^\infty(\mathbb{R}^d)} \leq M\|f\|_{L^1(\mathbb{R}^d)}$ if $\mathrm{Re}(s) = 1$.

Once (a) and (b) have been established, an interpolation via Proposition 4.4 yields

$$\|T_0(f)\|_{L^q} \leq M\|f\|_{L^p},$$

with $p = \frac{d+1}{d}$ and $q = d + 1$, and this leads to the desired conclusion.

[Hint: Part (a) follows from the estimates (70) and (71) for $\alpha = 0$, and the almost-orthogonality argument, Proposition 7.4. To prove (b) note that it suffices to prove that $(1 - 2^{1-s})e^{s^2} \sum_{k=0}^{r} 2^{-ks} \eta(2^{-k}u)$ has a bounded Fourier transform if $\mathrm{Re}(s) = 1$. Let v be the dual variable to u. We assume first $|v| \leq 1$. Let k_0 be the integer for which $2^{k_0} \leq 1/|v| \leq 2^{k_0+1}$. Now

$$\sum_{k=1}^{r} 2^{-ks} \int \eta(2^{-k}u) e^{2\pi i u v}\, du = \sum_{k \leq k_0} + \sum_{k > k_0}.$$

In the first sum, write $e^{2\pi i u v} = 1 + O(|u||v|)$, and recall that $\eta(\gamma)$ is supported in $1/2 \leq |\gamma| \leq 2$, thus

$$\sum_{k \leq k_0} = O\left(c \sum_{k \leq k_0} 2^{-ks} 2^k \right) + O\left(\sum_{k \leq k_0} 2^{-ks} \int \eta(2^{-k}u)\, |v|\ |u|\, du \right),$$

where $c = \int \eta$. However $\sum_{k \leq k_0} 2^{-ks} 2^k$ is $O(1/|1 - 2^{1-s}|)$ if $\mathrm{Re}(s) = 1$, while the second term above is (when $\mathrm{Re}(s) = 1$)

$$= O(|v|) \left(\sum_{k \leq k_0} 2^{-k} \int |\eta(2^{-k}u)||u|\, du \right) = O(|v|) \sum_{k \leq k_0} 2^k = O(1).$$

Finally for the second sum, $\sum_{k > k_0}$, integrate by parts, writing $e^{2\pi i u v}$ as $\frac{1}{2\pi i v} \frac{d}{du}(e^{2\pi i u v})$ to obtain a sum that is $O\left(\frac{1}{|v|} \sum_{k > k_0} 2^{-k}\right) = O(2^{-k_0}/|v|) = O(1)$.

If $|v| > 1$, take $k_0 = 0$, and argue similarly.]

21. Suppose Ω is a bounded open convex set with $0 \in \Omega$ and with C^2 boundary. Then there is a constant $c > 0$ so that whenever $R \geq 1$ and $\delta \leq 1$, then $x \in R\Omega$ and $|y| \leq \delta$ implies $x + y \in (R + c\delta)\Omega$.

[Hint: One may reduce to the case $R = 1$ by rescaling. To see, for example, that there is a μ so that $x + y = (1 + \mu\delta)\Omega$ whenever $x \in \partial\Omega$ and $|y| < \delta$ for δ sufficiently small, proceed as follows. By a Euclidean change of variables, introduce new coordinates so that x has been moved to $(0, 0) \in \mathbb{R}^{d-1} \times \mathbb{R}$, and near that point Ω is given by $x_d > \varphi(x')$, with $\phi(0) = 0$ and $\nabla_{x'}\varphi(0) = 0$. Then by convexity of Ω, the point corresponding to the initial origin is given by (z', z_d) with $z_d \geq c_1 > 0$. Also $x + y \in (1 + \mu\delta)\Omega$ is equivalent with

$$\frac{y_d + \mu\delta z_d}{1 + \mu\delta} > \varphi\left(\frac{y' + \mu\delta z'}{1 + \mu\delta}\right).$$

Since $|y_d| < \delta$, the left-hand side is $\geq \frac{c_1}{2}\frac{\mu\delta}{1+\mu\delta}$, as soon as $\mu \geq 2/c_1$. Fix such a μ. Now the right-hand side is dominated by

$$A\left|\frac{y' + \mu\delta z'}{1 + \mu\delta}\right|^2 \leq A'\left(\frac{\delta^2 + (\mu\delta)^2}{1 + \mu\delta}\right),$$

and we need only choose $\delta \leq c_2/\mu$, for appropriately small c_2.]

22. Prove the following two estimates for $r \to \infty$:

(a) $\sum_{1 \le k \le r} d(k) k^{\alpha} = O\left(r^{\alpha+1} \log r\right)$ if $\alpha > -1$.

(b) $\sum_{r < k} d(k) k^{\alpha} = O\left(r^{\alpha+1} \log r\right)$ if $\alpha < -1$.

[Hint: Write

$$\begin{aligned} \sum_{k>r} d(k) k^{\alpha} &= \sum \sum_{mn>r} (mn)^{\alpha} = \sum_{n} n^{\alpha} \left(\sum_{m>r/n} m^{\alpha} \right) \\ &= O\left(\sum_{n} n^{\alpha} \min(1, (r/n)^{\alpha+1}) \right). \end{aligned}$$

23. Prove that $rJ_1(r) = \int_0^r \sigma J_0(\sigma)\, d\sigma$, by verifying the following:

(a) $J_1'(r) = \frac{1}{2}\,(J_0(r) - J_2(r))$.

(b) $J_1(r) = \frac{r}{2}\,(J_0(r) + J_2(r))$.

The above shows that $rJ_1'(r) + J_1(r) = rJ_0(r)$, so $\frac{d}{dr}\,(rJ_1(r)) = rJ_0(r)$, proving the assertion.

[Hint: Recall that $J_m(r) = \frac{1}{2\pi}\int_0^{2\pi} e^{ir\sin\theta} e^{-im\theta}\, d\theta$. For (a), differentiate in r under the integral sign. For (b), write $e^{i\theta} = -\frac{1}{i}\frac{d}{d\theta}(e^{-i\theta})$ and integrate by parts.]